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POLICY-BASED AVERAGE-REWARD AND ROBUST MARKOV DECISION PROCESSES
AND REINFORCEMENT LEARNING

BY

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DISSERTATION

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ABSTRACT

This thesis addresses critical challenges in applying Reinforcement Learning (RL) to complex, real-world control problems by developing and analyzing policy-based algorithms for Markov Decision Processes (MDPs) under average-reward, robust (risk-sensitive), and countable-space settings. Traditional RL methods often fall short due to assumptions of finite state/action spaces, bounded costs, or reliance on discounted reward criteria, which may not align with long-run performance objectives in dynamic systems.

The research presented herein makes several key contributions. First, for average-reward MDPs, we establish rigorous finite-time performance bounds for Approximate Policy Iteration (API) and prove global convergence with $O(\log(T))$ regret for Projected Policy Gradient (PPG) methods, notably by proving the smoothness of the average-reward function. We also demonstrate global convergence for Natural Policy Gradient (NPG)/Mirror Descent Methods (MDM) in the tabular average-reward setting.

Second, to tackle problems with countably infinite state spaces and unbounded costs, common in queuing systems, we develop an NPG algorithm with state-dependent step sizes derived from novel policy-independent bounds on the relative value function, achieving non-trivial $O(\sqrt{T})$ regret bounds and relaxing learning error assumptions.

Third, for robust decision-making under model uncertainty and cost variability, we introduce Modified Policy Iteration (MPI) for risk-sensitive exponential cost average-cost MDPs, proving its finite-time convergence. This is extended to an Approximate MPI (AMPI) framework and an Approximate PI framework, providing the first convergence guarantees for approximate methods in this risk-sensitive, robust setting and deriving a unified

stability condition for error propagation across risk-neutral and risk-sensitive regimes.

Collectively, this work advances the theoretical understanding and practical applicability of policy-based RL, providing new algorithms, convergence guarantees, and analytical tools for optimizing long-run performance in large-scale, uncertain, and risk-aware environments.

*To every teacher who taught me with kindness &
to the loving memory of my grandmother and aunt.*

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INTRODUCTION

It is not knowledge, but the act of learning, not possession but the act of getting there, which grants the greatest enjoyment.

Carl Friedrich Gauss

1.1 Data–Driven Control of Large-Scale Systems

Modern engineering systems—cloud computing clusters, large-scale logistics networks, ride-sharing and matching platforms—are exploding in complexity and scale. Their sheer size forces operators to make *millions* of tightly coupled decisions per second: which job to run on which server, which rider to match with which driver, or how much inventory to route through a global supply chain. Efficient resource allocation in these environments is indispensable for sustaining latency guarantees, containing energy costs and meeting service-level agreements. Traditional rule-based controllers, crafted for far smaller systems, can no longer exploit the *avalanche of data* now streaming from sensors and telemetry logs. A new generation of *learning-enabled* control is urgently required.

Reinforcement learning (RL) as a principled framework. RL supplies exactly such a framework: agents interact with an uncertain environment, observe feedback and iteratively refine their policies to maximize long-run performance. Crucially, RL allows us

to embed domain knowledge—stochastic dynamics, queuing structure or physical constraints—directly into the learning loop while still capturing rich data-driven patterns unavailable to first-principles models. Yet deploying RL at industrial scale exposes three fundamental road-blocks

1. *Huge or countably infinite state spaces and unbounded costs.* Standard algorithms often assume a small, finite state space and bounded rewards. Real systems violate both assumptions: queue-lengths, delay counters and inventory levels can grow arbitrarily large. Without new theory and algorithms, RL loses its guarantees and its practicality in these settings.
2. *Model uncertainty and distributional shift.* When the environment encountered in deployment drifts from the conditions seen during training, classical RL can suffer drastic performance collapse. Robust RL, grounded in *robust* Markov Decision Processes, is therefore essential to deliver reliable behaviour in the wild.
3. *Short-term bias of discounted objectives.* Much of the canonical RL literature optimizes a *discounted* reward, emphasizing immediate gains. In high-stakes infrastructure—for instance smart-grid energy dispatch or long-horizon financial planning—what matters is *sustained, steady-state performance*. The **average-reward** formulation, while mathematically more delicate, better captures these objectives and aligns with the metrics operators already track (throughput, average delay, cost per unit time).

Towards scalable and robust RL algorithms. This thesis tackles these challenges head-on. By redesigning RL algorithms to (i) handle large or countably infinite state spaces and unbounded costs, (ii) embed robustness against model misspecification, and (iii) optimize *average* rather than discounted performance, we aim to close the gap between elegant theory and the messy realities of industrial-scale control systems. The subsequent chapters develop the mathematical foundations, algorithms and finite-time guarantees needed to realize this vision.

1.2 Thesis Outline

The thesis is outlined as follows:

1. Chapter 1: Introduction

This chapter introduces the challenges in controlling modern large-scale engineering systems, which often involve vast state spaces, uncertainty, and the need for long-run optimal performance. It posits Reinforcement Learning (RL) as a principled framework but highlights roadblocks: huge/infinite state spaces with unbounded costs, model uncertainty, and the limitations of discounted reward objectives. The thesis aims to address these by developing RL algorithms for average-reward objectives, incorporating robustness, and handling large/countable state spaces, focusing on policy-based methods.

2. Chapter 2: Approximate Policy Iteration for Average-Reward MDPs

This chapter lays foundational work for analyzing policy-based methods in the average-reward setting. It addresses a key deficiency in prior literature: performance bounds for discounted-reward Approximate Policy Iteration (API) become vacuous when extended to the average-reward case. The chapter introduces an API framework for average-reward MDPs, proving finite-time error bounds that gracefully handle policy evaluation and improvement errors, even when these errors are iteration-dependent or random. This is achieved by making an irreducibility assumption and employing a Schweitzer transformation. The framework's utility is demonstrated by applying it to derive convergence guarantees for several common policy-based RL algorithms operating under the average-reward criterion. This chapter is based on [1], published at NeurIPS, 2023.

3. Chapter 3: Global Convergence of Projected Policy Gradient for Average Reward

Building on the theme of average-reward optimization, this chapter focuses on the Projected Policy Gradient (PPG) method. A significant hurdle in analyzing policy gradient methods is often the assumption of smoothness of the objective function. This chapter overcomes this by rigorously proving the smoothness of the average

reward function using a projection technique and deriving an explicit expression for it. This theoretical groundwork enables the derivation of the first global convergence bounds for PPG in the average-reward setting with $O(\log(T))$ regret. Notably, when extended to the discounted setting, these bounds depend on the inherent complexity of the MDP rather than just the size of state and action spaces, offering a more nuanced understanding of performance. This chapter is based on [2], published at ICLR, 2025.

4. **Chapter 4: Natural Policy Gradient and Mirror Descent for Average-Reward MDPs**

This chapter investigates the Natural Policy Gradient (NPG) algorithm, specifically its connection to Mirror Descent Methods (MDM), within the average-reward framework. While NPG/MDM are well-understood in discounted settings, their average-reward counterparts lacked global convergence guarantees even with perfect value function knowledge. This work fills that gap by proving global convergence for MDM in the tabular, average-reward setting, assuming perfect value function estimation. A key element of the proof involves using a weighted KL divergence as a Lyapunov function, establishing an $O(1/T)$ convergence rate for the suboptimality gap under an assumption about stationary distributions. This chapter is based on [3], published at CDC, 2023.

5. **Chapter 5: Reinforcement Learning in Countable State Spaces with Unbounded Costs**

Many real-world problems, particularly in queuing theory and resource allocation, feature countably infinite state spaces and unbounded costs, posing significant challenges for standard RL algorithms. This chapter tackles this by developing an NPG algorithm tailored for such environments. The core innovation is a state-dependent step size, derived from novel policy-independent bounds on the solution to Poisson's equation (the relative value function). This approach yields non-trivial regret bounds and offers a practical heuristic for step-size selection. The analysis also relaxes assumptions on learning errors, accommodating more realistic scenarios where errors might be larger for less frequently visited states. This chapter is based on [4], currently under Major Revision at Operations Research.

6. Chapter 6: Modified Policy Iteration for Risk-Sensitive MDPs

The focus shifts towards robust decision-making by considering risk-sensitive MDPs with an exponential cost criterion, which penalizes variability in costs. Standard dynamic programming tools are often ill-suited due to the multiplicative nature of the Bellman operator. This chapter introduces Modified Policy Iteration (MPI)—a hybrid of value and policy iteration that uses partial policy evaluation—specifically for these risk-sensitive average-cost problems. It establishes both asymptotic and finite-time convergence guarantees for this MPI algorithm, employing novel proof techniques that manage the complexities arising from partial evaluation and the multiplicative operator, including a crucial normalization step to ensure boundedness of iterates. This chapter is based on [5], currently under Major Revision at Operations Research. A preliminary version [6] was published at L4DC, 2023.

7. Chapter 7: Approximate Policy Iteration for Risk-Sensitive MDPs

Extending the work from the previous chapter, this chapter addresses the practical limitation that exact policy evaluation or improvement is often infeasible in risk-sensitive MPI. It develops a unified framework for Approximate Modified Policy Iteration (AMPI) and its special case, Approximate Policy Iteration (API), for risk-sensitive robust MDPs with exponential costs. This is the first work to provide finite-time convergence guarantees for such approximate methods in this challenging setting. A key theoretical contribution is an explicit stability condition under which approximation errors contract geometrically. This condition also gracefully recovers the classical risk-neutral error bounds as the risk-sensitivity parameter diminishes, thereby unifying the analysis across risk attitudes. This work is currently under preparation for submission.

8. Chapter 8: Conclusion

This chapter summarizes the key findings of the thesis, reiterating the advancements made in policy-based algorithms for average-reward, robust (risk-sensitive), and countable-space MDPs. It highlights the theoretical contributions, such as new convergence proofs, finite-time bounds, and methods for handling previously intractable

settings, and discusses potential avenues for future research.

Each chapter stands alone, with its own notation introduced at the beginning of the chapter.

1.3 Contributions

The following contains a brief list of main contributions from each chapter:

1. Chapter 2: Approximate Policy Iteration (API) for Average Reward
 - Finite-time error bounds for average-reward API that converge to zero with vanishing policy evaluation/improvement errors.
 - Extension of API analysis to accommodate iteration-dependent and random errors.
 - Derivation of performance bounds for average-reward TD-learning with common policy update rules.
2. Chapter 3: Projected Policy Gradient (PPG) for Average Reward
 - Rigorous proof of smoothness for the average reward function, removing prior assumptions.
 - Derivation of an explicit expression for the smooth average cost.
 - Establishment of $O(\log(T))$ regret (sublinear convergence) bounds for PPG, dependent on MDP complexity.
3. Chapter 4: Natural Policy Gradient (NPG) / Mirror Descent Method (MDM) for Average Reward
 - Proof of global convergence and an $O(1/T)$ rate for MDM (equivalent to NPG) in tabular average-reward MDPs assuming perfect value function estimation.
 - Utilization of a weighted KL divergence as a Lyapunov function to establish convergence.
4. Chapter 5: Countable Space Reinforcement Learning (NPG)

- Development of an NPG algorithm with state-dependent step sizes suitable for countable state spaces and unbounded instantaneous costs.
- Derivation of $O(\sqrt{T})$ regret bounds through policy-independent, uniform bounds on the relative value function (solution to Poisson's equation).
- Relaxation of standard learning error assumptions to accommodate non-uniform errors across states.

5. Chapter 6: Modified Policy Iteration (MPI) for Risk-Sensitive MDPs

- Introduction of a Modified Policy Iteration (MPI) algorithm for risk-sensitive exponential cost average-cost MDPs.
- Proof of asymptotic and finite-time convergence for this risk-sensitive MPI.
- Development of novel proof techniques and a crucial normalization step for the multiplicative Bellman operator setting.

6. Chapter 7: Approximate Policy Iteration (AMPI/API) for Risk-Sensitive MDPs

- A unified framework for Approximate Modified Policy Iteration (AMPI) and Approximate Policy Iteration (API) in risk-sensitive robust MDPs with exponential costs.
- First finite-time convergence guarantees for AMPI/API in this combined risk-sensitive and robust setting.
- Derivation of an explicit stability condition that unifies error propagation analysis for both risk-sensitive and risk-neutral regimes.

I

AVERAGE REWARD MDPs AND RL

APPROXIMATE POLICY ITERATION

It is by logic that we prove, but by intuition that we discover.

Henri Poincaré

2.1 Introduction

Reinforcement Learning algorithms can be broadly classified into value-based methods and policy-based methods¹. In the case of discounted-reward Markov Decision Processes (MDPs), value-based methods such as Q-learning [8]–[12], fitted value iteration [13] and target network Q-learning [14] can be viewed as approximately solving the fixed point of the Bellman optimality equation to find the value function and an approximately optimal control policy. In other words value iteration is approximately implemented for discounted reward MDPs [15]–[17]. In policy-gradient methods [18], a gradient step is used to improve the policy, somewhat similar to the policy improvement step in policy iteration for discounted-reward MDPs. In a recent paper [19], it has been shown that many policy-based methods for discounted-reward MDPs can be viewed as special cases of approximate policy iteration. The classical results on approximate policy iteration [12] assume that the error in the policy evaluation and improvement steps are constant, independent of the iteration. The key idea in [19] is to show that a simple modification of the proof in [12] can be used

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to allow for iteration-dependent policy evaluation and improvement errors, which is then used to make the connection to policy-based methods. Our goal in this chapter is to derive similar results for average-reward problems. Average-reward problems [20]–[26] are, in some sense, harder to study than their discounted-reward counterparts. We now discuss why this is so by recalling the error bounds for discounted-reward problems and examining them in an appropriate limit to study their applicability to average-reward problems.

The fundamental result on approximate policy iteration for discounted reward MDPs which allows the connection to policy-based methods is the following [12]:

$$\limsup_{k \rightarrow \infty} \|J_{\mu_k} - J_*^\alpha\|_\infty \leq H_\alpha^2(\epsilon + 2\alpha\delta), \quad (2.1)$$

where α is the discount factor, $H_\alpha = 1/(1 - \alpha)$, J_*^α is the optimal value function, J_{μ_k} is the value function associated with the policy obtained after k iterations of approximate policy iteration, ϵ is the policy improvement error, and δ is the policy evaluation error. Thus, as $\epsilon, \delta \rightarrow 0$, we recover the result that standard policy iteration converges. On the other hand, to understand whether the above bound is useful to study average-reward problems, we write Equation (2.1) as

$$\limsup_{k \rightarrow \infty} \frac{1}{H_\alpha} \|J_{\mu_k} - J_*^\alpha\|_\infty \leq H_\alpha(\epsilon + 2\alpha\delta).$$

Under mild conditions [15], [27], it is well known that each element of the value function vector approaches the average-reward for each fixed policy and for the optimal policy, in the limit $\alpha \rightarrow 1$. Hence, the left-hand side of the above equation is an approximation to the error in approximate policy iteration for average-reward MDPs; the right-hand side which gives an upper bound on this error blows up to infinity, i.e., when $\alpha \rightarrow 1$. Thus, unlike in the discounted-reward case, the above bound fails to even recover the well-known convergence of standard policy iteration in average-reward case in the limit $\epsilon, \delta \rightarrow 0$ (since we have to let $\alpha \rightarrow 1$ before letting $\epsilon, \delta \rightarrow 0$). However, as mentioned in [12], it is also well known that approximate policy iteration performs much better than the bound suggests. The main goal of this chapter is to resolve this discrepancy between theory and practice.

2.1.1 Contributions

It is well known that the error bound for discounted-reward, approximate policy iteration is tight for unichain MDPs [12]. Thus, it is impossible to improve upon the bound in general. However, as will be explained in a later section, it is easy to convert most reasonable unichain MDPs to MDPs where every stationary policy results in an irreducible Markov chain with an arbitrarily small loss of reward. So the natural question to ask is whether the bound can be dramatically improved for MDPs where every stationary policy results in an irreducible Markov chain. To the best of our knowledge, no such bound was available in the prior literature.

Our main contributions are as follows:

- Under the assumption that every stationary policy induces an irreducible Markov chain, we first perform a Schweitzer transformation of the MDP [28] and obtain finite-time error bounds for average-reward approximate policy iteration.
- Using the above finite-time error bounds, we prove an error bound of the form

$$\limsup_{k \rightarrow \infty} J^* - J_{\mu_k} \leq f(\epsilon, \delta),$$

for average-reward approximate policy iteration for a function f such that $f(\epsilon, \delta) \rightarrow 0$ in the limit as $\epsilon, \delta \rightarrow 0$. Note that this is in sharp contrast to the result that one can obtain from the discounted-reward analysis where the error bound blows up to infinity when applied to discount factors approaching 1, which is the appropriate limit to convert discounted-reward problems to average-reward problems.

- The main difficulty in obtaining the above bound when compared to the discounted-reward case is the lack of infinity-norm contraction property for the Bellman operator in the average-reward case. In most analysis of average-reward problems, this difficulty is circumvented by the use of a span-contraction property [17]. However, the span contraction property is insufficient for our purpose and therefore, we use a technique due to [29] for the study of a different algorithm called modified policy iteration in [17].

- Next, we extend the above analysis to obtain finite-iteration error bounds for the case where the policy evaluation and improvement errors can potentially vary from iteration to iteration. Further, we allow these errors to be random (as would be the case, for example, when one uses TD learning for policy evaluation and soft policy improvement) and obtain expected error bounds on approximate policy iteration after a finite number of iterations.
- Our error bounds are in a form such that one can then apply them to many policy-based RL algorithms, i.e., we can simply plug in error bounds for specific policy evaluation or policy improvement algorithms. We illustrate the applicability of the results to several different policy-based RL methods (softmax update, greedy update and mirror descent update) studied in the literature [12], [18], [30], [31].
- While the main focus of this chapter is on offline learning, we provide connections to regret bounds for online learning in average-reward MDPs as in [31], [32].

2.1.2 Related Work

Our work in this chapter is most closely related to the work in [19]. The main contribution there was to recognize that an extension of the approximate policy iteration results in [12] can be used to derive finite-time error bounds for many RL-based algorithms. The idea of considering iteration-dependent policy evaluation and improvement bounds was also used to study policy iteration with function approximation in discounted-reward MDPs in [33]. Our contribution here is to derive the first error bound for approximate policy iteration for average-reward MDPs and then write it in a form which can be applied to policy-based RL methods for average-reward TD learning. To the best of our knowledge, no known error bounds existed in prior literature for average-reward approximate policy iteration due to the fact that the corresponding bounds for discounted MDPs increase proportional to the square of length of the horizon. Thus, in the limit as the horizon increases to infinity, we have essentially reduced the dependence of the error from the square of the length of the horizon to linear in the length of the horizon.

RL algorithms have not been studied as extensively for the average-reward case as they are

for discounted-reward MDPs. There is recent work [34] on average-reward TD-learning which we leverage to illustrate the applicability of our results. There are also asymptotic convergence results for policy gradient and actor-critic methods for average-reward MDPs [35]–[38]. However, these results only prove convergence to a stationary point and there are no global convergence results. The original paper on natural policy gradient (NPG) is written for average-reward MDPs, but the extensive performance analysis of NPG in subsequent work such as [18] seems to only consider the discounted-reward case. Recent work [31], [32] considers mirror descent in average reward RL but do not provide performance results for other RL algorithms. In a later section, we compare our results to [31], [32] in the special case of mirror descent-type policy updates.

2.2 Model and Preliminaries

2.2.1 Average Reward Formulation

We consider the class of infinite horizon MDPs with finite state space $\mathcal{S}$, finite action space $\mathcal{A}$, and transition kernel $\mathbb{P}$, where $|\mathcal{S}| = n$ and $|\mathcal{A}| = m$. Let $\Delta\mathcal{A}$ denote the probability simplex over actions. We consider the class of randomized policies $\Pi = \{\mu : \mathcal{S} \rightarrow \Delta\mathcal{A}\}$, so that a policy μ assigns a probability vector over actions to each state. Given a policy μ , the transition kernel for the underlying Markov process is denoted by $\mathbb{P}_\mu : \mathcal{S} \rightarrow \mathcal{S}$, where $\mathbb{P}_\mu(s'|s) := \sum_{a \in \mathcal{A}} \mu(a|s) \mathbb{P}(s'|s, a)$ is the probability of moving to state $s' \in \mathcal{S}$ from $s \in \mathcal{S}$ upon taking action $\mu(s) \in \mathcal{A}$. Associated with each state-action pair (s, a) , is a one-step reward which is denoted by $r_\mu(s) := \mathbb{E}_{a \sim \mu(s)} r(s, a)$.

Let $J_\mu \in \mathbb{R}$ be the average reward associated with the policy $\mu \in \Pi$, i.e. J_μ is defined as:

$$J_\mu = \lim_{T \rightarrow \infty} \frac{\mathbb{E}_\mu \left(\sum_{i=0}^{T-1} r_\mu(s_i) \right)}{T}.$$

Here the expectation is taken with respect to the measure $\mathbb{P}_\mu$ associated with the policy μ . Let $h_\mu \in \mathbb{R}^n$ be the relative value function associated with the policy μ . Defining $\mathbf{1} \in \mathbb{R}^n$ to be the vector of all ones, the pair (J_μ, h_μ) satisfies the following average reward Bellman

equation:

$$J_\mu \mathbf{1} + h_\mu = r_\mu + \mathbb{P}_\mu h_\mu. \quad (2.2)$$

Let $\pi_\mu \in \mathbb{R}^n$ denote the stationary distribution associated with the kernel $\mathbb{P}_\mu$. Let $\mathbb{P}_\mu^* = \mathbf{1}\pi_\mu^\top \in \mathbb{R}^{n \times n}$ denote the matrix whose rows are $\pi_\mu^\top$. Using $\mathbb{P}_\mu^* = \mathbb{P}_\mu^* \mathbb{P}_\mu$, we get the following characterization of the average reward by multiplying both sides of Equation (2.2) with $\mathbb{P}_\mu^*$: $J_\mu \mathbf{1} = \mathbb{P}_\mu^* r_\mu$.

Let $J^* := \max_{\mu \in \Pi} J_\mu$ be the optimal average reward. From standard MDP theory, there exists $h^* \in \mathbb{R}^n$, for which the pair (J^*, h^*) satisfies the following Bellman optimality equation:

$$J^* \mathbf{1} + h^* = \max_{\mu \in \Pi} r_\mu + \mathbb{P}_\mu h^*. \quad (2.3)$$

Let μ^* be the optimizing policy in Equation (2.3). Then, similar reasoning shows that $J^* \mathbf{1} = \mathbb{P}_{\mu^*}^* r_{\mu^*}$.

The goal of dynamic programming and reinforcement learning is to determine this optimal average reward J^* and its corresponding optimal policy μ^* . Policy iteration is one of the most widely used dynamic programming algorithms for this purpose. It consists of two steps: (i) evaluation of the value function associated with a policy, and (ii) determining a greedy policy with respect to this evaluation. There are primarily two challenges that arise when applying such an algorithm: (i) high memory and time complexity when the state space is too large, and (ii) the unknown transition probability kernel that governs the dynamics of the MDP.

When the state space is too large, it may be computationally infeasible to perform policy iteration exactly. Existing literature suggests that approximate policy iteration in the context of discounted reward MDPs yields good performance. However, such an algorithm has not been studied in the context of average reward MDPs. In fact, the results obtained for the discounted reward MDPs yield vacuous performance bounds for average reward MDPs given the standard relationship between discounted reward and average reward MDPs. We explore this further in Section 2.2.2.

2.2.2 Relationship to the Discounted Reward MDP

To provide some background, we first discuss the approximate policy iteration algorithm in the context of discounted reward MDPs. Let $\alpha \in [0, 1)$ denote the discount factor. The discounted reward associated with a deterministic policy $\mu \in \Pi$ starting from some state $s \in \mathcal{S}$ is denoted $J_\mu^\alpha(s)$ and is defined as:

$$J_\mu^\alpha(s) = \mathbb{E}_\mu \left[\sum_{i=0}^{\infty} \alpha^i r(s_i, \mu(s_i)) \mid s_0 = s \right].$$

The value function $J_\mu^\alpha \in \mathbb{R}^n$ is the unique fixed point of the Bellman operator $T_\mu^\alpha : \mathbb{R}^n \rightarrow \mathbb{R}^n$ associated with the policy μ , which is defined as $T_\mu^\alpha J = r_\mu + \alpha \mathbb{P}_\mu J$. For each state $s \in \mathcal{S}$, let $J_*^\alpha(s)$ denote the optimal discounted reward, i.e., $J_*^\alpha(s) = \max_{\mu \in \Pi} J_\mu^\alpha(s)$. Similarly, $J_*^\alpha \in \mathbb{R}^n$ is the unique fixed point of the optimality Bellman operator $T^\alpha : \mathbb{R}^n \rightarrow \mathbb{R}^n$, which is defined as $T^\alpha J = \max_{\mu \in \Pi} (r_\mu + \alpha \mathbb{P}_\mu J)$. Algorithm 1 is Approximate Policy Iteration for the discounted reward MDP:

Algorithm 1 Approximate Policy Iteration: Discounted Reward

Require $J_0 \in \mathbb{R}^n$

for $k = 0, 1, 2, \dots$ **do**

 1. Compute $\mu_{k+1} \in \Pi$ such that $\|T^\alpha J_k - T_{\mu_{k+1}}^\alpha J_k\|_\infty \leq \epsilon$ ▷ Approximate Policy Improvement

 2. Choose J_{k+1} such that $\|J_{k+1} - J_{\mu_{k+1}}\|_\infty \leq \delta$ ▷ Approximate Policy Evaluation

 where $J_{\mu_{k+1}} = T_{\mu_{k+1}}^\alpha J_{\mu_{k+1}}$

end for

Theorem 2.1. *Let μ_k be the sequence of policies generated from the approximate policy iteration algorithm (Algorithm 1). Then the performance error is bounded as:*

$$\limsup_{k \rightarrow \infty} \|J_{\mu_k} - J_*^\alpha\|_\infty \leq \frac{\epsilon + 2\alpha\delta}{(1 - \alpha)^2}. \quad (2.4)$$

Proof. The proof of this theorem can be found in [15] □

From literature [15], [27], we know that the average reward J_μ associated with any policy μ is related to its discounted reward $J_\mu^\alpha(s)$ counterpart as follows:

$$J_\mu = \lim_{\alpha \rightarrow 1} (1 - \alpha) J_\mu^\alpha(s).$$

Note that the above relation is independent of the state $s \in \mathcal{S}$, as the average reward does not depend on the initial state. Multiplying Equation (2.4) with $(1 - \alpha)$, and letting $\alpha \rightarrow 1$, still yields an approximation performance bound with $(1 - \alpha)$ in the denominator. This term blows up to infinity as $\alpha \rightarrow 1$. Note that this bound is known to be tight under unichain Markov structure of the probability transition kernel [15]. However, in practice it is observed that approximate policy iteration works well in the average reward MDP case although these theoretical bounds are not representative of this performance. To the best of our knowledge, we are unaware of an average reward approximate policy iteration performance bound, when the policies induce an irreducible Markov chain. We bridge this gap between theory and practice by providing a theoretical analysis of approximate policy iteration in the average reward MDP setting, with non trivial performance bounds.

2.3 Approximate Policy Iteration for Average Reward

A crucial component of the proof of convergence of approximate policy iteration in the context of discounted reward MDPs is the contraction due to the discount factor α which is absent in the average reward setting (since $\alpha = 1$). To get some source of contraction, we have to make some assumptions on the MDP.

Assumption 2.1. We make the following assumptions:

- (a) Every deterministic policy $\mu \in \Pi$ induces an irreducible Markov Chain $\mathbb{P}_\mu$.
- (b) For all policies, the diagonal elements of the probability transition matrix are positive, i.e., the Markov chain stays in the same state with non-zero probability.

Assumption (a) need not be satisfied by all MDPs. However, in order to satisfy this assumption, we consider a modified MDP where at every time step with probability ε , an action is chosen from the set of all possible actions with equal probability. Simultaneously,

with probability $1 - \varepsilon$, we choose an action dictated by some policy. The problem then is to choose this policy optimally. For most MDPs of interest, this small modification will satisfy our assumption with a small $O(\varepsilon)$ loss in performance. It is straightforward to show the $O(\varepsilon)$ loss, but we include the proof in the appendix for completeness. Assumption (b) is without loss of generality in the following sense: there exists a simple transformation which essentially leaves the MDP unchanged but ensures that this assumption is satisfied. This transformation known as the aperiodicity transformation or Schweitzer transformation was introduced in [28]. For more details, the reader is referred to the appendix. In the remainder of this chapter, we assume that all MDPs are transformed accordingly to ensure the assumptions are satisfied. One consequence of Assumption (b) is the following lemma which will be useful to us later.

Lemma 2.1. *There exists a $\gamma > 0$ such that, for any policy $\mu \in \Pi$,*

$$\min_{i \in \mathcal{S}} \pi_\mu(i) \geq \gamma,$$

where π_μ is stationary distribution over states associated with the policy μ .

Proof. This lemma has been proved in a more general sense in the context of deterministic policies in [29]. Since the aperiodicity transformation ensures a non-zero probability of staying in the same state, the expected return times are bounded and the lemma holds true for randomized policies as well. $\square$

Prior to presenting the algorithm, consider the following definitions. The average-reward Bellman operator $T_\mu : \mathbb{R}^n \rightarrow \mathbb{R}^n$ corresponding to a policy μ is defined as $T_\mu h = r_\mu + \mathbb{P}_\mu h$. The average-reward optimal Bellman operator $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is defined as $Th = \max_{\mu \in \Pi} r_\mu + \mathbb{P}_\mu h$. The value function h_μ associated with a policy μ satisfies the Bellman Equation (2.2) with single step reward r_μ and the transition kernel $\mathbb{P}_\mu$. Note that h_μ is unique up to an additive constant.

Solving for (J_μ, h_μ) using Equation (2.2) involves setting the value function for some fixed state x^* as 0 (since this reduces the system from being underdetermined to one with a unique solution). Further, the value function h_μ can be alternatively expressed as the $\lim_{m \rightarrow \infty} T_\mu^m h_0$

for any $h_0 \in \mathbb{R}^n$ as is the case with discounted reward MDPs. However, unlike the discounted reward Bellman Equation, there is no discount factor $\alpha < 1$ preventing the value function from exploding to infinity. Hence, we consider value function computed using the relative Bellman operator $\tilde{T}_\mu$ defined as

$$\tilde{T}_\mu h = r_\mu + \mathbb{P}_\mu h - r_\mu(x^*)\mathbf{1} - (\mathbb{P}_\mu h)(x^*)\mathbf{1}.$$

We now state the algorithm for Approximate Policy Iteration for Average Reward MDPs.

2.3.1 Approximate Policy Iteration Algorithm for Average Reward MDPs

Algorithm 2 Approximate Policy Iteration: Average Reward

- 1: Require $h_0 \in \mathbb{R}^n$
 - 2: **for** $k = 0, 1, 2, \dots$ **do**
 - 3: 1. Compute $\mu_{k+1} \in \Pi$ such that $\|Th_k - T_{\mu_{k+1}}h_k\|_\infty \leq \epsilon$ ▷ Approx. Policy Improvement
 - 4: 2. Compute h_{k+1} such that $\|h_{k+1} - h_{\mu_{k+1}}\|_\infty \leq \delta$ ▷ Approx. Policy Evaluation
 - 5: where $h_{\mu_{k+1}} = \lim_{m \rightarrow \infty} \tilde{T}_{\mu_{k+1}}^m h_k$
 - 6: **end for**
-

2.3.2 Performance bounds for Average Reward Approximate Policy Iteration

We are now ready to present the main result of this work.

Theorem 2.2. *Let J^* be the optimal average reward of an MDP that satisfies Assumption 2.1. The sequence of policies μ_k generated by Algorithm 2 and their associated average rewards J_{μ_k} satisfy the following bound:*

$$(J^* - J_{\mu_{k+1}}) \leq \underbrace{\frac{(1 - (1 - \gamma)^k)}{\gamma} (\epsilon(\gamma + 1) + 2\delta)}_{\text{approximation error}} + \underbrace{(1 - \gamma)^k \left(J^* - \min_i (Th_0 - h_0)(i) + \epsilon \right)}_{\text{initial condition error}}.$$

Interpretation of the Bound: Since $0 < \gamma < 1$, as $k \rightarrow \infty$, the error due to initial condition h_0 drops to zero. The approximation error consists of two components, ϵ and δ . Here, ϵ represents the approximation error due to a suboptimal policy and δ represents the approximation error associated with evaluating the value function corresponding to a policy. The limiting behavior of the average reward associated with the policies obtained as a consequence of the algorithm is captured in the following corollary.

Corollary 2.1. *The asymptotic performance of the policies obtained from Algorithm 2 satisfies:*

$$\limsup_{k \rightarrow \infty} (J^* - J_{\mu_k}) \leq \frac{(1 + \gamma)\epsilon + 2\delta}{\gamma}. \quad (2.5)$$

Note that the asymptotic bound in Equation (2.5) is in a very similar form as the asymptotic performance bound for the discounted reward MDP in Equation (2.4) where γ plays the role of $(1 - \alpha)^2$. However, when ϵ and δ got to zero, this bound also goes to zero which is not the case if we try to approximate average-reward problems by using the results for discounted-reward problems in the limit where the horizon goes to infinity.

2.4 Application to Reinforcement Learning

Approximate policy iteration is closely related to RL. In general, approximate policy improvement and approximate policy evaluation in Algorithm 2 are executed through approximations to policy improvement (such as greedy update, mirror descent, softmax) and TD learning for value function approximation. First, we present a generic framework to analyze policy-based RL algorithms by building upon the theory presented in the last section. For this purpose, we define the state-action relative value function Q (instead of the relative state value function h as before) to evaluate a policy. Let $T_\mu^Q Q$ denote the Bellman operator with respect to Q and policy μ where $(T_\mu^Q Q)(s, a) = r(s, a) + (Q_\mu Q)(s, a)$, where $Q_\mu(s', a'|s, a) = \mu(a'|s')\mathbb{P}(s'|s, a)$. The relative state action value function corresponding to policy μ is represented by Q_μ and is the solution to the following Bellman equation: $J_\mu + Q_\mu(s, a) = r(s, a) + \sum_{s' \in \mathcal{S}, a' \in \mathcal{A}} Q_\mu(s', a'|s, a)Q_\mu(s', a')$, for all state action pairs $(s, a) \in (\mathcal{S}, \mathcal{A})$.

We present a generic policy-based algorithm for average-reward problem below.

Algorithm 3 Generic Policy Based Algorithm: Q-function Average Reward

Require $Q_0 \in \mathbb{R}^{|S||A|}$

for $k = 0, 1, 2, \dots, T$ **do**

1. Determine $\mu_{k+1} \in \Pi$ as a function of Q_k using a possibly random policy improvement step

2. Compute Q_{k+1} as an approximation to $Q_{\mu_{k+1}}$ using (state, action, reward) samples from a trajectory generated by policy μ_{k+1}

end for

Note that the error in Steps 1 (policy improvement) and 2 (policy evaluation) of the above algorithm could be random. Thus, the analysis for Theorem 2.2 has to be adapted to Algorithm 3. The resulting expected deviation from optimality is characterized in the lemma below.

Lemma 2.2. *Let μ_k and Q_k be the sequence of policies and relative state-action value function iterates generated by Algorithm 3. For all $k \in 0, \dots, T$, we have:*

$$\begin{aligned} \left(J^* - \left(\min_{(s,a)} \left(T^Q Q_k - Q_k \right) (s, a) \right) \right) &\leq (1 - \gamma) \left(J^* - \left(\min_{(s,a)} \left(T^Q Q_{k-1} - Q_{k-1} \right) (s, a) \right) \right) \\ &\quad + \left\| T^Q Q_{k-1} - T_{\mu_k}^Q Q_{k-1} \right\|_{\infty} + 2 \|Q_k - Q_{\mu_k}\|_{\infty}, \end{aligned}$$

where

$$\gamma = \min_{\substack{\mu \in \mu_1, \mu_2, \dots, \mu_T \\ (s,a) \in S \times A}} Q_{\mu}^*(s, a) > 0,$$

and Q_{μ}^* is a matrix with identical rows corresponding to the invariant distribution of Q_{μ} .

Iteratively applying Lemma 2.2 yields the following proposition for a generic policy-based algorithm.

Proposition 2.1. *For any $T > 0$, the iterates of Algorithm 3 satisfy*

$$\begin{aligned} \mathbb{E}[J^* - J_{\mu_{T+1}}] &\leq \underbrace{(1-\gamma)^T \mathbb{E}\left[J^* - \min_i \left(T^Q Q_0 - Q_0\right)(i)\right]}_{\text{Initial condition error}} + \underbrace{2 \sum_{\ell=0}^{T-1} (1-\gamma)^\ell \mathbb{E}\left[\|Q_{T-\ell} - Q_{\mu_{T-\ell}}\|_\infty\right]}_{\text{Policy evaluation error}} \\ &\quad + \underbrace{\sum_{\ell=1}^{T-1} (1-\gamma)^{\ell-1} \mathbb{E}\left[\|T_{\mu_{T+1-\ell}}^Q Q_{T-\ell} - T^Q Q_{T-\ell}\|_\infty\right]}_{\text{Policy improvement error}} + \mathbb{E}\left[\|T_{\mu_{T+1}}^Q Q_T - T^Q Q_T\|_\infty\right]. \end{aligned}$$

We now illustrate the applicability of the above result to specific RL algorithms. More specifically, we characterize finite time performance bounds when state-action relative value functions are learnt through TD-Learning [34], while several different update rules are employed for policy improvement.

2.4.1 Performance bounds for RL algorithms

For every iteration of Algorithm 3, it is necessary to observe trajectories of length τ in order to learn the state action value function associated with any policy. Using linear function approximations for value functions, the error in policy evaluation at iteration k in Algorithm 3 can be expressed in the following form:

$$\|Q_k - Q_{\mu_k}\|_\infty \leq \|\Phi(\theta_k - \theta_{\mu_k}^*)\|_\infty + \|Q_{\mu_k} - \Phi\theta_{\mu_k}^*\|_\infty, \quad (2.6)$$

where $\Phi \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}| \times d}$ corresponds to the feature vector matrix and $\theta \in \mathbb{R}^d$ is the parameter vector. Further, θ_k is the parameter vector obtained as a result of the TD learning algorithm. The projection of Q_{μ_k} onto the span of Φ is given by $\Phi\theta_{\mu_k}^*$. The last term in Equation (2.6) corresponds to the function approximation error which depends on the span of the feature matrix Φ . We denote this error term by $\delta_{0,k}$. The error due to TD learning (first term in Equation (2.6)) is denoted by $\delta_{\text{TD},k}$. Assuming $\max_{(s,a) \in \mathcal{S} \times \mathcal{A}} \|\phi(s,a)\|_2 \leq 1$, and $\|\theta_{\mu_k}^*\|_2$ is bounded uniformly in k , [34] shows that there exists $C > 0$ such that the length of the sample trajectory τ and $\delta_{\text{TD},k}$ are related as follows:

$$\mathbb{E}[\delta_{\text{TD},k}] = \frac{C}{\sqrt{\tau}}. \quad (2.7)$$

Let $c_0 = \left(J^* - \min_{i \in \mathcal{S}} (T^Q Q_0 - Q_0)(i) \right)$ be the error due to initial condition and $\bar{\delta}_0 = \max_t \delta_{0,t}$ be the maximum function approximation error across all iterations. Then we obtain the following performance bounds when TD learning is used for policy evaluation.

Corollary 2.2. (*Greedy policy update*) Let $\tilde{a}(s) = \operatorname{argmax}_{a \in \mathcal{A}} Q_k(s, a')$. Let $\beta > 0$ be such that the sequence of policies in Algorithm 3 corresponding to all state action pairs (s, a) are obtained using the following algorithm, which we call the greedy policy update.

$$\mu_{k+1}(a|s) = \begin{cases} \frac{1}{\beta|\mathcal{A}|}, & \text{if } a \neq \tilde{a}(s) \\ \frac{1}{\beta|\mathcal{A}|} + 1 - \frac{1}{\beta}, & \text{if } a = \tilde{a}(s), \end{cases} \quad (2.8)$$

Let $\eta = \max_{\substack{s \in \mathcal{S}, a \in \mathcal{A} \\ k \in 1 \dots T}} |Q_k(s, a)|$. Using TD Learning with linear value function approximation for policy evaluation, we get the following finite time performance bound:

$$\mathbb{E} [J^* - J_{\mu_{T+1}}] \leq (1 - \gamma)^T c_0 + \left(\frac{1 + \gamma}{\gamma} \right) \frac{2\eta}{\beta} + \frac{2}{\gamma} \left(\bar{\delta}_0 + \frac{C}{\sqrt{\tau}} \right). \quad (2.9)$$

Corollary 2.3. (*Softmax policy update*) Let $\beta > 0$ be such that the sequences of policies in Algorithm 3 corresponding to all state action pairs (s, a) are obtained using the following algorithm, which we call the softmax policy update.

$$\mu_{k+1}(a|s) = \frac{\exp(\beta Q_k(s, a))}{\sum_{a' \in \mathcal{A}} \exp(\beta Q_k(s, a'))}. \quad (2.10)$$

Using TD Learning with linear value function approximation for policy evaluation, we get the following finite time performance bound:

$$\mathbb{E} [J^* - J_{\mu_{T+1}}] \leq (1 - \gamma)^T c_0 + \left(\frac{1 + \gamma}{\gamma} \right) \frac{\log(|\mathcal{A}|)}{\beta} + \frac{2}{\gamma} \left(\bar{\delta}_0 + \frac{C}{\sqrt{\tau}} \right). \quad (2.11)$$

Corollary 2.4. (*Mirror descent update*) Let $\beta > 0$ be such that the sequences of policies in Algorithm 3 corresponding to all state action pairs (s, a) are obtained using the following algorithm,

which we call the mirror descent policy update.

$$\mu_{k+1}(\mathbf{a}|s) \propto \mu_k(\mathbf{a}|s) \exp\{\beta Q_k(s, \mathbf{a})\}. \quad (2.12)$$

Let $\omega = \min_{\substack{k \in 1 \dots T \\ s \in \mathcal{S}}} \mu_k(\mathbf{a}^*(s)|s)$ where $\mathbf{a}^*(s)$ is the optimal action at state s . Using TD Learning with linear value function approximation for policy evaluation, we get the following finite time performance bound:

$$\mathbb{E} [J^* - J_{\mu_{T+1}}] \leq (1 - \gamma)^T c_0 + \left(\frac{1 + \gamma}{\gamma \beta} \right) \log \left(\frac{1}{\omega} \right) + \frac{2}{\gamma} \left(\bar{\delta}_0 + \frac{C}{\sqrt{\tau}} \right). \quad (2.13)$$

Remark 2.1. The following remarks are in order:

- In each of the above performance bounds, there is an inevitable constant error ($\bar{\delta}_0$) due to function approximation. If the value function lies in the class of functions chosen, then this error will be zero.
- The error due to policy update (characterized by β) can be made arbitrarily small by choosing β large. However, choosing a large β may increase C due to the fact that C is a function of the mixing time of each policy, which is affected by the choice of β . However, C cannot be arbitrarily large due to Assumption (a).
- The error due to the initial condition decays exponentially fast but the rate of decay can be slow because γ can be very small. In this regard, it is interesting to compare our work with the results in [31], [32]. We do so in the next subsection.

2.4.2 Comparison with [31], [32]

Our model for approximate policy iteration is intended to be a general framework to study policy-based RL algorithms. A specific class of such algorithms has been studied in [31], [32], where regret bounds are presented for a mirror descent like algorithm referred to as POLITEX. Although our model is intended for offline algorithms, it is possible to adapt the techniques in [31], [32] to obtain regret bounds for our model as well.

Let $r(s_t, \mathbf{a}_t)$ be the reward obtained at time t . Let K be the total number of time units for

which Algorithm 3 is executed. Hence $K = T\tau$, where τ is the length of the sample trajectory for every iteration of policy evaluation. The regret, defined as $R(K) = \sum_{t=1}^K (J^* - r(s_t, a_t))$ can be further decomposed as:

$$R(K) = \sum_{t=1}^K (J^* - J_{\mu_t}) + (J_{\mu_t} - r(s_t, a_t)). \quad (2.14)$$

The first term i.e., $\sum_{t=1}^K (J^* - J_{\mu_t})$ is referred to as the pseudo regret $R_{PS}(K)$. The second term is essentially bounding the average value from its estimate and can be bounded using a martingale analysis identical to [31], [32]. Here, we focus on obtaining an upper bound on the pseudo regret with mirror descent update and TD learning. The pseudo regret can be further decomposed as:

$$R_{PS}(K) = \tau \sum_{t=1}^T (J^* - J_{\mu_t}).$$

For the mirror descent policy update in Corollary 2.4, we get the following expected regret bound

$$R_{PS}(K) \leq \tau \sum_{t=1}^T (1 - \gamma)^T c_0 + \frac{1 + \gamma}{\gamma \beta} \log \left(\frac{1}{\omega} \right) + \frac{2}{\gamma} \left(\bar{\delta}_0 + \frac{c_3}{\sqrt{\tau}} \right). \quad (2.15)$$

Let $\beta = \sqrt{\tau}$, then the above regret can be expressed as:

$$R_{PS}(K) \leq \frac{1}{\gamma} \left(\tau c_0 + 2K \bar{\delta}_0 + \frac{K c_5}{\sqrt{\tau}} \right).$$

Optimizing for the regret yields $\tau = O(K^{\frac{2}{3}})$ and the following pseudo regret upper bound,

$$R_{PS}(K) \leq \frac{1}{\gamma} \left(2K \bar{\delta}_0 + c_6 K^{\frac{2}{3}} \right). \quad (2.16)$$

We now compare our regret bounds to the ones in [31], [32]. Since all of the bounds will have an inevitable error ($\bar{\delta}_0$) due to function approximation, the comparison is with respect to the other terms in the regret bound.

- We get a better order-wise bound on the regret compare to [31] which obtains a bound of $O(K^{3/4})$. However, they have better constants which are in terms of mixing-time coefficients rather than in terms of γ .

- [32] gets a better regret bound ($O(\sqrt{K})$) than us.

The results in [31], [32] use Monte Carlo policy evaluation, which allows for a closed-form expression for the value function estimate. The structure of the closed-form expressions allows them to exploit the fact that the policy varies slowly from one iteration to the next (which is specific to the mirror descent policy update) and hence γ does not appear in their analysis. However, the Monte Carlo policy evaluation algorithms use matrix inversion which is often too expensive to implement in practice unlike the TD learning algorithm in our analysis which is widely used. Nevertheless, the ideas in [31], [32] provide a motivation to investigate whether one can extend their analysis to more general policy evaluation and policy improvement algorithms. This is an interesting topic for future research.

2.5 Conclusions

We present a general framework for analyzing policy-based reinforcement learning algorithms for average-reward MDPs. Our main contribution is to obtain performance bounds without using the natural source of contraction available in discounted reward formulations. We apply our general framework to several well-known RL algorithms to obtain finite-time error bounds on the average reward. We conclude with a comparative regret analysis and an outline of possible future work.

3

GLOBAL CONVERGENCE OF PROJECTED POLICY GRADIENT

The more you learn the more you realize what you know is a measure zero set.

Anonymous

3.1 Introduction

Average reward Markov Decision Processes (MDPs) find applications in domains where decisions are made over time to optimize long-term performance¹. Some of these applications include resource allocation, portfolio management in finance, healthcare, and robotics [23], [39]–[42]. Approaches for determining the optimal policy can be broadly categorized into dynamic programming algorithms (such as value and policy iteration [25], [31], [43]) and gradient-based algorithms. Although gradient-based algorithms are heavily used in practice [44], [45], the theoretical analysis of their global convergence is a relatively recent undertaking.

While extensive research has been conducted on the global convergence of policy gradient methods in the context of discounted reward MDPs [46], [47], comparatively less attention

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has been given to its average reward counterpart. Contrary to average reward MDPs, the presence of a discount factor ($\gamma < 1$) serves as a source of contraction that alleviates the technical challenges involved in analyzing the performance of various algorithms in the context of discounted reward MDPs. Consequently, many algorithms designed for average reward MDPs are evaluated using the framework of discounted MDPs, where the discount factor approaches one [48].

In the context of discounted reward MDPs, the projected policy gradient (PPG) algorithm converges as

$$\rho_{\gamma}^* - \rho_{\gamma}^{\pi_k} \leq O\left(\frac{1}{(1-\gamma)^5}\right) \quad (3.1)$$

where γ is the discount factor, π_k is the policy obtained at the k -th iteration of the PPG algorithm, ρ^* represents the optimal value function, and $\rho_{\gamma}^{\pi_k}$ represents the value function iterates obtained through projected gradient ascent [49]. Let ρ^{π} denote the average reward associated with some policy π . It is well known that $\rho^{\pi} = \lim_{\gamma \rightarrow 1} (1-\gamma)\rho_{\gamma}^{\pi}$ under some mild conditions [50], [51]. Utilizing this relationship in (3.1), we observe the upper bound tends to infinity in the limit $\gamma \rightarrow 1$. Hence, it is necessary to devise an alternate approach to study the convergence of policy gradient in the context of average reward MDPs.

3.1.1 Related Work

There is a wealth of literature on discounted reward MDPs. [52] were among the first to establish the global convergence of policy gradients, specifically within the domain of linear quadratic regulators. [53] established a connection between the policy gradient and policy iteration objectives, determining conditions under which policy gradient algorithms converge to the globally optimal solution. [46] offer convergence bounds of $O(\frac{1}{\epsilon^2(1-\gamma)^6})$ for policy gradient and $O(\frac{1}{\epsilon(1-\gamma)^2})$ for natural policy gradient, where ϵ represents the suboptimality. It is noteworthy that while their convergence bounds for policy gradient rely on the cardinality of the state and action space, the convergence bounds for natural policy gradient are independent of them. [49] enhance the $O(\frac{1}{\epsilon^2(1-\gamma)^6})$ policy gradient bounds by refining the dependency on the discount factor, yielding improved bounds of $O(\frac{1}{\epsilon(1-\gamma)^5})$. [54] prove that variance-reduced versions of stochastic policy gradient also converge to the

global optimal solution. They achieve this through a gradient truncation mechanism. [30] analyze global convergence of softmax-based gradient methods and prove exponential rejection of suboptimal policies.

The global convergence of policy gradient methods has been extensively studied in the context of planning for average reward Markov Decision Processes (MDPs). [55] and [3] provide foundational results for natural policy gradient methods, proving their global convergence for finite state and action spaces. Extending this work, [56] analyze the more challenging case of infinite state spaces, establishing theoretical guarantees for convergence in this general setting. A critical contribution of our work lies in proving that the smoothness of the average reward holds for a large class of MDPs in the tabular setting. This smoothness property allows us to establish the first global convergence bounds for policy gradient methods in average reward MDPs, eliminating the need for previously unverified and restrictive smoothness assumptions.

In the learning context, actor-critic and gradient-based methods have been a central focus for analyzing average reward MDPs. [57] investigate two-time-scale actor-critic algorithms, employing linear function approximation for value functions and demonstrating asymptotic local convergence to a stationary point. Similarly, [37] analyze four gradient-based methods, including natural policy gradients, and establish asymptotic local convergence using an ODE-based framework. Addressing global convergence, [58] study policy gradient methods under the assumption that the average reward is smooth with respect to the policy parameters, achieving a regret bound of $O(T^{1/4})$. [59] consider the average reward natural actor-critic algorithm which does not rely on knowledge of mixing time, and provide a regret bound of $O(\sqrt{T})$. However, the results in both of these works rely on unverified smoothness assumptions, leaving open questions about their applicability to general MDPs. Our contributions further strengthen the understanding of when these assumptions might indeed be true.

3.1.2 Contributions

In this subsection, we outline the key contributions of this chapter.

- **Elimination of Smoothness Assumption:** Unlike previous work which assumes the underlying average cost function is smooth, we prove its smoothness by introducing a new analysis technique. This technique addresses the key difficulty of the lack of uniqueness of the value function in average-reward problems. We overcome this challenge by using a projection technique to ensure uniqueness and leveraging the properties of the projection to prove smoothness. This removes a significant assumption and strengthens the theoretical foundations of policy gradient methods in average reward MDPs.
- **Expression for Smooth Average Cost:** We derive an explicit expression for the average cost that is shown to be smooth in the policy π . This contribution is critical as it provides a deeper understanding and new insights into the behavior of the average reward MDPs under policy gradient methods.
- **Sublinear Convergence Bounds:** Using the above smoothness property, we present finite time bounds on the optimality gap over time, showing that the iterates approach the optimal policy with an overall regret of $O(\log(T))$. In contrast, the regret bounds in [58] are at least $O(T^{\frac{1}{4}})$ without learning error and with tabular parametrization. In place of the discount factor and the cardinality of the state and action spaces in the discounted setting, our finite-time performance bounds involves a different parameter which characterizes the complexity of the underlying MDP.
- **Extension to Discounted Reward MDPs:** Our analysis can also be applied to the discounted reward MDP problem to provide stronger results than the state of the art. In particular, we show that our performance bounds for discounted MDPs can be expressed in terms of a problem complexity parameter, which can be independent of the size of the state and action spaces in some problems.
- **Experimental Validation:** We simulate the performance of policy gradient across a simple class of MDPs to empirically evaluate its performance. The simulations illustrate the impact of MDP complexity on convergence rates. Unlike previous results, where the bounds depend solely on the size of the state and action spaces, these simulations demonstrate how the underlying structure of the MDP can result

in significantly different convergence rates, even with fixed state and action spaces. These observations further validate the theoretical bounds derived for the convergence of projected policy gradient in average reward MDPs.

3.2 Preliminaries

In this section, we introduce our model, address the limitations of applying the optimality gap bounds from discounted reward MDPs to the average reward scenario, present the gradient ascent update, and discuss the assumptions underlying our analysis.

3.2.1 Average Reward MDP Formulation

We consider the class of infinite horizon average reward MDPs with finite state space $\mathcal{S}$ and finite action space $\mathcal{A}$. The environment is modeled as a probability transition kernel denoted by $\mathbb{P}$. We consider a class of randomized policies $\Pi = \{\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})\}$, where a policy π maps each state to a probability vector over the action space. The transition kernel corresponding to a policy π is represented by $\mathbb{P}^\pi : \mathcal{S} \rightarrow \mathcal{S}$, where $\mathbb{P}^\pi(s'|s) = \sum_{a \in \mathcal{A}} \pi(a|s) \mathbb{P}(s'|s, a)$ denotes the single step probability of moving from state s to s' under policy π . Let $r(s, a)$ denote the single step reward obtained by taking action $a \in \mathcal{A}$ in state $s \in \mathcal{S}$. The single-step reward associated with a policy π at state $s \in \mathcal{S}$ is defined as $r^\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) r(s, a)$.

The infinite horizon average reward objective ρ^π associated with a policy π is defined as:

$$\rho^\pi = \lim_{N \rightarrow \infty} \frac{\mathbb{E}_\pi \left[\sum_{n=0}^{N-1} r^\pi(s_n) \right]}{N}, \quad (3.2)$$

where the expectation is taken with respect to $\mathbb{P}^\pi$. The average reward is independent of the initial state distribution under some mild conditions [51], [60] and can be alternatively expressed as $\rho^\pi = \sum_{s \in \mathcal{S}} d^\pi(s) r^\pi(s)$, where $d^\pi(s)$ is the stationary measure corresponding to state s under the transition kernel $\mathbb{P}^\pi$, ensuring that $d^\pi \mathbb{P}^\pi = d^\pi$. Associated with a policy is a relative state value function $v^\pi \in \mathbb{R}^{|\mathcal{S}|}$ that satisfies the following

average reward Bellman equation

$$\rho^\pi \mathbb{1} + v^\pi = r^\pi + \mathbb{P}^\pi v^\pi, \quad (3.3)$$

where $\mathbb{1}$ is the all ones vector [50], [51]. Note that v^π is unique up to an additive constant. Setting $\sum_{s \in \mathcal{S}} d^\pi(s) v^\pi(s) = 0$ imposes an additional constraint over v^π , providing a unique value function vector denoted by v_0^π , known as the basic differential reward function [35]. It can be shown that v_0^π can alternatively be expressed as $v_0^\pi(s) = \mathbb{E}_\pi \left[\sum_{n=0}^{\infty} (r^\pi(s_n) - \rho^\pi) \mid s_0 = s \right]$. Hence any element in the set $\{v_0^\pi + c\mathbb{1} : c \in \mathbb{R}\}$ is a solution v^π to the Bellman equation (3.3).

The relative state action value function $Q^\pi \in \mathbb{R}^{\mathcal{S} \times \mathcal{A}}$ associated with a policy π is defined as:

$$Q^\pi(s, a) = r(s, a) + \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} \mathbb{P}(s' \mid s, a) \pi(a' \mid s') Q^\pi(s', a') - \rho^\pi \quad \forall (s, a) \in \mathcal{S} \times \mathcal{A} \quad (3.4)$$

Similar to v^π , Q^π is also unique up to an additive constant. Analogously, every solution Q^π of (3.4) can be expressed as an element in the set $\{Q_0^\pi(s, a) + c\mathbb{1} : c \in \mathbb{R}\}$ where $Q_0^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{n=0}^{\infty} (r^\pi(s_n) - \rho^\pi) \mid s_0 = s, a_0 = a \right]$. Upon averaging (3.4) with policy π , it follows that $v^\pi(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) Q^\pi(s, a)$. The average reward policy gradient theorem [61] for policies parameterized by θ is given by:

$$\frac{\partial \rho}{\partial \theta} = \sum_{s \in \mathcal{S}} d^\pi(s) \sum_{a \in \mathcal{A}} \frac{\partial \pi(s, a)}{\partial \theta} Q^\pi(s, a)$$

As we focus on tabular policies in this chapter, our parameterization aligns with the tabular policy, where θ is equivalent to π . The policy gradient update considered is defined below.

$$\pi_{k+1} := \mathbf{Proj}_\Pi \left[\pi_k + \eta \frac{\partial \rho^\pi}{\partial \pi} \Big|_{\pi=\pi_k} \right] \quad \forall k \geq 0, \quad (3.5)$$

where $\mathbf{Proj}_\Pi$ denotes the orthogonal projection in the Euclidean norm onto the space of randomized policies Π and η denotes the step size of the update. In the following subsection, we recall the policy gradient result within the framework of discounted reward MDPs and

address why it cannot be directly applied to the average reward scenario.

3.2.2 Relationship to discounted reward MDPs

Let $\rho_{\mu,\gamma}^\pi := \mu^\top (1 - \gamma \mathbb{P}^\pi)^{-1} \mathbf{r}^\pi$ represent the discounted reward value function associated with policy π , under the initial distribution $\mu \in \Delta \mathcal{S}$ and where γ represents the discount factor [51]. Consider the projected policy gradient update given below.

$$\pi_{k+1} := \mathbf{Proj}_\Pi \left[\pi_k + \eta \frac{\partial \rho_{\mu,\gamma}^\pi}{\partial \pi} \Big|_{\pi=\pi_k} \right] \quad \forall k \geq 0. \quad (3.6)$$

When the step size $\eta = \frac{(1-\gamma)^3}{2\gamma|\mathcal{A}|}$, the iterates π_k generated from projected gradient ascent (3.6) satisfy the following equation:

$$\rho_{\mu,\gamma}^* - \rho_{\mu,\gamma}^{\pi_k} \leq \frac{256|\mathcal{S}||\mathcal{A}|}{k(1-\gamma)^5} \left\| \frac{d_{\mu,\gamma}^{\pi^*}}{\mu} \right\|_\infty^2, \quad (3.7)$$

where $\rho_{\mu,\gamma}^*$ represents the optimal value function under initial distribution μ , and $d_{\mu,\gamma}^{\pi^*} := (1-\gamma)\mu^\top (1-\gamma\mathbb{P}^{\pi^*})^{-1}$ represents the state occupancy measure under optimal policy π^* [49]. Under some mild conditions the average reward ρ^π associated with a policy π and the value function $\rho_{\mu,\gamma}^\pi(s)$ are related as below:

$$\rho^\pi = \lim_{\gamma \rightarrow 1} (1-\gamma) \rho_{\mu,\gamma}^\pi(s). \quad (3.8)$$

Note that the above relation [51], [60] holds for all $s \in \mathcal{S}$ and all $\mu \in \Delta \mathcal{S}$ since the average reward is independent of the initial state distribution. Upon leveraging the relation in (3.8) and multiplying (3.7) with $(1-\gamma)$, it is apparent that the upper bound of (3.7) in the limit of $\gamma \rightarrow 1$ tends to infinity. This is due to $(1-\gamma)^4$ that remains in the denominator of (3.7) upon multiplying with $(1-\gamma)$. Therefore it is necessary to devise an alternative proof technique in order to analyze the global convergence of policy gradient in the context of average reward MDPs. Prior to presenting the main result and its proof, we state the assumption used in our analysis.

Assumption 3.1. For every policy $\pi \in \Pi$, the transition matrix $\mathbb{P}^\pi$ associated with the

induced Markov chain is irreducible and aperiodic. This assumption also means that there exist constants $C_e < \infty$ and $\lambda \in [0, 1)$ such that for any $k \in \mathbb{N}$ and any $\pi \in \Pi$, the Markov chain corresponding to $\mathbb{P}^\pi$ is geometrically ergodic i.e., $\|(\mathbb{P}^\pi)^k - \mathbb{1}(\mathbf{d}^\pi)^\top\|_\infty \leq C_e \lambda^k$.

3.3 Main Results

Theorem 3.1. *Let ρ^{π_k} be the average reward corresponding to the policy iterates π_k , obtained through the policy gradient update (3.5). Let ρ^* represent the optimal average reward, that is, $\rho^* = \max_{\pi \in \Pi} \rho^\pi$. There exist constants L_2^Π and C_{PL} , determined by the underlying MDP, such that when the step size $\eta < \frac{1}{L_2^\Pi}$, the following holds:*

- For all MDPs it is true that,

$$\rho^* - \rho^{\pi_k} \leq \frac{1}{\frac{1}{\rho^* - \rho^{\pi_0}} + \nu k}, \quad \forall k \geq 0.$$

$$\text{where } \nu := \left(\frac{1}{32C_{PL}^2|\mathcal{S}|L_2^\Pi} \right) \left(1 + 4 \left(\frac{1}{32C_{PL}^2|\mathcal{S}|L_2^\Pi} \right) \right)^{-\frac{3}{2}}$$

- For simple MDPs (i.e. $L_2^\Pi \ll 1$) we obtain exponential convergence, that is

$$\rho^* - \rho^{\pi_k} \leq c^{-\frac{k}{2}} \left(\rho^* - \rho^{\pi_0} \right)^{\frac{1}{2k}}, \quad \forall k \geq 0$$

$$\text{where } \frac{1}{c} = 32|\mathcal{S}|L_2^\Pi C_{PL}^2 < 1.$$

Remark 3.1. It is worth noting that the above bounds correspond to a regret of $O(\log(T))$. While regret is typically a concept associated with online learning settings, we present it here to facilitate the development of learning algorithms inspired by the findings of this work. All previous results on convergence for discounted MDPs are of the form $\frac{\sigma}{k^p}$ [46], [62], [63], where σ is a large constant. However, since the worst sub-optimality is 1, the bound $\frac{\sigma}{k^p}$ becomes less meaningful for initial k . In contrast, our bound $\frac{1}{\frac{1}{\rho^* - \rho^{\pi_0}} + \nu k}$ is meaningful from the very first iteration. The maximum sub-optimality is $\rho^* - \rho^{\pi_0}$ at $k = 0$, and it decreases monotonically thereafter.

Our second result, an observation that is novel to this work, shows that simple MDPs

exhibit much faster (linear) convergence rates. This explains why techniques like reward shaping, which simplify the MDP, can be highly effective.

3.3.1 Key Ideas and Proof outline

A similar result was proved for discounted reward MDPs in [46], [63]. An important property pivotal to the global convergence analysis of the projected policy gradient is the smoothness of the discounted reward value function. Demonstrating the smoothness of the discounted reward value function is relatively straightforward due to the contractive properties of the discount factor. However, this poses a significant challenge in the context of average reward MDPs. Here, the absence of a discount factor as a source of contraction, coupled with the lack of uniqueness in the average reward value function, complicates the task of proving the smoothness of the average reward. Therefore, the first important property we prove is the smoothness of average reward.

Smoothness of average reward

A differentiable function $f : \mathcal{C} \rightarrow \mathbb{R}$ is called L -smooth if it satisfies

$$\|\nabla f(y) - \nabla f(x)\|_2 \leq L\|y - x\|_2 \quad \forall y, x \in \mathcal{C}.$$

where $\mathcal{C}$ is some subset of $\mathbb{R}^n$. Further if the function is L -smooth, it satisfies the following property.

$$\left| f(y) - f(x) - \langle \nabla f(x), y - x \rangle \right| \leq \frac{L}{2} \|y - x\|^2 \quad \forall y, x \in \mathcal{C},$$

From the above definition, it is apparent that if f is L -smooth then cf is $|c|L$ -smooth for any $c \in \mathbb{R}$. It can be shown that the infinite horizon discounted reward $V_{\mu, \gamma}^\pi$ is $\frac{2\gamma|\mathcal{A}|}{(1-\gamma)^3}$ -smooth. Leveraging the result in (3.8), one can see that the smoothness constant of ρ^π is $\lim_{\gamma \rightarrow 1} \frac{2\gamma|\mathcal{A}|}{(1-\gamma)^2} \rightarrow \infty$. Hence, the smoothness of the discounted reward cannot be leveraged to show the smoothness of the average reward.

In this chapter, we establish the smoothness of the infinite horizon average reward by first establishing the smoothness of the associated relative value function. We then leverage the

average reward Bellman Equation (3.3) to establish the smoothness of the average reward in terms of the smoothness of the relative value function. However, since the relative value function is unique up to an additive constant, we consider the projection of the value function onto the subspace orthogonal to the $\mathbb{1}$ vector. This provides us with a unique representation of the value function whose smoothness can be evaluated.

Let $\Phi \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{S}|}$ be the projection matrix that maps any vector to its orthogonal projection in the Euclidean norm onto the subspace perpendicular to the $\mathbb{1}$ vector. Then the following lemma holds.

Lemma 3.1. *Let I be the identity matrix and $\mathbb{1}$ be the all ones vector, both of dimension $|\mathcal{S}|$. Then the orthogonal projection matrix is given by $\Phi = \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{|\mathcal{S}|}\right)$. The unique value function v_Φ^π is obtained as a solution to the following fixed point equation,*

$$v_\Phi^\pi = \Phi \left(r^\pi + \mathbb{P}^\pi v_\Phi^\pi - \rho^\pi \mathbb{1} \right)$$

and can be alternatively represented as

$$v_\Phi^\pi = (I - \Phi \mathbb{P}^\pi)^{-1} \Phi r^\pi. \quad (3.9)$$

Since $\mathbb{P}^\pi$ has 1 as its Perron Frobenius eigenvalue, $(I - \mathbb{P}^\pi)$ is a singular matrix. It can be verified that $\Phi \mathbb{1} = 0$, hence $\mathbb{1}$ is an eigenvector of $\Phi \mathbb{P}^\pi$ for all π with a corresponding eigenvalue of 0. It can subsequently be proven that the rest of the eigenvalues of $\Phi \mathbb{P}^\pi$ are all less than one in terms of their absolute value and hence Equation (3.9) is well defined.

With a unique closed form for the average reward value function established, the subsequent task is to determine its smoothness constant. Given that the smoothness constant of a function f corresponds to the largest eigenvalue of its Hessian, we adopt an analytical approach similar to that presented in [46]. This involves utilizing directional derivatives and evaluating the maximum rate of change of derivatives across all directions within the policy space. It's important to note that since we are maximizing over directions expressible as differences between any two policies within the policy space, the resulting Lipschitz and smoothness constants are referred to as the restricted Lipschitz and smoothness constants,

respectively. The restricted smoothness of the average reward value function is stated below.

Lemma 3.2. *For any policy $\pi \in \Pi$, there exist constants $C_m, C_p, C_r, \kappa_r \in \mathbb{R}^+$ which are determined by the underlying MDP, such that the value function v_ϕ^π is $4 \left(2C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r \right)$ -smooth.*

Since the average reward value function is Lipschitz and smooth with respect to its policy, one can directly utilize this property to establish the Lipschitzness and smoothness of the average reward. These results are characterized in the following lemmas.

Lemma 3.3. *For any policy $\pi \in \Pi$, there exist constants $C_m, C_p, C_r, \kappa_r \in \mathbb{R}^+$ which are determined by the underlying MDP, such that the average reward ρ^π is L_1^Π -Lipschitz.*

$$\left| \left\langle \frac{\partial \rho^\pi}{\partial \pi}, \pi' - \pi \right\rangle \right| \leq L_1^\Pi \|\pi' - \pi\|_2, \quad \forall \pi, \pi' \in \Pi,$$

where $L_1^\Pi = 2(C_r + C_p C_m \kappa_r + 2(C_m^2 C_p \kappa_r + C_m C_r))$

The restricted Lipschitzness of the average reward is utilized to prove its restricted smoothness.

Lemma 3.4. *For any policy $\pi \in \Pi$, there exist constants $C_m, C_p, C_r, \kappa_r \in \mathbb{R}^+$ which are determined by the underlying MDP, such that the average reward ρ^π is L_2^Π -smooth.*

$$\left| \left\langle \pi' - \pi, \frac{\partial^2 \rho^\pi}{\partial \pi^2} (\pi' - \pi) \right\rangle \right| \leq \frac{L_2^\Pi}{2} \|\pi' - \pi\|_2^2 \quad \forall \pi, \pi' \in \Pi,$$

where $L_2^\Pi = 4(C_p^2 C_m^2 \kappa_r + C_p C_m C_r + (C_p + 1)(C_m^2 C_p \kappa_r + C_m C_r) + 4(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r))$.

Note that the restricted Lipschitz constant of the average reward is upper bounded by its general Lipschitz constant:

$$\max_{\pi' \in \Pi, \|\pi' - \pi\|_2 \leq 1} \left| \left\langle \frac{\partial \rho^\pi}{\partial \pi}, \pi' - \pi \right\rangle \right| \leq \max_{u \in \mathbb{R}^{\mathcal{S} \times \mathcal{A}}, \|u\|_2 \leq 1} \left| \left\langle \frac{\partial \rho^\pi}{\partial \pi}, u \right\rangle \right|$$

By confining our analysis of the smoothness constants to the policy class, we introduce a dependency of our convergence bounds on MDP-specific constants, including C_r, C_p, C_m, C_e

Table 3.1: Constants capturing the MDP Complexity

	Definition	Range	Remark
C_m	$\max_{\pi \in \Pi} \ (I - \Phi P^\pi)^{-1}\ $	$\frac{2C_e S }{1-\lambda}$ See Assumption 3.1 for definition of C_e and λ	Lowest rate of mixing
C_p	$\max_{\pi, \pi' \in \Pi} \frac{\ \mathbb{P}^{\pi'} - \mathbb{P}^\pi\ }{\ \pi' - \pi\ _2}$	$[0, \sqrt{ \mathcal{A} }]$	Diameter of transition kernel
C_r	$\max_{\pi, \pi'} \frac{\ \mathbf{r}^{\pi'} - \mathbf{r}^\pi\ _\infty}{\ \pi' - \pi\ _2}$	$[0, \sqrt{ \mathcal{A} }]$	Diameter of reward function
κ_r	$\max_{\pi} \ \Phi \mathbf{r}^\pi\ _\infty$	$[0, 2)$	Variance of reward function
$\frac{L_1^\Pi}{2}$	$C_r + C_p C_m \kappa_r + 2(C_m^2 C_p \kappa_r + C_m C_r)$	$[0, k_1 \sqrt{ \mathcal{A} } C_m^2]$	Restricted Lipschitz constant
$\frac{L_2^\Pi}{4}$	$C_p^2 C_m^2 \kappa_r + C_p C_m C_r + (C_p + 1)(C_m^2 C_p \kappa_r + C_m C_r) + 4(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r)$	$[0, k_2 \mathcal{A} C_m^3]$	Restricted smoothness constant

C_p, C_m are defined using operator norm w.r.t. L_∞ norm. Precisely,

$$C_m = \max_{\pi} \max_{\|v\|_\infty \leq 1} \|(I - \Phi P^\pi)^{-1} v\|_\infty, \text{ and } C_p = \max_{\pi, \pi' \in \Pi} \max_{\|v\|_\infty \leq 1} \frac{\|(\mathbb{P}^{\pi'} - \mathbb{P}^\pi)v\|_\infty}{\|\pi' - \pi\|_2}$$

and κ_r . These constants capture the complexity of the underlying MDP and are exclusive to the analysis presented in this chapter, as there appears to be no such dependency observed in the global convergence bounds of [46]. A more detailed description of these constants can be found in Table 3.1, where k_1 and k_2 represent MDP-independent numeric constants. These constants, which rely on the characteristics of the MDP, suggest that the projected policy gradient may achieve faster convergence in MDPs with lower complexity as opposed to those with higher complexity. The range of these constants can be found in Lemma B.10. We now proceed to analyze the convergence of projected policy gradient utilizing the smoothness of the average reward.

Convergence of policy gradient

Using the smoothness property of the average reward, it is possible to show that the improvement in the successive average reward iterates is bounded from below by the

product of the smoothness constant and the difference in the policy iterates, as described in the lemma below.

Lemma 3.5. *Let ρ^{π_k} be the average reward corresponding to the policy iterate π_k obtained from (3.5). Let L_2^Π be as in Lemma 3.4. Then,*

$$\rho^{\pi_{k+1}} - \rho^{\pi_k} \geq \frac{L_2^\Pi}{2} \|\pi_{k+1} - \pi_k\|^2, \quad \forall k \in \mathbb{N}.$$

Successively increasing iterates are not sufficient to guarantee finite time global convergence bounds. It is therefore necessary to bound the suboptimality associated with each iterate. We do so by leveraging the performance difference lemma stated below.

Lemma 3.6. *Let ρ^* be the globally optimal average reward. Then for any $\pi \in \Pi$, the suboptimality of ρ^π can be expressed as:*

$$\rho^* - \rho^\pi = \sum_s d^{\pi^*}(s) \sum_a Q^\pi(s, a) [\pi^*(a|s) - \pi(a|s)]. \quad (3.10)$$

Proof. The proof can be found in [64]. □

In the next lemma, we upper bound the right-hand side of (3.10) in terms of the gradient of ρ^π .

Lemma 3.7. *The suboptimality of any $\pi \in \Pi$ satisfies:*

$$\rho^* - \rho^\pi \leq C_{PL} \max_{\pi' \in \Pi} \left\langle \pi' - \pi, \frac{\partial \rho^\pi}{\partial \pi} \right\rangle, \quad \forall \pi \in \Pi,$$

where $C_{PL} = \max_{\pi \in \Pi} \max_{s \in \mathcal{S}} \frac{d^{\pi^*}(s)}{d^\pi(s)}$.

Note that C_{PL} is a constant that is proportional to the size of the state space. We do not know if the appearance of such a constant is inevitable or not; however, it should be noted such a constant appears in prior works on discounted reward problems as well [46], [63].

It is possible to further upper bound the expression in Lemma 3.7 using the smoothness property of the average reward.

Lemma 3.8. *Let π_k be the policy iterates generated by (3.5). Then for all $\pi' \in \Pi$ it is true that,*

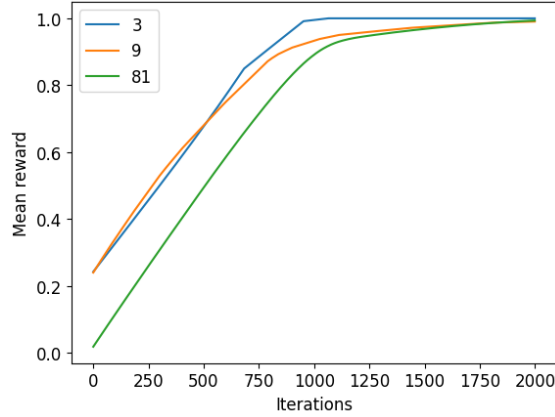
$$\left\langle \frac{\partial \rho_{\pi_{k+1}}}{\partial \pi_{k+1}}, \pi' - \pi_{k+1} \right\rangle \leq 4\sqrt{|\mathcal{S}|} L_2^\Pi \|\pi_{k+1} - \pi_k\|,$$

Lemmas 3.5 and 3.7 and Lemma 3.8 are combined to prove the result in Theorem 3.1.

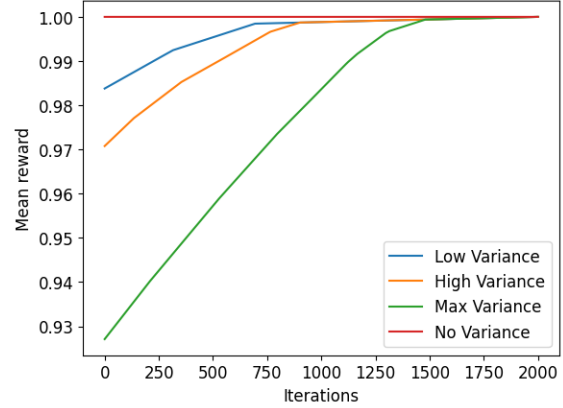
3.3.2 Extension to Discounted Reward MDPs

Existing performance bounds in the context of discounted reward MDPs require an iteration complexity of $O\left(\frac{|\mathcal{S}||\mathcal{A}|}{(1-\gamma)^5\epsilon}\right)$ to achieve policies with suboptimality of ϵ [63]. These bounds are independent of the hardness of the underlying MDP. Our approach improves on this bound, yielding an $O\left(\frac{|\mathcal{S}|L_2^\Pi}{\epsilon}\right)$ iteration complexity, where $L_2^\Pi = C_p^2 \hat{C}_m^2 \kappa_r + C_p \hat{C}_m C_r + (C_p + 1)(\hat{C}_m^2 C_p \kappa_r + \hat{C}_m C_r) + 4(\hat{C}_m^3 C_p^2 \kappa_r + \hat{C}_m^2 C_p C_r)$, where $\hat{C}_m := \|(I - \gamma \mathbb{P}^\pi)^{-1}\|$. It is straightforward to see that $\hat{C}_m \leq \frac{1}{1-\gamma}$. Hence, the iteration complexity improves to $O\left(\frac{L_2^\Pi |\mathcal{S}|}{(1-\gamma)^5\epsilon}\right)$, as the constants satisfy $\kappa_r \leq 2$ and $C_p, C_r \leq \sqrt{|\mathcal{A}|}$. Further, the approach considered in this chapter provides faster convergence rates for MDPs with low complexity, i.e., MDPs that have low values of C_p or C_r . The exact performance bounds can be obtained from an approach similar to the one outlined in [65], where L_2^Π represents the restricted smoothness constant of the discounted return ρ_γ^π . This constant can be derived through a process analogous to the one described in this chapter.

For instance, consider a trivial MDP for which $C_p = 0$ or $\kappa_r = 0$ (implies $C_r = 0$), i.e., an MDP where the transition kernel is independent of the action enacted. For this trivial MDP every policy is an optimal policy. The state of the art convergence guarantees [63], still requires $O(|\mathcal{S}||\mathcal{A}|\epsilon^{-1})$ iterations for ϵ close optimal policy. Whereas, the performance bounds presented in this chapter predict $O(|\mathcal{S}|\epsilon^{-1})$ iterations for convergence. Thus the constant L_2^Π captures the hardness of the MDP. Therefore, MDPs with lower complexity, i.e., lower values of L_2^Π , converge faster than MDPs with higher complexity, thus improving on current complexity-independent bounds.



(a) Convergence as a function of state space cardinality



(b) Convergence as a function of reward variance

Figure 3.1: Improvement in average reward as a function of MDP complexity

3.4 Simulations

Here, we present simulations corresponding to two MDP complexity measures². In order to study the convergence of projected policy gradient in the context of average reward MDPs and its dependence on the underlying MDP complexity, we present simulation results corresponding to two complexity measures: the cardinality of state and action spaces, and the diameter of the reward function.

Figure 3.1(a) considers MDPs with $(|\mathcal{S}|, |\mathcal{A}|) = \{(3, 3), (9, 9), (81, 81)\}$. We construct the transition kernel and the reward function in the same manner for all MDPs, which we discuss in the Appendix B.3. Projected policy gradient was implemented for 2000 iterations and the overall average reward is plotted as a function of iteration number. As expected, the convergence rate is slower when $(|\mathcal{S}|, |\mathcal{A}|)$ are larger due to the fact that the reward smoothness constant is larger. This reduction stems from small values of C_M, C_r, C_p , which are characteristic of MDPs with smaller state and action space cardinalities when the transition kernel and reward structures are similar. A less obvious result is that, even for MDPs with a fixed cardinality of state and action spaces, the rate of convergence can be considerably different as shown in Figure 3.1(b). For this simulation, we fix the state and action space cardinality at $(|\mathcal{S}|, |\mathcal{A}|) = (16, 16)$. We randomly generate a transition kernel,

²These simulations have been produced by other co-authors in [2].

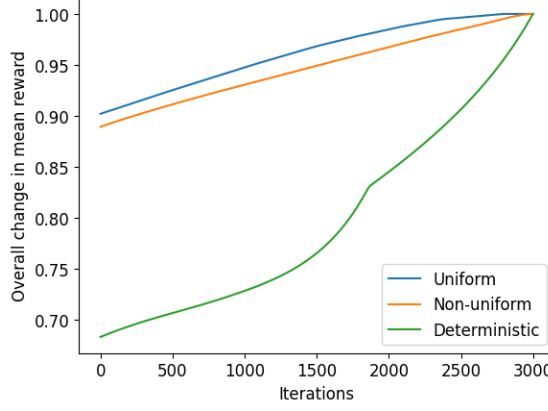


Figure 3.2: Convergence as a function of C_p

which remains constant across different single-step reward functions corresponding to varying reward variances. In particular, we consider four different reward variances - no variance, low, high and maximal variance. We recall the definition of C_r found in Table 3.1. We see that C_r scales with reward variance. Specifically, C_r is large when small changes to the policy result in significant modifications to the mean reward. Therefore, we anticipate that higher reward variance will lead to slower convergence. Additional details are found in the Appendix B.3. The observed convergence trend aligns with the theoretical bounds obtained, indicating that MDPs with small values of C_r tend to converge relatively faster.

Next, we discuss the impact of C_p on convergence of policy gradient. We consider MDPs of size 16, i.e., $(|\mathcal{S}|, |\mathcal{A}|) = (16, 16)$. We generate three different transition kernels. The first is uniform, so the actions do not change the transition probabilities, the second is deterministic (i.e., there exists some $s' \in \mathcal{S}$ such that $\mathbb{P}(s'|s, a) = 1$ for all $(s, a) \in (\mathcal{S}, \mathcal{A})$), and the last is non-uniform but stochastic transition kernel. We recall the definition of C_p from Table 3.1. We see that C_p is larger when the transition probabilities change by a greater amount with small changes to the policy. Thus, deterministic MDPs should have higher C_p values than ones that are more stochastic. Additional details are provided in the Appendix B.3. We run the policy gradient algorithm considered in this chapter for each MDP setting and plot the overall change in average reward as a function of iterations, for 3000 iterations. Figure 3.2 indicates that policy gradient in MDPs corresponding to small values of C_p converges relatively faster than for MDPs corresponding to large values of C_p . Hence, the performance bounds obtained in Theorem 3.1 are in some sense, more

representative of the empirical convergence trend of the policy gradient algorithm.

Note on Limitations and Future Work: Although we study tabular policies, this approach can be generalized to parametric class of policies. This study uses the exact value of the gradient and does not account for learning errors in the analysis. Nonetheless, we highlight that this is the first comprehensive proof of global convergence for policy gradient methods in average reward MDPs. Future work will focus on incorporating learning errors into this framework.

3.5 Conclusion

In this chapter, we presented the first comprehensive finite-time global convergence analysis of policy gradient for infinite horizon average reward MDPs. Key contributions include eliminating the smoothness assumption from previous work, deriving an explicit expression for the smooth average cost, and proving sublinear convergence with a regret of $O(\log(T))$. Our findings offer a more general and robust understanding of policy gradient methods in average reward MDPs, addressing long-standing challenges such as the lack of uniqueness in value functions. We also extended our analysis to discounted reward MDPs, providing stronger performance bounds by incorporating a complexity parameter beyond state and action space sizes. The theoretical results were further supported by simulations that highlighted how the structure of the underlying MDP influences convergence rates. These insights open avenues for refining performance bounds and exploring real-world applications in both average and discounted reward MDPs.

4

NATURAL POLICY GRADIENT AND MIRROR DESCENT

There are two modes of acquiring knowledge, namely, by reasoning and by experience.

Roger Bacon

4.1 Introduction

Popular algorithms to solve for optimal policies in the context of Markov decision processes (MDPs) include dynamic programming based algorithms such as value iteration, policy iteration, and modified policy iteration [16], [17]¹. While these algorithms require knowledge of the underlying MDP, reinforcement learning algorithms such as policy gradient algorithms can be implemented without the knowledge of an underlying model [11]. A variant of policy gradient methods, called Natural Policy Gradient (NPG) [66], [67] utilizes the Fisher information associated with a policy to condition the gradient. Such a conditioning is known to have good properties, including global convergence when the gradient can be calculated exactly or good performance bounds otherwise [18]. In the tabular setting, NPG has been shown to be closely related to mirror descent [68]–[71].

¹Material from our CDC 2023 paper[3] is reproduced here without copyright conflict.

Mirror descent is a generalization of gradient descent, which uses Bregman divergences as its distance metric and has been studied in various contexts [68], [72]–[77]. In the rest of the chapter, we study one specific mirror-descent based method (MDM) which coincides with MDM in the tabular setting.

In the discounted-reward tabular setting, when the value function can be computed exactly, MDM has been shown to converge to the optimal policy with a sublinear rate of $O(\frac{1}{T})$ [18], [78] that is, the MDM update yields the following error bound

$$V_{\eta}^{\pi^*} - V_{\eta}^{\pi_T} \leq O\left(\frac{1}{T(1-\alpha)^3}\right),$$

where η is the initial distribution over the state space, π^* is the optimal policy (in fact it can be any arbitrary policy; it is possible to show that the policies obtained through MDM perform better than any arbitrarily chosen policy), V_{η}^{π} is the discounted reward associated with policy π . It is well known that the average reward associated with a policy π can be expressed in terms of its discounted reward counterpart as $J^{\pi} = \lim_{\alpha \rightarrow 1} (1-\alpha)V_{\eta}^{\pi}(s)$, independent of the state s and initial distribution η [16], [79], [80]. However, multiplying the above bound by $(1-\alpha)$ on both sides and letting $\alpha \rightarrow 1$ yields ∞ on the right-hand side. Therefore, the above bound is not useful to understand the behavior of MDM in average-reward MDPs.

In the study of MDMs, two types of results have been obtained for discounted-reward MDPs: (i) global convergence in the tabular setting with perfect value function estimation, and (ii) performance bounds in the non-tabular setting with function approximations. In this chapter, our main contribution is to show the analog of (i) for average-reward MDPs. For completeness, we also discuss the non-tabular setting by leveraging our recent results for approximate policy iteration in [1].

4.1.1 Related Work

Average reward MDPs have been well studied in the context of reinforcement learning [11], [23], [26], [37], [81]–[83]. It is employed to model scenarios where the importance associated with the rewards does not decay with time. Unlike the discounted reward MDPs

formulation, average reward MDPs do not possess a discount factor $\alpha < 1$ to aid in the convergence of standard learning and dynamic programming techniques [84], [85]. Hence in many situations, it is necessary to devise different proof techniques to study performance of well-known algorithms in the average reward setting [1], [34], [86].

Some of the very early works on average reward TD-Learning include [35], [87]. In [37], four natural actor-critic algorithms in the context of average reward are considered, and their asymptotic convergence to a neighbourhood around the local maxima is proven. In [36], actor critic methods with function approximation are studied, where the feature vectors are expressed as a span of functions of the parametrized class of policies. Episodic model free reinforcement learning algorithms are presented in [88]; however the episodic setting is very different from the setting we study in this chapter. In [1], policy based reinforcement learning algorithms are modelled as variants of approximate policy iteration and the corresponding performance bounds are provided.

To the best of our knowledge, no global convergence results are available in the average reward case, even when the value function can be estimated precisely. On the other hand such global convergence results are available for NPG [47], [18] and for also for MDMs [72], [73] in the case of discounted reward MDPs .

The outline of the rest of the chapter is as follows: Section II contains model and preliminaries where the relationship between average reward and discounted reward MDPs is also discussed. Section III contains the main body of the chapter. It begins with the irreducibility assumption necessary for our proof of convergence of average reward MDMs, followed by the algorithm and its finite time convergence analysis. After a discussion on our assumption, Section III ends with the finite time bounds for non-tabular MDMs with function approximation in average reward MDPs. Section IV contains concluding remarks.

4.2 Model and Preliminaries

We consider infinite horizon Markov Decision Processes (MDPs) with finite state space $\mathcal{S}$, finite action space $\mathcal{A}$ and class of randomized policies $\pi \in \Pi$. Let $\nabla(\mathcal{A})$ indicate the class of probability distributions over $\mathcal{A}$. Then $\Pi : \mathcal{S} \rightarrow \nabla(\mathcal{A})$. The underlying environment is

modeled by a transition kernel $\mathbb{P} : \mathcal{S} \times \mathcal{A} \rightarrow \nabla(\mathcal{S})$. The probability of transition from s to s' under policy π is given by $\mathbb{P}(s'|s, \pi(s)) = \sum_{a \in \mathcal{A}} \pi(a|s) \mathbb{P}(s'|s, a)$. The single step reward associated with a policy π and state s is denoted by $r(s, \pi(s)) = \sum_{a \in \mathcal{A}} \pi(a|s) r(s, a)$. We assume the single step reward associated with any state and action is bounded.

The average reward associated with policy π is denoted by J^π and is defined as:

$$J^\pi = \lim_{T \rightarrow \infty} \frac{\mathbb{E}_\pi \left[\sum_{i=0}^{T-1} r(x_i, \pi(x_i)) \right]}{T},$$

where x_i is the state at time i and the expectation is taken with respect to the transition kernel $\mathbb{P}_\pi$. Under standard assumptions [16], [17], J^π is independent of the distribution of the initial state x_0 . Assuming π induces an ergodic Markov chain, J^π can be alternatively expressed as, $J^\pi = \sum_{s \in \mathcal{S}} d^\pi(s) \sum_{a \in \mathcal{A}} \pi(a|s) r(s, a)$, where $d^\pi(s)$ is the stationary distribution over $\mathcal{S}$ under the policy π . The state-action value function $Q^\pi(s, a)$ associated with a policy π is defined as the solution to the average reward Bellman equation given by:

$$J^\pi \mathbf{1} + Q^\pi(s, a) = r(s, a) + \sum_{s' \in \mathcal{S}} \mathbb{P}(s'|s, a) Q^\pi(s', \pi(s')), \quad (4.1)$$

where $\mathbf{1}$ is the all ones vector. In order to obtain the state value function $V^\pi(s)$ associated with policy π , Equation (4.1) is averaged with the policy vector π to obtain,

$$J^\pi \mathbf{1} + V^\pi(s) = r(s, \pi(s)) + \sum_{s' \in \mathcal{S}} \mathbb{P}(s'|s, \pi(s)) V^\pi(s'), \quad (4.2)$$

where $V^\pi(s') = \sum_{a \in \mathcal{A}} \pi(a|s') Q^\pi(s', a)$. We note that V^π is the relative value function, even though we refer to it as the value function for compactness. The advantage function A with respect to policies π and π' , for all states $s \in \mathcal{S}$, is defined as

$$A^\pi(s, \pi'(s)) = Q^\pi(s, \pi'(s)) - V^\pi(s). \quad (4.3)$$

The advantage function $A^\pi(s, a)$ is analogous to the definition in the discounted-reward case [89], but the interpretation is a bit trickier in the average-reward case due to the fact that V^π is the relative value function. Let the optimal average reward be denoted as

$J^{\pi^*} = \max_{\pi \in \Pi} J^{\pi}$ where $\pi^* = \operatorname{argmax}_{\pi \in \Pi} J^{\pi}$. Then under mild regularity conditions, there exists a value function V^{π^*} that satisfies the average reward Bellman optimality equation given by:

$$\begin{aligned} J^{\pi^*} \mathbf{1} + V^{\pi^*}(s) &= \max_{\pi \in \Pi} r(s, \pi(s)) + \sum_{s' \in \mathcal{S}} \mathbb{P}(s'|s, \pi(s)) V^{\pi^*}(s') \\ &= r(s, \pi^*(s)) + \sum_{s' \in \mathcal{S}} \mathbb{P}(s'|s, \pi^*(s)) V^{\pi^*}(s'). \end{aligned}$$

Note that the value functions Q^{π} and V^{π} corresponding to any policy π are unique up to an additive constant, by virtue of Equation (4.1) and Equation (4.2). In the next section we discuss the relationship between average reward MDPs and discounted reward MDPs.

4.2.1 Relationship to Discounted Reward MDP

The value function associated with a state $s \in \mathcal{S}$, policy π and discount factor $0 < \alpha < 1$ is defined as: $V_{\pi}^{\alpha}(s) = \mathbb{E}_{\pi} \left[\sum_{i=0}^{\infty} \alpha^i r(s_i, \pi(s_i)) \middle| s_0 = s \right]$. Similarly, the state-action value function associated with (s, a) is defined as:

$$Q_{\pi}^{\alpha}(s, a) = \mathbb{E}_{\pi} \left[r(s_0, a_0) + \sum_{i=1}^{\infty} \alpha^i r(s_i, \pi(s_i)) \middle| s_0 = s, a_0 = a \right].$$

The advantage $A_{\pi}^{\alpha}(s, a)$ of using action a instead of policy π is defined as

$$A_{\pi}^{\alpha}(s, a) = Q_{\pi}^{\alpha}(s, a) - V_{\pi}^{\alpha}(s). \quad (4.4)$$

The associated discounted reward $J_{\pi}^{\alpha}(\eta)$ is the value function weighted with the initial distribution η over the state space, that is $J_{\pi}^{\alpha}(\eta) = \sum_{s \in \mathcal{S}} \eta(s) V_{\pi}^{\alpha}(s)$.

Mirror Descent-Based Policy methods

Since the objective of most reinforcement learning algorithms is to determine the optimal policy yielding maximum discounted reward, a popular algorithm to achieve that is natural policy gradient. NPG is closely related to mirror descent in the tabular setting whose

objective at the k th iteration is to maximize:

$$\pi_{k+1} = \operatorname{argmax}_{\pi \in \Pi} \nabla_{\pi}^T J_{\pi_k}^{\alpha}(\eta)(\pi - \pi_k) - \frac{1}{\beta} \sum_{s \in \mathcal{S}} d_{\eta}^{\pi_k}(s) D_{\text{KL}}(\pi(\cdot|s) \parallel \pi_k(\cdot|s)), \quad (4.5)$$

where $d_{\eta}^{\pi_k}(s) = (1 - \alpha) \sum_{i=0}^{\infty} \alpha^i \mathbb{P}_{\pi}(s_k = s | s_0 \sim \eta)$ is the normalized state visitation measure corresponding to state s and $D_{\text{KL}}(\pi(\cdot|s) \parallel \pi_k(\cdot|s))$ is the Kullback-Leibler divergence between policies π and π_k . State visitation measure is the stationary distribution analog of the discounted reward MDP. Under tabular policy class, it can be shown that

$$\frac{\partial J_{\pi}^{\alpha}}{\partial \pi(a|s)} = \frac{1}{1 - \alpha} d_{\eta}^{\pi}(s) A_{\pi}^{\alpha}(s, a). \quad (4.6)$$

From Equation (4.5) and Equation (4.6), we get the update equation

$$\pi_{k+1}(a|s) = \frac{\pi_k(a|s) e^{\beta A_{\pi_k}^{\alpha}(s, a)}}{\sum_{a' \in \mathcal{A}} \pi_k(a'|s) e^{\beta A_{\pi_k}^{\alpha}(s, a')}}. \quad (4.7)$$

Given the definition of the advantage function in Equation (4.4), the above update equation can be equivalently written as $\pi_{k+1}(a|s) = \frac{\pi_k(a|s) e^{\beta Q_{\pi_k}^{\alpha}(s, a)}}{\sum_{a' \in \mathcal{A}} \pi_k(a'|s) e^{\beta Q_{\pi_k}^{\alpha}(s, a')}}$. In the limit as $\beta \rightarrow \infty$, the policy obtained under such an update is identical to the one obtained through policy iteration. Hence MDM can be interpreted as soft policy iteration. A finite β ensures that all actions are explored with some non-zero probability which is crucial in order to effectively utilize algorithms such as TD learning to determine $Q_{\pi}^{\alpha}(s, a)$ and $A_{\pi}^{\alpha}(s, a)$.

Let $V_{\pi_*}^{\alpha}(s) = \max_{\pi \in \Pi} V_{\pi}^{\alpha}(s)$. Assuming the initial policy π_0 is uniformly randomized across all actions and the bounded rewards (with a maximum value of 1), under the update rule Equation (4.7), previous literature has shown convergence of the policy updates to the optimal policy as below:

$$V_{\pi_*}^{\alpha}(s) \leq V_{\pi_T}^{\alpha}(s) + \frac{1}{T(1 - \alpha)^3} + \frac{\log(|\mathcal{A}|)}{\beta T}, \quad \forall s \in \mathcal{S}. \quad (4.8)$$

However this performance bound is vacuous in the context of average reward MDP as explained below.

Average Reward MDPs

Note that under mild regularity conditions standard MDP theory establishes the following relation between discounted reward MDPs and average reward MDPs:

$$J^\pi = \lim_{\alpha \rightarrow 1} (1 - \alpha) V_\pi^\alpha(s). \quad (4.9)$$

Since the average cost is independent of the initial state, the above relation is true for all states $s \in \mathcal{S}$. Upon multiplying Equation (4.8) with $(1 - \alpha)$ and setting $\alpha \rightarrow 1$, we still are left with a $(1 - \alpha)^2$ in the denominator on the RHS, which leaves us with vacuous bounds for convergence of MDM for average reward MDPs. But since this is only an upper bound, it is interesting to understand whether the algorithm converges in the average-reward case. In the next section, we outline under what conditions we obtain convergence of MDM for average reward MDPs as well, and subsequently provide a proof of the same.

4.3 Average Reward MDM

We now present the average reward MDM in the tabular setting with perfect estimates of the value functions. The algorithm is a natural extension of the corresponding algorithm in the discounted-reward case.

Algorithm 4 Average Reward MDM

Input: $\beta > 0$, $\pi_0(\cdot|s) \in \Delta(\mathcal{A})$, $\forall s \in \mathcal{S}$

for $k = 0, \dots, T - 1$

- 1: Compute $A^{\pi_k}(s, a) \forall (s, a) \in \mathcal{S} \times \mathcal{A}$
- 2: Update for all $(s, a) \in \mathcal{S} \times \mathcal{A}$

$$\pi_{k+1}(a|s) = \frac{\pi_k(a|s) e^{\beta A^{\pi_k}(s, a)}}{\sum_{a' \in \mathcal{A}} \pi_k(a'|s) e^{\beta A^{\pi_k}(s, a')}} \quad (4.10)$$

endfor

Output: π_T

We prove the global convergence of the above algorithm under the following assumption.

Assumption 4.1. Let $d^\pi(s)$ be the stationary probability of being in state s under $\mathbb{P}_\pi$. Let π^* be the optimal policy. We assume that

$$\Delta = \inf_{\substack{\pi \in \Pi \\ s \in \mathcal{S}}} \frac{d^\pi(s)}{d^{\pi^*}(s)} > 0. \quad (4.11)$$

Later, we will discuss how most practical MDPs can be made to satisfy this assumption for a small loss in performance.

4.3.1 Convergence Analysis of Algorithm 4

Our convergence analysis follows the same outline as in the case of discounted-reward MDPs: we first utilize the average reward performance-difference lemma, then show the monotonicity of the average reward and finally, we use a weighted KL distance as a Lyapunov function to establish convergence. However, each of the steps has to be appropriately modified to get rid of the dependence on the discount factor and replace it with other relevant quantities.

Lemma 4.1. (*Performance Difference Lemma*) Recall the definition of advantage function in Equation (4.3). Let J^π and $J^{\pi'}$ be the average rewards associated with policies π and π' respectively. Then it is true that,

$$J^\pi - J^{\pi'} = \sum_{s \in \mathcal{S}} d^\pi(s) A^{\pi'}(s, \pi(s)), \quad (4.12)$$

where d^π is the stationary distribution induced over $\mathcal{S}$ by policy π .

As in the discounted-reward case, the performance difference lemma plays a key role in the proof of convergence of average reward MDM, especially to establish the monotonicity of the sequence of average rewards obtained through Algorithm 4. However, in our average-reward case, it is also utilized to study the drift of the Lyapunov function used to prove the convergence of the algorithm.

Monotonicity of the MDM Update

We now present a key lemma used to prove the convergence of the MDM update to the optimal policy.

Lemma 4.2. *Given π_k which are generated according to Equation (4.10), the corresponding sequence of average rewards J^{π_k} are increasing, that is,*

$$J^{\pi_{k+1}} - J^{\pi_k} \geq 0. \quad (4.13)$$

Now that we have established the monotonicity of the average reward associated with the policy iterates, we now compare the policy iterates with the optimal policy. In order to prove the convergence of the MDM iterates, consider the following Lyapunov function whose argument is a policy vector.

$$W(\pi) = \sum_{s \in \mathcal{S}} d^{\pi^*}(s) D_{\text{KL}}(\pi^*(\cdot|s) \parallel \pi(\cdot|s)). \quad (4.14)$$

The idea is to show that with the subsequent iterates, the value of the associated Lyapunov function reduces, which implies that the NPG iterates approach the optimal policy. A lemma important to prove the result is stated below.

Lemma 4.3. *Consider the sequence of policies π_k obtained from Algorithm 4. It is true that*

$$W(\pi_{k+1}) - W(\pi_k) = \sum_{s \in \mathcal{S}} d^{\pi^*}(s) \log(z_k(s)) - \beta (J^{\pi^*} - J^{\pi_k}),$$

where $z_k(s) = \sum_{a' \in \mathcal{A}} \pi_k(a'|s) e^{\beta A^{\pi_k}(s, a')}$.

The above lemma captures the impact of consecutive policy iterates on the chosen Lyapunov function. The main result follows by telescoping the expression from the previous lemma as proved in the theorem below.

Theorem 4.1. *The sequence of policies π_k generated through the update rule in Equation (4.10),*

correspond to MDPs with average reward J^{π_k} , which approach the optimal average reward as below:

$$J^{\pi^*} - J^{\pi_T} \leq \frac{W(\pi_0) - W(\pi_T)}{\beta T} + \frac{J^{\pi_T} - J^{\pi_0}}{\Delta T}.$$

4.3.2 Discussion: Assumption 1

Assumption 1 requires the stationary distribution induced over the state space by all policies within the randomized policy function class to be uniformly bounded away from zero. Depending on the underlying transition kernel, this assumption need not always be satisfied. However, it is often the case that there exists a policy (possibly randomized) under which the resulting Markov chain is irreducible. An example of such a policy is one where each action is chosen uniformly at random in each state. Let us denote such a policy by π' . We can modify the original MDP as follows: in each state, with probability ϵ , pick an action specified by policy π' and with probability $1 - \epsilon$, choose an action specified by another policy π . Now, the MDP can be viewed as optimizing over policy π with π' being fixed. This induces an $O(\epsilon)$ suboptimality in the average reward that MDM converges to and thus, our convergence result should be interpreted as a near optimality result.

4.3.3 Discussion: Non-Tabular Average Reward MDM

While performance bounds for non-tabular average-reward MDM have not been provided in prior literature, they can be easily inferred from the results in [1], [19].

Algorithm 5 Non Tabular Average Reward MDM

Input: $\beta > 0$, $\pi_0(\cdot|s) \in \Delta(\mathcal{A})$, $\forall s \in \mathcal{S}$

for $k = 0, \dots, T - 1$

- 1: Use TD-Learning to compute $Q_k(s, a)$ as an approximation to $Q^{\pi_k}(s, a) \forall (s, a) \in \mathcal{S} \times \mathcal{A}$
- 2: Update for all $(s, a) \in \mathcal{S} \times \mathcal{A}$

$$\pi_{k+1}(a|s) = \frac{\pi_k(a|s)e^{\beta Q_k(s,a)}}{\sum_{a' \in \mathcal{A}} \pi_k(a'|s)e^{\beta Q_k(s,a')}} \quad (4.15)$$

endfor

Output: π_T

Note that in general TD-Learning algorithms are used to learn the state-action value function $Q^{\pi_k}(s, a)$. However, since we know the policy π_k , one can average the learnt approximation Q_k to obtain V_k and consequently the advantage function A_k as given by Equation (4.3).

TD Learning algorithms in the context of average reward MDPs are presented in [34], where function approximation is utilized for state-action value function $Q^{\pi_k}(s, a)$. More precisely, $Q_k(s, a) = \Phi(s, a)^\top \theta_k$ where $\Phi(s, a) \in \mathbb{R}^d$ is the feature vector, and θ_k is the parameter vector that requires estimation at every iteration. The TD learning error at each time step can then be expressed as follows:

$$\|Q_k - Q^{\pi_k}\|_\infty \leq \underbrace{\|\Phi(\theta_k - \theta_k^*)\|_\infty}_{\text{policy evaluation error}} + \underbrace{\|\Phi\theta_k^* - Q^{\pi_k}\|_\infty}_{\text{function approximation error}}, \quad (4.16)$$

where $\theta_k^* = \operatorname{argmin}_{\theta \in \mathbb{R}^d} \|\Phi\theta - Q^{\pi_k}\|_2^2$.

The policy evaluation error depends on the number of samples utilized to learn the best feature vector θ_k whereas the function approximation error is a function of the richness of the feature vector Φ (and does not depend on the number of samples).

Given a policy π_k and state s , since $V_k(s)$ does not depend on action, the policy update equation in Algorithm 5 is equivalent to Equation (4.10).

Define the Bellman operator $T_\pi : \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|} \rightarrow \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$ with respect to a policy π and optimal Bellman operator $T : \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|} \rightarrow \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$ as follows:

$$(T_\pi Q)(s, a) = r(s, a) + \sum_{s' \in \mathcal{S}} \hat{P}_\pi(s', a' | s, a) Q(s', a'),$$

and

$$(TQ)(s, a) = r(s, a) + \max_{a' \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \hat{P}(s', a' | s, a) Q(s', a').$$

The policy improvement error obtained as a result of using MDM policy update π_k as opposed to the action maximizing TQ_k is given by:

$$0 \leq (TQ_k - T_{\pi_{k+1}}Q_k)(s, a) \leq \frac{1}{\beta} \log \frac{1}{\min_{s \in \mathcal{S}} \pi_{k+1}(a^*(s) | s)} \quad (4.17)$$

where $\alpha^*(s)$ is the optimal action at state s obtained as the maximizing argument of the optimal Bellman operator acting on Q_k . The error is quantified in terms of the probability of choosing the optimal action under the policy obtained through MDM. The details of the above bound can be found in [19].

Let J^{π^*} be the optimal average reward associated with $\hat{\mathbb{P}}$. Given the policy evaluation error in Equation (4.16) and the policy improvement error in Equation (4.17) as a result of the MDM update π_k given by Equation (4.15), we obtain the following finite time performance bound:

$$\begin{aligned}
\mathbb{E}[J^{\pi^*} - J_{\pi_{T+1}}] &\leq \underbrace{(1 - \gamma)^T \mathbb{E} \left[J^{\pi^*} - \min_i (\mathcal{T}Q_0 - Q_0)(i) \right]}_{\text{Initial condition error}} \\
&\quad + \underbrace{\sum_{\ell=1}^{T-1} (1 - \gamma)^{\ell-1} \mathbb{E} \left[\|\mathcal{T}_{\pi_{T+1-\ell}} Q_{T-\ell} - \mathcal{T}Q_{T-\ell}\|_{\infty} \right]}_{\text{Policy improvement error}} \\
&\quad + \underbrace{\mathbb{E} \left[\|\mathcal{T}_{\pi_{T+1}} Q_T - \mathcal{T}Q_T\|_{\infty} \right]}_{\text{Policy improvement error}} \\
&\quad + \underbrace{2 \sum_{\ell=0}^{T-1} (1 - \gamma)^{\ell} \mathbb{E} \left[\|Q_{T-\ell} - Q_{\pi_{T-\ell}}\|_{\infty} \right]}_{\text{Policy evaluation error}}.
\end{aligned}$$

where $\gamma = \inf_{\substack{s \in \mathcal{S} \\ \pi \in \Pi}} d^{\pi}(s)$ where $d^{\pi}(s)$ is the stationary measure associated with s under policy π . More details on this bound can be found in [1].

Note that in the tabular case, with perfect value function information, we were able to prove global convergence. However, in the non tabular version of average reward MDM, relative value function estimates are obtained through TD-Learning, an algorithm that requires each (state, action) pair to be visited infinitely often. As $\beta \rightarrow \infty$, it is easy to see that some state action pairs will never be visited, which leads to a significant error in their relative value function estimation. However, from Equation (4.17), we know that as $\beta \rightarrow \infty$, the policy improvement error associated with the MDM update tends to zero. Hence the performance bound for non-tabular average reward MDM crucially depends on the choice of β , i.e., there is a tradeoff between policy improvement error and policy evaluation error.

This raises an interesting question about the learning algorithms utilized for the purpose of relative value function evaluation. If there is a way to estimate the relative function in a manner that is independent of β , then it may be possible to get better performance bounds depending upon the choice of the behavioral policy. This is an interesting question to explore in the future.

4.4 Conclusion

In this chapter we prove the global convergence of MDM, along with finite time bounds, in the tabular setting of average reward MDPs with perfect value function estimates. We then proceed to leverage some results from recent literature to present finite time performance bounds for non-tabular MDM with approximate policy evaluations and policy improvements. This work extends some of the very well known core results in the context of discounted reward MDPs to the average reward MDP domain, thus opening up more avenues for research in average reward MDMs.

We note that one can use connection between learning from expert advice and tabular MDM as in [90] to show that the difference between the average reward and the reward obtained by MDM decays as $O(1/\sqrt{T})$, instead of $O(1/T)$ as in this chapter. We note that both results are useful in different respects: while we establish a faster rate of convergence in terms of T , the result in [90] is independent of Δ and thus, independent of the size of the state space.

REINFORCEMENT LEARNING IN COUNTABLE STATE SPACES WITH UNBOUNDED COSTS

*He who loves practice without theory is like the sailor who boards ship
without a rudder and compass and never knows where he may cast.*

Leonardo Da Vinci

5.1 Introduction

We are motivated by control problems in queuing models of resource allocation, such as those arising in communication networks, cloud computing systems, and riding hailing services. Examples of such systems include the following:

- (a) The switch fabric in Internet routers and data centers where packets have to be transported (or switched) from one of many input ports to one of many output ports [91]: the system is modeled as a bipartite graph with input ports on one side and output ports on the other side. Technological constraints dictate that at each time slot, a matching must be selected in the bipartite graph, and packets are transferred along the edges of the matching from each input to the corresponding output. The goal is to find a sequence of matchings to minimize either the average delay experienced by the packets in the switch or the probability that the delay exceeds some threshold.

- (b) Scheduling problems at base stations in 5G networks [91]: at a central controller (typically the base station associated with a cell in a cellular network), packets arrive and are queued in a separate queue for each receiver. The goal is to schedule these packets over different frequencies and time slots to minimize the average delay of the packets in the system, while taking into account the time-varying channel conditions in a wireless network due to fading and other wireless medium effects.
- (c) Scheduling workloads in cloud computing systems [92]: a workload in such systems takes the form of a collection directed acyclic graphs, where each DAG represents a job, the nodes in the graphs represent tasks in the job and the directed edges represent precedence relationships among the tasks in the graph. The goal is to allocate resources to tasks from a sequence of arriving jobs, while respecting the precedence relationships of the tasks within each job and minimizing the average delay experienced by the jobs.
- (d) Customer-driver matching in ride hailing platforms such as Uber and Lyft [93], [94]: the role of such platforms can be modeled as controlling the number of nodes in a bipartite graph, where one side is the set of waiting customers and the other side in the set of available drivers. The goal of a ride hailing platform is to choose a set of prices and match customers to drivers so that a weighted combination of the average delay experienced by customers and the average profit is optimized.

The above problems exhibit several common features:

- (i) The state space of these problems is discrete, typically consisting of the queue lengths of the various entities waiting in the system such as packets, customers, drivers, jobs and tasks, depending on the context. Discrete state spaces are commonly studied in the reinforcement learning (RL) literature; however, in our applications, the state space is also countably infinite for all practical purposes, since queue lengths can become unbounded. In some applications, such as communication networks, the packet buffers may be finite but it is well known that modeling them as infinite buffers leads to good scheduling algorithms [91]. It should be noted that even if one were to model the finiteness of the buffers explicitly in our model, our results will still hold,

and our performance guarantees would not depend on the size of the buffers.

- (ii) Because we are dealing with a vector of queue lengths as the state of the system, the problems have some limited amount of structure that can and should be exploited to design good algorithms. In particular, it is relatively straightforward to design algorithms that ensures that the system is stable, i.e., the queue length is finite with probability one [91]. On the other hand, algorithms to optimize performance objectives such as average delay are unknown except in limited regimes [95], [96]. Therefore, data-driven approaches such as reinforcement learning (RL) are natural candidates to solve such problems.
- (iii) Due to a well-known result called Little’s law, minimizing average delay is equivalent to minimizing average queue lengths [97]. Thus, the natural instantaneous cost in such problems is the current total queue length. Note that unlike many RL models, this cost is unbounded and results which assume that the costs (or rewards) at each (state, action) pair are uniformly bounded do not hold for our problems.

Given the above background, our goal in this chapter is to study policy optimization algorithms for such countable state space models with discrete, finite action spaces where the cost is proportional to the total queue length in the system, and can thus grow in an unbounded fashion. For this purpose, we study the natural policy gradient (NPG) algorithm. Our main contributions are the following:

- (1) **Algorithmic Contribution:** A standard regret analysis for NPG relies on its connection to a classic learning theory problem known as the best-experts problem. However, we demonstrate that this analysis doesn’t hold in our case due to the unbounded nature of the instantaneous cost in our setting. By making a small but crucial adjustment to the step size used in the best-experts algorithm and leveraging bounds on the relative value function (also known as the solution to Poisson’s equation in applied probability), we establish nontrivial regret bounds. In addition to ensuring the convergence of NPG in countable state-space MDPs, this algorithmic adjustment significantly accelerates convergence in finite state MDPs compared to the fixed step-size NPG algorithm. Notably, prior work offered no heuristic for selecting an optimal

fixed step size, often relying on hyperparameter tuning. Our approach, grounded in Poisson’s equation, provides an effective heuristic for selecting both fixed and adaptive step sizes without the need for extensive parameter tuning. This improvement streamlines the application of NPG and enhances its overall performance.

- (2) **Theoretical Contribution:** An important component of our work is to obtain bounds on the solution of Poisson’s equation that are uniform across all policies. To the best of our knowledge, prior works on obtaining bounds on the solution of Poisson’s equation are limited to specific policies. A key contribution of this chapter is to show that uniform bounds can be obtained by exploiting certain structural properties of the mathematical models for the motivating applications mentioned earlier. These bounds are essential for achieving final regret bounds that are independent of the state space cardinality.
- (3) **Relaxation of learning error assumption:** Policy evaluation using temporal difference learning and Monte Carlo methods have been well studied in the literature, so we do not consider them explicitly in this chapter. However, we do consider the error due to function approximation. Traditionally, for analytical purposes, it is assumed that there is a uniform bound on the function approximation error of the value function. We argue that this assumption is not reasonable for countable state space models, particularly queuing models with unbounded instantaneous costs. Instead, we propose a more general model for function approximation, where the error bounds in learning are relaxed for states that are less frequently visited. Existing mathematical tools for the study of convergence of RL algorithms cannot handle our proposed model for the function approximation error. However, we show that, by exploiting the special structure of our queuing models and the associated bounds on the solution to Poisson’s equation, we can obtain non-trivial regret bounds for policy optimization.
- (4) **Empirical Evaluation:** We evaluate the performance of our algorithmic modification in finite state space applications, focusing specifically on cloud computing scenarios driven by autoscaling. We conduct two sets of experiments where $TD(\lambda)$ is used to learn the value functions. Utilizing our bounds on the solution to Poisson’s Equation, we determine the fixed step size according to established theory and compare the

performance of our algorithm against one that employs this fixed step size. Additionally, we conduct experiments in a noiseless environment to evaluate the robustness of our algorithm against learning errors. We empirically show the vast improvement in convergence when utilizing an adaptive state dependent step size to that of a fixed step size, where the former rate of convergence remains independent of the underlying state space cardinality. By demonstrating similar iteration complexities with and without noise, we validate the robustness of our algorithm within our framework of relaxed assumptions over learning error, which accommodates greater noise in the value function for less frequently visited states.

5.1.1 Related Work

The Natural Policy Gradient algorithm is a well-known and extensively studied algorithm for MDP optimization, in both the average-reward and discounted-reward settings [1], [3], [18], [66], [78]. An important line of research on the NPG algorithm treats the MDP-optimization problem as many parallel instances of the expert advice problem, and treats the NPG algorithm as many parallel instances of the weighted majority algorithm. [55] use this approach to prove the first convergence result for NPG in the finite-state average-reward tabular setting, and [31] expand upon that result to incorporate function approximation. Our result uses the same “parallel weighted majority” framing, but generalizes the result to the infinite-state-space setting by incorporating state-dependent learning rates.

Policy gradient algorithms have been studied in certain specialized settings with average-reward uncountably-infinite state spaces [52], [98]: the Linear Quadratic Regulator and the base-stock inventory control problem, demonstrating rapid convergence to the optimal policy. However, follow-up study of these settings has demonstrated that they exhibit additional structure which is critical to these results, causing these policy-gradient algorithms to act like policy improvement algorithms [53]. Our result is the first to handle an infinite state-space setting without the specialized structure of these prior results.

Key to our result are novel bounds on the relative value function, building off of our drift assumption for the policy space. This drift assumption is reasonable in a queuing setting, as we discuss in Section 5.3.2 [91]. Prior drift-based bounds on the relative-value function

exist [99], but are policy-dependent. In contrast, we prove policy-independent bounds on the relative-value function using reasonable assumptions on the MDP structure, which are motivated by the structure of MDPs in queuing networks. Our policy-independent bounds are critical to implement our state-dependent learning rates, allowing us to generalize the NPG algorithm to the infinite-state setting.

A variety of papers have studied applications of reinforcement learning to queuing problems, including policy-gradient-based algorithms. Several such results focus on the problem of learning the relative value function from samples, including variance reduction techniques [100] and sample augmentation techniques [101]. Our results complement these results, as we focus on the function approximation step, and prove results on overall algorithmic performance, while these results focus on the policy evaluation step, and empirically demonstrate performance improvements. [100] in particular empirically demonstrate that with variance reduction techniques in use, policy gradient algorithms with function approximation rapidly converge to the optimal policy in an infinite-state-space queuing setting. Our results theoretically justify this empirical observation.

5.2 Model and Preliminaries

We consider the class of Markov Decision Processes (MDP) with countably infinite states $\mathcal{S}$, finite actions $\mathcal{A}$ and the infinite horizon average cost objective. We consider a randomized class of policies Π , where a policy $\pi \in \Pi$ maps each state to a probability vector over actions $\mathcal{A}$, that is, $\pi : \mathcal{S} \rightarrow \Delta\mathcal{A}$. The state and action at time t are denoted by $(\mathbf{q}_t, a_t)$ respectively. The underlying probability transition kernel is denoted by $\mathbb{P} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ and the transition kernel corresponding to any policy π is denoted by $\mathbb{P}_\pi$, where $\mathbb{P}_\pi(\mathbf{q}'|\mathbf{q}) = \sum_{a \in \mathcal{A}} \pi(a|\mathbf{q})\mathbb{P}(\mathbf{q}'|\mathbf{q}, a)$ is the probability of transitioning from $\mathbf{q}$ to $\mathbf{q}'$ under policy π in a single step. Associated with each state $\mathbf{q}$ and action a is a single step cost $c(\mathbf{q}, a)$ which is modeled as non-negative. Let $\underline{c}(\mathbf{q}) = \min_{a \in \mathcal{A}} c(\mathbf{q}, a)$ and $\bar{c}(\mathbf{q}) = \max_{a \in \mathcal{A}} c(\mathbf{q}, a)$ be the minimum and maximum instantaneous cost respectively in state $\mathbf{q}$ across all actions $a \in \mathcal{A}$. The single step cost under policy π at state $\mathbf{q}$ is thus denoted by $c_\pi(\mathbf{q})$, where $c_\pi(\mathbf{q}) = \sum_{a \in \mathcal{A}} \pi(a|\mathbf{q})c(\mathbf{q}, a)$. When dealing with queuing systems this single step cost can correspond to total queue

length, which can be unbounded thus yielding unbounded instantaneous costs. Hence this formulation relaxes the bounded single step costs assumption which is a common feature of many algorithms previously studied in literature [1], [31]. The infinite horizon average cost associated with a policy π is denoted by J_π , and is defined as follows:

$$J_\pi = \lim_{T \rightarrow \infty} \frac{\mathbb{E}_\pi \left[\sum_{t=0}^{T-1} c_\pi(\mathbf{q}_t) \right]}{T} \quad (5.1)$$

where the expectation is taken with respect to the trajectory generated by $\mathbb{P}_\pi$. If the transition kernel $\mathbb{P}_\pi$ admits a unique stationary distribution $\mathbf{d}_\pi$ over the state space, then the infinite horizon average reward can be reformulated as $J_\pi = \sum_{\mathbf{q} \in \mathcal{S}} \mathbf{d}_\pi(\mathbf{q}) c_\pi(\mathbf{q})$. If a function $V_\pi : \mathcal{S} \rightarrow \mathbb{R}$ associated with a policy π is absolutely integrable, that is it satisfies:

$$\sum_{\mathbf{q}' \in \mathcal{S}} \mathbb{P}_\pi(\mathbf{q}'|\mathbf{q}) |V_\pi(\mathbf{q}')| < \infty \quad (5.2)$$

and is a solution to the Poisson's equation:

$$J_\pi + V_\pi(\mathbf{q}) = c_\pi(\mathbf{q}) + \sum_{\mathbf{q}' \in \mathcal{S}} \mathbb{P}_\pi(\mathbf{q}'|\mathbf{q}) V_\pi(\mathbf{q}'), \quad \forall \mathbf{q} \in \mathcal{S} \quad (5.3)$$

then $V_\pi(\mathbf{q})$ is defined as the relative value function associated with the policy π [102]. Since $V_\pi(\mathbf{q})$ is unique upto an additive constant, any function of the form $V_\pi(\mathbf{q}) + C$, where C is a constant is also a solution to the Poisson's equation. However, the most frequently used representation of the value function, which is also unique, is given by:

$$V_\pi(\mathbf{q}) = \mathbb{E}_\pi \left[\sum_{i=0}^{\tau_{\mathbf{q}^*}^\pi - 1} (c_\pi(\mathbf{q}_i) - J_\pi) \middle| \mathbf{q}_0 = \mathbf{q} \right] \quad (5.4)$$

where $\tau_{\mathbf{q}^*}^\pi$ represents the first time to hit state $\mathbf{q}^*$ starting from any state $\mathbf{q}$ under policy π . Hence, from definition it follows that $V_\pi(\mathbf{q}^*) = 0$. The value function associated with a state $\mathbf{q}$ represents the expected difference between the total cost and the expected total cost obtained under policy π when starting from state $\mathbf{q}$ until state $\mathbf{q}^*$ is reached for the first time. The relative state action value function $Q_\pi(\mathbf{q})$ is analogously defined as the solution

to the following equation:

$$J_\pi + Q_\pi(\mathbf{q}, \mathbf{a}) = c(\mathbf{q}, \mathbf{a}) + \sum_{\mathbf{q}' \in \mathcal{S}} \mathbb{P}(\mathbf{q}' | \mathbf{q}, \mathbf{a}) V_\pi(\mathbf{q}') \quad (5.5)$$

The state action value function $Q_\pi(\mathbf{q}, \mathbf{a})$ has a similar interpretation as state value function $V_\pi(\mathbf{q})$ except the action enacted at time 0 is $\mathbf{a}$ and not dictated by the policy π .

The goal of reinforcement learning is to determine the policy $\pi^* \in \Pi$, such that the infinite horizon average cost is minimized. That is, to solve for

$$J^* = \min_{\pi \in \Pi} J_\pi \quad (5.6)$$

where $\pi^* = \operatorname{argmin}_{\pi \in \Pi} J_\pi$. The focus of this chapter is to analyze the performance of Natural Policy Gradient in determining the optimal policy that minimizes the infinite horizon average cost.

5.2.1 Natural Policy Gradient Algorithm

Natural Policy Gradient algorithm is related to the mirror descent algorithm in the context of tabular policies. The objective of mirror descent involves minimizing the first order approximation of the average cost with KL regularizer. In the context of tabular policies, the NPG policy update we consider is of the form below:

$$\pi_{i+1}(\mathbf{a} | \mathbf{q}) \propto \pi_i(\mathbf{a} | \mathbf{q}) \exp \left(-\eta_{\mathbf{q}} \hat{Q}_{\pi_i}(\mathbf{q}, \mathbf{a}) \right) \quad (5.7)$$

where $\eta_{\mathbf{q}} > 0$ is the state dependent step size and $\hat{Q}_{\pi_i}$ is the estimate of state action value function Q_{π_i} learnt using policy evaluation algorithms. Since in the limit as $\eta_{\mathbf{q}} \rightarrow \infty$, the above update picks the action with the lowest state action value function, NPG is also considered to be a form of soft policy iteration. The magnitude of $\eta_{\mathbf{q}}$ determines the greediness of the policy.

In finite state spaces, previous literature [1], [31] generally considers a state-independent

step size η , where theoretical convergence guarantees require

$$\eta \leq \min_{\substack{\pi \in \Pi \\ \mathbf{q} \in \mathcal{S}, \mathbf{a} \in \mathcal{A}}} \frac{1}{\widehat{Q}_\pi(\mathbf{q}, \mathbf{a})}.$$

However, as we will demonstrate, in the context of unbounded instantaneous costs, such as those encountered in queuing systems with infinite buffers, the value of Q_π and its estimate $\widehat{Q}_\pi$ increases with the cardinality of the state space. This creates two issues with using a fixed step size: (i) Even in the context of finite state spaces, there are currently no clear guidelines for selecting this η , leading most practical applications to rely on hyperparameter tuning to find a value of η that achieves some level of convergence; (ii) As the state space grows larger (even if finite), the value of η required for algorithm convergence decreases, resulting in a sharp increase in the iteration complexity of NPG. In the following sections, we address how both of these issues can be mitigated by identifying policy-independent bounds on $\widehat{Q}_\pi$ and using them to determine a state-dependent step size $\eta_{\mathbf{q}}$. This approach yields an NPG iteration complexity that is independent of the state space cardinality, providing non-trivial convergence bounds for countable state spaces as well.

With the state space being infinitely large, a common approach to evaluate value functions is through linear function approximations. This simplifies the complexity from infinity to the dimension of the parameter vector, although with some loss in accuracy. A popular method involves using neural networks, where the weights act as the parameter vector and the network itself serves as the feature space. For each policy π , the estimate $\widehat{Q}_\pi$ of the state-action value function Q_π is then computed using overparametrized neural networks and samples gathered from trajectories under policy π . For further details, please see Subsection 5.3.2.

Algorithm 6 Natural Policy Gradient Algorithm

Input: $T, \pi_0 \in \Delta\mathcal{A}$

for $i = 0, 1, 2, 3, \dots, T - 1$ **do**

- 1: Generate trajectory $\{\mathbf{q}_0, \mathbf{a}_0, \mathbf{q}_1, \mathbf{a}_1, \dots, \mathbf{q}_n, \mathbf{a}_n\}$ using policy π_i .
- 2: Evaluate $\hat{Q}_{\pi_i}$ using policy evaluation algorithms (for instance through neural network linear function approximation).
- 3: Update policy as:

$$\pi_{i+1}(\mathbf{a}|\mathbf{q}) = \frac{\pi_i(\mathbf{a}|\mathbf{q}) \exp\left(-\eta_{\mathbf{q}} \hat{Q}_{\pi_i}(\mathbf{q}, \mathbf{a})\right)}{\sum_{\mathbf{a}' \in \mathcal{A}} \pi_i(\mathbf{a}'|\mathbf{q}) \exp\left(-\eta_{\mathbf{q}} \hat{Q}_{\pi_i}(\mathbf{q}, \mathbf{a}')\right)} \quad (5.8)$$

end for

Output: π_T

To aid our analysis, we make the following assumptions, which are typically met by queuing systems. The irreducibility of the Markov chain under any policy is a standard assumption in reinforcement learning. This ensures adequate exploration and visitation of all state-action pairs, which is crucial for learning policies with reasonable confidence.

Assumption 5.1. For all policies $\pi \in \Pi$, the induced Markov Chain $\mathbb{P}_{\pi}$ is irreducible.

In countable state Markov chains, irreducibility together with positive recurrence ensures the existence of the stationary distribution which aids in the proof of convergence of NPG. The next assumption ensures that the underlying Markov chain is positive recurrent (see Lemma D.1).

Assumption 5.2. There exists a function $f : \mathcal{S} \rightarrow [0, \infty)$ and constants $\epsilon > 0, g, D$ independent of policy π such that for every policy $\pi \in \Pi$ and every $\mathbf{q} \in \mathcal{S}$,

1. The drift equation

$$\mathbb{E}_{\pi} \left[f^2(\mathbf{q}_{k+1}) - f^2(\mathbf{q}_k) | \mathbf{q}_k = \mathbf{q} \right] \leq -\epsilon \bar{c}(\mathbf{q}) + g. \quad (5.9)$$

is satisfied.

2. Single step transitions are uniformly bounded, i.e.,

$$|f(\mathbf{q}') - f(\mathbf{q})| \leq D \quad \forall \mathbf{q}' \in \mathcal{S} : \mathbb{P}_\pi(\mathbf{q}'|\mathbf{q}) > 0. \quad (5.10)$$

3. The set

$$B := \left\{ \mathbf{q} \in \mathcal{S} : \underline{c}(\mathbf{q}) \leq \frac{2g}{\epsilon} \right\}, \quad (5.11)$$

is finite and $f(\mathbf{q}) > 0$ if $\mathbf{q} \in B^c$.

We will call the function f in the above assumption a Lyapunov function. In addition to ensuring positive recurrence, the drift equation (5.9) gives a uniform bound on the average cost of any policy.

It turns out that we also need policy independent bounds on the value function, which is ensured by our next assumption.

Assumption 5.3. We assume that there exist constants T_B and p_B , independent of policy π , such that

$$\mathbb{P}_\pi^{T_B}(\mathbf{q}'|\mathbf{q}) \geq p_B \quad \forall \mathbf{q} \in B, \forall \mathbf{q}' \in B, \forall \pi \in \Pi, \quad (5.12)$$

where $\mathbb{P}_\pi^{T_B}$ is the T_B -step probability transition matrix.

Equation (5.12) requires that any state $\mathbf{q} \in B$ can be reached from any state $\mathbf{q}' \in B$ in at most T_B transitions with atleast p_B probability under any policy $\pi \in \Pi$. Equation 5.10 states that it is not possible to move arbitrarily far away from the current state in a single transition under any policy.

5.3 Main Result and Discussion

We now present the main result, which is the performance of NPG in the context of infinite state MDPs within the learning framework. We then contextualize Assumptions 5.1, 5.2 and 5.3 and elaborate on how they can be satisfied in the context of queuing systems.

5.3.1 Main Result

Theorem 5.1. Consider the sequence of policies $\pi_1, \pi_2, \dots, \pi_T$ obtained from Algorithm 6 with a state-dependent step size $\eta_{\mathbf{q}} = \sqrt{\frac{8 \log |\mathcal{A}|}{T}} \frac{1}{M_{\mathbf{q}}}$, where

$$M_{\mathbf{q}} = \left(2\delta(\mathbf{q}) + \frac{2}{\epsilon} f^2(\mathbf{q}) + \frac{4D}{\epsilon} + \bar{c}(\mathbf{q}) + g_1 \right)$$

and $\delta(\mathbf{q}) := \sup_{\pi \in \Pi} \left\| \widehat{Q}_{\pi}(\mathbf{q}, \mathbf{a}) - Q_{\pi}(\mathbf{q}, \mathbf{a}) \right\|_{\infty}$. Let J_{π_k} be the average cost associated with policy π_k and let J_* be the minimum average cost across policy class Π . Let the learning error satisfy the following:

$$\mathbb{E}_{\pi} [\delta(\mathbf{q})] \leq \kappa(\mathbf{q}) \quad \forall \mathbf{q} \in \mathcal{S}, \pi \in \Pi \quad (5.13)$$

Then, under Assumptions 5.1, 5.2 and 5.3, there exist constants c', c'' not depending on T or $\pi_1, \pi_2, \dots, \pi_T$ such that:

$$\sum_{k=1}^T \mathbb{E} (J_{\pi_k} - J_*) \leq c' \sqrt{T} + c'' T \quad (5.14)$$

where $c' = \sqrt{\frac{\log |\mathcal{A}|}{2}} \left(2\beta + \beta_1 + \beta_2 + \frac{g}{\epsilon} + g_1 \right)$, $c'' = 2\beta$, $\beta := \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} [\kappa(\mathbf{q})]$, $\beta_1 = \frac{4D}{\epsilon} \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} [f(\mathbf{q})]$, $\beta_2 = \frac{2}{\epsilon} \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} [f^2(\mathbf{q})]$ and $g_1 = \frac{2D^2}{\epsilon} + (K + C_B + \frac{g}{\epsilon}) \left(\frac{T_B}{P_B} \right)$.

Proof. The proof is in Appendix D.4. An outline is provided in Section 5.4. □

5.3.2 Discussion on Assumptions

In this section we discuss how the assumptions can be satisfied in the context of stochastic networks, a broad class of applications. We focus on three main categories of these applications: (i) large but finite state spaces, (ii) countable state spaces with abandonments, and (iii) scheduling in switches.

Finite but large state spaces

Consider MDPs with finite state and action spaces, where the state $\mathbf{q} \in \{0, 1, \dots, S\}^K$ is a vector of length K , with each element representing the number of jobs in the corresponding queue. Here, S denotes the buffer size, making it a finite-state problem. The instantaneous

costs are frequently modeled as linear in $\|\mathbf{q}\|$, so both $\bar{c}(\mathbf{q}) = O(\|\mathbf{q}\|)$ and $\underline{c}(\mathbf{q}) = O(\|\mathbf{q}\|)$, given that the number of actions is finite. In these applications, choosing $f(\mathbf{q}) = \|\mathbf{q}\|_1$ automatically satisfies Assumption 5.2. Due to finiteness of the state space, choosing a sufficiently large g ensures that Equation (5.9) is trivially satisfied.

If the policy and transition kernels ensure a non-zero probability of no job arrivals and no departures across the policy class (which is typical in most queuing systems), it is possible to transition from any state to $\mathbf{q}^*$ (which corresponds to the zero state $\mathbf{0}$, representing empty queues) and from $\mathbf{q}^*$ to any other state. This guarantees irreducibility as per Assumption 5.1 and satisfies Assumption 5.3. Since we are working with finite Markov chains, irreducibility is sufficient to ensure the existence of a unique stationary distribution.

In previous literature that utilized a state-independent fixed step size η , the theoretical bound for η is:

$$\eta \propto \min_{\substack{\pi \in \Pi \\ \mathbf{q} \in \mathcal{S}, \mathbf{a}, \mathbf{a}' \in \mathcal{A}}} \frac{1}{|\hat{Q}_\pi(\mathbf{q}, \mathbf{a}) - \hat{Q}_\pi(\mathbf{q}, \mathbf{a}')|} \quad (5.15)$$

It is practically not possible to estimate the value η from Equation (5.15). Hence, a broad hyperparameter tuning without any guideline was necessary. For instance, consider the case of perfect policy evaluation, that is $Q_\pi(\mathbf{q}, \mathbf{a}) = \hat{Q}_\pi(\mathbf{q}, \mathbf{a})$. Then Equation (5.15) suggests,

$$\eta \propto \min_{\substack{\pi \in \Pi \\ \mathbf{q} \in \mathcal{S}, \mathbf{a}, \mathbf{a}' \in \mathcal{A}}} \frac{1}{|Q_\pi(\mathbf{q}, \mathbf{a}) - Q_\pi(\mathbf{q}, \mathbf{a}')|} \quad (5.16)$$

Our theory suggests (and later experimentally verified) that $Q_\pi(\mathbf{q}, \mathbf{a}) = O(\|\mathbf{q}\|^2)$ when $c(\mathbf{q}) = O(\|\mathbf{q}\|)$. Hence this leads to $\eta \propto \frac{1}{\|\mathbf{q}_{\max}\|^2}$. Hence as the size of the state space increases, the value of η reduces, vastly increasing the iteration complexity of NPG as the size of the state space increases.

In prior literature [1], [31] the assumption over policy evaluation error is a high probability bound as below:

$$\sup_{\pi \in \Pi} \|Q_\pi - \hat{Q}_\pi\|_{d_\pi} \leq \epsilon \quad (5.17)$$

where $\epsilon > 0$ is a constant. From this, a very loose upper bound on

$\sup_{\pi \in \Pi} \|Q_\pi(\mathbf{q}, \cdot) - \hat{Q}_\pi(\mathbf{q}, \cdot)\|_\infty$ is obtained by assuming a lower bound on $d_\pi(\mathbf{q})$ and in

theory, the stepsize η has to be chosen inversely proportional to this quantity. However, this stepsize can lead to very slow convergence as the size of the state space increases. Potentially, one can search for the best stepsize by treating it as a hyperparameter and tuning it experimentally. However, there are no easy guidelines for this hyperparameter tuning.

On the other hand, the assumption on policy evaluation in Theorem 5.1 models the value function estimation error as

$$\sup_{\pi \in \Pi} \left\| Q_{\pi}(\mathbf{q}, \cdot) - \widehat{Q}_{\pi}(\mathbf{q}, \cdot) \right\|_{\infty} \leq \delta(\mathbf{q}). \quad (5.18)$$

The value function error consists of two parts: one a function approximation error and another a learning error associated with learning the parameters of the function approximation. First, let us consider the tabular case, i.e., one where the value function is directly estimated for each (state, action) pair without using function approximation. Then, our bounds on the value function indicate an upper bound on Q which is quadratic in state $\|\mathbf{q}\|$. This bound can thus be leveraged to ensure that $\delta(\mathbf{q}) \approx \|\mathbf{q}\|^2$. Moreover, since the approach to obtaining performance bounds of NPG in prior literature [1], [31] does not explicitly characterize upper bounds on the solution to the Poisson's equation, the provable error bounds i.e., ϵ in Equation (5.17) is thus agnostic of the structure of the state action value function and consequently loose.

When dealing with learning using linear function approximations, since Lemma D.7 indicates $Q_{\pi}(\mathbf{q}, \mathbf{a}) \leq O(\|\mathbf{q}\|^2)$ for all $\mathbf{a} \in \mathcal{A}$ (as $f(\mathbf{q}) = \|\mathbf{q}\|_1$), choosing a feature space representation Φ , where the largest element of $\phi(\mathbf{q}, \mathbf{a})$ is quadratic in $\|\mathbf{q}\|$ yields a learning error $\delta(\mathbf{q}) \leq O(\|\mathbf{q}\|^2)$. Since all moments of $\|\mathbf{q}\|$ exist, this provides a guideline regarding the choice of a state dependent step size i.e., $\eta(\mathbf{q}) \propto \frac{1}{\|\mathbf{q}\|^2}$. The choice of η in prior literature explicitly relied on the knowledge of $\sup_{\pi \in \Pi} \max_{\substack{\mathbf{q} \in \mathcal{S} \\ \mathbf{a} \in \mathcal{A}}} \widehat{Q}_{\pi}(\mathbf{q}, \mathbf{a})$, hence necessitating a broad hyperparameter search without any prior knowledge to aid with this search.

In the context of really large spaces, a powerful tool employed to approximate Q functions are large scale neural networks, which can be potentially utilized to learn in countable state spaces as well (as elaborated in Section 5.3.2). Since neural networks of sufficient width can approximate continuous functions arbitrarily well, $\delta(\mathbf{q}) \approx O(\|\mathbf{q}\|^2)$ is a reasonable

error bound as $Q_\pi(\mathbf{q}, \mathbf{a}) \leq O(\|\mathbf{q}\|^2)$ for all $\pi \in \Pi, \mathbf{q} \in \mathcal{S}, \mathbf{a} \in \mathcal{A}$.

Countable State Spaces with Abandonments

Abandonments occur when the wait time for service of a job is too long. For instance, in two sided queuing systems, such as those encountered in ride hailing apps such as Uber/Lyft, a person might leave a queue if not serviced sufficiently quickly. Suppose that at each time instant, an individual abandons the queue independently of others with probability ν . In this case, Equation (5.9) exhibits a strong negative drift as the queue length grows. This can be demonstrated by choosing the Lyapunov function $f(\mathbf{q}) := \|\mathbf{q}\|^2$, similar to the finite state case. The reasoning is that as the queue length increases, the likelihood of abandonments rises accordingly. Thus, in the presence of abandonments and with bounded arrivals and departures within a single time slot, Assumption 5.2 is naturally satisfied. Assumption 5.1 and Assumption 5.3 are satisfied as long as there is a non-zero probability of no job arrivals and no service, in a fashion identical to as described in the finite state setting.

Scheduling in Switches

Switch scheduling is encountered in a wide array of applications such as wireless networks, cloud computing, data centres, etc. Here the state space denotes a vector of job lengths corresponding to different queues. The action space in such a setting corresponds to different possible bipartite matchings from the input queues to the output queues. In such a scenario, the cost associated with a state is independent of the matching chosen and hence can be modeled as $c(\mathbf{q}) := \|\mathbf{q}\|_1$. Consequently the Lyapunov function chosen is also identical i.e., $f(\mathbf{q}) = \|\mathbf{q}\|_1$. As in most applications, the number of jobs that can arrive and depart in a single time instant is uniformly bounded. In such systems, the drift Equation (5.9) in Assumption 5.2 can be satisfied as described below.

In a large class of queuing systems, the MaxWeight policy is known to ensure stability, i.e., positive recurrence (see Chapter 4, [91]). Assumption 5.2 is inspired by the so-called MaxWeight policy, which is known to satisfy the drift equation below:

$$\mathbb{E}_{\pi_{\text{MW}}} \left[\|\mathbf{q}_{k+1}\|^2 - \|\mathbf{q}_k\|^2 | \mathbf{q}_k = \mathbf{q} \right] \leq -\epsilon \|\mathbf{q}\|_1 + d_1 \quad (5.19)$$

where the expectation is taken with respect to π_{MW} and ϵ, d_1 are some positive constants independent of policy. Assumption 5.2 is designed so that we explore a family of randomized policies that inherit stability from MaxWeight, while also enabling us to learn policies that outperform MaxWeight.

In particular, we consider policies obtained by using a combination of MaxWeight and arbitrary randomized actions by transforming the underlying MDP as follows. Let the policies obtained from update Equation 5.8 be referred to as π_{NPG} . Modify the underlying MDP such that the probability transition kernel corresponds to a policy π defined below:

$$\pi(a|\mathbf{q}) = \begin{cases} \pi_{\text{NPG}}(a|\mathbf{q}), & \text{w.p. } \min\left(1, \frac{1}{\lambda\|\mathbf{q}\|}\right) \\ \pi_{\text{MW}}(a|\mathbf{q}), & \text{w.p. } 1 - \min\left(1, \frac{1}{\lambda\|\mathbf{q}\|}\right) \end{cases} \quad (5.20)$$

where $\lambda > 0$ is a fixed parameter with a very small positive value.

As the queue length grows larger, the above transformed MDP enacts the Max-Weight policy with greater probability at higher queue lengths. The value of λ decides the threshold at which Max-Weight policy starts influencing the transition dynamics. Once queue lengths exceed $\frac{1}{\lambda}$, this soft thresholding compromises some optimality to prioritize stability. This differs from the hard thresholding approach taken in [103].

We will now illustrate that this family of soft-thresholded policies satisfy Assumption 5.2. First note that from the bounded arrivals and departures assumption in Equation (5.10), it is easy to show that π_{NPG} satisfies

$$\mathbb{E}_{\pi_{\text{NPG}}} \left[\|\mathbf{q}_{k+1}\|^2 - \|\mathbf{q}_k\|^2 | \mathbf{q}_k = \mathbf{q} \right] \leq d_2 \|\mathbf{q}\|_1 + d_3 \quad (5.21)$$

where the expectation is taken with respect to π_{NPG} and d_2, d_3 are some positive constants independent of policy. Thus the drift equation corresponding to policy π in Equation 5.20

is as follows:

$$\mathbb{E}_\pi \left[\|\mathbf{q}_{k+1}\|^2 - \|\mathbf{q}_k\|^2 | \mathbf{q}_k = \mathbf{q} \right] \leq \begin{cases} \frac{d_2}{\lambda} + d_3, & \|\mathbf{q}\|_1 \leq \frac{1}{\lambda} \\ \frac{1}{\lambda \|\mathbf{q}\|_1} (d_2 \|\mathbf{q}\|_1 + d_3) + \left(1 - \frac{1}{\lambda \|\mathbf{q}\|_1}\right) (-\epsilon \|\mathbf{q}\|_1 + d_1), & \|\mathbf{q}\|_1 > \frac{1}{\lambda} \end{cases} \quad (5.22)$$

Combining the cases in Equations 5.22, we obtain the following drift relation for policy π for all $\mathbf{q} \in \mathcal{S}$:

$$\mathbb{E}_\pi \left[\|\mathbf{q}_{k+1}\|^2 - \|\mathbf{q}_k\|^2 | \mathbf{q}_k = \mathbf{q} \right] \leq -\epsilon \|\mathbf{q}\|_1 + D_1 \quad (5.23)$$

where D_1 is a constant independent of policy π but is a function of constants d_1, d_2, d_3, ϵ and λ . Note that the constant ϵ remains the same in both (5.19) and (5.23). This constitutes one such class of policies that satisfies the required the drift equation (5.9) for our analysis.

Policy Evaluation in Stochastic Networks with Countable States

The theory for performing policy evaluation using learning is not well developed for countable state spaces. We present some speculative ideas in this regard, but this requires considerable further work which is beyond the scope of this chapter. However, in practice, there has been experimental work for countable state spaces, which seems to indicate learning based control is possible [100].

It is a well-known fact that neural networks with at least one hidden layer of sufficient width and a non-linear activation function can approximate any continuous function on a compact domain arbitrarily well [104]–[106]. A potential technique to evaluate value functions associated with infinite state spaces can be through neural network temporal difference learning. In order to do so, consider the following transformation to compactify the domain of the problem. Recall that the system comprises of K queues that is, $\mathbf{q} \in \mathbb{N}^K$. Let q_i represent the number of jobs in the i^{th} queue. Then define a vector $\mathbf{x} \in [0, 1]^K$ such that the i^{th} element is $x_i = \frac{1}{1+q_i}$. Given a policy π , consider a linear function approximation

$\widehat{Q}_\pi(\mathbf{q}, a)$ of the state-action value function $Q_\pi(\mathbf{q}, a)$ as below:

$$\frac{\widehat{Q}_\pi(\mathbf{q}, a)}{\|\mathbf{q}\|^2} = \theta_\pi^\top \phi(\mathbf{x}(\mathbf{q}), a) \quad (5.24)$$

where the feature vector ϕ is defined as below,

$$\phi(\mathbf{x}(\mathbf{q}), a) = \begin{bmatrix} \mathbb{I}_{w_1^\top(\mathbf{x}(\mathbf{q}), a) \geq 0}(\mathbf{x}(\mathbf{q}), a) \\ \vdots \\ \mathbb{I}_{w_m^\top(\mathbf{x}(\mathbf{q}), a) \geq 0}(\mathbf{x}(\mathbf{q}), a) \end{bmatrix}. \quad (5.25)$$

Here, $w_i \sim \mathcal{N}(0, I)$ and $I \in \mathbb{R}^{(K+1) \times (K+1)}$ is the identity matrix. This linearized model is well-studied approximation to a neural network and is called the Neural Tangent kernel (NTK) approximation; see [107], for example. We will not discuss the merits of the NTK approximation here since that is irrelevant to our analysis, but we only introduce the NTK approximation to discuss why we chose our model for function approximation. In the NTK approximation, $w_i \in \mathbb{R}^{K+1}$ is random initialization which chooses a random set of features. Each feature vector $\phi(\mathbf{x}(\mathbf{q}), a)$ is of length $m|\mathcal{A}|K$, where m represents the width of the hidden layer in the neural network. Finally, θ_π^* represents the optimal parameter vector, i.e., the parameter that best estimates $Q_\pi(\mathbf{q}, a)$.

The state action value function $Q_\pi(\mathbf{q}, a)$ can be approximated arbitrarily well if Q_π is a continuous function. This is indeed the case for some simple contexts such as the M/M/1 queue, where the value function is a quadratic function in queue length ([108]). More generally, Equation (5.37) indicates that the $Q_\pi(\mathbf{q}, a)$ can be upper bounded by a quadratic function. Therefore, under the assumption that Q_π is continuous, the learning error due to policy evaluation using the neural network can be characterized as follows:

$$\|Q_\pi(\mathbf{q}, a) - \widehat{Q}_\pi(\mathbf{q}, a)\| = \|Q_\pi(\mathbf{q}, a) - \theta_\pi^\top \phi(\mathbf{x}(\mathbf{q}), a)\| \|\mathbf{q}\|^2 \quad (5.26)$$

$$\leq \left\| Q_\pi(\mathbf{q}, a) - \theta_\pi^{*\top} \phi(\mathbf{x}(\mathbf{q}), a) \right\| \|\mathbf{q}\|^2 \quad (5.27)$$

$$+ \left\| \theta_\pi^{*\top} \phi(\mathbf{x}(\mathbf{q}), a) \|\mathbf{q}\|^2 - \theta_\pi^\top \phi(\mathbf{x}(\mathbf{q}), a) \|\mathbf{q}\|^2 \right\| \quad (5.28)$$

The function approximation error is captured in Equation (5.27) as follows:

$$\left\| Q_\pi(\mathbf{q}, \mathbf{a}) - \theta_\pi^{*\top} \phi(\mathbf{x}(\mathbf{q}), \mathbf{a}) \right\| \|\mathbf{q}\|^2 = \left\| \frac{Q_\pi(\mathbf{q}, \mathbf{a})}{\|\mathbf{q}\|^2} - \theta_\pi^{*\top} \phi(\mathbf{x}(\mathbf{q}), \mathbf{a}) \right\| \|\mathbf{q}\|^2 \leq \delta_1(m) \|\mathbf{q}\|^2 \quad (5.29)$$

where $\delta(m)$ is a constant that is independent of the underlying policy but depends on the width of the hidden layer. In fact, it is shown in [109] that when approximating polynomials, as $m \rightarrow \infty$, $\delta_1(m) \rightarrow 0$. The temporal difference (TD) learning error is captured in Equation (5.28) and is a function of the number of samples available and can be quantified as $\delta_2 \|\mathbf{q}\|^2$ with high probability. And thus, with high probability, the overall state dependent error can be quantified as follows:

$$\|Q_\pi(\mathbf{q}, \mathbf{a}) - \hat{Q}_\pi(\mathbf{q}, \mathbf{a})\| \leq \delta \|\mathbf{q}\|^2 \quad (5.30)$$

where $\delta = \delta_1(m) + \delta_2$.

5.4 Proof outline and Key Insights

The difference in average cost associated with a policy π and the optimal average cost is linked to the Q_π function through the performance difference lemma ([110]) as below:

$$J_\pi - J^* = \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} \left[Q_\pi(\mathbf{q}, \pi(\mathbf{q})) - Q_\pi(\mathbf{q}, \pi^*(\mathbf{q})) \right]. \quad (5.31)$$

Hence, the regret in LHS of Equation (5.14) can be captured in terms of difference in the state action value function Q_π . However, in practise it is not possible to determine Q_π exactly since the model might be unknown or the state space is infinite. Hence, we incorporate the estimates $\hat{Q}_\pi$ of the value function Q_π . If the estimates satisfy Equation (5.13), then from Equation (5.31) we obtain the following regret formulation:

$$\sum_{k=1}^T \mathbb{E} (J_{\pi_k} - J^*) \leq \underbrace{2T \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} \kappa(\mathbf{q}) + \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} \left[\mathbb{E} \left(\sum_{k=1}^T \hat{Q}_{\pi_k}(\mathbf{q}, \pi^*(\mathbf{q})) - \hat{Q}_{\pi_k}(\mathbf{q}, \pi_k(\mathbf{q})) \right) \right]}_{(a)} \quad (5.32)$$

The term linear in T , i.e., $2\mathbb{E}_{\mathbf{q} \sim \mathbf{d}^{\pi^*}} \kappa(\mathbf{q})$ is a consequence of function approximation and is generally unavoidable [31]. The primary task is to bound (a) in Equation (5.32). We approach this in four steps: (i) examining the link between NPG and prediction through expert advice as highlighted in prior literature, and identifying challenges specific to our countable state-space model and cost structure, (ii) deriving policy-independent bounds on the value functions, i.e., the solution to Poisson's Equation (5.3), (iii) accounting for policy evaluation errors and establishing policy-independent bounds on the estimates of the value function, and (iv) integrating all these steps to achieve the final result. We now proceed with the proof outline.

5.4.1 Step 1 (Connection to Weighted Averaging):

This step involves connecting learning within Markov Decision Processes (MDPs) to prediction through expert advice. This connection was initially identified in [55] for MDPs and later extended to the learning setting in [31]. We now discuss this connection in some detail and explain why we need our proof techniques to adapt this connection to the countable state-space setting. In the framework of prediction through expert advice, the agent selects an action a_t at time t , and the environment responds with a corresponding loss $l_t(a_t)$. Concurrently, an expert follows a predetermined strategy, which in our context can be simplified to a single action a^* taken at each time step, also experiencing a loss of $l_t(a^*)$. The agent's objective is to minimize the overall loss by considering all its past observations when choosing an action. If the expert opts for a fixed strategy π^* over the available actions, the following holds true.

Theorem 5.2. (Section 4.2, Corollary 4.2, [111].) *Consider the exponentially weighted average forecaster problem. Let the set of actions possible at each time step and each instance be denoted by $\mathcal{A} := \{1, \dots, n\}$. For a fixed instance s , let $l_t(s, i)$ be the loss associated with action $i \in \mathcal{A}$ at time t such that for any pair of actions $i, i' \in \mathcal{A}$,*

$$|l_t(s, i) - l_t(s, i')| \leq M(s) \quad (5.33)$$

Consider the action strategy below:

$$\pi_t(i|s) = \frac{\pi_{t-1}(i|s) \exp(-\eta_s l_{t-1}(s, i))}{\sum_{k=1}^n \pi_{t-1}(k|s) \exp(-\eta_s l_{t-1}(s, k))} \quad (5.34)$$

Then, for any fixed policy π^* , setting $\eta_s = \sqrt{\frac{8 \log n}{T}} \frac{1}{M(s)}$ yields the following overall regret corresponding to instance s .

$$\sum_{k=1}^T \left(l_k(s, \pi_k(s)) - l_k(s, \pi^*(s)) \right) \leq M(s) \sqrt{\frac{T \log n}{2}} \quad (5.35)$$

The NPG algorithm can be interpreted as applying the weighted averaging algorithm to each state $\mathbf{q}$ in the state space, with the goal of learning the optimal policy for each state. In this context, the loss function associated with an action a in state $\mathbf{q}$ at time k is the estimate $\hat{Q}_{\pi_k}(\mathbf{q}, a)$ of the state-action value function, where the policy in use at time k is π_k . However, as indicated by Equation (5.33), the loss function— $\hat{Q}_{\pi_k}(\mathbf{q}, a)$ —must be bounded for any given state $\mathbf{q}$. In finite-dimensional MDPs, a state-independent uniform bound on the state-action value function is typically assumed [31]. This is due to the fact that the step-size η is assumed to be independent of s . Note that, compared to [31], [55], we have made a small, but critical, change to the best-experts algorithm by allowing the step-size η to be a function of s . When the state-space is countable, the state-action value function Q_π cannot be uniformly bounded and hence, a constant step-size cannot be assumed. With the introduction of a state-dependent step-size, we can choose a different step-size for each state using bounds on the solution to Poisson's equation, i.e., $Q_\pi(s, a)$, which depends on the state, but is uniform over all policies. Obtaining such bounds is one of the key contributions of the chapter.

5.4.2 Step 2 (Value Function Bounds):

To establish bounds on Poisson's Equation 5.5, we initially rely on Assumptions 5.1 and 5.2. In dealing with countable state space MDPs, along with irreducibility, we require the Markov chain to be positive recurrent for a unique stationary distribution to exist. The drift equation 5.9 along with the rest of Assumption 5.2 ensures the positive recurrence

of the underlying Markov chain. Since Q_π is related to the state value function V_π (see Equation 5.5), we initially constrain V_π using Assumptions 5.1 and 5.2. This leads to an upper bound on $V_\pi(\mathbf{q})$ for all $\mathbf{q} \in \mathcal{S}$,

$$V_\pi(\mathbf{q}) \leq \frac{2}{\epsilon} f^2(\mathbf{q}) + \underbrace{O\left(\mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi} \mathbb{I}(\mathbf{q}_k \in B) \mid \mathbf{q}_0 = \mathbf{q} \right]\right)}_{(b)}, \quad (5.36)$$

where B is defined in Equation 5.11. Recall Equation (5.33) in the context of weighted expert averaging. The constraint on the loss function's bound ($M(s)$) must be independent of time. When applied to the NPG framework, this implies the necessity of a policy-independent upper bound on the state-action function Q_π , which, in turn, necessitates a policy-independent bound on the state value function V_π . For (b) to be well-defined, the drift alone is insufficient, as indicated in previous studies [99], [102]. Addressing this is the second challenge in our analysis, which we navigate by introducing a mild structural Assumption 5.3 commonly satisfied in stochastic networks.

These structural assumptions yield a uniform upper bound on the time spent inside B before hitting state $\mathbf{q}^*$, defined in Equation (D.13), when starting from any point within B . This uniform upper bound on hitting time aids in bounding the state value function V_π from below. The drift inequality (5.9) along with a bound on this hitting time assists in bounding the value function V_π from above. As a consequence, we obtain the following,

$$|Q_\pi(\mathbf{q}, a) - Q_\pi(\mathbf{q}, a')| \leq O(f^2(\mathbf{q})) \quad \forall \pi \in \Pi, \forall a, a' \in \mathcal{A} \text{ and } \forall \mathbf{q} \in \mathcal{S} \quad (5.37)$$

As a result, we establish policy-independent bounds on the value function Q_π . While the drift assumption 5.2 played a crucial role in deriving policy-dependent bounds on the value function V_π , for the purpose of NPG, we need these bounds to be independent of the policy. The structural assumption 5.3 eliminates this policy dependence. Consequently, from Equation 5.5, this translates into policy-independent bounds on Q_π .

5.4.3 Step 3 (Handling Estimation Errors):

Since our loss function in the context of Theorem 5.2 is $\widehat{Q}_\pi$, we need uniform bounds on $\widehat{Q}_\pi$. We leverage the bounds on Q_π obtained in Equation 5.37 and in conjunction with the evaluation error as modeled in Theorem 5.1, we obtain the following:

$$\left| \widehat{Q}_\pi(\mathbf{q}, \mathbf{a}) - \widehat{Q}_\pi(\mathbf{q}, \mathbf{a}') \right| \leq O(f^2(\mathbf{q})) + O(\delta(\mathbf{q})) \quad \forall \pi \in \Pi, \forall \mathbf{a}, \mathbf{a}' \in \mathcal{A} \text{ and } \forall \mathbf{q} \in \mathcal{S} \quad (5.38)$$

Adapting Equation 5.33 to the context of context of infinite state NPG, implies that $M_{\mathbf{q}} = O(f^2(\mathbf{q})) + O(\delta(\mathbf{q}))$.

5.4.4 Step 4 (Piecing it all together):

The upper bound $M_{\mathbf{q}}$ on $\widehat{Q}_\pi$ in Step 3 is utilized to determine the state dependent step size as $\eta_{\mathbf{q}} = \sqrt{\frac{8 \log |\mathcal{A}|}{T} \frac{1}{M_{\mathbf{q}}}}$. With bounds over $\widehat{Q}$ quantified in Equation 5.38, (a) of Equation (5.32) is upper bounded by leveraging the connection to the prediction through expert advice Theorem 5.2. This yields the final result.

The detailed proof of all steps and the main theorem can be found in Appendix D.

5.5 Simulations

In this section, we empirically evaluate the performance of the algorithmic change proposed in the convergence of natural policy gradient. We consider tabular policies and finite state spaces. Motivated by autoscaling in cloud computing, we consider the following two settings.

5.5.1 Single Queue System

Setting

We consider a single queue system of finite buffer size B . Jobs arrive as a Poisson process with rate $\Lambda = 0.45$. There are two service rates $\mu_1 = 0.5$ and $\mu_2 = 0.8$ available, where time taken to service a job under μ_i is distributed as $\text{Exp}(\mu_i)$. The state space of this system

corresponds to the number of jobs q in the buffer, waiting to be serviced. The action a at each state is the choice of the service rate. Hence $|\mathcal{S}| = B + 1$ and $\mathcal{A} = 2$. The policy is a probability vector over these two actions corresponding to each state. The discrete time probability transition matrix is the one obtained through uniformization of the CTMC corresponding to this system. The instantaneous cost at state q under action $a = \mu_i$ is $c(q, a) = q + c_i$, where $c_1 = 1$ and $c_2 = 10$. Utilizing a higher service rate incurs a higher cost but ensures faster job completion, thereby reducing the overall queue length.

Policy Evaluation

We use the $TD(\lambda)$ algorithm to evaluate the state-action value function Q_π for each policy. For further details on the algorithm, please refer to [34]. First, we generate a trajectory of length n according to the transition kernel described earlier. The average cost is estimated by averaging the instantaneous costs obtained from the trajectory, and this estimate is then used to evaluate the state-action value function Q . In these simulations, the learning rate is set to $\beta = 0.1$ and $\lambda = 0.95$.

Policy Improvement

The policy improvement step is as in Equation (5.8). The initial policy is chosen to be uniform across all actions. Our theory on bounding the solution to the Poisson's Equation suggests that $Q(q, a)$ is of the order of $\frac{1}{q^2}$, with constants that may depend on the problem parameters. Therefore, to test the robustness of our algorithm, we choose $\eta_q = \frac{k}{q^2}$, independent of the problem parameters. To the best of our knowledge, there are no guidelines given for how to choose a fixed step size η in prior literature. But based on our theory, since $Q_{\max}$ is of the order B^2 , where B is the buffer size, we chose $\eta = \frac{1}{B^2}$ for fixed step size NPG. Note that previously there was no guideline to even choose a fixed η in prior work, but our bounds on the solution to the Poisson's equation can be used to choose a state-independent η as analyzed in prior work.

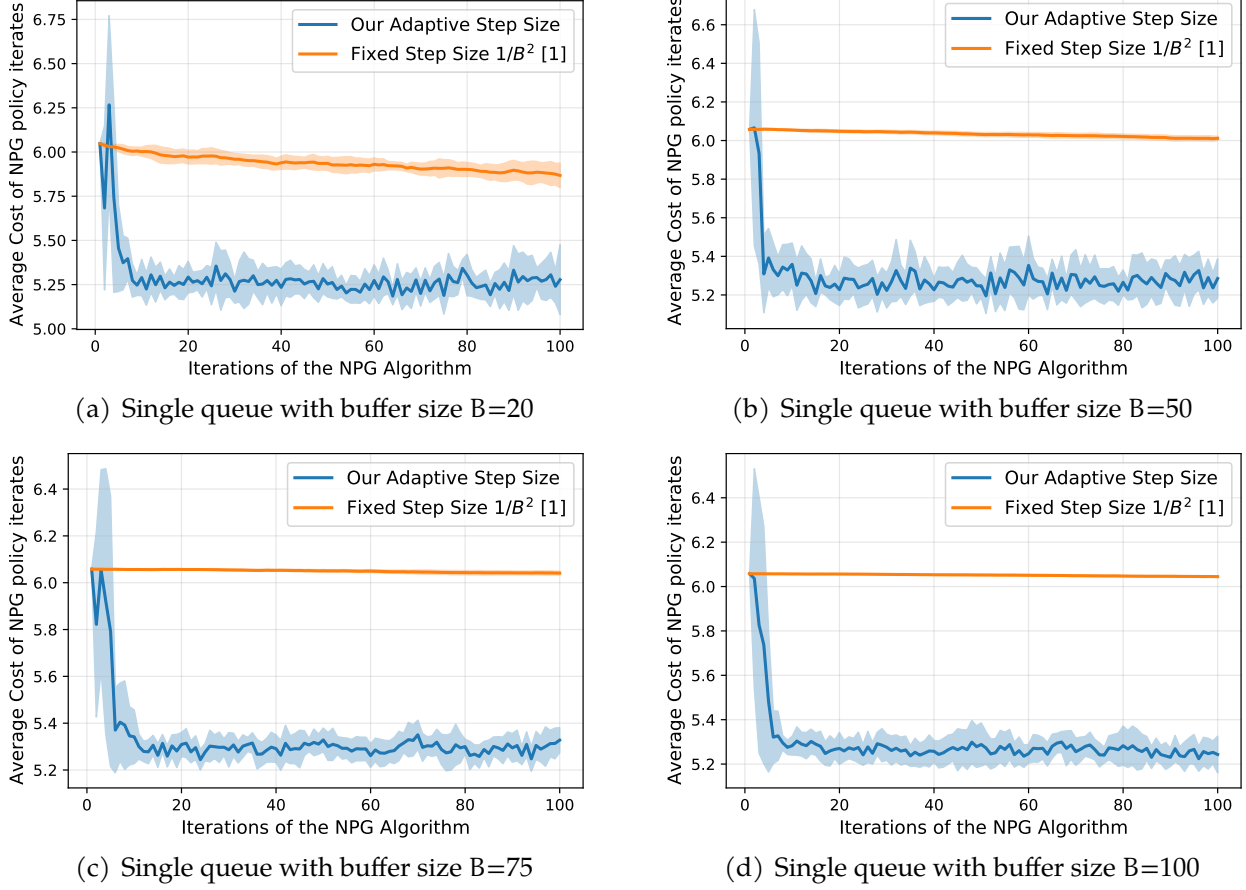


Figure 5.1: NPG in a single queue system

Observations

The simulations for this setting are depicted in Figure 5.1. The blue lines represent the performance of NPG with an adaptive step size, while the orange lines indicate the performance with a fixed step size. The y-axis represents the average cost of the policies generated through the two NPG algorithms. The x-axis represents the iteration number. Figure 5.1(a) corresponds to a queuing system with a single queue with a maximum buffer size of 20 (jobs that arrive after the buffer is full are dropped). The length of the trajectory for policy evaluation i.e., $n = 3000$. The performance is averaged over 15 runs of both algorithms. Figure 5.1(b) corresponds to a buffer capacity of 50 jobs, with $n = 5000$, averaged over 15 runs. The step size is set as $\eta_q = \frac{1}{q^2}$ for both these instances. Figure 5.1(c) corresponds to a buffer capacity of 75 jobs, with $n = 8000$, averaged over 10 runs. Similarly, Figure 5.1(d) corresponds to a buffer capacity of 100 jobs, with $n = 10000$, averaged over 15 runs. For

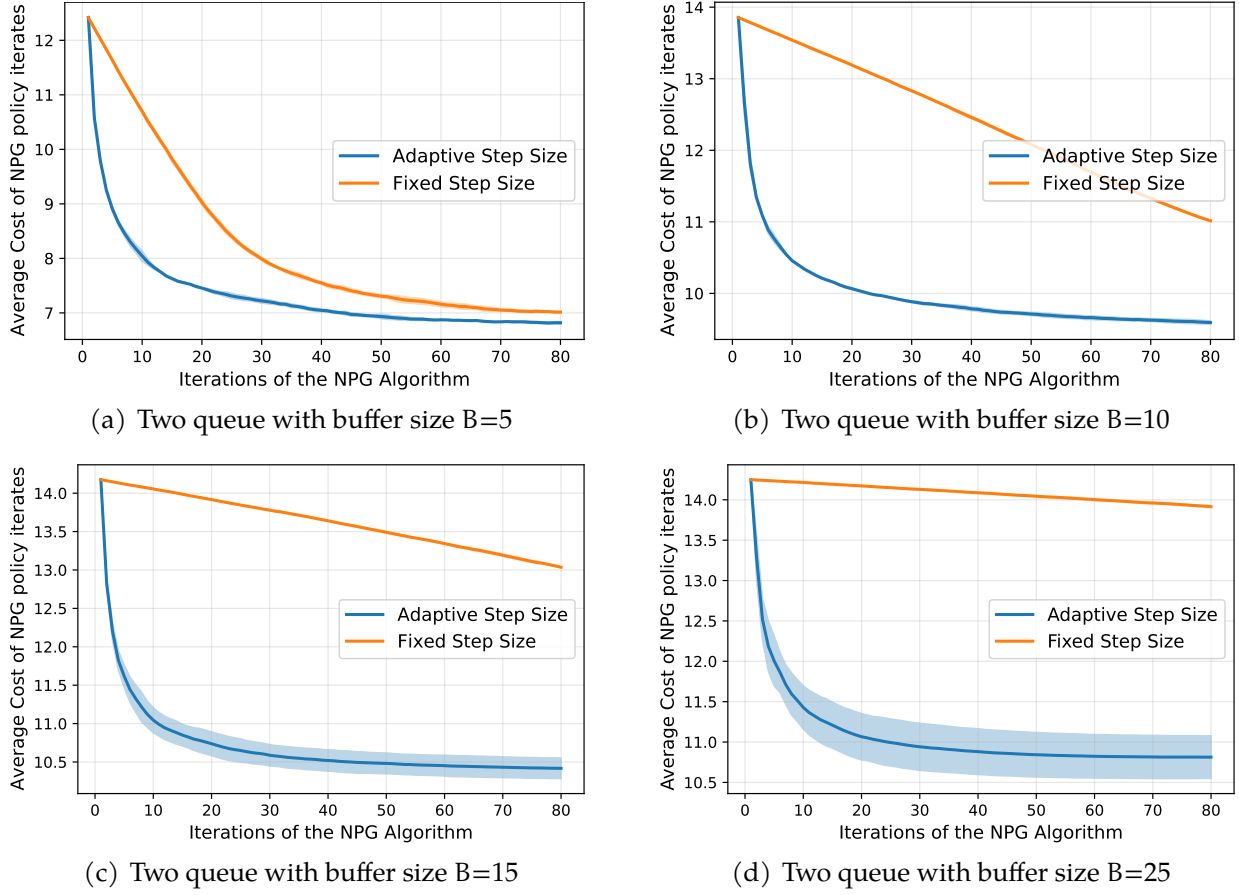


Figure 5.2: NPG in a two queue system

the latter two cases $\eta_q = \frac{0.5}{q^2}$.

5.5.2 Two Queue System

Setting

We consider a system with two queues each with buffer size B . Jobs arrive as a Poisson process at rate $\Lambda = 0.45$ and are routed according to the JSQ (join the shortest queue) policy. Each queue has two service rate options $\mu_1 = 0.25$ and $\mu_2 = 0.3$ with c_i as in Setting 1 in Section 5.5.1. The state of the system is now a vector $\mathbf{q} = (q_1, q_2)$ representing the number of jobs in both queues. The action is the choice of service rates for both queues. The cost when employing $\mathbf{a} = (\mu_i, \mu_j)$ in state $\mathbf{q}$ is $c(\mathbf{q}, \mathbf{a}) = q_1 + q_2 + c_i + c_j$. Higher service rate incurs a higher cost but ensures faster job completion.

Policy Evaluation

We use $TD(\lambda)$ algorithm for average cost MDPs as in the previous setting. We first generate a trajectory of length n , estimate the average cost and use this estimate to learn the Q function. We set the learning rate $\beta = 0.1$ and $\lambda = 0.1$.

Policy Improvement

We compare the NPG algorithm with two different learning rates η namely the adaptive stepsize and the fixed step size. The policy improvement is as in Equation (5.8) with $\eta_q = \frac{k}{(q_1 + q_2)^2}$, which is chosen based on our theory and the fixed step size is set as $\eta = \frac{1}{Q_{\max}}$. Since $Q_{\max}$ is of the order $4B^2$ where B is the buffer capacity for both queues, the fixed step size is thus chosen to be $\frac{1}{4B^2}$.

Observations

Figure 5.2(a), Figure 5.2(b), Figure 5.2(c) and Figure 5.2(d) corresponds to a buffer capacity of $(5, 5)$, $(10, 10)$, $(15, 15)$ and $(25, 25)$ jobs respectively. The length of the trajectory for policy evaluation for all four settings is $n = 1000$. The step size for the first two cases is $\eta_q = \frac{1}{(q_1 + q_2)^2}$ whereas for the latter two it is $\eta_q = \frac{0.5}{(q_1 + q_2)^2}$. The performance is averaged over 3 runs for the first three cases and over 5 runs for the last case.

5.5.3 Noiseless Setting

We also examine the case with no learning error, i.e., exact evaluation, to determine convergence rates for both step sizes in the previously described settings. Due to the NPG policy improvement update, the sequence of average costs is monotonic. In the single-queue scenario, where the optimal average cost is approximately 4.89 across all buffer sizes, we plot the number of iterations needed to for the cost to fall below 4.96. In Figure 5.3(a), the y-axis shows the number of iterations required to reach this cost threshold. In the case of two queues, with a buffer size of 5, the optimal average cost is approximately 6, increasing to 10.17 when the buffer size grows to 25. In the plots, we compare the performance of our algorithm to the fixed step size algorithm by comparing the number of iterations needed for the average cost to fall below a threshold: for a buffer size of 5, the threshold we choose

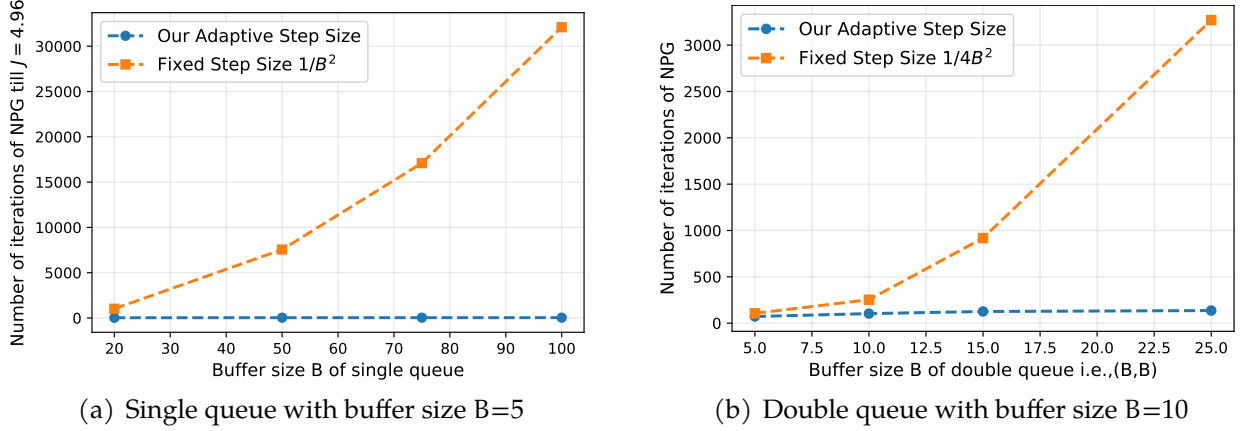


Figure 5.3: NPG with perfect policy evaluation

was 6; for a buffer size of 10, the threshold was 9.2; and for buffer sizes of 15 and 25, the threshold was set to be 10.26. In the case of single queues, using our adaptive step size $\eta_q = \frac{1}{q}$, the cutoff criterion is reached in roughly 35 iterations, regardless of buffer size. In contrast, with a fixed step size, the number of iterations required to meet the cutoff criterion increases by orders of magnitude as the buffer size grows, as shown in Figure 5.3(a). Similarly, for two queues, our adaptive step size $\eta_q = \frac{1}{q_1 + q_2}$ achieves the cutoff criterion in approximately 100 iterations, independent of the state space size. However, when using a fixed step size, the number of iterations required to reach the cutoff criterion follows the pattern illustrated in Figure 5.3(b).

5.5.4 Key Takeaways

- The NPG algorithm with the adaptive learning rate seems to converge to the near optimal policy in a state-space cardinality independent manner. The magnitude of the slope of the orange line in Figure 5.1 and Figure 5.2 reduces as the buffer size increases indicating larger number of NPG iterations as the state space grows where as an adaptive step size doesn't face this issue, thus confirming the observations from our theoretical analysis.
- The number of iterations to converge to near optimal policy is more or less similar in the context of perfect information and with learning. This suggests that the algorithm is robust to greater errors in the value function estimates of states not visited frequently

enough. Hence, the proposed learning rate accommodates realistic learning errors.

- Previous literature lacked a heuristic for selecting an effective step size for the NPG algorithm. In contrast, our analysis, based on bounds from Poisson’s Equation, offers a state-dependent rule of thumb that significantly improves upon prior step size choices and requires minimal knowledge of the specific MDP instance.

II

RISK SENSITIVE MDPs AND RL

6

MODIFIED POLICY ITERATION

If you can't solve a problem, then there is an easier problem you can solve: find it.

George Pólya

6.1 Introduction

We consider stochastic control problems over finite state and action spaces, also known as MDPs¹. Various applications, including communication systems, autonomous car navigation, and queuing systems, are expressed as MDPs for the purpose of control. Conventional methods for determining optimal policies in these MDPs typically employ dynamic programming techniques such as policy iteration, value iteration, or linear programming [112], [113], [114].

Reinforcement learning (RL) attempts to solve the control problem when the probability transition matrix is either unknown or the probability transition matrix is known but the state space is very large to obtain exact solutions [115]. A considerable portion of earlier research in this domain is centered on discounted-cost problems or average-cost MDPs, without accounting for potential distributional differences in the underlying probabil-

¹Material from our L4DC 2023 paper[6] is reproduced here without copyright conflict.

ity transition kernel that characterizes the environment in which the agent operates. In this chapter, our focus is on the robust version of the average-cost problem. While this chapter employs the KL divergence metric to model distributional uncertainties, existing literature captures robustness through various other metrics, including the Wasserstein distance [116]–[118] and state-dependent uncertainty sets [119], [120].

Robust control problems with linear state-space and quadratic costs have been well studied in the control theory literature [121], [122], [123]. It is also well-known that these robust control problems are closely related to the control of systems with a risk-sensitive exponential cost [124]. Here, we consider the finite-state, finite-action counterpart of such robust/risk-sensitive control problems [125]–[127]. Unlike the Linear Quadratic Gaussian (LQG) setting in [124], the risk-sensitive MDP does not admit a closed-form solution even when the system model is known.

Within the domain of risk-sensitive MDPs, the focus in dynamic programming has largely revolved around value iteration and policy iteration. The RL counterparts of these algorithms have also undergone thorough investigation. This chapter presents an alternative dynamic programming algorithm, termed MPI, which incorporates favorable characteristics from both value iteration and policy iteration algorithms. Value iteration involves iteratively applying the optimal Bellman operator, whereas policy iteration involves successive iterations of policy improvement followed by perfect policy evaluation. In MPI, rather than relying on perfect policy evaluation, we approximately evaluate the value function by repeatedly applying the Bellman operator corresponding to the improved policy a finite number of times.

The approximate nature of MPI can be used to obtain finite-time performance bounds for policy-based RL, where each iteration of the RL algorithm can be viewed as an approximate policy evaluation and an approximate policy improvement [128], [129]. However, such results have been obtained only for the classical, non-robust versions of MDPs. While there has been work on the asymptotic theory of RL algorithms for risk-sensitive MDPs [7], [125], [126], there has not been much work on the finite-time performance of such algorithms except in the case of value-based algorithms [130]. While we do not study RL algorithms in this chapter, we hope that the results in this chapter can be used as a first step to analyze

RL algorithms for robust MDPs.

Main Contributions:

- We introduce a new dynamic programming algorithm, referred to as modified policy iteration (MPI), within the framework of risk-sensitive exponential cost MDPs, which leverages properties of both value and policy iteration.
- We analyze the MPI algorithm and provide asymptotic as well as finite-time convergence bounds for the sequence of policies derived from this algorithm towards the globally optimal robust policy.
- The presence of partial policy evaluation means that the properties facilitating the convergence of policy and value iteration no longer apply in the context of this algorithm. Consequently, we devise novel proof techniques in order to establish the convergence analysis mentioned above. Additionally, we introduce a form of normalization within the algorithm and demonstrate its crucial role in ensuring the boundedness of consecutive value function iterates.

chapter Outline:

- Section 6.2: We discuss prior work on risk-sensitive MDPs and MPI in risk-neutral MDPs and their RL applications.
- Section 6.3: We introduce the essential preliminaries, state our model and assumptions, and present the algorithm studied in this chapter.
- Section 6.4: We present our convergence result for risk-sensitive MPI and outline its proof.
- Appendix E: We present the proof of convergence, including all the supporting lemmas used in the convergence analysis of risk-sensitive MPI.
- Appendix E.5: We present simulation results in this section.

6.2 Prior Work

In this section we discuss previous work in the domain of robust MDPs and the use of MPI in risk-neutral MDPs.

6.2.1 Risk-Sensitive MDPs

The focus of dynamic programming techniques in the context of risk-sensitive MDPs has predominantly revolved around policy and value iteration. In [131], the authors provide convergence analysis for policy and value iteration in the context of countably infinite state space and finite action space. The asymptotic bounds leverage the fact that consecutive iterations of these algorithms yield monotonically decreasing risk-sensitive average costs. [132] and [133] consider finite state-action space value iteration. The analysis in [133] depends crucially on value function iterates being related through the optimal risk-sensitive Bellman Operator. They provide asymptotic convergence bounds under relaxed assumptions, i.e., they assume a solution to the optimal multiplicative Bellman operator exists but do not explicitly rely on irreducibility of the associated Markov chains. The monotonicity associated with the optimal Bellman operator, in addition to irreducibility and aperiodicity of the Markov chain is leveraged in [132] to provide finite-time bounds for convergence of value iteration. This is achieved through a span seminorm contraction analysis with the Birkhoff coefficient serving as the contraction factor. However, neither of these analyses can be seamlessly extended to address the convergence of MPI. This limitation arises from the introduction of partial policy evaluation, leading to the breakdown of monotonicity arguments central to the approaches presented in [133] and [132]. In recent literature, [130] consider risk-sensitive value iteration in the context of finite horizon MDPs. They focus on optimizing the total reward over episodes of finite length and do not take into account the objective of the infinite horizon risk-sensitive average reward, which is the primary goal of this chapter. [134] utilizes time-dependent risk factors and derives dynamic programming techniques to solve for the risk-sensitive optimal policy when the objective is finite horizon discounted reward.

Outside the exponential risk formulation, there are several other notions of robustness

employed to account for model uncertainties in literature. In the context of discounted MDPs, a common approach is to model the ambiguity in the system parameters through state-dependent rectangular sets, where these sets are assumed to be mutually independent [119]. However, this leads to extremely conservative solutions as the adversary can leverage the worst case situation for every state. [120], [135], [136] propose alternative modeling techniques which alleviate the aforementioned conservative nature of the solution. Apart from the KL divergence metric considered in the present work, the distributional difference is also quantified through the Wasserstein metric [116], [118]. A broader framework to model robustness is provided in [117].

6.2.2 Modified Policy Iteration in risk-neutral MDPs

Recently, several papers have studied RL problems using versions of dynamic programming techniques that are computationally more tractable compared to traditional value iteration or policy iteration [137], [128]. These algorithms use two key ideas: (i) MPI: some version of policy iteration is used, where instead of exact policy evaluation, a few iterations of fixed-point iterations are performed [114], and (ii) approximate policy iteration: both the policy evaluation and the few iterations of fixed-point iterations mentioned in (i) are performed approximately [113]. As shown in [128], [137], MPI and approximate policy iteration can be used to model the concepts used in practical RL algorithms such as tree search, rollout, lookahead, and function approximation. However, all the known results in this context are for the discounted-cost or average-cost risk-neutral infinite-horizon problem.

To develop an analog of the rich theory that exists for discounted-cost problems, one has to first develop a theory for MPI and approximate policy iteration in the context of risk-sensitive exponential cost MDPs. For risk-neutral average cost problems, there exists a theory of MPI [29]. For risk-sensitive MDPs, we are unaware of any results for either MPI or approximate policy iteration. In this chapter, as a first step towards developing a theory of RL for risk-sensitive problems with known but large probability transition matrices, we define the equivalent of MPI in the case of risk-sensitive MDPs and prove that it converges. In the case of discounted-cost problems and average-cost problems, the

proof of convergence relies on the properties of the Bellman operator which is additive in nature. Our main contribution in this chapter is to develop risk-sensitive counterparts of the results in [29], which deals with the risk-neutral setting. Specifically, the fact that the Bellman operator has multiplicative terms, instead of additive terms, means that we have to deal with potential unboundedness in the iterates of the MPI algorithm. We will detail the differences in the proof techniques when we present the mathematical results later in the chapter.

6.3 Preliminaries

In this section, we introduce risk-sensitive average cost MDPs and the associated multiplicative Bellman Operator. We also discuss our assumptions and present the MPI algorithm that is studied in this chapter.

6.3.1 Risk-Sensitive Markov Decision Process

We consider an infinite horizon MDP with finite-state space $\mathcal{S}$, finite action space $\mathcal{A}$, and transition kernel $\mathbb{P}$. The class of deterministic policies is denoted by $\Pi = \{f: \mathcal{S} \rightarrow \mathcal{A}\}$, where each policy assigns an action to each state. Given a policy $f \in \Pi$, the underlying Markov process is denoted by $\mathbb{P}_f: \mathcal{S} \rightarrow \mathcal{S}$, where $\mathbb{P}_f(s'|s) := \mathbb{P}(s'|s, f(s))$ is the probability of moving to state $s' \in \mathcal{S}$ from state $s \in \mathcal{S}$ upon taking action $f(s) \in \mathcal{A}$. Associated with each state-action pair $(s, f(s))$, is a one-step cost which is denoted by $c_f(s) := c(s, f(s)) \in [\underline{c}, \bar{c}]$ where $\bar{c} := \max_{i \in \mathcal{S}, a \in \mathcal{A}} c(i, a)$ and $\underline{c} := \min_{i \in \mathcal{S}, a \in \mathcal{A}} c(i, a)$.

To motivate our risk-sensitive cost formulation in the next subsection, we first briefly recall that, in the traditional average cost MDP the average cost J_f associated with a deterministic policy $f \in \Pi$ is given by

$$J_f = \lim_{t \rightarrow \infty} \frac{1}{t} \mathbb{E} \left[\sum_{k=0}^{t-1} c_f(s_k) \right].$$

Here s_k represents the state at time k and the expectation is taken with respect to the transition probability $\mathbb{P}_f$ associated with the policy f . The limit mentioned above is well-

defined and is independent of the initial state distribution, subject to certain mild conditions on the underlying MDP [113], [114].

Let the stationary distribution over the states under policy f be represented by η_f . Equivalently, the average cost can be written in terms of the η_f as:

$$J_f = \mathbb{E}_{s \sim \eta_f} [c_f(s)].$$

However, this formulation is predicated on the assumption that the nominal transition kernel representing the underlying environment is error-free. One way to handle model uncertainty in robust MDPs is to minimise the worst-case average cost over a KL neighbourhood of the nominal model:

$$\sup_{Q: \mathbb{E}_{s \sim \eta_Q} (D_{\text{KL}}(Q(s, \cdot) \| \mathbb{P}_f(s, \cdot))) \leq \beta} \mathbb{E}_{s \sim \eta_Q} [c_f(s)],$$

where D_{KL} denotes the Kullback-Leibler divergence, η_Q represents the stationary distribution over the state space corresponding to the probability transition kernel Q and $\beta > 0$ is the radius of the KL-ball. The above objective considers the largest average cost across a ball of distributions centered around the nominal transition kernel corresponding to each policy. This is the robust MDP objective considered in this chapter. The dual formulation of the robust MDP objective is:

$$\sup_{Q \ll \mathbb{P}_f} \mathbb{E}_{s \sim \eta_Q} [c_f(s)] - \frac{1}{\alpha} \mathbb{E}_{s \sim \eta_Q} [D_{\text{KL}}(Q(s, \cdot) \| \mathbb{P}_f(s, \cdot))],$$

where the constant $\alpha = \alpha(\beta) > 0$ depends on β and $\ll$ denotes absolute continuity. Using the Donsker-Varadhan variational formula and Collatz–Wiendt formula, it can be shown that optimizing the robust MDP objective is equivalent to minimizing (see [7], [138] for details)

$$\Lambda_f(\alpha) = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \left(\mathbb{E} \left[\exp \left(\sum_{k=0}^{t-1} \alpha c_f(s_k) \right) \right] \right), \quad (6.1)$$

where the expectation is taken with respect $\mathbb{P}_f$. The existence of the above limit is a consequence of the Perron-Frobenius Theorem, whose details can be found in [7], [139]. $\Lambda_f(\alpha)$ is known as the risk-sensitive average cost. Similar to J_f , the value of $\Lambda_f(\alpha)$ does not depend on the initial state s_0 . The level of risk aversion is quantified by α , often referred to as the risk factor [124]. Higher values of α correspond to increased risk aversion. Note that in the limit as $\alpha \rightarrow 0$, the risk-sensitive average cost converges to the risk-neutral average cost, i.e.,

$$\lim_{\alpha \rightarrow 0} \Lambda_f(\alpha) = J_f.$$

For simplicity, from now on, we fix $\alpha > 0$ and write Λ_f instead of $\Lambda_f(\alpha)$.

Analogous to the risk-neutral average cost formulation, the objective (6.1) can be expressed as the solution to a Bellman Equation (6.2), but one that is in a multiplicative form [131]:

$$e^{\Lambda_f} e^{V_f(i)} = e^{\alpha c_f(i)} \sum_{j \in \mathcal{S}} \mathbb{P}_f(j|i) e^{V_f(j)}, \quad \forall i \in \mathcal{S}. \quad (6.2)$$

Note that the relative value function e^{V_f} is the eigenvector corresponding to the Perron-Frobenius eigenvalue e^{Λ_f} associated with the non negative matrix $[e^{\alpha c_f(i)} \mathbb{P}_f(j|i)]_{i,j}$ (see [140] for the statement of the Perron-Frobenius theorem, which we have also restated in Appendix E). Since it is a Perron-Frobenius eigenvector, the value function is unique up to a multiplicative constant, thus being referred to as the relative value function. The multiplicative Bellman operator corresponding to a policy f , is an operator $T_f : \mathbb{R}_+^{|\mathcal{S}|} \rightarrow \mathbb{R}_+^{|\mathcal{S}|}$ defined as [131]:

$$(T_f e^V)(i) = e^{\alpha c_f(i)} \sum_{j \in \mathcal{S}} \mathbb{P}_f(j|i) e^{V(j)}.$$

The optimal multiplicative Bellman operator $T : \mathbb{R}_+^{|\mathcal{S}|} \rightarrow \mathbb{R}_+^{|\mathcal{S}|}$ is defined as [131]

$$(T e^V)(i) = \min_{f \in \Pi} (T_f e^V)(i), \quad \forall i \in \mathcal{S}. \quad (6.3)$$

The optimal risk-sensitive average cost is defined as the minimum risk-sensitive average

cost across all policies, i.e.,

$$\Lambda^* = \min_{f \in \Pi} \Lambda_f = \min_{f \in \Pi} \lim_{t \rightarrow \infty} \frac{1}{t} \ln \left(\mathbb{E} \left[\exp \left(\sum_{k=0}^{t-1} \alpha c_f(s_k) \right) \middle| s_0 = i \right] \right). \quad (6.4)$$

Let $f \in \Pi$ denote the deterministic policy for which $\Lambda_f = \Lambda^*$, and let $e^{V^*} = e^{V_f}$ denote its relative value function. It can be shown that [139] the pair (Λ^*, e^{V^*}) is the unique solution (up to multiplicative constant of e^{V^*}) to the following equation:

$$e^{\Lambda^*} e^{V^*(i)} = \min_{f \in \Pi} e^{\alpha c_f(i)} \sum_{j \in \mathcal{S}} \mathbb{P}_f(j|i) e^{V^*(j)}, \quad \forall i \in \mathcal{S}. \quad (6.5)$$

6.3.2 Assumptions

Our assumptions on the model and policy class are summarized below.

Assumption 6.1. We assume that:

- The state space and the action space are finite.
- The one-step cost associated with each state-action pair $(s, a) \in \mathcal{S} \times \mathcal{A}$ is deterministic and bounded.
- The Markov process associated with each deterministic policy $f \in \Pi$ is irreducible.

The irreducibility condition is somewhat restrictive. However, in MDPs where this condition is not satisfied, one can replace $\mathbb{P}$ with $\tilde{\mathbb{P}} = (1 - \epsilon)\mathbb{P} + \epsilon \mathbf{1}\mathbf{1}^\top$ where $\mathbf{1}$ is the all-one column vector and $\epsilon > 0$ is a fixed constant. This leads to an $O(\epsilon)$ suboptimality. While aperiodicity of the Markov chain associated with each policy is not an assumption, it is a relatively straightforward to transform the original MDP into a new MDP where every policy induces an aperiodic Markov chain. In the literature, such a transformation is referred to as the aperiodicity or the lazy-chain transformation. This transformation is essential to guarantee the convergence of the algorithm under consideration in this chapter and is therefore described below.

Fix a constant $\kappa \in (0, 1)$ and transform the dynamics of the MDP as follows:

- The transformed single-step cost is given by:

$$d_f(i) = \frac{1}{\alpha} \log((1 - \kappa)e^{\alpha c_f(i)} + \kappa), \quad \forall i \in \mathcal{S}.$$

- The transformed transition probabilities are given by:

$$\mathbb{Q}(j|i, a) = \frac{(1 - \kappa)e^{\alpha c(i, a)} \mathbb{P}(j|i, a) + \kappa \mathbf{1}(i = j)}{(1 - \kappa)e^{\alpha c(i, a)} + \kappa}, \quad \forall (i, a) \in \mathcal{S} \times \mathcal{A},$$

where $\mathbf{1}(i = j)$ is the indicator function. For any policy $f \in \Pi$, $\mathbb{Q}_f(j|i)$ denotes the new probability of moving to state $j \in \mathcal{S}$ from state $i \in \mathcal{S}$ upon taking action $f(i)$.

Notice that for all $(i, a) \in \mathcal{S} \times \mathcal{A}$, we have $\mathbb{Q}(i|i, a) \geq \frac{\kappa}{(1 - \kappa)e^{\alpha c} + \kappa} > 0$. In particular, the probability of staying in the same state under all policies is non-zero. This aids in diminishing the mixing time of the MDP. The following theorem from [133] establishes a one-to-one correspondence between the optimal risk-sensitive average cost and the associated relative value function in the original MDP and the transformed MDP. Hence, finding an optimal policy for the transformed dynamics is equivalent to finding an optimal policy for the original MDP.

Theorem 6.1. *Given $\kappa \in (0, 1)$, we have the following:*

1. *Given (Λ^*, e^{V^*}) satisfies (6.5), define*

$$\tilde{\Lambda}^* = \log((1 - \kappa)e^{\Lambda^*} + \kappa).$$

Then $(\tilde{\Lambda}^, e^{V^*})$ solves the following multiplicative Bellman equation:*

$$e^{\tilde{\Lambda}^*} e^{V^*(i)} = \min_{f \in \Pi} e^{\alpha d_f(i)} \sum_{j \in \mathcal{S}} \mathbb{Q}_f(j|i) e^{V^*(j)}, \quad \forall i \in \mathcal{S}. \quad (6.6)$$

2. Conversely, given $(\tilde{\Lambda}^*, e^{V^*})$ satisfies (6.6), then

$$e^{\tilde{\Lambda}^*} \geq \kappa.$$

Define

$$\Lambda^* = \log \left(\frac{e^{\tilde{\Lambda}^*} - \kappa}{1 - \kappa} \right). \quad (6.7)$$

Then the pair (Λ^*, e^{V^*}) satisfies (6.5).

3. Further, the transformed and original problems possess the same optimal policies.

6.3.3 Algorithm

The MPI algorithm for risk-sensitive exponential cost MDPs which have been transformed using the aperiodicity transformation is stated in Algorithm 7. The algorithm takes as input a sequence of natural numbers $(m_i : i \in \mathbb{N})$ such that $m_i \geq 1$ and a vector $V'_0 \in \mathbb{R}^n$ such that $\sum_{i \in \mathcal{S}} e^{V'_0(i)} = 1$. The m_i that are input in the algorithm can vary for every iteration i but are predetermined and not random. The accuracy of the policy-evaluation step improves as m_i increases.

Algorithm 7 Risk-Sensitive Modified Policy Iteration

Require: $(m_i : i \in \mathbb{N})$, $m_i \in \mathbb{N}$, $m_i < M$, V'_0 .

- 1: **for** $k = 0, 1, \dots$ **do**
 - 2: Set $f_{k+1}(i) = \operatorname{argmin}_{a \in \mathcal{A}} e^{\alpha d(i,a)} \sum_{j \in \mathcal{S}} Q(j | i, a) e^{V'_k(j)} \quad \forall i \in \mathcal{S} \triangleright \text{Policy Improvement}$
 Define $\left(T_{f_{k+1}} e^{V'_k} \right)(i) = e^{\alpha d_{f_{k+1}}(i)} \sum_{j \in \mathcal{S}} Q_{f_{k+1}}(j | i) e^{V'_k(j)}$
 - 3: $e^{V_{k+1}(i)} \leftarrow \left(T_{f_{k+1}}^{m_k} e^{V'_k} \right)(i)$ for all $i \in \mathcal{S}$. $\triangleright \text{Partial Policy Evaluation}$
 - 4: $e^{V'_{k+1}(i)} \leftarrow \frac{e^{V_{k+1}(i)}}{\sum_i e^{V_{k+1}(i)}}$ for all $i \in \mathcal{S}$ $\triangleright \text{Normalization}$
 - 5: **end for**
 - 6: **return** $(e^{V'_k} : k \geq 0)$ and $(f_k : k \geq 1)$
-

6.4 Results

The main result of this chapter shows that Algorithm 7 converges and also provides finite-time performance guarantees.

Theorem 6.2. *Recall the definition of the optimal multiplicative Bellman operator in (6.3). Let $e^{V'_n}$ and f_n , for all $n \in \mathbb{N}$ be the outputs of Algorithm 7. Then there exists $\gamma \in (0, 1)$ such that for each n , the iterates of risk-sensitive MPI satisfy:*

$$\left(\max_{i \in \mathcal{S}} \frac{(\mathcal{T}e^{V'_n})(i)}{e^{V'_n(i)}} - e^{\tilde{\lambda}^*} \right) \leq (1 - \gamma) \left(\max_{i \in \mathcal{S}} \frac{(\mathcal{T}e^{V'_{n-R}})(i)}{e^{V'_{n-R}(i)}} - e^{\tilde{\lambda}^*} \right), \quad (6.8)$$

where $\gamma \geq \frac{e^2 e^{2R\alpha d}}{e^{2MR\alpha d} |\mathcal{S}|}$, $e^{\tilde{\lambda}^*}$ is defined in (6.6) and R is defined in Lemma 6.3. Consequently, the risk-sensitive average cost iterates converge to $e^{\tilde{\lambda}^*}$, i.e.

$$\lim_{n \rightarrow \infty} e^{\tilde{\lambda}_{f_n}} = e^{\tilde{\lambda}^*}. \quad (6.9)$$

The proof can be found in Appendix E.4. We will now outline the motivation behind our proof, highlight its distinctions from the proofs related to the convergence of other conventional dynamic programming techniques, and provide an overview of its key components.

6.5 Key Ideas and Proof Outline

Compared to dynamic programming techniques such as value iteration or policy iteration, the convergence analysis of MPI in risk-sensitive MDPs faces additional challenges due to partial policy evaluation in step 3 of algorithm 7. In this section we present the key ideas in our technique that circumvent these challenges.

In the context of value iteration, it is well known that the consecutive value function iterates possess a span-seminorm contraction property [132], [131]. More precisely, let $g_0, h_0 \in \mathbb{R}^n$.

Then there exist constants τ, k, r such that $0 < \tau < 1$, and $\mathbb{N} \ni k, r < \infty$ such that

$$\text{sp}(g_k - h_k) \leq \tau^r \text{sp}(g_0 - h_0), \quad (6.10)$$

where the span of a vector v is defined as $\text{sp}(v) = \max_i v(i) - \min_i v(i)$ and

$$g_k(i) = \min_{f \in \Pi} \left\{ \alpha d_f(i) + \ln \left(\sum_{j \in \mathcal{S}} Q_f(j|i) e^{g_{k-1}(j)} \right) \right\}, \quad (6.11)$$

$$h_k(i) = \min_{f \in \Pi} \left\{ \alpha d_f(i) + \ln \left(\sum_{j \in \mathcal{S}} Q_f(j|i) e^{h_{k-1}(j)} \right) \right\}. \quad (6.12)$$

Note that (6.11) and (6.12) are value iteration updates corresponding to value function vectors g_{k-1} and h_{k-1} respectively. A similar contraction in the sup norm is satisfied in the discounted-cost setting, where the discount factor serves as the source of contraction. The span contraction factor τ in (6.10) is the Birkhoff coefficient of a matrix, which is dependent on the risk-sensitivity factor α , the single-step cost vector and the underlying probability transition kernel. It is worth noting that irreducible and aperiodic Markov chain under any policy are needed to ensure $\tau < 1$. To the best of our knowledge, all prior works which obtain finite-time bounds for dynamic programming algorithms for risk-sensitive MDPs seem to use these assumptions. When h_0 is the optimal risk-sensitive value function, (6.10) ensures convergence of the iterates g_k , since h_k are all identical up to a multiplicative constant [132]. The inequality (6.10) is a consequence of consecutive iterates of g_k, h_k being related through the optimal multiplicative Bellman operator, which by virtue of its definition possesses a monotonicity property. This property is also central to the convergence analysis of risk-sensitive value iteration discussed in [133], even though they provide asymptotic convergence bounds by proving that the difference between two consecutive iterates tends to zero.

However, such monotonicity fails to hold when partial policy evaluation is involved. In the context of risk-sensitive MPI, at each iteration k , the iterate $e^{V'_{k+1}}$ is derived by applying the

multiplicative Bellman operator associated with the policy f_{k+1} to the preceding iterate $e^{V'_k}$, a total of m_k times. In this computation, only the first application of the Bellman operator corresponds to the optimal Bellman operator acting on $e^{V'_k}$, since $T_{f_{k+1}} e^{V'_k} = T e^{V'_k}$ (from step 2 of algorithm 7). This computation is followed by applying $T_{f_{k+1}}$ repeatedly for a total of $m_k - 1$ times - which is the step that distinguishes the algorithm from value iteration. Hence the monotonicity arguments leveraged for the finite-time convergence analysis of value iteration break down in the context of MPI. As a corollary to this observation, one can even view the MPI algorithm as a broader form of value iteration, since setting $m_k = 1$ for all $k \in \mathbb{N}$ yields the value iteration algorithm. Consequently, the convergence analysis presented in this chapter also implies finite-time convergence bounds for value iteration

The risk-sensitive policy iteration algorithm can be obtained by replacing step 3 in Algorithm 7 with complete policy evaluation (this step involves determining the Perron Frobenius eigenvector of a function of the transition kernel). In this setting, there exists a state i such that,

$$\begin{aligned} \left(T_{f_{k+1}} e^{V_{f_k}} \right) (i) &:= e^{\alpha d_{f_{k+1}}(i)} \sum_{j \in \mathcal{S}} Q_{f_{k+1}}(j | i) e^{V_k(j)} \\ &\stackrel{(a)}{<} e^{\alpha d_{f_k}(i)} \sum_{j \in \mathcal{S}} Q_{f_k}(j | i) e^{V_k(j)} \\ &= e^{\tilde{\Lambda}_{f_k}} e^{V_{f_k}(i)}, \end{aligned}$$

where (a) follows from policy improvement. The above strict inequality (a) is true for some state $i \in \mathcal{S}$ if the algorithm is yet to converge. Upon repeatedly applying $T_{f_{k+1}}$ operator in the above inequality, in the infinite limit of its application, we obtain that $e^{\tilde{\Lambda}_{f_{k+1}}} < e^{\tilde{\Lambda}_{f_k}}$. In the setting of finite state-action spaces with deterministic policies, this monotonicity of risk-sensitive average costs is sufficient to ensure convergence. However, this proof technique relies on complete policy evaluation and once again is ineffective in the context of risk-sensitive MPI due to step 3 in Algorithm 7. Despite MPI lacking value function iterates that are related through contracting span seminorms, or risk-sensitive average cost iterates that are monotonically decreasing, the ratio $\frac{(T e^{V'_k})(i)}{e^{V'_k(i)}}$ exhibits interesting properties that can be used to circumvent the above mentioned issues. The inspiration for our proof

comes from a related proof of convergence of MPI for average cost MDPs in [29]. However, unlike in [29] the fact that we have to deal with ratios instead of differences presents new challenges. We will soon show that a function of the ratio $\frac{(\tau e^{V'_k})_{(i)}}{e^{V'_k(i)}}$ possesses monotonicity properties and contraction properties which aid in the convergence analysis. We now present an outline of the proof below.

6.5.1 Step 1: Characterizing the impact of applying the optimal multiplicative Bellman operator once

To exploit the contraction and monotonicity of $\frac{(\tau e^{V'_k})_{(i)}}{e^{V'_k(i)}}$, it is necessary to relate this ratio (i) to the risk-sensitive average cost of the policy induced by $e^{V'_k}$ and (ii) to the optimal risk-sensitive cost obtained by solving the multiplicative Bellman optimality equation (6.6). This result is captured in Lemma 6.1, which can be utilized to extend convergence properties of $\frac{(\tau e^{V'_k})_{(i)}}{e^{V'_k(i)}}$ to the associated risk-sensitive average costs.

Before presenting the supporting lemmas and the main theorem, we introduce the following notation. Let g_n be defined as

$$g_n(i) := \frac{(\tau e^{V'_n})_{(i)}}{e^{V'_n(i)}}, \quad (6.13)$$

and set u_n and ℓ_n as

$$u_n := \max_{i \in \mathcal{S}} (g_n(i)) \quad \ell_n := \min_{i \in \mathcal{S}} (g_n(i)). \quad (6.14)$$

Define $\tilde{Q}_f(i, j)$ corresponding to some policy $f \in \Pi$, as follows:

$$\tilde{Q}_f(i, j) := e^{\alpha d_f(i)} Q_f(j|i) \quad (6.15)$$

Lemma 6.1. *Let $\tilde{\Lambda}^*$ be the optimal risk-sensitive average cost associated with the MDP considered*

in Algorithm 7. Then $\forall n > 0$:

$$\ell_n \leq e^{\tilde{\Lambda}^*} \leq e^{\tilde{\Lambda}_{f_{n+1}}} \leq u_n. \quad (6.16)$$

The above Lemma is crucial to the proof of convergence of MPI. A similar relation would hold for the reward maximization problem: $\ell_n \leq e^{\tilde{\Lambda}_{f_{n+1}}} \leq e^{\tilde{\Lambda}^*} \leq u_n$. Such a relation can be obtained in the context of risk-neutral average cost as well [29]. The multiplicative nature of Bellman operator combined with the exponential cost formulation, necessitates a different proof idea which hinges on the careful utilization of the Perron-Frobenius Theorem.

6.5.2 Step 2: Establishing an appropriate monotonicity

We show that $\max_{i \in \mathcal{S}} \frac{(\tau e^{V'_k})_{(i)}}{e^{V'_k(i)}}$ is a monotonically decreasing sequence. Note that from (6.6) we know that $\frac{(\tau e^{V^*})_{(i)}}{e^{V^*(i)}} = e^{\tilde{\Lambda}^*}$ for all $i \in \mathcal{S}$. Even though $e^{V'_k}$ are not value functions in the sense of (6.2), the monotonically decreasing $\max_{i \in \mathcal{S}} \frac{(\tau e^{V'_k})_{(i)}}{e^{V'_k(i)}}$ alludes to convergence of the algorithm. Lemma 6.2 contains the proof of this observation. It is important to note that this monotonicity is observed only for $\max_{i \in \mathcal{S}} \frac{(\tau e^{V'_k})_{(i)}}{e^{V'_k(i)}}$ and not for $\min_{i \in \mathcal{S}} \frac{(\tau e^{V'_k})_{(i)}}{e^{V'_k(i)}}$. If $\min_{i \in \mathcal{S}} \frac{(\tau e^{V'_k})_{(i)}}{e^{V'_k(i)}}$ were to be monotonically increasing as well, we would have a span contraction for these iterates which would have simplified the proof.

Lemma 6.2. *The sequence u_n is non-increasing, i.e. $u_n \leq u_{n-1}$ for all n .*

Value Iteration leads to monotonicity in u_n (non-increasing) and ℓ_n (non-decreasing). This is a consequence of improving the policy at every iteration without any partial policy evaluation. This symmetric monotonicity leads to an overall span contraction in the value function. However, due to partial policy evaluation in MPI, such a monotonicity is observed only for the maximum of the ratio of iterates, i.e., u_n (or ℓ_n in case risk-sensitive reward maximization). Consequently, there need not be a span contraction for the value functions. Hence it is necessary to rely on arguments independent of span in order to prove algorithm convergence. This approach is delineated in the theorem below.

6.5.3 Step 3: Identifying a source of contraction

In order to establish finite-time convergence bounds it is necessary to identify a source of contraction between subsequent iterates. In discounted cost MDPs, it is achieved by the discount factor. In average cost MDPs it is achieved by the least value of the stationary measure across all states and policies. In the risk-sensitive value iteration setting, it is the Birkhoff's coefficient of a model-dependent matrix.

Here, two conditions need to be satisfied in order to obtain a valid source of contraction: (i) aperiodicity and irreducibility of the transition kernel under all policies and (ii) uniform boundedness of all iterates $e^{V_k'}$ away from zero. Condition (i) is satisfied by virtue of our modeling assumptions. To ensure condition (ii), we use a specific form of normalization (step 4 of Algorithm 7.) In the case of risk-neutral average-cost MPI, the normalization step generally involves subtracting the value function at some fixed state from the rest of the states. This ensures that the value function iterates do not diverge with repeated execution of the algorithm. However, a similar normalization trick where one divides the value function for all states by the value function of one state does not seem to work for risk-sensitive MPI; therefore we divide by the sum of the value function of all states. This allows us to ensure the value functions do not diverge and that they are uniformly bounded away from zero.

Lemma 6.3 utilizes condition (i) and Lemma 6.4 proves the boundedness of $e^{V_k'(i)}$ to establish a source of contraction necessary for convergence analysis in the main proof.

Lemma 6.3. *There exists a finite natural number R such that $\forall R' > R$ for any sequence of policies $f_1, f_2, \dots, f_{R'} \in \Pi$, there exists a constant $\epsilon_{R'}$ such that*

$$\min_{i,j \in \mathcal{S}} Q_{f_1} Q_{f_2} \dots Q_{f_{R'}}(j|i) \geq \epsilon_{R'} > 0. \quad (6.17)$$

Lemma 6.4. *Recall that $\max_k m_k < M$, where m_k corresponds to the number of fixed point iterations performed during partial policy evaluation during the k th execution of the algorithm.*

Then, there exists β such that $0 < \beta < 1$,

$$e^{V'_k(i)} > \beta > 0 \quad \forall i \in \mathcal{S} \quad \forall k \geq R. \quad (6.18)$$

6.5.4 Step 4: Convergence of risk-sensitive MPI

Lemma 6.1 and the monotonicity of $\max_{i \in \mathcal{S}} \frac{(\tau e^{V'_k})^{(i)}}{e^{V'_k(i)}}$ along with the source of contraction are utilized to prove the exponential convergence of $\max_{i \in \mathcal{S}} \frac{(\tau e^{V'_k})^{(i)}}{e^{V'_k(i)}}$ to $e^{\tilde{\Lambda}^*}$. Lemma 6.1 is further leveraged to prove the convergence of $e^{\tilde{\Lambda}_{f_k}}$ to $e^{\tilde{\Lambda}^*}$ and $\min_{i \in \mathcal{S}} \frac{(\tau e^{V'_k})^{(i)}}{e^{V'_k(i)}}$ to $e^{\tilde{\Lambda}^*}$ thereby proving the convergence of MPI.

From Theorem 6.1 we can equivalently obtain the original optimal risk-sensitive average cost Λ^* and the corresponding value function associated with it. Note that if $\frac{(\tau e^V)^{(i)}}{e^{V(i)}} = \delta > 0$, then the transformation in Equation (6.7) provides a Λ which is in a δ -scaled neighbourhood of Λ^* . More details can be found in [133].

Remark 6.1 (Comment on γ). Recall that the contraction factor is $1 - \gamma$, where $\gamma \geq \frac{e^2 e^{2R\alpha d}}{e^{2MR\alpha d} |\mathcal{S}|}$. It is an interesting question as to whether this bound is tight. To understand this, we compare our result to the risk-neutral case considered in [29] by letting $\alpha \rightarrow 0$. In this case, the contraction factor we have is $1 - \epsilon^2$ whereas [29] obtains a factor of $1 - \epsilon$. The additional ϵ in our analysis is due to the multiplicative nature of the Bellman equation and we are unable to show that this bound can be tightened. The bound is likely loose (both for the risk-neutral and risk-sensitive cases) for specific MDPs since the contraction factor is obtained without taking into account the specific nature of each MDP. Preliminary simulations indicate that the algorithm does indeed converge quite fast in practice, see the appendix for more details. However, it is unlikely that it can be tightened significantly beyond the ϵ factor due to the result for the risk-neutral case. This is an interesting topic for further research.

6.6 Conclusion

We presented an MPI algorithm to reduce the computational burden of standard policy iteration for risk-sensitive MDPs. We present the proof of convergence for this algorithm and characterize finite-time performance bounds. As in prior work for discounted-cost problems, it would be interesting to investigate if our results can be used to provide performance guarantees for RL algorithms for risk-sensitive MDPs.

7

APPROXIMATE POLICY ITERATION

The art of doing mathematics consists in finding that special case which contains all the germs of generality.

David Hilbert

7.1 Introduction

Risk-sensitive control aims to protect a decision maker against variability in long-run costs, not just their expectation. A principled way to encode that attitude is the exponential cost criterion, whose logarithm recovers a certainty-equivalent risk-sensitive average cost. When the underlying Markov Decision Process is also robust—i.e., transition kernels lie in an ambiguity set—one obtains a model that simultaneously hedges against rare but costly events and misspecified dynamics. Such risk-sensitive robust MDPs capture cyber-physical systems facing adversarial disturbances, asset-allocation under market shocks, and queuing or inventory networks that must survive demand surges.

Unfortunately, the tools that make discounted or risk-neutral MDPs tractable break down here: the Bellman operator is multiplicative, the value function is defined only up to a positive scaling, and policy evaluation involves matrix exponentials. Modified Policy Iteration (MPI) offers a practical compromise—interleaving partial policy evaluation with

improvement—but even in the exact setting, convergence results are quite new. However, in practice, it is not possible to determine the value function or the best next-step policy exactly. Since the computational complexity of standard dynamic programming algorithms scales poorly with the state and action spaces, its application is limited to tractable problems. When encountering large cardinalities for either state or action space, the evaluation and improvements are performed somewhat approximately. It is, thus, important to analyze the performance bounds of these algorithms in the context of inexact evaluations and improvements – the focus of this chapter.

Understanding the performance of approximate dynamic programming techniques is typically the first step in modeling reinforcement learning algorithms. Reinforcement learning algorithms are employed when the agent doesn’t have access to the underlying environment. These algorithms are broadly classified into policy based and value based methods. Samples from the environment are collected to glean information about the transition matrix in different forms pertaining to each method. Many policy-based reinforcement-learning algorithms—from Actor–Critic and Natural Policy Gradient to TRPO and PPO—can be interpreted as stochastic or sample-based instantiations of (approximate) policy iteration. Establishing convergence in the risk-sensitive robust setting therefore creates a template for analysing robust, risk-aware RL methods that are increasingly demanded in safety-critical applications.

7.1.1 Contributions

This chapter makes three principal contributions:

- (i) **Unified approximation framework.** We develop an algorithmic formulation of *approximate modified policy iteration* (AMPI) that subsumes *approximate policy iteration* (API) as the special case $m \rightarrow \infty$. The scheme is analysed for risk-sensitive *robust* MDPs with exponential cost, providing the first convergence guarantees in this setting.
- (ii) **Stability condition and risk-neutral recovery.** An explicit condition is derived under which the evaluation and improvement errors contract geometrically, thereby leading to a stable propagation of approximation errors. When the risk parameter $\alpha \rightarrow 0$

the threshold reduces to the classical risk-neutral bound, thereby unifying the two regimes.

- (iii) **Error constants for policy-improvement oracles.** We quantify the policy-improvement error for a class of update rules—including soft-max and mirror-descent steps —aiding with overall algorithm design.

7.2 Related Work

Risk-sensitive MDP foundations. Early studies by [141] and [142] introduced the exponential-utility criterion as a principled way to penalise variability in long-run returns. Modern surveys (e.g., [143], [144]) formalise existence and optimality for discounted and average-cost models, but note that the multiplicative Bellman operator breaks standard contraction arguments.

Robust MDPs. [119] and [145] framed transition-kernel ambiguity as a min-max control problem, establishing value- and policy-iteration convergence in finite MDPs. Subsequent work extended robustness to unbounded costs, linear-programming formulations, and distributional sets such as Wasserstein balls [146].

Linking risk and robustness. [147] proved that an exponential-cost MDP is equivalent to a Kullback–Leibler robust MDP, motivating algorithms that handle both effects simultaneously and interpreting risk sensitivity as *soft* robustness.

Dynamic-programming algorithms under risk sensitivity. For exponential criteria, value iteration and full policy iteration enjoy convergence proofs in finite and countable spaces (see [143]). In contrast, *Modified Policy Iteration* (MPI) received little attention until [5] analysed exact risk-sensitive MPI. No prior work tackles *approximate* MPI or combines risk sensitivity with robustness.

Approximate dynamic programming. In the risk-neutral setting, [148] and [149] provide error-propagation bounds for approximate MPI and API, relying on additive Bellman

operators—an assumption invalid for exponential costs.

Risk-sensitive and robust reinforcement learning. Policy-gradient and actor-critic variants for exponential and CVaR objectives appear in [150] and [151], while distributionally robust RL optimises worst-case returns over Wasserstein or KL balls [146]. These studies consider *either* robustness *or* risk sensitivity, and most adopt value-based updates rather than the policy-based MPI lens.

Position of the present work. To our knowledge, the attached study is the *first* to provide finite-time and asymptotic convergence guarantees for *approximate* MPI under *simultaneous* exponential risk sensitivity and transition-kernel robustness, thereby bridging risk-sensitive control, robust MDPs, and approximate policy-based DP, and laying groundwork for analysing robust, variance-aware RL algorithms.

7.3 Preliminaries

We study a finite-state, finite-action Markov decision process (MDP) with state set $\mathcal{S}$, action set $\mathcal{A}$, and one-step cost $c : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$. A (stationary, possibly randomised) policy is a mapping $f : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ where $\Delta(\mathcal{A})$ denotes the probability simplex on $\mathcal{A}$; the collection of all such policies is written Π . Let $(s_t, a_t)_{t \geq 0}$ be the state-action process obtained when the system evolves under f and the prescribed transition kernel $P(\cdot | s_t, a_t)$.

To model risk aversion (or distributional robustness) we introduce an *exponential* performance index with sensitivity parameter $\alpha > 0$ —larger α corresponds to greater risk aversion. The infinite-horizon *risk-sensitive average cost* of policy f is

$$\Lambda_f := \lim_{T \rightarrow \infty} \frac{1}{T} \log \left(\mathbb{E}_f \left[\exp \left(\sum_{i=0}^{T-1} \alpha c(s_i, a_i) \right) \right] \right).$$

Similar to the risk neutral setting, Λ_f is independent of the initial state distribution under some mild conditions.

Associated with each policy f is a multiplicative Bellman Equation defined below:

$$e^{\Lambda_f} e^{V_f(s)} = \sum_{a \in \mathcal{A}} f(a|s) \left[e^{\alpha c(s,a)} \sum_{s' \in \mathcal{S}} \mathbb{P}(s'|s, a) e^{V_f(s')} \right]$$

where $e^{V_f(s)}$ is the relative state value function associated with state s and policy f . From the above equation it is apparent that the value function is unique only up to a multiplicative constant. Similarly the relative state action value function associated with a policy f and a state action pair (s, a) is defined as:

$$e^{Q_f(s,a)} := \frac{e^{\alpha c(s,a)} \sum_{s' \in \mathcal{S}} \mathbb{P}(s'|s, a) e^{V_f(s')}}{\Lambda_f} \quad (7.1)$$

A short calculation yields the following result:

$$\begin{aligned} e^{\Lambda_f} e^{Q_f(s,a)} &= e^{\alpha c(s,a)} \sum_{s' \in \mathcal{S}} \mathbb{P}(s' | s, a) e^{V_f(s')} \\ e^{\Lambda_f} \sum_{a \in \mathcal{A}} f(a | s) e^{Q_f(s,a)} &= \sum_{a \in \mathcal{A}} f(a | s) e^{\alpha c(s,a)} \sum_{s' \in \mathcal{S}} \mathbb{P}(s' | s, a) e^{V_f(s')} \\ &= \sum_{j \in \mathcal{S}} \left(\sum_{a \in \mathcal{A}} f(a | s) e^{\alpha c(s,a)} \mathbb{P}(s' | s, a) \right) e^{V_f(j)} \\ &= e^{\Lambda_f} e^{V_f(s)}. \end{aligned}$$

Hence,

$$e^{V_f(s)} = \sum_{a \in \mathcal{A}} f(a | s) e^{Q_f(s,a)} \quad \forall s \in \mathcal{S} \ a \in \mathcal{A} \quad (7.2)$$

Hence the Bellman equation corresponding to the state-action value function $e^{Q_f(s,a)}$ is given by:

$$e^{\Lambda_f} e^{Q_f(s,a)} = e^{\alpha c(s,a)} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} \mathbb{P}_f(s', a' | s, a) e^{Q_f(s', a')} \quad (7.3)$$

where $\mathbb{P}_f(s', a' | s, a) := \mathbb{P}(s'|s, a)f(a'|s')$. We analogously define the following two operators. The Bellman operator T_f associated with a policy f acts on a vector $h \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$ as

follows:

$$(T_f h)(s, a) := e^{\alpha c(s, a)} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} \mathbb{P}_f(s', a' | s, a) h(s', a') \quad (7.4)$$

Similarly the optimal Bellman operator T is defined as follows:

$$(Th)(s, a) := e^{\alpha c(s, a)} \sum_{s' \in \mathcal{S}} \mathbb{P}(s' | s, a) \min_{a' \in \mathcal{A}} h(s', a') \quad (7.5)$$

The goal of dynamic programming is to identify the policy f^* which yields the lowest Λ_f . The optimal risk sensitive cost is defined as:

$$\Lambda_{f^*} := \min_{f \in \Pi} \Lambda_f \quad (7.6)$$

The optimal cost Λ_{f^*} , is the solution to the following multiplicative Bellman optimality equation:

$$e^{\Lambda_{f^*}} e^{Q_{f^*}(s, a)} = e^{\alpha c(s, a)} \sum_{s' \in \mathcal{S}} \left(\mathbb{P}(s' | s, a) \min_{a' \in \mathcal{A}} e^{Q_{f^*}(s', a')} \right) \quad (7.7)$$

where $e^{Q_{f^*}}$ is the optimal state action value function. In the previous chapter, we proved the convergence of modified policy iteration. Here we consider approximate modified policy iteration and provide finite time performance bounds capturing its convergence. Analogously, we can also obtain the corresponding performance bounds for approximate policy iteration as a corollary to this analysis. We present the approximate modified policy iteration algorithm below.

7.3.1 Algorithm

To establish geometric convergence of the iterates, a contraction mapping is required. We therefore apply the aperiodicity transformation from Chapter 6 and use Theorem 6.1 to connect the original and transformed optimal costs. All subsequent results are stated for the transformed MDP given in Theorem 6.1.

Algorithm 8 Approximate Risk Sensitive Modified Policy Iteration

Require: $(m_i : i \in \mathbb{N})$, Q_0 such that $\sum_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} e^{Q_0(s,a)} = 1$.

- 1: Set $k = 0$
- 2: Compute $f_{k+1} \in \Pi$ such that:

$$1 \leq \left\| \frac{T_{f_{k+1}} e^{Q_k}}{T e^{Q_k}} \right\|_{\infty} \leq e^{\alpha \epsilon}.$$

▷ Approximate Policy Improvement

- 3: Define $e^{Q_{f_{k+1}}(s,a)} \leftarrow \left(T_{f_{k+1}}^{m_k} e^{Q_k} \right) (s, a) \quad \forall s \in \mathcal{S}, \forall a \in \mathcal{A}.$ ▷ Partial Policy Evaluation

- 4: $e^{Q'_{f_{k+1}}(s,a)} \leftarrow \frac{e^{Q_{f_{k+1}}(s,a)}}{\sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_{k+1}}(s',a')}} \quad \forall s \in \mathcal{S}, \forall a \in \mathcal{A}.$ ▷ Normalization

- 5: Compute $e^{Q_{k+1}}$ such that,

$$e^{\alpha \delta_1} \leq \left\| \frac{e^{Q'_{f_{k+1}}}}{e^{Q_{k+1}}} \right\|_{\infty} \leq e^{\alpha \delta_2}.$$

▷ Approximate Policy Evaluation

7.4 Main Result

Theorem 7.1. Let $\gamma \in (0, 1)$. Let $n \in \mathbb{N}$ such that $m_k + m_{k-1} + \dots + m_{k-n} \geq R$. Suppose the approximation errors $e^{\alpha \delta_1}$, $e^{\alpha \delta_2}$ and $e^{\alpha \epsilon}$ are such that

$$\left(\frac{e^{\alpha \delta_2} e^{\alpha \epsilon}}{e^{\alpha \delta_1}} \right)^n (1 - \gamma) < 1.$$

Then, the iterates Q_k generated by Algorithm 8 satisfy:

$$\left(e^{\tilde{\Lambda}_{f_{k+1}}} - e^{\tilde{\Lambda}^*} \right) \leq (\gamma')^{\frac{k}{n}} \left(u_0 e^{\alpha \epsilon} - e^{\tilde{\Lambda}^*} \right) + \frac{\sigma e^{\tilde{\Lambda}^*}}{1 - \gamma'},$$

where $\sigma = \left(\left(\frac{e^{\alpha \delta_2} e^{\alpha \epsilon}}{e^{\alpha \delta_1}} \right)^n (1 + (e^{\alpha \epsilon} - 1)\gamma) - 1 \right)$, $u_0 = \max_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} \frac{(T e^{Q_0})(s,a)}{e^{Q_0(s,a)}}$ and $\gamma' = \left(\frac{e^{\alpha \delta_2} e^{\alpha \epsilon}}{e^{\alpha \delta_1}} \right)^n (1 - \gamma)$.

7.4.1 Some Remarks on Theorem 7.1

- Note that as the approximation becomes perfect, that is $e^{\alpha\delta_1} = 1$, $e^{\alpha\delta_2} = 1$ and $e^{\alpha\epsilon} = 1$, the risk sensitive average cost $e^{\hat{\lambda}_{f_{k+1}}}$ obtained from the algorithm converges exactly to the optimal risk sensitive average cost. In other words σ will be 0.
- Note the condition on γ' for it to constitute a valid source of contraction is a consequence of the multiplicative Bellman equation associated with the risk sensitive average cost. In the case of risk neutral average cost, such a condition disappears and the approximation results hold more generally for all magnitudes of $e^{\alpha\delta_1}$, $e^{\alpha\delta_2}$ and $e^{\alpha\epsilon}$. For more details refer [1].
- Since the condition on γ' requires more accurate policy evaluations and policy improvements as n increases, a possible way to circumvent this issue is by ensuring that $m_i \geq R$ for all iterations of the algorithm. This would ensure n to be equal to one. In other words, better the policy evaluation, lesser the approximation loss.

7.5 Key Ideas and Proof Outline

Consider the following definitions required for the proof of convergence of the algorithm iterates:

$$g_k(s, a) = \frac{(Te^{Q_k})(s, a)}{e^{Q_k(s, a)}}$$

$$l_k = \min_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} g_k(s, a)$$

$$u_k = \max_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} g_k(s, a)$$

In order to obtain the finite time and asymptotic performance of the iterates obtained as a consequence of the above algorithm, it is necessary to prove the lemma characterising a single step of approximate policy improvement below.

Lemma 7.1. Given the iterates l_k, u_k obtained from Algorithm 8, $\forall k$ it is true that,

$$l_k \leq e^{\tilde{\lambda}^*} \leq e^{\tilde{\lambda}_{f_{k+1}}} \leq u_k e^{\alpha\epsilon} \quad (7.8)$$

Lemma 7.2. Let $\max_k m_k < C_1$, where m_k corresponds to the number of fixed point iterations performed during partial policy evaluation during the k th execution of the algorithm 8. Then, there exists τ' such that

$$e^{Q_k(s,a)} \geq \tau' > 0. \quad \forall s \in \mathcal{S}, \forall a \in \mathcal{A}$$

7.5.1 Proof Outline for Theorem 7.4.1

Sketch of Proof. The argument proceeds in four structured steps and relies on Lemmas 7.5.1–7.5.2.

Step 1: One-step monotonicity (Lemma 7.5.1). Lemma 7.5.1 establishes that the *normalised* value-function iterates $\{Q_k\}_{k \geq 0}$ produced by Algorithm 8 satisfy

$$l_k \leq e^{\tilde{\lambda}^*} \leq e^{\tilde{\lambda}_{f_{k+1}}} \leq u_k e^{\alpha\epsilon}, \quad \forall k \geq 0,$$

where l_k and u_k are, respectively, lower- and upper-envelope sequences updated during the proof of the lemma. The inequality couples *consecutive* iterates through the policy-improvement error $e^{\alpha\epsilon}$ and provides a *monotone squeezing* of the optimal cost $e^{\tilde{\lambda}^*}$. This monotonic-in- k behaviour is crucial because it allows us to eventually capture the overall suboptimality between the policy obtained from Algorithm 8 and the optimal risk sensitive cost $e^{\tilde{\lambda}^*}$.

Step 2: Uniform positive lower bound (Lemma 7.5.2). Lemma 7.5.2 shows that if the number of fixed-point iterations used during partial evaluation is uniformly bounded, $\max_k m_k < C_1$, then

$$e^{Q_k(s,a)} \geq \tau' > 0, \quad \forall s \in \mathcal{S}, \forall a \in \mathcal{A}, k \geq 0.$$

The result guarantees that the approximations of the value functions that we obtain at each iteration of the algorithm is uniformly lower bounded away from zero. Unlike the exact evaluation setting, this lower bound consists of the approximation errors as well. The difference between the consecutive risk sensitive costs and the optimal cost contracts in a nontrivial fashion ensured by this lower bound. Consequently, the contraction factor we build in Step 3 is well defined and strictly less than one.

Step 3: n-step error recursion and contraction factor. Because Algorithm 8 performs only m_k Bellman iterates between control updates, we analyse the algorithm in blocks of n outer iterations. Combining the triangular inequality implicit in Lemma 7.5.1 with the uniform bound of Lemma 7.5.2 yields

$$(e^{\tilde{J}_{f_{k+n}}} - e^{\tilde{J}^*}) \leq \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (1 - \gamma) (e^{\tilde{J}_{f_k}} - e^{\tilde{J}^*}) + \sigma e^{\tilde{J}^*},$$

where σ collects second-order terms and $\gamma \in (0, 1)$ encodes the mixing time of the underlying Markov chain under each policy. The prefactor $\left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (1 - \gamma)$ serves as the *contraction factor*. The sufficient condition in the theorem, $\left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (1 - \gamma) < 1$, ensures this factor is strictly less than one even in the presence of approximation errors.

Step 4: Geometric convergence and steady-state error bound. Iterating the n -step recursion from Step 3, we obtain for any $k \geq 0$:

$$e^{\tilde{J}_{f_{k+1}}} - e^{\tilde{J}^*} \leq (\gamma')^{k/n} (u_0 e^{\alpha\epsilon} - e^{\tilde{J}^*}) + \frac{\sigma e^{\tilde{J}^*}}{1 - \gamma'}, \quad \gamma' = \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (1 - \gamma),$$

which is exactly the inequality claimed in Theorem 7.4.1. The first term decays geometrically thanks to $\gamma' < 1$; the second term gives the residual bias floor induced by the approximation errors and vanishes as $e^{\alpha\delta_1} \rightarrow e^{\alpha\delta_2} \rightarrow 1$ and $e^{\alpha\epsilon} \rightarrow 1$.

Conclusion. Lemma 7.5.1 provides a monotone relation that connects successive iterates via the improvement error; Lemma 7.5.2 ensures a uniform positive bound necessary to define a legitimate contraction factor. Together they yield the n -step recursion in Step 3, from which the geometric convergence bound of Theorem 7.4.1 follows. $\square$

7.6 Approximate Policy Iteration

Approximate *Policy* Iteration (API) is obtained from Algorithm 8 by taking the policy evaluation horizon to infinity, i.e. setting $m_k \rightarrow \infty$ for every outer iteration k . In this limit the Bellman iterate in Step 3 converges to the exact value function for policy f_{k+1} ; any practical implementation therefore re-introduces error only through an *approximate evaluator* such as TD(λ), LSTD, or fitted value iteration. These errors are captured by the same multiplicative condition of Step 5 in the algorithm, so the analysis of Theorem 7.4.1 applies verbatim after replacing the block length n by 1.

Single-step contraction. Because policy evaluation is now exact in the idealised algorithm, the n -step recursion used in the proof of Theorem 7.4.1 collapses to a *one-step* inequality:

$$\left(e^{\tilde{\lambda}_{f_{k+1}}} - e^{\tilde{\lambda}^*} \right) \leq \left(\frac{e^{\alpha \delta_2} e^{\alpha \epsilon}}{e^{\alpha \delta_1}} \right) (1 - \gamma) \left(e^{\tilde{\lambda}_{f_k}} - e^{\tilde{\lambda}^*} \right) + \sigma e^{\tilde{\lambda}^*},$$

where $\gamma \in (0, 1)$ is the ergodic coefficient of the transition kernel under f_k and σ collects second-order terms as before. Consequently, API converges geometrically whenever

$$\left(\frac{e^{\alpha \delta_2} e^{\alpha \epsilon}}{e^{\alpha \delta_1}} \right) (1 - \gamma) < 1. \quad (7.9)$$

Lemma 7.5.2 remains essential but manifests differently in this setting: provided the policy class is nice enough, the uniform positive lower bound on the value-function iterates guarantees that the contraction factor in (7.9) is well defined even in the presence of evaluation noise.

7.6.1 Dependence on the risk parameter α

For a fixed stationary policy π the *multiplicative Bellman equation* is

$$e^{\alpha \tilde{\lambda}_\pi} e^{\alpha Q_\pi(s, a)} = e^{\alpha d(s, a)} \sum_{s'} Q(s' | s, a) \sum_{a'} \pi(a' | s') e^{\alpha Q_\pi(s', a')} \quad (7.10)$$

for every state–action pair (s, a) . The analogous definition of the risk sensitive average cost corresponding to a policy π for the Bellman equation defined above, is as follows:

$$\tilde{\lambda}_\pi := \lim_{T \rightarrow \infty} \frac{1}{\alpha} \frac{1}{T} \log \left(\mathbb{E}_\pi \left[\exp \left(\sum_{i=0}^{T-1} \alpha d(s_i, a_i) \right) \right] \right).$$

First-order series expansion. With $e^{\alpha x} = 1 + \alpha x + \frac{1}{2} \alpha^2 x^2 + O(\alpha^3)$,

$$\begin{aligned} e^{\alpha \tilde{\lambda}_\pi} e^{\alpha Q_\pi(s, a)} &= 1 + \alpha [\tilde{\lambda}_\pi + Q_\pi(s, a)] + O(\alpha^2), \\ e^{\alpha d(s, a)} \sum_{s', a'} Q(s'|s, a) \pi(a'|s') e^{\alpha Q_\pi(s', a')} &= 1 + \alpha [d(s, a) + \mathbb{E}_{Q, \pi}[Q_\pi](s, a)] + O(\alpha^2), \end{aligned}$$

where

$$\mathbb{E}_{Q, \pi}[Q_\pi](s, a) := \sum_{s'} Q(s'|s, a) \sum_{a'} \pi(a'|s') Q_\pi(s', a').$$

Matching the $O(\alpha)$ terms yields the *additive* Bellman relation

$$\tilde{\lambda}_\pi + Q_\pi(s, a) = d(s, a) + \mathbb{E}_{Q, \pi}[Q_\pi](s, a) + O(\alpha), \quad \alpha \rightarrow 0. \quad (7.11)$$

Error terms in exponential form Suppose an approximate evaluator produces $\hat{Q}_\pi$ with additive supremum error $\|\hat{Q}_\pi - Q_\pi\|_\infty \leq \delta$. Because the algorithm works with $e^{\alpha Q}$, its *multiplicative* error is

$$\frac{e^{\alpha \hat{Q}_\pi(s, a)}}{e^{\alpha Q_\pi(s, a)}} = e^{\alpha [\hat{Q}_\pi(s, a) - Q_\pi(s, a)]}.$$

Thus

$$e^{-\alpha \delta} \leq \frac{e^{\alpha \hat{Q}_\pi(s, a)}}{e^{\alpha Q_\pi(s, a)}} \leq e^{\alpha \delta}, \quad \forall (s, a).$$

Exactly the same reasoning turns policy–improvement tolerance ϵ into multiplicative bounds $e^{\alpha \epsilon}$. Consequently the contraction condition in risk–sensitive AMPI/API naturally appears as

$$\left(\frac{e^{\alpha \delta_2} e^{\alpha \epsilon}}{e^{\alpha \delta_1}} \right) (1 - \gamma) < 1,$$

which reduces to the classical risk-neutral inequality $(1 - \gamma) < 1$ when $\alpha \rightarrow 0$.

Accordingly, approximation errors are expressed in exponential form. To analyse robustness against model uncertainty we must therefore work with the multiplicative Bellman formulation developed in this chapter. Because the multiplicative error factors satisfy $e^{\alpha\varepsilon}, e^{\alpha\delta_1}, e^{\alpha\delta_2} \rightarrow 1$ as $\alpha \rightarrow 0$, smaller values of α naturally produce tighter evaluation and improvement errors, making the contraction condition easier to satisfy. In the limit $\alpha \rightarrow 0$ the framework collapses to the familiar approximate-policy-iteration result for the risk-neutral case. Note, however, that the resulting contraction coefficient γ differs from the one that arises in the risk-neutral analysis; reconciling these two constants is non-trivial and remains an open problem for future work.

7.7 Characterization of Policy Improvement Errors

In this section we characterize the policy improvement errors corresponding to mirror descent update and softmax update. A supporting lemma for the analysis is below.

Lemma 7.3. Let $d \in \mathbb{N}$, fix $\beta > 0$, and let $\mathbf{q} = (q_1, \dots, q_d) \in \mathbb{R}^d$. For a probability vector $\mathbf{p} = (p_1, \dots, p_d) \in \Delta_d := \{\mathbf{p} \in [0, 1]^d : \sum_{i=1}^d p_i = 1\}$, define

$$q_{\min} := \min_{1 \leq i \leq d} q_i, \quad i_{\min} := \arg \min_{1 \leq i \leq d} q_i, \quad p_{\min} := p_{i_{\min}},$$

and assume $p_{\min} > 0$. Then

$$\frac{\sum_{i=1}^d p_i q_i e^{-\beta q_i}}{\sum_{i=1}^d p_i e^{-\beta q_i}} - q_{\min} \leq \frac{1}{\beta} \log\left(\frac{1}{p_{\min}}\right).$$

7.7.1 Policy–Improvement Error for the Mirror-Descent Update

Lemma 7.4 (Mirror-descent policy error). Fix $\beta > 0$ and let $\{Q_k(s, a)\}_{s,a}$ be the risk-sensitive Q -values at iteration k . Define the mirror-descent policy update¹

$$f_{k+1}(a | s) = \frac{f_k(a | s) \exp(-\beta Q_k(s, a))}{\sum_{a' \in \mathcal{A}} f_k(a' | s) \exp(-\beta Q_k(s, a'))}. \quad (7.12)$$

Let $(T_{f_{k+1}} e^{Q_k})(s, a)$ and $(Te^{Q_k})(s, a)$ denote, respectively, the controlled and optimal risk-sensitive Bellman operators²

For every $(s, a) \in \mathcal{S} \times \mathcal{A}$,

$$0 \leq \frac{(T_{f_{k+1}} e^{Q_k})(s, a)}{(Te^{Q_k})(s, a)} - 1 \leq \frac{1}{\beta} \max_{j \in \mathcal{S}} \frac{\log(1/f_k(a^*(j) | j))}{\min_{a'} e^{Q_k(j, a')}}, \quad (7.13)$$

where $a^(j) \in \arg \min_{a'} Q_k(j, a')$.*

7.7.2 Policy–Improvement Error for the Softmax Update

Lemma 7.5 (Soft-max policy error). Fix $\beta > 0$ and let $\{Q_k(s, a)\}_{s,a}$ be the current risk-sensitive Q -values. The soft-max (a.k.a. Boltzmann) policy update is

$$f_{k+1}(a | s) = \frac{\exp(-\beta Q_k(s, a))}{\sum_{a' \in \mathcal{A}} \exp(-\beta Q_k(s, a'))}, \quad \forall (s, a). \quad (7.14)$$

With the Bellman operators $T_{f_{k+1}}$ and T defined as in Lemma 7.4,

$$0 \leq \frac{(T_{f_{k+1}} e^{Q_k})(s, a)}{(Te^{Q_k})(s, a)} - 1 \leq \frac{1}{\beta \min_{a'} e^{Q_k(j, a')}} \log |\mathcal{A}|, \quad \forall (s, a). \quad (7.15)$$

¹The update can also be interpreted as one step of entropic mirror descent or soft-min regularisation with stepsize β^{-1} .

²Their definitions are the ones used in Sections 7.2–7.3; in particular $(T_{f_{k+1}} e^{Q_k})(s, a) = e^{c(s,a)} \sum_j P(j | s, a) \sum_{a'} f(a' | j) e^{Q_k(j, a')}$ and $(Te^{Q_k})(s, a) = e^{c(s,a)} \sum_j P(j | s, a) \min_{a'} e^{Q_k(j, a')}$.

8

CONCLUSION

Between two minds of equal logical power, the one that is superior is the one that is richer in intuitions.

Jacques Hadamard

This thesis has embarked on a comprehensive exploration of policy-based reinforcement learning methods, addressing significant challenges in their application to complex, real-world decision-making problems. The core focus has been on extending the capabilities of these algorithms to operate effectively under average-reward criteria, in the presence of model uncertainty and risk-sensitivity, and within large or countably infinite state spaces. The research presented has yielded novel theoretical insights, new algorithmic frameworks, and rigorous performance guarantees.

A primary contribution of this work lies in advancing the understanding and applicability of policy-based methods for average-reward Markov Decision Processes. We established finite-time error bounds for Approximate Policy Iteration in this setting, providing a crucial tool for analyzing RL algorithms that rely on inexact computations. Furthermore, we rigorously proved the smoothness of the average-reward function, a foundational result that enabled the derivation of global convergence guarantees for Projected Policy Gradient methods with $O(\log(T))$ regret. The convergence of Natural Policy Gradient, via its connection to Mirror Descent, was also formally established in the tabular average-reward context.

Recognizing the limitations of traditional RL in systems with unbounded state spaces and costs, such as those encountered in queuing networks, this thesis introduced a Natural Policy Gradient algorithm employing state-dependent step sizes. By deriving novel policy-independent bounds on the relative value function, we achieved non-trivial regret bounds and offered a more robust approach to learning in these challenging, countable-space environments. This work also relaxed common assumptions on learning errors, moving towards more realistic modeling of function approximation in practical scenarios.

In the domain of robust control, this thesis addressed risk-sensitive MDPs characterized by an exponential cost criterion. We developed a Modified Policy Iteration algorithm tailored for this setting, proving its convergence and providing finite-time performance bounds. Crucially, this framework was then extended to accommodate inexact computations through Approximate Modified Policy Iteration, for which we established the first convergence guarantees in the risk-sensitive, robust setting. A key outcome of this analysis is a unified stability condition for error propagation that bridges the gap between risk-neutral and risk-sensitive regimes.

In summary, this thesis has systematically tackled several fundamental limitations in policy-based reinforcement learning. By developing new theoretical tools, proposing novel algorithmic modifications, and providing rigorous analyses, this work significantly expands the scope and reliability of RL methods for a diverse range of complex decision-making problems characterized by long-run objectives, uncertainty, and large-scale state representations. The findings herein provide a stronger foundation for deploying intelligent agents in critical systems where sustained performance and robustness are paramount.

8.1 Future Work

Below we outline a few future directions pertaining to the work presented in this thesis.

1. **Improving Dependence on System Cardinality:** The convergence rate established for Approximate Policy Iteration (API) in the average-reward setting exhibits a dependence on the parameter γ , which can scale unfavorably with the cardinality of the state and action spaces. This may be an artifact of the proof techniques employed,

particularly the reliance on infinity-norm contractions. Future investigation could explore whether specific structural properties of well-behaved MDPs, or convergence analysis in alternative norms (e.g., weighted norms or norms related to the stationary distribution), could yield bounds with improved scaling, thereby enhancing the applicability of API to very large-scale systems.

2. **Mitigating Dependence on Concentrability Coefficients:** The global convergence bounds for Projected Policy Gradient (PPG) currently depend on concentrability coefficients, which can lead to poor scaling with system size. A significant research direction is to develop analytical techniques that can either relax or circumvent this dependence, potentially by exploiting specific MDP structures or alternative performance difference lemmas.
3. **Extension to Reinforcement Learning Settings:** The current convergence results for PPG are established in a planning (full information) context. Leveraging the novel smoothness properties and convergence insights from this work to develop and analyze policy gradient algorithms in the reinforcement learning setting (i.e., with sampled or estimated gradients) is a natural and important next step. This would involve addressing challenges related to sample complexity and stochastic approximation.
4. **Circumventing Concentrability Dependence in NPG/MDM:** Similar to PPG, the analysis of Natural Policy Gradient (NPG) and Mirror Descent Methods (MDM) for average reward also involves concentrability coefficients. While NPG in the exact discounted setting can achieve bounds independent of state/action space cardinality (though potentially dependent on the discount factor γ), devising alternative Lyapunov functions or proof techniques that can mitigate or remove the reliance on concentrability for average-reward NPG/MDM remains a compelling research avenue.
5. **Relaxing Stability Assumptions on Policy Classes:** The current NPG algorithm for countable state spaces assumes that all policies within the considered class satisfy a Lyapunov drift condition, ensuring uniform stability. A critical line of future work is

to extend these results to broader policy classes where such uniform stability might not hold, or where stability must be learned concurrently with control. This could involve adaptive drift conditions or techniques for robust policy search in potentially unstable regions.

6. **Data-Driven Control for Specific Queueing Applications:** The insights gained from developing RL algorithms for countable state spaces with unbounded costs can be directly applied to design specialized, data-driven control algorithms for specific queueing network applications. Examples include dynamic pricing and matching in ride-hailing platforms, optimal job routing and resource allocation in data centers, and scheduling in communication networks. This involves tailoring the feature representations, reward structures, and exploration strategies to the unique characteristics of these systems.
7. **Tightening Contraction Factor Bounds for Risk-Sensitive MPI:** The derived contraction factor for Modified Policy Iteration (MPI) in risk-sensitive MDPs appears to be conservative, particularly evidenced by a persistent gap of γ when the risk-sensitivity parameter approaches zero (the risk-neutral limit) compared to known risk-neutral contraction rates. Refining the analysis, possibly through more nuanced handling of the multiplicative Bellman operator or by exploiting finer structural properties of risk-sensitive value functions, to close this gap and establish tighter, potentially problem-dependent, contraction factors is an important objective.
8. **Sample Complexity Analysis for Risk-Sensitive TD-Learning:** With the established convergence framework for Approximate Policy Iteration in risk-sensitive MDPs, a natural progression is to develop and analyze Temporal Difference (TD) learning algorithms for estimating the exponential cost value functions. This would involve quantifying the sample complexity required to achieve a desired level of accuracy (i.e., relating the error parameter δ to the length of sample trajectories), thereby providing practical guidelines for implementing learning-based risk-sensitive control.
9. **Global Optimality and Finite-Time Bounds for Risk-Sensitive Policy Gradient in RL:** Current risk-sensitive policy gradient algorithms in the learning setting often

provide asymptotic convergence guarantees, typically to local optima. Future research should investigate whether the insights from risk-neutral policy gradient analysis and the AMPI framework can be leveraged to: (a) establish convergence to global optima for risk-sensitive policy gradient methods in the learning setting, and (b) derive corresponding finite-time convergence bounds, addressing the challenges of stochastic gradients and exploration in risk-sensitive contexts.

In conclusion, this thesis has presented a body of work aimed at extending the frontiers of policy-based reinforcement learning. Through novel theoretical developments and algorithmic designs for average-reward, risk-sensitive, and countable-space MDPs, we have provided new tools and insights for tackling complex control tasks. The findings reported here contribute to the ongoing effort to build more capable, robust, and efficient intelligent systems.

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A

PROOFS: APPROXIMATE POLICY ITERATION FOR AVERAGE REWARD MDPs

A.1 Discussion on Assumption 2.1

A.1.1 Assumption 2.1(a)

In order to ensure every policy induces an irreducible Markov chain, we modify the MDP where at every time step with probability ϵ , an action is chosen from the set of all possible actions with equal probability. Simultaneously, with probability $1 - \epsilon$, we choose an action dictated by some policy. Let the transition kernel of the MDP associated with policy μ be denoted as $\mathbb{P}_\mu$. We denote the modified kernel by $\hat{\mathbb{P}}_\mu$, where the transition probability from state i to j is given by:

$$\hat{\mathbb{P}}_\mu(j|i) = (1 - \epsilon)\mathbb{P}_\mu(j|i) + \epsilon \left(\frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \mathbb{P}(j|i, a) \right)$$

Let $\mathbb{P}_\rho(j|i) = \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \mathbb{P}(j|i, a)$, where ρ represents randomized policy, that is $\rho(a|i) = \frac{1}{|\mathcal{A}|}$. Let (J_μ, V_μ) be the average reward and state value function vector associated with the policy μ . Then they satisfy the following Bellman Equation:

$$J_\mu + V_\mu = r_\mu + \mathbb{P}_\mu V_\mu,$$

where $r_\mu(i) = \sum_{a \in \mathcal{A}} \mu(a|i)r(i, a)$. Similarly $(\hat{J}_\mu, \hat{V}_\mu)$ be the average reward and state value function vector that satisfy the following Bellman Equation,

$$\hat{J}_\mu + \hat{V}_\mu = \hat{r}_\mu + \hat{\mathbb{P}}_\mu \hat{V}_\mu$$

where $\hat{r}_\mu = (1 - \epsilon)r_\mu + \epsilon r_\rho$. Since $\hat{\mathbb{P}}_\mu = (1 - \epsilon)\mathbb{P}_\mu + \epsilon\mathbb{P}_\rho$, the above equation can be rewritten as:

$$\hat{J}_\mu = (1 - \epsilon) \left(r_\mu + \mathbb{P}_\mu \hat{V}_\mu - \hat{V}_\mu \right) + \epsilon \left(r_\rho + \mathbb{P}_\rho \hat{V}_\mu - \hat{V}_\mu \right)$$

Multiplying the above equation by the vector $\mathbb{P}_\mu^*$, which is the invariant distribution over the state space due to policy μ , we obtain,

$$\hat{J}_\mu = (1 - \epsilon) \left(\mathbb{P}_\mu^* r_\mu \right) + \epsilon \mathbb{P}_\mu^* \left(r_\rho + \mathbb{P}_\rho \hat{V}_\mu - \hat{V}_\mu \right)$$

since $\mathbb{P}_\mu^* \mathbb{P}_\mu = \mathbb{P}_\mu^*$. We also know that $J_\mu = \mathbb{P}_\mu^* r_\mu$. Hence we obtain,

$$\hat{J}_\mu = (1 - \epsilon) J_\mu + \epsilon \mathbb{P}_\mu^* \left(r_\rho + \mathbb{P}_\rho \hat{V}_\mu - \hat{V}_\mu \right)$$

Therefor we obtain the following result,

$$J_\mu - \hat{J}_\mu = \epsilon \left(J_\mu - \mathbb{P}_\mu^* \left(r_\rho + \mathbb{P}_\rho \hat{V}_\mu - \hat{V}_\mu \right) \right)$$

Since the rewards are bounded, so are J_μ, r_ρ and $\mathbb{P}_\rho \hat{V}_\mu - \hat{V}_\mu$ for all μ . Hence we obtain,

$$J_\mu - \hat{J}_\mu = O(\epsilon)$$

The difference in the average reward associated with the original MDP and the modified MDP vary by $O(\epsilon)$. Let the optimal policy corresponding to original MDP be μ^* and the optimal policy corresponding to modified MDP be $\hat{\mu}^*$. Then,

$$J_{\mu^*} - \hat{J}_{\hat{\mu}^*} = J_{\mu^*} - \hat{J}_{\mu^*} + \hat{J}_{\mu^*} - \hat{J}_{\hat{\mu}^*}$$

Since $\hat{\mu}^*$ is the optimizing policy for the modified MDP, we have $\hat{J}_{\mu^*} - \hat{J}_{\hat{\mu}^*} \leq 0$. Hence, we obtain,

$$J_{\mu^*} - \hat{J}_{\hat{\mu}^*} \leq J_{\mu^*} - \hat{J}_{\mu^*} = O(\epsilon).$$

Similarly,

$$J_{\mu^*} - \hat{J}_{\hat{\mu}^*} = J_{\mu^*} - J_{\hat{\mu}^*} + J_{\hat{\mu}^*} - \hat{J}_{\hat{\mu}^*}$$

Since μ^* is the optimizing policy for the original MDP, we have $J_{\mu^*} - J_{\hat{\mu}^*} \geq 0$. Hence we obtain,

$$J_{\mu^*} - \hat{J}_{\hat{\mu}^*} \geq J_{\hat{\mu}^*} - \hat{J}_{\hat{\mu}^*} = O(\epsilon).$$

Thus the optimal average reward of the original MDP and modified MDP differ by $O(\epsilon)$. That is,

$$|J_{\mu^*} - \hat{J}_{\hat{\mu}^*}| \leq O(\epsilon)$$

A.1.2 Assumption 2.1(b)

To ensure Assumption 3.1 (b) is satisfied, an aperiodicity transformation can be implemented. Under this transformation, the transition probabilities of the original MDP are modified such that the probability of staying in any state under any policy is non-zero. In order to compensate for the change in transition kernel, the single step rewards are analogously modified such that the Bellman equation corresponding to the original and transformed dynamics are scaled versions of one another. This thus ensures that the optimality of a policy remains the same irrespective of this transformation, along with yielding quantifiable convergence properties. Mathematically, this transformation is described below.

Definition A.1 (Aperiodicity Transformation). Let $\kappa \in (0, 1)$. For every policy $\mu \in \Pi$ and for all states $i, j \in \mathcal{S}$, consider the following transformation:

$$\hat{\mathbb{P}}_{\mu}(i|i) = \kappa + (1 - \kappa)\mathbb{P}_{\mu}(i|i) \tag{A.1}$$

$$\hat{\mathbb{P}}_{\mu}(j|i) = (1 - \kappa)\mathbb{P}_{\mu}(j|i), \quad j \neq i \tag{A.2}$$

$$\hat{r}(i, \mu(i)) = (1 - \kappa)r(i, \mu(i)) \tag{A.3}$$

Theorem A.1. Given the transformation in Equation (A.1)-Equation (A.3), for every $\mu \in \Pi$, let (J_μ, h_μ) and $(\hat{J}_\mu, \hat{h}_\mu)$ be the solution to the Bellman Equation (Equation 2) corresponding to the original MDP $(\mathbb{P}_\mu, r_\mu)$ and the transformed MDP $(\hat{\mathbb{P}}_\mu, \hat{r}_\mu)$. Then,

$$(\hat{J}_\mu, \hat{h}_\mu) = ((1 - \kappa)J_\mu, h_\mu)$$

Proof. The proof of this theorem can be found in [28]. □

Remark A.1. Due to this bijective relationship between the original and the transformed problem, it suffices to solve the transformed problem. Given that the trajectory used for algorithms such as TD Learning corresponds to the original system, such a transformation necessitates a mild change in how samples from a given policy are utilized in TD learning. More specifically, for each (state, action) sample, with probability κ , we have to add the same sample to the data set for the next time instant. With probability $1 - \kappa$ choose the next state in the trajectory. Hence, this transformation is easy to incorporate in RL algorithms.

A.2 Proof of Theorem 2.2

Prior to presenting the proof, we define

$$\begin{aligned} u_k &= \max_i (Th_k - h_k)(i), \\ l_k &= \min_i (Th_k - h_k)(i). \end{aligned} \tag{A.4}$$

A key lemma in the proof of convergence of approximate policy iteration in the context of average reward is:

Lemma A.1. Let J^* be the optimal average reward associated with the transformed MDP. For all $k \in \mathbb{N}$:

$$l_k - \epsilon \leq J_{\mu_{k+1}} \leq J^* \leq u_k$$

Proof. From definition,

$$l_k \mathbf{1} \leq T h_k - h_k$$

Since $\|T h_k - T_{\mu_{k+1}} h_k\|_{\infty} \leq \epsilon$,

$$\begin{aligned} l_k \mathbf{1} &\leq T_{\mu_{k+1}} h_k - h_k + \epsilon \mathbf{1} \\ &= r_{\mu_{k+1}} + \mathbb{P}_{\mu_{k+1}} h_k - h_k + \epsilon \mathbf{1} \\ \mathbb{P}_{\mu_{k+1}}^* l_k \mathbf{1} &\leq \mathbb{P}_{\mu_{k+1}}^* r_{\mu_{k+1}} + \mathbb{P}_{\mu_{k+1}}^* \mathbb{P}_{\mu_{k+1}} h_k - \mathbb{P}_{\mu_{k+1}}^* h_k + \mathbb{P}_{\mu_{k+1}}^* \epsilon \mathbf{1} \end{aligned}$$

where $\mathbb{P}_{\mu_{k+1}}^*$ is a matrix whose rows are identical and are the invariant distribution associated with the probability transition kernel $\mathbb{P}_{\mu_{k+1}}$. Since $\mathbb{P}_{\mu_{k+1}}^* \mathbb{P}_{\mu_{k+1}} = \mathbb{P}_{\mu_{k+1}}^*$,

$$\begin{aligned} l_k \mathbf{1} &\leq \mathbb{P}_{\mu_{k+1}}^* r_{\mu_{k+1}} + \epsilon \mathbf{1} \\ l_k \mathbf{1} - \epsilon \mathbf{1} &\leq J_{\mu_{k+1}} \mathbf{1} \\ l_k - \epsilon &\leq J_{\mu_{k+1}} \end{aligned}$$

From definition,

$$J^* = \max_{\mu \in \Pi} J_{\mu} \geq J_{\mu_{k+1}}$$

Let μ^* be the policy corresponding to the optimal average reward J^* and value function h^* , ie.,

$$\begin{aligned} J^* + h^* &= \max_{\mu} r_{\mu} + \mathbb{P}_{\mu} h^* \\ &= r_{\mu^*} + \mathbb{P}_{\mu^*} h^* \end{aligned}$$

Then,

$$\begin{aligned} u_k \mathbf{1} &\geq T h_k - h_k \\ &\stackrel{(a)}{\geq} T_{\mu^*} h_k - h_k \end{aligned}$$

$$\begin{aligned}
u_k \mathbf{1} &\geq r_{\mu^*} + \mathbb{P}_{\mu^*} h_k - h_k \\
\mathbb{P}_{\mu^*}^* u_k \mathbf{1} &\geq \mathbb{P}_{\mu^*}^* r_{\mu^*} + \mathbb{P}_{\mu^*}^* \mathbb{P}_{\mu^*} h_k - \mathbb{P}_{\mu^*}^* h_k \\
u_k \mathbf{1} &\geq \mathbb{P}_{\mu^*}^* r_{\mu^*} \\
u_k &\geq J^*
\end{aligned}$$

where (a) is due to the fact that T is the maximising Bellman operator.

Hence $\forall k \in \mathbb{N}$,

$$l_k - \epsilon \leq J_{\mu_{k+1}} \leq J^* \leq u_k$$

□

Without any approximation, the above lemma indicates that there exists a state whose value function changes by an amount less than the average reward associated with the optimizing policy at that iteration, and also that there exists a state whose value function increases by an amount larger than the optimal average reward.

The proof proceeds by showing a geometric convergence rate for l_k . More precisely, the following lemma is proved.

Lemma A.2. For all $k \in \mathbb{N}$, it is true that:

$$(J^* - l_k) \leq (1 - \gamma) (J^* - l_{k-1}) + \epsilon + 2\delta$$

Proof. Define:

$$g_k(i) = (Th_k - h_k)(i)$$

$$g_k(i) \geq (T_{\mu_k} h_k - h_k)(i)$$

Since T is the maximizing Bellman operator.

We know that:

$$1. \quad \|h_k - h_{\mu_k}\|_{\infty} \leq \delta$$

2. For all constant $p \in \mathbb{R}$:

$$\begin{aligned}
T_{\mu_k}(h_k + p\mathbf{1}) &= r_{\mu_k} + \mathbb{P}_{\mu_k}(h_k + p\mathbf{1}) \\
&= r_{\mu_k} + \mathbb{P}_{\mu_k}h_k + p\mathbf{1} \\
&= T_{\mu_k}h_k + p\mathbf{1}
\end{aligned}$$

We thus obtain,

$$g_k(i) \geq T_{\mu_k}(h_{\mu_k} - \delta) - (h_{\mu_k} + \delta) \quad (\text{A.5})$$

$$\geq (T_{\mu_k}h_{\mu_k} - h_{\mu_k})(i) - 2\delta \quad (\text{A.6})$$

Recall that $h_{\mu_k} = \lim_{m \rightarrow \infty} \tilde{T}_{\mu_{k+1}}^m h_k$

$$\begin{aligned}
\tilde{T}_{\mu_k}^m h_{k-1} &= \tilde{T}_{\mu_k}^{m-1} \tilde{T}_{\mu_k} h_{k-1} \\
&= \tilde{T}_{\mu_k}^{m-1} (r_{\mu_k} + \mathbb{P}_{\mu_k} h_{k-1} - r_{\mu_k}(x^*)\mathbf{1} - (\mathbb{P}_{\mu_k} h_{k-1})(x^*)\mathbf{1}) \\
&= \tilde{T}_{\mu_k}^{m-1} (T_{\mu_k} h_{k-1} - (T_{\mu_k} h_{k-1})(x^*)\mathbf{1}) \\
&= \tilde{T}_{\mu_k}^{m-2} \tilde{T}_{\mu_k} (T_{\mu_k} h_{k-1} - (T_{\mu_k} h_{k-1})(x^*)\mathbf{1}) \\
&= \tilde{T}_{\mu_k}^{m-2} (r_{\mu_k} + \mathbb{P}_{\mu_k} (T_{\mu_k} h_{k-1} - T_{\mu_k} h_{k-1}(x^*)) - r_{\mu_k}(x^*)\mathbf{1} \\
&\quad - (\mathbb{P}_{\mu_k} (T_{\mu_k} h_{k-1} - T_{\mu_k} h_{k-1}(x^*))(x^*)\mathbf{1}) \\
&= \tilde{T}_{\mu_k}^{m-2} (r_{\mu_k} + \mathbb{P}_{\mu_k} (T_{\mu_k} h_{k-1}) - T_{\mu_k} h_{k-1}(x^*) - r_{\mu_k}(x^*)\mathbf{1} \\
&\quad - \mathbb{P}_{\mu_k} (T_{\mu_k} h_{k-1})(x^*)\mathbf{1} + T_{\mu_k} h_{k-1}(x^*)) \\
&= \tilde{T}_{\mu_k}^{m-2} (T_{\mu_k}^2 h_{k-1} - T_{\mu_k}^2 h_{k-1}(x^*)\mathbf{1})
\end{aligned}$$

Iterating, we thus obtain,

$$h_{\mu_k} = \lim_{m \rightarrow \infty} T_{\mu_k}^m h_{k-1} - T_{\mu_k}^m h_{k-1}(x^*)\mathbf{1}$$

Substituting back in Equation (A.5),

$$\begin{aligned}
g_k(i) &\geq \left(T_{\mu_k} \left(\lim_{m \rightarrow \infty} T_{\mu_k}^m h_{k-1} - T_{\mu_k}^m h_{k-1}(x^*) \mathbf{1} \right) - \lim_{m \rightarrow \infty} T_{\mu_k}^m h_{k-1} - T_{\mu_k}^m h_{k-1}(x^*) \mathbf{1} \right) (i) - 2\delta \\
&= \left(T_{\mu_k} \left(\lim_{m \rightarrow \infty} T_{\mu_k}^m h_{k-1} \right) - \lim_{m \rightarrow \infty} T_{\mu_k}^m h_{k-1}(x^*) \mathbf{1} \right) (i) - \\
&\quad \left(\lim_{m \rightarrow \infty} T_{\mu_k}^m h_{k-1} + \lim_{m \rightarrow \infty} T_{\mu_k}^m h_{k-1}(x^*) \mathbf{1} \right) (i) - 2\delta \\
&= \left(T_{\mu_k} \left(\lim_{m \rightarrow \infty} T_{\mu_k}^m h_{k-1} \right) - \lim_{m \rightarrow \infty} T_{\mu_k}^m h_{k-1} \right) (i) - 2\delta
\end{aligned}$$

Since T_{μ_k} is a continuous operator, it is possible to take the limit outside. We thus obtain,

$$\begin{aligned}
g_k(i) &\geq \lim_{m \rightarrow \infty} \left(T_{\mu_k}^{m+1} h_{k-1} - T_{\mu_k}^m h_{k-1} \right) (i) - 2\delta \\
&= \lim_{m \rightarrow \infty} \left(T_{\mu_k}^m T_{\mu_k} h_{k-1} - T_{\mu_k}^m h_{k-1} \right) (i) - 2\delta \\
&= \lim_{m \rightarrow \infty} \left(\mathbb{P}_{\mu_k}^m (T_{\mu_k} h_{k-1} - h_{k-1}) \right) (i) - 2\delta \\
&= \mathbb{P}_{\mu_k}^* (T_{\mu_k} h_{k-1} - h_{k-1}) (i) - 2\delta
\end{aligned}$$

Since $\|T_{\mu_k} h_{k-1} - Th_{k-1}\|_{\infty} \leq \epsilon$,

$$\begin{aligned}
g_k(i) &\geq \mathbb{P}_{\mu_k}^* (Th_{k-1} - h_{k-1}) (i) - 2\delta - \epsilon \\
&= \left(\mathbb{P}_{\mu_k}^* g_{k-1} \right) (i) - 2\delta - \epsilon
\end{aligned}$$

Note that from Lemma 2.1 in the mainpaper, we know that

$$\min_{i,j \in \mathcal{S}} \mathbb{P}_{\mu_k}^* (j|i) > \gamma > 0$$

Also, since every policy is assumed to induce an irreducible Markov Process, we have a limiting distribution with all positive entries. We thus obtain the following crucial relationship,

$$g_k(i) \geq (1 - \gamma)l_{k-1} + \gamma u_{k-1} - 2\delta - \epsilon \tag{A.7}$$

Since Equation (A.7) is true for all states i , it has to also be true for $\text{argmin}_i (Th_k - h_k)(i)$. Hence, we obtain,

$$l_k \geq (1 - \gamma)l_{k-1} + \gamma u_{k-1} - 2\delta - \epsilon$$

Note that from Lemma A.1, we know that $u_k \geq J^*$ for all $k \in \mathbb{N}$. Hence we obtain,

$$l_k \geq (1 - \gamma)l_{k-1} + \gamma J^* - 2\delta - \epsilon$$

Upon rearranging we obtain the result, that is,

$$(J^* - l_k) \leq (1 - \gamma)(J^* - l_{k-1}) + 2\delta + \epsilon$$

□

We now present the proof of Theorem 2.2.

From Lemma A.2, we thus have,

$$(J^* - l_k) \leq (1 - \gamma)(J^* - l_{k-1}) + 2\delta + \epsilon \tag{A.8}$$

However, we know that,

$$l_k - \epsilon \leq J^*$$

In order to iterate Equation (A.8), need to ensure the terms are non-negative. Rearranging the terms thus yields,

$$(J^* - l_k + \epsilon) \leq (1 - \gamma)(J^* - l_{k-1} + \epsilon) + 2\delta + (1 + \gamma)\epsilon$$

Let

$$\omega = 2\delta + (1 + \gamma)\epsilon$$

and

$$a_k = J^* - l_k + \epsilon.$$

Then upon iterating, we obtain,

$$\begin{aligned}
a_k &\leq (1 - \gamma)a_{k-1} + \omega \\
&\leq (1 - \gamma)((1 - \gamma)a_{k-2} + \omega) + \omega \\
&\leq \omega \frac{1 - (1 - \gamma)^k}{\gamma} + (1 - \gamma)^k a_0
\end{aligned}$$

Substituting back, we obtain,

$$(J^* - l_k + \epsilon) \leq \left(\frac{1 - (1 - \gamma)^k}{\gamma} \right) (2\delta + (1 + \gamma)\epsilon) + (1 - \gamma)^k (J^* - l_0 + \epsilon) \quad (\text{A.9})$$

$$(J^* - l_k) \leq \left(\frac{1 - (1 - \gamma)^k}{\gamma} \right) (2\delta + (1 + \gamma)\epsilon) - \epsilon + (1 - \gamma)^k (J^* - l_0 + \epsilon)$$

Note that from Lemma A.1 we know that

$$l_k \leq J_{\mu_{k+1}} + \epsilon \leq J^* + \epsilon$$

Hence we get,

$$(J^* - J_{\mu_{k+1}}) \leq (J^* - l_k + \epsilon)$$

Thus we obtain,

$$(J^* - J_{\mu_{k+1}}) \leq \left(\frac{1 - (1 - \gamma)^k}{\gamma} \right) (2\delta + (1 + \gamma)\epsilon) + (1 - \gamma)^k (J^* - l_0 + \epsilon)$$

Theorem 2.2 presents an upper bound on the error in terms of the average reward. Recall that, for the optimal relative value function h , the following relationship holds: $Th - h = J^* \mathbf{1}$. Thus, it is also interesting to understand how $Th_k - h_k$ behaves under approximate policy iteration. The following proposition characterizes the behavior of this term.

Proposition A.1. The iterates generated from approximate policy iteration algorithm 2 satisfy the

following bound:

$$(\mathbf{u}_{k-1} - \mathbf{l}_{k-1}) \leq \underbrace{\left(\frac{1 - (1 - \gamma)^k}{\gamma^2} \right) (2\delta + (1 + \gamma)\epsilon) + \frac{2\delta + \epsilon}{\gamma}}_{\text{approximation error}} + \underbrace{\frac{(1 - \gamma)^k (J^* - \mathbf{l}_0 + \epsilon)}{\gamma}}_{\text{initial condition error}},$$

where $\mathbf{u}_k, \mathbf{l}_k$ are defined in Equation (A.4).

Proof. It is known from Equation (A.7) that,

$$g_k(\mathbf{i}) \geq (1 - \gamma)\mathbf{l}_{k-1} + \gamma\mathbf{u}_{k-1} - 2\delta - \epsilon$$

This further yields,

$$\mathbf{l}_k \geq (1 - \gamma)\mathbf{l}_{k-1} + \gamma\mathbf{u}_{k-1} - 2\delta - \epsilon$$

From Lemma A.1, we know that,

$$\mathbf{l}_k \leq J^* + \epsilon$$

$$J^* + \epsilon \geq (1 - \gamma)\mathbf{l}_{k-1} + \gamma\mathbf{u}_{k-1} - 2\delta - \epsilon$$

$$J^* - \mathbf{l}_k + \epsilon \geq \gamma(\mathbf{u}_{k-1} - \mathbf{l}_{k-1}) - 2\delta - \epsilon$$

$$\gamma(\mathbf{u}_{k-1} - \mathbf{l}_{k-1}) \leq (J^* - \mathbf{l}_k + \epsilon) + 2\delta + \epsilon$$

From Equation (A.9), we thus obtain,

$$\gamma(\mathbf{u}_{k-1} - \mathbf{l}_{k-1}) \leq \left(\frac{1 - (1 - \gamma)^k}{\gamma} \right) (2\delta + (1 + \gamma)\epsilon) + (1 - \gamma)^k (J^* - \mathbf{l}_0 + \epsilon) + 2\delta + \epsilon$$

We thus obtain the result in Proposition 3.11,

$$(\mathbf{u}_{k-1} - \mathbf{l}_{k-1}) \leq \left(\frac{1 - (1 - \gamma)^k}{\gamma^2} \right) (2\delta + (1 + \gamma)\epsilon) + \frac{2\delta + \epsilon}{\gamma} + \frac{(1 - \gamma)^k (J^* - \mathbf{l}_0 + \epsilon)}{\gamma}$$

□

Corollary A.1. The asymptotic behavior of the relative value function iterates is given by

$$\limsup_{k \rightarrow \infty} (u_k - l_k) \leq \frac{\epsilon (1 + 2\gamma) + 2\delta (1 + \gamma)}{\gamma^2}$$

Comment of the Novelty of our Proof Technique: As mentioned in the main body of the paper, our proof is inspired by the proof of convergence of modified policy iteration in [29]. However, since our algorithm is quite different from modified policy iteration due to the presence of errors in each step of the algorithm, special care is needed to obtain useful performance bounds. Specifically, the impact of the errors at each step have to be carefully bounded to ensure that the performance bounds do not blow up to infinity as it does when we obtain a similar result for the discounted-reward case and let the discount factor go to one.

A.3 Proofs from Section 2.4

Algorithm 2 and its analysis, adapted to the context of Q function with time dependent approximation errors is presented in this section. Most RL algorithms use the state-action relative value function Q instead of the relative state value function h to evaluate a policy. The corresponding Bellman Equation in terms of Q associated with any state-action pair (s, a) is given by

$$J_\mu + Q_\mu(s, a) = r(s, a) + (\mathbb{Q}_\mu Q_\mu)(s, a)$$

where $\mathbb{Q}_\mu(s', a'|s, a) = \mu(a'|s')\mathbb{P}(s'|s, a)$. Since we are interested in randomized policies for exploration reasons, the irreducibility assumptions imposed on $\mathbb{P}_\mu$ also hold true for the transition kernel $\mathbb{Q}_\mu$. The state relative value function h_μ and state-action relative value function Q_μ are related as $h_\mu(s) = \sum_a \mu(a|s)Q_\mu(s, a)$. Consider the following similar definitions of the Bellman operators corresponding to the state-action value function:

$$(T_\mu^Q Q)(s, a) = r(s, a) + (\mathbb{Q}_\mu Q)(s, a)$$

and

$$(T^Q Q)(s, a) = r(s, a) + \max_{\mu \in \Pi} (\mathbb{Q}_\mu Q)(s, a).$$

Let (s^*, a^*) represent some fixed state. Then the relative Bellman operator, relative to (s^*, a^*) is defined as:

$$\left(\tilde{T}_\mu^Q Q\right)(s, a) = r(s, a) + (Q_\mu Q)(s, a) - r(s^*, a^*) - (Q_\mu Q)(s^*, a^*).$$

The algorithm, thus adapted to state action relative value function is given below:

Algorithm 9 Approximate Policy Iteration: Average Reward

- 1: Require $Q_0 \in \mathbb{R}^n$
 - 2: **for** $k = 0, 1, 2, \dots$ **do**
 - 3: 1. Compute $\mu_{k+1} \in \Pi$ such that $\|TQ_k - T_{\mu_{k+1}}Q_k\|_\infty \leq \epsilon_k$
 - 4: 2. Compute Q_{k+1} such that $\|Q_{k+1} - Q_{\mu_{k+1}}\|_\infty \leq \delta_{k+1}$
 - 5: where $Q_{\mu_{k+1}} = \lim_{m \rightarrow \infty} \left(\tilde{T}_{\mu_{k+1}}^Q\right)^m Q_k$
 - 6: **end for**
-

Define

$$g_k(s, a) = \left(T^Q Q_k - Q_k\right)(s, a), \quad \forall (s, a) \in \mathcal{S} \times \mathcal{A},$$

and set

$$l_k = \min_{(s, a) \in \mathcal{S} \times \mathcal{A}} g_k(s, a), \quad u_k = \max_{(s, a) \in \mathcal{S} \times \mathcal{A}} g_k(s, a).$$

We prove the convergence in three parts:

1. We prove that $l_k - \epsilon_k \leq J_{\mu_{k+1}} \leq J^* \leq u_k$.
2. We use the result from 1 to prove Lemma 2.2.
3. We iteratively employ the result in Lemma 2.2 to prove Proposition 2.1.

A.3.1 Proof of part 1

By definition, we have

$$l_k \mathbf{1} \leq T^Q Q_k - Q_k.$$

Since $\|T^Q Q_k - T_{\mu_{k+1}}^Q Q_k\|_\infty \leq \epsilon_k$, it follows that

$$\begin{aligned} l_k \mathbf{1} &\leq T_{\mu_{k+1}}^Q Q_k - Q_k + \epsilon_k \mathbf{1} \\ &= r + Q_{\mu_{k+1}} Q_k - Q_k + \epsilon_k \mathbf{1}. \end{aligned}$$

Multiplying by $Q_{\mu_{k+1}}^*$, we get

$$\begin{aligned} Q_{\mu_{k+1}}^* l_k \mathbf{1} &\leq Q_{\mu_{k+1}}^* r + Q_{\mu_{k+1}}^* Q_{\mu_{k+1}} Q_k - Q_{\mu_{k+1}}^* Q_k + Q_{\mu_{k+1}}^* \epsilon_k \mathbf{1} \\ &= Q_{\mu_{k+1}}^* r + \epsilon_k \mathbf{1} \end{aligned} \tag{A.10}$$

where $Q_{\mu_{k+1}}^*$ is a matrix whose rows are identical and are the invariant distribution associated with the probability transition kernel $Q_{\mu_{k+1}}$. Note that $Q_{\mu_{k+1}}^* \mathbf{1} = \mathbf{1}$, and that $Q_{\mu_{k+1}}^* Q_{\mu_{k+1}} = Q_{\mu_{k+1}}^*$.

Consider the Bellman Equation corresponding to the state-action relative value function associated with the policy μ_{k+1} :

$$J_{\mu_{k+1}} + Q_{\mu_{k+1}} = r + Q_{\mu_{k+1}} Q_{\mu_{k+1}}$$

It follows that

$$Q_{\mu_{k+1}}^* J_{\mu_{k+1}} + Q_{\mu_{k+1}}^* Q_{\mu_{k+1}} = Q_{\mu_{k+1}}^* r + Q_{\mu_{k+1}}^* Q_{\mu_{k+1}} Q_{\mu_{k+1}}.$$

Since $Q_{\mu_{k+1}}^* Q_{\mu_{k+1}} = Q_{\mu_{k+1}}^*$, we have that

$$J_{\mu_{k+1}} \mathbf{1} = Q_{\mu_{k+1}}^* r.$$

Hence, Equation (A.10) yields

$$l_k \leq J_{\mu_{k+1}} + \epsilon_k.$$

Equivalently, we have

$$l_k - \epsilon_k \leq J_{\mu_{k+1}}.$$

From definition,

$$J^* = \max_{\mu \in \Pi} J_{\mu} \geq J_{\mu_{k+1}}.$$

Let μ^* be the policy corresponding to the optimal average reward J^* . Let Q^* denote the state-action relative value function associated with the policy μ^* . Note that (J^*, Q^*) is the solution of the Bellman optimality equation, i.e.,

$$\begin{aligned} J^* + Q^* &= r + \max_{\mu \in \Pi} Q_{\mu} Q^* \\ &= r + Q_{\mu^*} Q^*. \end{aligned}$$

Then,

$$\begin{aligned} u_k \mathbf{1} &\geq T^Q Q_k - Q_k \\ &\stackrel{(a)}{\geq} T_{\mu^*}^Q Q_k - Q_k \\ &= r + Q_{\mu^*} Q_k - Q_k, \end{aligned}$$

where (a) is due to the fact that T^Q is the Bellman optimality operator. Therefore, we have

$$Q_{\mu^*}^* u_k \mathbf{1} \geq Q_{\mu^*}^* r + Q_{\mu^*}^* Q_{\mu^*} Q_k - Q_{\mu^*}^* Q_k$$

Equivalently,

$$u_k \mathbf{1} \geq Q_{\mu^*}^* r$$

Therefore, we conclude that

$$u_k \geq J^*$$

Hence, for all $k \in \mathbb{N}$,

$$l_k - \epsilon_k \leq J_{\mu_{k+1}} \leq J^* \leq u_k \quad (\text{A.11})$$

A.3.2 Proof of Lemma 2.2

Recall that

$$g_k(s, a) = \left(T^Q Q_k - Q_k \right) (s, a) \geq \left(T_{\mu_k}^Q Q_k - Q_k \right) (s, a),$$

where the inequality follows by the fact that T^Q is the Bellman optimality operator. Note that the Bellman operator $T_{\mu_k}^Q$ is shift-invariant, i.e., for all $p \in \mathbb{R}$:

$$\begin{aligned} T_{\mu_k}^Q (Q_k + p\mathbf{1}) &= r + Q_{\mu_k} (Q_k + p\mathbf{1}) \\ &= r + Q_{\mu_k} Q_k + p\mathbf{1} \\ &= T_{\mu_k}^Q Q_k + p\mathbf{1} \end{aligned}$$

Recall that $\|Q_k - Q_{\mu_k}\|_\infty \leq \delta_k$. Hence, we have

$$\begin{aligned} g_k(s, a) &\geq T_{\mu_k}^Q (Q_{\mu_k} - \delta_k \mathbf{1})(s, a) - (Q_{\mu_k}(s, a) + \delta_k) \\ &\geq \left(T_{\mu_k}^Q Q_{\mu_k} - Q_{\mu_k} \right) (s, a) - 2\delta_k \end{aligned} \quad (\text{A.12})$$

Recall that $Q_{\mu_k} = \lim_{m \rightarrow \infty} \left(\tilde{T}_{\mu_{k+1}}^Q \right)^m Q_k$. Therefore,

$$\begin{aligned} \left(\tilde{T}_{\mu_k}^Q \right)^m Q_{k-1} &= \left(\tilde{T}_{\mu_k}^Q \right)^{m-1} \left(\tilde{T}_{\mu_k}^Q \right) Q_{k-1} \\ &= \left(\tilde{T}_{\mu_k}^Q \right)^{m-1} \left(r + Q_{\mu_k} Q_{k-1} - r(s^*, a^*)\mathbf{1} - (Q_{\mu_k} Q_{k-1})(s^*, a^*)\mathbf{1} \right) \end{aligned}$$

$$\begin{aligned}
(\tilde{T}_{\mu_k}^Q)^m Q_{k-1} &= (\tilde{T}_{\mu_k}^Q)^{m-1} \left(T_{\mu_k}^Q Q_{k-1} - (T_{\mu_k}^Q Q_{k-1})(s^*, a^*) \mathbf{1} \right) \\
&= (\tilde{T}_{\mu_k}^Q)^{m-2} (\tilde{T}_{\mu_k}^Q) \left(T_{\mu_k}^Q Q_{k-1} - (T_{\mu_k}^Q Q_{k-1})(s^*, a^*) \mathbf{1} \right) \\
&= (\tilde{T}_{\mu_k}^Q)^{m-2} \left(r + Q_{\mu_k} \left(T_{\mu_k}^Q Q_{k-1} - (T_{\mu_k}^Q Q_{k-1})(s^*, a^*) \mathbf{1} \right) \right. \\
&\quad \left. - r(s^*, a^*) \mathbf{1} - \left(Q_{\mu_k} \left(T_{\mu_k}^Q Q_{k-1} - (T_{\mu_k}^Q Q_{k-1})(s^*, a^*) \mathbf{1} \right) \right) (s^*, a^*) \mathbf{1} \right) \\
&= (\tilde{T}_{\mu_k}^Q)^{m-2} \left(r + Q_{\mu_k} (T_{\mu_k}^Q Q_{k-1}) - (T_{\mu_k}^Q Q_{k-1})(s^*, a^*) \mathbf{1} \right. \\
&\quad \left. - r(s^*, a^*) \mathbf{1} - Q_{\mu_k} (T_{\mu_k}^Q Q_{k-1})(s^*, a^*) \mathbf{1} + (T_{\mu_k}^Q Q_{k-1})(s^*, a^*) \mathbf{1} \right) \\
&= (\tilde{T}_{\mu_k}^Q)^{m-2} \left((T_{\mu_k}^Q)^2 Q_{k-1} - (T_{\mu_k}^Q)^2 Q_{k-1}(s^*, a^*) \mathbf{1} \right)
\end{aligned}$$

Iterating, we thus obtain,

$$Q_{\mu_k} = \lim_{m \rightarrow \infty} (T_{\mu_k}^Q)^m Q_{k-1} - (T_{\mu_k}^Q)^m Q_{k-1}(s^*, a^*) \mathbf{1}$$

Substituting back in Equation (A.12),

$$\begin{aligned}
g_k(s, a) &\geq \left(T_{\mu_k}^Q \left(\lim_{m \rightarrow \infty} (T_{\mu_k}^Q)^m Q_{k-1} - (T_{\mu_k}^Q)^m Q_{k-1}(s^*, a^*) \mathbf{1} \right) \right. \\
&\quad \left. - \lim_{m \rightarrow \infty} (T_{\mu_k}^Q)^m Q_{k-1} - (T_{\mu_k}^Q)^m Q_{k-1}(s^*, a^*) \mathbf{1} \right) (s, a) - 2\delta_k \\
&= \left(T_{\mu_k}^Q \left(\lim_{m \rightarrow \infty} (T_{\mu_k}^Q)^m Q_{k-1} \right) - \lim_{m \rightarrow \infty} (T_{\mu_k}^Q)^m Q_{k-1}(s^*, a^*) \mathbf{1} \right. \\
&\quad \left. - \lim_{m \rightarrow \infty} (T_{\mu_k}^Q)^m Q_{k-1} - \lim_{m \rightarrow \infty} (T_{\mu_k}^Q)^m Q_{k-1}(s^*, a^*) \mathbf{1} \right) (s, a) - 2\delta_k \\
&= \left(T_{\mu_k}^Q \left(\lim_{m \rightarrow \infty} (T_{\mu_k}^Q)^m Q_{k-1} \right) - \lim_{m \rightarrow \infty} (T_{\mu_k}^Q)^m Q_{k-1} \right) (s, a) - 2\delta_k
\end{aligned}$$

Since $T_{\mu_k}^Q$ is a continuous operator, it is possible to take the limit outside. We thus obtain,

$$\begin{aligned}
g_k(s, a) &\geq \lim_{m \rightarrow \infty} \left(\left(T_{\mu_k}^Q \right)^{m+1} Q_{k-1} - \left(T_{\mu_k}^Q \right)^m Q_{k-1} \right) (s, a) - 2\delta_k \\
&= \lim_{m \rightarrow \infty} \left(\left(T_{\mu_k}^Q \right)^m T_{\mu_k}^Q Q_{k-1} - \left(T_{\mu_k}^Q \right)^m Q_{k-1} \right) (s, a) - 2\delta_k \\
&= \lim_{m \rightarrow \infty} \left(Q_{\mu_k}^m \left(T_{\mu_k}^Q Q_{k-1} - Q_{k-1} \right) \right) (s, a) - 2\delta_k \\
&= Q_{\mu_k}^* \left(T_{\mu_k}^Q Q_{k-1} - Q_{k-1} \right) (s, a) - 2\delta_k.
\end{aligned}$$

Since $\left\| T_{\mu_k}^Q Q_{k-1} - T^Q Q_{k-1} \right\|_{\infty} \leq \epsilon_{k-1}$,

$$\begin{aligned}
g_k(s, a) &\geq Q_{\mu_k}^* \left(T^Q Q_{k-1} - Q_{k-1} \right) (s, a) - 2\delta_k - \epsilon_{k-1} \\
&= \left(Q_{\mu_k}^* g_{k-1} \right) (s, a) - 2\delta_k - \epsilon_{k-1}.
\end{aligned}$$

Recall the definition of $\gamma > 0$ in Lemma 2.2. Note that all entries of $Q_{\mu_k}^*$ are bounded from below by γ . Hence, we have

$$g_k(s, a) \geq (1 - \gamma)l_{k-1} + \gamma u_{k-1} - 2\delta_k - \epsilon_{k-1}.$$

Since the above inequality is true for all states $(s, a) \in \mathcal{S} \times \mathcal{A}$, we obtain,

$$l_k \geq (1 - \gamma)l_{k-1} + \gamma u_{k-1} - 2\delta_k - \epsilon_{k-1}.$$

The above inequality combined with Equation (A.11), yields

$$l_k \geq (1 - \gamma)l_{k-1} + \gamma J^* - 2\delta_k - \epsilon_{k-1}.$$

Upon rearranging the above inequality, we obtain the result, that is,

$$(J^* - l_k) \leq (1 - \gamma) (J^* - l_{k-1}) + 2\delta_k + \epsilon_{k-1}.$$

Subsequently,

$$\begin{aligned} \left(J^* - \min_{(s,a)} \left(T^Q Q_k - Q_k \right) (s, a) \right) &\leq (1 - \gamma) \left(J^* - \min_{(s,a)} \left(T^Q Q_{k-1} - Q_{k-1} \right) (s, a) \right) \\ &\quad + 2 \|Q_k - Q_{\mu_k}\|_\infty + \|T^Q Q_{k-1} - T_{\mu_k}^Q Q_{k-1}\|_\infty \end{aligned}$$

A.3.3 Proof of Proposition 2.1

We now prove Proposition 2.1. From Lemma 2.2, we have

$$(J^* - l_k) \leq (1 - \gamma) (J^* - l_{k-1}) + 2\delta_k + \epsilon_{k-1}. \quad (\text{A.13})$$

To iterate over Equation (A.13), we need to ensure the terms are non-negative. Note that by the first part,

$$l_k - \epsilon_k \leq J^*.$$

Hence, rearranging the terms thus yields,

$$(J^* - l_k + \epsilon_k) \leq (1 - \gamma) (J^* - l_{k-1} + \epsilon_{k-1}) + \epsilon_k + 2\delta_k + \epsilon_{k-1} - (1 - \gamma)\epsilon_{k-1}.$$

This is equivalent to

$$(J^* - l_k + \epsilon_k) \leq (1 - \gamma) (J^* - l_{k-1} + \epsilon_{k-1}) + (\epsilon_k + \gamma\epsilon_{k-1}) + 2\delta_k.$$

Let

$$a_k = J^* - l_k + \epsilon_k.$$

Then, we obtain,

$$\begin{aligned} a_k &\leq (1 - \gamma)a_{k-1} + \epsilon_k + \gamma\epsilon_{k-1} + 2\delta_k \\ &\leq (1 - \gamma) [(1 - \gamma)a_{k-2} + \epsilon_{k-1} + \gamma\epsilon_{k-2} + 2\delta_{k-1}] + \epsilon_k + \gamma\epsilon_{k-1} + 2\delta_k \\ &= (1 - \gamma)^k a_0 + \sum_{\ell=0}^{k-1} (1 - \gamma)^\ell \epsilon_{k-\ell} + \gamma \sum_{\ell=0}^{k-1} (1 - \gamma)^\ell \epsilon_{k-1-\ell} + 2 \sum_{\ell=0}^{k-1} (1 - \gamma)^\ell \delta_{k-\ell} \end{aligned}$$

Since $\epsilon_0 = 0$,

$$a_k \leq (1 - \gamma)^k a_0 + \epsilon_k + \sum_{\ell=1}^{k-1} (1 - \gamma)^{\ell-1} \epsilon_{k-\ell} + 2 \sum_{\ell=0}^{k-1} (1 - \gamma)^\ell \delta_{k-\ell}$$

Substituting for a_k , we get

$$(J^* - l_k) \leq (1 - \gamma)^k (J^* - l_0) + \sum_{\ell=1}^{k-1} (1 - \gamma)^{\ell-1} \epsilon_{k-\ell} + 2 \sum_{\ell=0}^{k-1} (1 - \gamma)^\ell \delta_{k-\ell}.$$

Since from part 1, we know that $J^* - J_{\mu_{k+1}} \leq J^* - l_k + \epsilon_k$, we have

$$\begin{aligned} J^* - J_{\mu_{k+1}} &\leq (1 - \gamma)^k \left[J^* - \min_i \left(T^Q Q_0 - Q_0 \right) (i) \right] \\ &\quad + \sum_{\ell=1}^{k-1} (1 - \gamma)^{\ell-1} \left[\left\| T_{\mu_{k+1-\ell}}^Q Q_{k-\ell} - T^Q Q_{k-\ell} \right\|_\infty \right] + \left\| T_{\mu_{k+1}}^Q Q_k - T^Q Q_k \right\|_\infty \\ &\quad + 2 \sum_{\ell=0}^{k-1} (1 - \gamma)^\ell \left[\left\| Q_{k-\ell} - Q_{\mu_{k-\ell}} \right\|_\infty \right]. \end{aligned}$$

Taking expectation, we have

$$\begin{aligned} \mathbb{E} [J^* - J_{\mu_{k+1}}] &\leq (1 - \gamma)^k \mathbb{E} \left[J^* - \min_i \left(T^Q Q_0 - Q_0 \right) (i) \right] \\ &\quad + \sum_{\ell=1}^{k-1} (1 - \gamma)^{\ell-1} \mathbb{E} \left[\left\| T_{\mu_{k+1-\ell}}^Q Q_{k-\ell} - T^Q Q_{k-\ell} \right\|_\infty \right] + \mathbb{E} \left[\left\| T_{\mu_{k+1}}^Q Q_k - T^Q Q_k \right\|_\infty \right] \\ &\quad + 2 \sum_{\ell=0}^{k-1} (1 - \gamma)^\ell \mathbb{E} \left[\left\| Q_{k-\ell} - Q_{\mu_{k-\ell}} \right\|_\infty \right]. \end{aligned}$$

A.3.4 TD Learning for Q function

One RL algorithm to estimate the state-action relative value function with linear function approximation, associated with a policy μ is $TD(\lambda)$. Finite-time sample complexity of $TD(\lambda)$ has been recently studied in [34]. Invoking their results, restating it in terms of the state-action relative value function rather than the state relative value function, we characterize the sample complexity required to achieve certain accuracy in the approximation.

We estimate $Q_\mu(s, a)$ linearly by $\phi(s, a)^\top \theta_\mu^*$, for some $\theta_\mu^* \in \mathbb{R}^d$, where $\phi(s, a) = [\phi_1(s, a), \dots, \phi_d(s, a)]^\top \in \mathbb{R}^d$ is the feature vector associated with $(s, a) \in \mathcal{S} \times \mathcal{A}$. Here, θ_μ^* is a fixed point of $\Phi\theta = \Pi_{D, W_\phi} T_\mu^{Q, \lambda} \Phi\theta$, where Φ is an $|\mathcal{S}| \times d$ matrix whose k -th column is ϕ_k , $T_\mu^{Q, \lambda} = (1 - \lambda) \sum_{m=0}^{\infty} \lambda^m \left(\hat{T}_\mu^Q \right)^{m+1}$ where $\hat{T}_\mu^Q = T_\mu^Q - J_\mu \mathbf{1}$, D is the diagonal matrix with diagonal elements given by the stationary distribution of the policy μ on $\mathcal{S} \times \mathcal{A}$, and $\Pi_{D, W_\phi} = \Phi(\Phi^\top D \Phi)^{-1} \Phi^\top D$ is the projection matrix onto $W_\phi = \{\Phi\theta | \theta \in \mathbb{R}^d\}$ with respect to the norm $\|\cdot\|_D$. Note that this fixed point equation may have multiple solutions. In particular, if $\Phi\theta_e = \mathbf{1}$, then $\theta_\mu^* + p\theta_e$ is also a solution for any $p \in \mathbb{R}$. Hence, we focus on the set of equivalent classes E where we say $\theta_1 \sim \theta_2$ if $\Phi(\theta_1 - \theta_2) = \mathbf{1}$. The value of $\mathbb{E}[\|\Pi_{2,E}(\theta - \theta_\mu^*)\|_2^2]$ characterizes the accuracy of our approximation, where $\Pi_{2,E}$ is the projection onto E with respect to $\|\cdot\|_2$. Suppose that $\{\phi_1, \phi_2, \dots, \phi_d\}$ are linearly independent and that $\max_{(s,a) \in \mathcal{S} \times \mathcal{A}} \|\phi(s, a)\|_2 \leq 1$.

Upon making a new observation (s_{t+1}, a_{t+1}) for $t \geq 0$, the average reward TD(λ) uses the following update equations:

$$\begin{aligned}
&\text{Eligibility trace: } z_t = \lambda z_{t-1} + \phi(s_t, a_t) \\
&\text{TD error: } d_t = r(s_t, a_t) - J_t + \phi(s_{t+1}, a_{t+1})^\top \theta_t \\
&\quad - \phi(s_t, a_t)^\top \theta_t \\
&\text{average-reward update: } J_{t+1} = J_t + c_\alpha \beta_t (r(s_t, a_t) - J_t), \\
&\text{parameter update: } \theta_{t+1} = \theta_t + \beta_t \delta_t(\theta) z_t,
\end{aligned} \tag{A.14}$$

where β_t is the scalar step size, $c_\alpha > 0$ is a constant, and z_{-1} is initialized to be zero. Following the same argument as in [34], we get the following theorem which characterizes the number of samples required to get a certain accuracy in the approximation.

Theorem A.2. Let $\beta_t = \frac{c_1}{t+c_2}$, and suppose that the positive constants c_1 , c_2 , and c_α are properly chosen. Then, the number of samples required to ensure an approximation accuracy of $\mathbb{E}[\|\Pi_{2,E}(\theta - \theta_\mu^)\|_2^2] \leq \delta$, is given by*

$$\tau = O\left(\frac{K \log(\frac{1}{\delta}) \|\theta_\mu^*\|_2^2}{\Delta^4 (1 - \lambda)^4 \delta^2}\right),$$

where K is the minimum mixing time constant across all probability transition matrices induced by the policies,

$$\Delta = \min_{\|\theta\|_2=1, \theta \in E} \theta^\top \Phi^\top D (I - \mathbb{Q}^\lambda) \Phi \theta > 0,$$

$$\text{and } \mathbb{Q}^\lambda = (1 - \lambda) \sum_{m=0}^{\infty} \lambda^m \mathbb{Q}^{m+1}.$$

Proof. The proof is a simple adaptation of the proof in [34] for the state relative value function. $\square$

Let $Q_k = \Phi \theta_k$, where θ_k is the output of TD(λ) algorithm with T samples. Then, we have

$$\|Q_k - Q_{\mu_k}\|_\infty \leq \|\Phi(\theta_k - \theta_{\mu_k}^*)\|_\infty + \|\Phi \theta_{\mu_k}^* - Q_{\mu_k}\|_\infty,$$

where the second term above is the error associated with the choice of Φ .

Since $\max_{(s,a) \in \mathcal{S} \times \mathcal{A}} \|\phi(s, a)\|_2 \leq 1$, we get

$$\delta_k = d \|\theta_k - \theta_{\mu_k}^*\|_\infty + \|\Phi \theta_{\mu_k}^* - Q_{\mu_k}\|_\infty.$$

Using Theorem A.2, we get a bound on the value of $\|\theta_k - \theta_{\mu_k}^*\|_\infty$. Note that if $\Phi \theta_e = \mathbf{1}$ for some $\theta_e \in \mathbb{R}^d$, then we can pick θ_{k+1} and $\theta_{\mu_{k+1}}^*$ so that $\phi(s^*, a^*)^\top \theta_{k+1} = \phi(s^*, a^*)^\top \theta_{\mu_{k+1}}^* = 0$. Otherwise, there is a unique choice for $\theta_{\mu_{k+1}}^*$, and we have $E = \mathbb{R}^d$.

The expected value of learning error can thus be expressed as:

$$\mathbb{E}[\delta_k] = \mathbb{E}[\delta_{TD,k}] + \mathbb{E}[\delta_{0,k}]$$

where $\delta_{TD,k} = d \|\theta_{k+1} - \theta_{\mu_{k+1}}^*\|_\infty$ represents the TD learning error of the parameter vector $\theta_{\mu_{k+1}}^*$ and $\delta_{0,k} = \|\Phi \theta_{\mu_{k+1}}^* - Q_{\mu_{k+1}}\|_\infty$ represents the function approximation error associated with the span of the feature vector matrix Φ .

$\delta_{TD,k}$ has a direct dependence on the number of samples utilized for TD learning and the mixing time of the corresponding Markov Chain induced by the policy μ_k . More precisely,

from Theorem A.2,

$$\mathbb{E} [\delta_{\text{TD},k}] = O \left(\sqrt{\frac{K \log(\frac{1}{\Delta}) \|\theta_{\mu_k}^*\|_2^2}{\Delta^4 (1-\lambda)^4 \tau}} \right),$$

Hence as long as the mixing constant Δ is uniformly greater than zero, and the feature vectors $\theta_{\mu_k}^*$ are such that they are uniformly upper bounded in k , it is true that

$$\mathbb{E} [\delta_{\text{TD},k}] = \frac{C}{\sqrt{\tau}},$$

for some constant $C > 0$.

Next, we prove the corollaries for specific policy improvement algorithms. Since the policy improvement part does not depend on whether the problem is a discounted-reward problem or an average reward problem, we can borrow the results from [19] to identify ϵ_k in each of the corollaries.

A.3.5 Proof of Corollary 2.2

Recall the greedy update rule. Let $a^* = \operatorname{argmax}_a Q_k(s, a')$. Given a parameter $\beta > 0$, at any time instant k , the greedy policy update μ_{k+1} is given by:

$$\mu_{k+1}(a|s) = \begin{cases} \frac{1}{\beta|\mathcal{A}|}, & \text{if } a \neq a^* \\ \frac{1}{\beta|\mathcal{A}|} + 1 - \frac{1}{\beta}, & \text{if } a = a^* \end{cases} \quad (\text{A.15})$$

Let $\eta_k = \max_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} |Q_k(s', a')|$. The policy improvement approximation can be shown to be the following:

$$\begin{aligned} \epsilon_k &= \left(T^Q Q_k - T_{\mu_{k+1}}^Q Q_k \right) (s, a) \\ &\leq \sum_{s' \in \mathcal{S}} \mathbb{P}(s'|s, a) \frac{2}{\beta} \max_{a' \in \mathcal{A}} |Q_k(s', a')| \\ &\leq \frac{2\eta_k}{\beta}. \end{aligned}$$

Recall the error due to TD Learning

$$\mathbb{E} [\delta_k] = \mathbb{E} [\delta_{\text{TD},k}] + \mathbb{E} [\delta_{0,k}].$$

Let $\bar{\delta}_0 = \max_t \delta_{0,t}$. Then we obtain the following:

$$\mathbb{E} [\delta_k] = \bar{\delta}_0 + \frac{C}{\sqrt{\tau}}.$$

Substituting for expressions in Proposition 2.1, we obtain,

$$\begin{aligned} \mathbb{E} [J^* - J_{\mu_{T+1}}] &\leq (1 - \gamma)^T \mathbb{E} \left[J^* - \min_i \left(T^Q Q_0 - Q_0 \right) (i) \right] \\ &\quad + 2 \sum_{\ell=0}^{T-1} (1 - \gamma)^\ell \left(\bar{\delta}_0 + \frac{C}{\sqrt{\tau}} \right) + \sum_{\ell=1}^{T-1} (1 - \gamma)^{\ell-1} \frac{2\eta_{T-\ell}}{\beta} + \frac{2\eta_T}{\beta}. \end{aligned}$$

Let $c_0 = \mathbb{E} \left[J^* - \min_i \left(T^Q Q_0 - Q_0 \right) (i) \right]$ be the error associated with the initial condition. Let $\bar{\eta} = \max_t \eta_t$ ($\bar{\eta}$ can be the uniform upper bound of the estimates Q_k of relative state action value function Q_{μ_k} over k .) Then we obtain the result in the corollary,

$$\mathbb{E} [J^* - J_{\mu_{T+1}}] \leq (1 - \gamma)^T c_0 + \frac{2}{\gamma} \left(\bar{\delta}_0 + \frac{C}{\sqrt{\tau}} \right) + \left(\frac{1 + \gamma}{\gamma} \right) \frac{2\bar{\eta}}{\beta}. \quad (\text{A.16})$$

A.3.6 Proof of Corollary 2.3

Recall the softmax policy update. Given a parameter $\beta > 0$, at any time instant k , the Softmax update policy μ_{k+1} is:

$$\mu_{k+1}(a|s) = \frac{\exp(\beta Q_k(s, a))}{\sum_{a' \in \mathcal{A}} \exp(\beta Q_k(s, a'))}. \quad (\text{A.17})$$

The policy improvement approximation for this update rule turns out to be time independent and is given by:

$$\epsilon_k = \left(T^Q Q_k - T_{\mu_{k+1}}^Q Q_k \right) (s, a) \leq \frac{\log |\mathcal{A}|}{\beta}.$$

Given $\bar{\delta}_0 = \max_t \delta_{0,t}$ the following is true,

$$\mathbb{E} [\delta_k] = \bar{\delta}_0 + \frac{C}{\sqrt{\tau}}.$$

Substituting for expressions in Proposition 2.1, we obtain,

$$\begin{aligned} \mathbb{E} [J^* - J_{\mu_{\tau+1}}] &\leq (1 - \gamma)^T \mathbb{E} \left[J^* - \min_i \left(T^Q Q_0 - Q_0 \right) (i) \right] \\ &\quad + 2 \sum_{\ell=0}^{T-1} (1 - \gamma)^\ell \left(\bar{\delta}_0 + \frac{C}{\sqrt{\tau}} \right) + \sum_{\ell=1}^{T-1} (1 - \gamma)^{\ell-1} \frac{\log |\mathcal{A}|}{\beta} + \frac{\log |\mathcal{A}|}{\beta}. \end{aligned}$$

Let $c_0 = \mathbb{E} \left[J^* - \min_i \left(T^Q Q_0 - Q_0 \right) (i) \right]$ be the error associated with the initial condition. Then we obtain the result in the corollary,

$$\mathbb{E} [J^* - J_{\mu_{\tau+1}}] \leq (1 - \gamma)^T c_0 + \frac{2}{\gamma} \left(\bar{\delta}_0 + \frac{C}{\sqrt{\tau}} \right) + \left(\frac{1 + \gamma}{\gamma} \right) \frac{\log (|\mathcal{A}|)}{\beta}. \quad (\text{A.18})$$

A.3.7 Proof of Corollary 2.4

Recall the mirror descent update. Given $\beta > 0$ the mirror descent update is given by:

$$\mu_{k+1}(a|s) = \frac{\mu_k(a|s) e^{\beta Q_k(s,a)}}{\sum_{a' \in \mathcal{A}} \mu_k(a'|s) e^{\beta Q_k(s,a')}} \quad (\text{A.19})$$

The policy improvement error for this update rule and is given by:

$$\begin{aligned} \epsilon_k &= \left(T^Q Q_k - T_{\mu_{k+1}}^Q Q_k \right) (s, a) \\ &\leq \frac{1}{\beta} \log \frac{1}{\min_{s \in \mathcal{S}} \mu_{k+1} (a^*(s)|s)}. \end{aligned}$$

where $a^*(s)$ is the optimal action at state s . Let $\omega_{k+1} = \min_{s \in \mathcal{S}} \mu_{k+1} (a^*(s)|s)$ Given that the TD learning error is of the form

$$\mathbb{E} [\delta_k] = \bar{\delta}_0 + \frac{C}{\sqrt{\tau}},$$

Substituting for expressions in Proposition 2.1, we obtain,

$$\begin{aligned} \mathbb{E} [J^* - J_{\mu_{T+1}}] &\leq (1 - \gamma)^T \mathbb{E} \left[J^* - \min_i \left(\tau^Q Q_0 - Q_0 \right) (i) \right] \\ &\quad + 2 \sum_{\ell=0}^{T-1} (1 - \gamma)^\ell \left(\bar{\delta}_0 + \frac{C}{\sqrt{\tau}} \right) + \sum_{\ell=1}^{T-1} (1 - \gamma)^{\ell-1} \frac{2}{\beta} \log \frac{1}{\omega_{T-\ell}} + \frac{2}{\beta} \log \frac{1}{\omega_T}. \end{aligned}$$

Let $c_0 = \mathbb{E} \left[J^* - \min_i \left(\tau^Q Q_0 - Q_0 \right) (i) \right]$ be the error associated with the initial condition.

Let $\underline{\omega} = \min_t \omega_t$. Then we obtain the result in the corollary,

$$\mathbb{E} [J^* - J_{\mu_{T+1}}] \leq (1 - \gamma)^T c_0 + \frac{2}{\gamma} \left(\bar{\delta}_0 + \frac{C}{\sqrt{\tau}} \right) + \left(\frac{1 + \gamma}{\gamma \beta} \right) \log \left(\frac{1}{\underline{\omega}} \right). \quad (\text{A.20})$$

A.3.8 Regret Analysis in Section 2.4.2

Recall the pseudo regret defined as follows:

$$R_{PS}(K) = \sum_{t=1}^K (J^* - J_{\mu_t}), \quad (\text{A.21})$$

where μ_t is the policy utilized at time t and obtained through mirror descent, and K is the time horizon. Since $K = \tau T$, we have

$$R_{PS}(K) = \tau \sum_{t=1}^T (J^* - J_{\mu_t}). \quad (\text{A.22})$$

Substituting for $J^* - J_{\mu_t}$, it is true that

$$R_{PS}(K) = \tau \left(\sum_{t=1}^T (1 - \gamma)^T c_0 + \frac{1 + \gamma}{\gamma \beta} \log \left(\frac{1}{\underline{\omega}} \right) + \frac{2}{\gamma} \left(\bar{\delta}_0 + \frac{c_3}{\sqrt{\tau}} \right) \right). \quad (\text{A.23})$$

Let $\beta = \sqrt{\tau}$. The above regret can be simplified as:

$$R_{PS}(K) = \frac{\tau c_0}{\gamma} + \frac{K}{\gamma \sqrt{\tau}} \left((1 + \gamma) \log \left(\frac{1}{\underline{\omega}} \right) + 2c_3 \right) + \frac{2K \bar{\delta}_0}{\gamma} \quad (\text{A.24})$$

Let $c_5 = (1 + \gamma) \log(1/\omega) + 2c_3$. We then have

$$R_{PS}(K) = \frac{\tau c_0}{\gamma} + \frac{K c_5}{\gamma \sqrt{\tau}} + \frac{2K \bar{\delta}_0}{\gamma}. \quad (A.25)$$

Optimizing for regret involves equating $\frac{\tau c_0}{\gamma}$ and $\frac{K c_5}{\gamma \sqrt{\tau}}$. This yields $\tau = \left(\frac{K c_5}{c_0}\right)^{2/3}$. Further substituting for τ yields

$$R_{PS}(K) = \left(\frac{K c_5}{c_0}\right)^{2/3} \cdot \frac{c_0}{\gamma} + \frac{K c_5 c_0^{1/3}}{\gamma (K c_5)^{1/3}} + \frac{2K \bar{\delta}_0}{\gamma} \quad (A.26)$$

$$= \frac{(K c_5)^{2/3} \cdot c_0^{1/3}}{\gamma} + \frac{(K c_5)^{2/3} \cdot c_0^{1/3}}{\gamma} + \frac{2K \bar{\delta}_0}{\gamma} \quad (A.27)$$

$$= \frac{2c_5^{2/3} \cdot c_0^{1/3} \cdot K^{2/3}}{\gamma} + \frac{2K \bar{\delta}_0}{\gamma}. \quad (A.28)$$

Let $c_6 = 2c_5^{2/3} c_0^{1/3}$. We have

$$R_{PS}(K) = \frac{K^{2/3} c_6}{\gamma} + \frac{2K \bar{\delta}_0}{\gamma}. \quad (A.29)$$

B

PROOFS: GLOBAL CONVERGENCE OF PROJECTED POLICY GRADIENT FOR AVERAGE REWARD

B.1 Smoothness of Average Reward

B.1.1 Proof of Lemma 3.1

Consider the subspace orthogonal E to the all ones vector $\mathbb{1} \in \mathbb{R}^{|\mathcal{S}|}$ defined below:

$$E = \text{span} \left\{ \theta \in \mathbb{R}^{|\mathcal{S}|} : \theta^\top \mathbb{1} = 0. \right\}$$

The orthogonal projection v_ϕ of a vector v in the Euclidean norm onto the subspace E is defined as:

$$v_\phi = \underset{u \in E}{\text{argmin}} \|v - u\|_2$$

It can be checked that the closed form expression for v_ϕ is given by:

$$v_\phi = \left(I - \frac{\mathbb{1} \mathbb{1}^\top}{|\mathcal{S}|} \right) v$$

where $I \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{S}|}$ is the identity matrix.

Consider the projection of the vector $r^\pi + \mathbb{P}^\pi v^\pi - \rho^\pi \mathbb{1}$ onto E for any policy $\pi \in \Pi$. The

above projection is identical to the projection of $\mathbf{r}^\pi + \mathbb{P}^\pi \mathbf{v}^\pi$ onto $\mathcal{E}$, since $\rho^\pi \mathbb{1}$ lies in the nullspace of Φ .

$$\begin{aligned}
\Phi(\mathbf{r}^\pi - \rho^\pi \mathbb{1} + \mathbb{P}^\pi \mathbf{v}) &= (\mathbf{r}^\pi - \rho^\pi \mathbb{1} + \mathbb{P}^\pi \mathbf{v}) - \langle \mathbb{1}, \mathbf{r}^\pi - \rho^\pi \mathbb{1} + \mathbb{P}^\pi \mathbf{v} \rangle \frac{\mathbb{1}}{|\mathcal{S}|} \\
&= (\mathbf{r}^\pi + \mathbb{P}^\pi \mathbf{v}) - \langle \mathbb{1}, \mathbf{r}^\pi + \mathbb{P}^\pi \mathbf{v} \rangle \frac{\mathbb{1}}{|\mathcal{S}|} \\
&= \mathbf{r}^\pi - \langle \mathbf{r}^\pi, \mathbb{1} \rangle \frac{\mathbb{1}}{|\mathcal{S}|} + \mathbb{P}^\pi \mathbf{v} - \langle \mathbb{1}, \mathbb{P}^\pi \mathbf{v} \rangle \frac{\mathbb{1}}{|\mathcal{S}|} \\
&= \mathbf{r}^\pi - \langle \mathbf{r}^\pi, \mathbb{1} \rangle \frac{\mathbb{1}}{|\mathcal{S}|} + \mathbb{P}^\pi \mathbf{v} - \frac{\mathbb{1}}{|\mathcal{S}|} (\mathbb{1}^\top \mathbb{P}^\pi \mathbf{v}) \\
&= (\mathbf{I} - \frac{\mathbb{1} \mathbb{1}^\top}{|\mathcal{S}|}) \mathbf{r}^\pi + (\mathbf{I} - \frac{\mathbb{1} \mathbb{1}^\top}{|\mathcal{S}|}) \mathbb{P}^\pi \mathbf{v} \\
&= \Phi [\mathbf{r}^\pi + \mathbb{P}^\pi \mathbf{v}].
\end{aligned}$$

Consider the average reward Bellman equation corresponding to policy $\pi \in \Pi$:

$$\rho^\pi \mathbb{1} + \mathbf{v}^\pi = \mathbf{r}^\pi + \mathbb{P}^\pi \mathbf{v}^\pi \quad (\text{B.1})$$

Imposing an additional constraint $\mathbf{v}^{\pi^\top} \mathbb{1} = 0$ yields a unique average reward value function denoted by $\mathbf{v}_\phi^\pi$. Moreover, it is true that,

$$\begin{aligned}
\Phi \mathbf{v}_\phi^\pi + \Phi \rho^\pi \mathbb{1} &= \Phi \mathbf{r}^\pi + \Phi \mathbb{P}^\pi \mathbf{v}_\phi^\pi \\
\implies \Phi \mathbf{v}_\phi^\pi &= \Phi \mathbf{r}^\pi + \Phi \mathbb{P}^\pi \mathbf{v}_\phi^\pi, \\
&\stackrel{(a)}{\implies} \mathbf{v}_\phi^\pi = \Phi [\mathbf{r}^\pi + \mathbb{P}^\pi \mathbf{v}_\phi^\pi],
\end{aligned}$$

where (a) is true because $\mathbf{v}^{\pi^\top} \mathbb{1} = 0 \implies \Phi \mathbf{v}^\pi = \mathbf{v}^\pi$. Thus the projected value function with an unique representation is given by:

$$\mathbf{v}_\phi^\pi = [\mathbf{I} - \Phi \mathbb{P}^\pi]^{-1} \Phi \mathbf{r}^\pi, \quad (\text{B.2})$$

and the existence of the inverse is proven in Subsection B.1.2, Lemma B.4. An alternate expression for the projected value function is given by: $\mathbf{v}_\phi^\pi = \left(\mathbf{I} + \mathbb{1} \mathbb{1}^\top \mathbf{D} - \frac{\mathbb{1} \mathbb{1}^\top}{|\mathcal{S}|} \right) \mathbf{v}_0^\pi$, where $\mathbf{D} \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{S}|}$ is a diagonal matrix whose entries correspond to the stationary measure over

the states associated with policy π . See [35] for more details.

B.1.2 Proof that Eigenvalues of $(I - \Phi^{\mathbb{P}^\pi})$ are Non-zero

In this subsection, we introduce the lemmas required to establish the proof of the eigenvalues of $(I - \frac{\mathbb{1}\mathbb{1}^\top}{|\mathcal{S}|}) \mathbb{P}^\pi$ being nonzero. We use the following notation: $\mathbb{1} \in \mathbb{R}^n$ represents the all ones vector and $I \in \mathbb{R}^{n \times n}$ is the identity matrix.

Lemma B.1. Let $A \in \mathbb{R}^{n \times n}$ be a stochastic matrix. It is true that

$$\left(\left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A \right)^k = \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A^k.$$

Proof. For any $k \in \mathbb{N}$, consider,

$$\begin{aligned} \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A^k \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A &= \left(A^k - \frac{\mathbb{1}\mathbb{1}^\top}{n} A^k \right) \left(A - \frac{\mathbb{1}\mathbb{1}^\top}{n} A \right) \\ &= A^{k+1} - \frac{\mathbb{1}\mathbb{1}^\top}{n} A^{k+1} - A^k \frac{\mathbb{1}\mathbb{1}^\top}{n} A + \frac{\mathbb{1}\mathbb{1}^\top}{n} A^k \frac{\mathbb{1}\mathbb{1}^\top}{n} A \\ &\stackrel{(a)}{=} A^{k+1} - \frac{\mathbb{1}\mathbb{1}^\top}{n} A^{k+1} - \frac{\mathbb{1}\mathbb{1}^\top}{n} A + \frac{\mathbb{1}\mathbb{1}^\top}{n} \frac{\mathbb{1}\mathbb{1}^\top}{n} A, \\ &\stackrel{(b)}{=} A^{k+1} - \frac{\mathbb{1}\mathbb{1}^\top}{n} A^{k+1} - \frac{\mathbb{1}\mathbb{1}^\top}{n} A + \frac{\mathbb{1}\mathbb{1}^\top}{n} A, \\ &= A^{k+1} - \frac{\mathbb{1}\mathbb{1}^\top}{n} A^{k+1}. \\ &= \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A^{k+1} \end{aligned}$$

where (a) is true because $A^k \mathbb{1} = \mathbb{1}$ and (b) follows from the fact that $\frac{\mathbb{1}^\top \mathbb{1}}{n} = \mathbb{1}$. From mathematical induction it thus follows that,

$$\left(\left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A \right)^k = \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A^k \quad \forall k \in \mathbb{N}.$$

□

Lemma B.2. For any irreducible and aperiodic stochastic matrix $A \in \mathbb{R}^{n \times n}$, it is true that

$$\lim_{k \rightarrow \infty} \left(\left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A \right)^k = 0$$

Proof. From Lemma B.1 we have,

$$\left(\left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A \right)^k = \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A^k \quad \forall k \in \mathbb{N}$$

Since A is irreducible and aperiodic, the following limit converges to the stationary distribution $\mathbf{d} \in \mathbb{R}_+^n$ associated with A .

$$\lim_{k \rightarrow \infty} A^k = \mathbb{1}\mathbf{d}^\top \quad (\text{B.3})$$

Consider the following,

$$\begin{aligned} \lim_{k \rightarrow \infty} \left(\left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A \right)^k &= \lim_{k \rightarrow \infty} \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A^k \\ &= \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) \lim_{k \rightarrow \infty} A^k \\ &= \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) \mathbb{1}\mathbf{d}^\top \quad (\text{from Equation (B.3)}), \\ &= \mathbb{1}\mathbf{d}^\top - \frac{\mathbb{1}\mathbb{1}^\top}{n} \mathbb{1}\mathbf{d}^\top \\ &\stackrel{(a)}{=} \mathbb{1}\mathbf{d}^\top - \mathbb{1}\mathbf{d}^\top \\ &= 0. \end{aligned}$$

where (a) is true because $\frac{\mathbb{1}^\top \mathbb{1}}{n} = 1$. □

Lemma B.3. Let $A \in \mathbb{R}^{n \times n}$ be a matrix such that $\lim_{k \rightarrow \infty} A^k = 0$. Then $(I - A)^{-1} = \sum_{k=0}^{\infty} A^k$.

Proof. For any $K \in \mathbb{N}$, consider the following,

$$\begin{aligned} (I - A) \left(\sum_{k=0}^K A^k \right) &= I - A^{K+1}, \\ \implies (I - A) \left(\lim_{K \rightarrow \infty} \sum_{k=0}^K A^k \right) &= \lim_{K \rightarrow \infty} (I - A^{K+1}) \stackrel{(a)}{=} I, \end{aligned}$$

where (a) follows from the fact that $\lim_{k \rightarrow \infty} A^k = 0$. Hence the inverse of $(I - A)$ can be expressed as $(I - A)^{-1} = \sum_{k=0}^{\infty} A^k$. $\square$

Lemma B.4. Let $A \in \mathbb{R}^{n \times n}$ be an irreducible and aperiodic stochastic matrix. Then the matrix $\left(I - \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A \right)$ is invertible and its inverse is given by:

$$\left(I - \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A \right)^{-1} = \sum_{k=0}^{\infty} \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A^k$$

Proof. Let λ_i be eigenvalues of $\left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A$.

Then λ_i^k represents the eigenvalues of $\left(\left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A \right)^k$. But from Lemma B.2, we know that

$$\lim_{k \rightarrow \infty} \left(\left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A \right)^k = 0$$

Since eigenvalues are continuous functions of their corresponding matrices and all eigenvalues of a zero matrix are zero, we thus have,

$$\lim_{k \rightarrow \infty} \lambda_i^k = 0 \quad \forall i \in \{1, \dots, n\} \quad (\text{B.4})$$

Equation B.4 thus implies that $|\lambda_i| < 1, \forall i \in \{1, \dots, n\}$.

Hence the matrix $\left(I - \left(\left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A \right) \right)$ has all non zero eigenvalues and is thus invertible.

From Lemma B.3, we know that

$$(I - A)^{-1} = \sum_{k=0}^{\infty} A^k$$

when $\lim_{k \rightarrow \infty} A^k = 0$. Since, $\lim_{k \rightarrow \infty} \left(\left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A \right)^k = 0$ from Lemma B.2, we have the following result,

$$\left(I - \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A \right)^{-1} = \sum_{k=0}^{\infty} \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right) A^k$$

□

From definition we have $\Phi = \left(I - \frac{\mathbb{1}\mathbb{1}^\top}{n} \right)$. Hence the inverse $(1 - \Phi \mathbb{P}^\pi)^{-1} = \sum_{k=0}^{\infty} \Phi (\mathbb{P}^\pi)^k$ exists and is well defined for all $\pi \in \Pi$.

B.1.3 Smoothness of Average Reward Value Function v_ϕ^π – Lemma 3.2

In order to prove the smoothness of the average reward value function and the infinite horizon average reward, we consider an analysis inspired by [46], where instead of computing the maximum eigenvalue of the associated Hessian matrices, we consider the maximum value of the directional derivative across all directions within the policy class.

Let $\pi, \pi' \in \Pi$ be any policies within the policy class. Then define π_α as a convex combination of policies π and π' . That is

$$\begin{aligned} \pi_\alpha &:= (1 - \alpha)\pi + \alpha\pi' \\ &= \pi + \alpha(\pi' - \pi) \\ &= \pi + \alpha u \end{aligned}$$

where $u = \pi' - \pi$.

Since π_α is linear in α , it is true that

$$\nabla_\alpha \pi_\alpha = \frac{d(\pi + \alpha u)}{d\alpha} = u, \quad \text{and} \quad \nabla_\alpha^2 \pi_\alpha = 0.$$

This thus implies,

$$\|\nabla_{\alpha}\pi_{\alpha}\|_2 = \|\mathbf{u}\|_2 \leq \|\pi' - \pi\|_1 \leq 2S, \quad \text{and} \quad \left\|\nabla_{\alpha}^2\pi_{\alpha}\right\|_2 = 0,$$

Thus, π_{α} is both $\|\mathbf{u}\|_2$ -Lipschitz and 0-smooth with respect to α , for all $\mathbf{u}$ that can be represented as the difference of any two policies.

From the definition of $\mathbb{P}^{\pi}$, we have

$$\begin{aligned} \mathbb{P}^{\pi_{\alpha}}(s'|s) &= \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\alpha}(\mathbf{a}|s) \mathbb{P}(s'|s, \mathbf{a}) \\ &= \sum_{\mathbf{a} \in \mathcal{A}} [\pi(\mathbf{a}|s) + \alpha \mathbf{u}(\mathbf{a}|s)] \mathbb{P}(s'|s, \mathbf{a}) \\ \implies \frac{\partial \mathbb{P}^{\pi_{\alpha}}(s'|s)}{\partial \alpha} &= \sum_{\mathbf{a} \in \mathcal{A}} \mathbf{u}(\mathbf{a}|s) \mathbb{P}(s'|s, \mathbf{a}). \end{aligned}$$

That is,

$$\nabla_{\alpha} \mathbb{P}^{\pi_{\alpha}} = \mathbb{P}^{\mathbf{u}}, \quad \text{consequently} \quad \nabla_{\alpha}^2 \mathbb{P}^{\pi_{\alpha}} = 0. \quad (\text{B.5})$$

From the definition of $\mathbf{r}^{\pi}$, we have

$$\begin{aligned} \mathbf{r}^{\pi_{\alpha}}(s) &= \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\alpha}(\mathbf{a}|s) \mathbf{r}(s, \mathbf{a}) \\ &= \sum_{\mathbf{a} \in \mathcal{A}} [\pi(\mathbf{a}|s) + \alpha \mathbf{u}(\mathbf{a}|s)] \mathbf{r}(s, \mathbf{a}) \\ \implies \frac{\partial \mathbf{r}^{\pi_{\alpha}}(s)}{\partial \alpha} &= \sum_{\mathbf{a} \in \mathcal{A}} \mathbf{u}(\mathbf{a}|s) \mathbf{r}(s, \mathbf{a}). \end{aligned}$$

That is,

$$\nabla_{\alpha} \mathbf{r}^{\pi_{\alpha}} = \mathbf{r}^{\mathbf{u}}, \quad \text{consequently} \quad \nabla_{\alpha}^2 \mathbf{r}^{\pi_{\alpha}} = 0. \quad (\text{B.6})$$

Hence the policy π_{α} , the associated reward $\mathbf{r}^{\pi_{\alpha}}$ and the transition kernel $\mathbb{P}^{\pi_{\alpha}}$ are all Lipschitz and smooth with respect to α .

Lemma B.5. Let $A(\alpha) \in \mathbb{R}^{n \times n}$ be a matrix such that $(I - A(\alpha))$ is invertible for all $\alpha \in [0, 1]$.

Define $M(\alpha) := (I - A(\alpha))^{-1}$. Then it is true that,

$$\frac{\partial^2 M(\alpha)}{\partial \alpha^2} = \frac{\partial M(\alpha)}{\partial \alpha} \frac{\partial A(\alpha)}{\partial \alpha} M(\alpha) + M(\alpha) \frac{\partial^2 A(\alpha)}{\partial \alpha^2} M(\alpha) + M(\alpha) \frac{\partial A(\alpha)}{\partial \alpha} \frac{\partial M(\alpha)}{\partial \alpha}.$$

Proof.

$$\begin{aligned} M(\alpha) (I - A(\alpha)) &= I \\ \frac{\partial M(\alpha)}{\partial \alpha} (I - A(\alpha)) - M(\alpha) \frac{\partial A(\alpha)}{\partial \alpha} &= 0 \\ \frac{\partial M(\alpha)}{\partial \alpha} &= M(\alpha) \frac{\partial A(\alpha)}{\partial \alpha} M(\alpha) \\ \frac{\partial^2 M(\alpha)}{\partial \alpha^2} &= \frac{\partial}{\partial \alpha} \left(M(\alpha) \frac{\partial A(\alpha)}{\partial \alpha} M(\alpha) \right), \end{aligned}$$

The last equation can be expanded as

$$\frac{\partial^2 M(\alpha)}{\partial \alpha^2} = \frac{\partial M(\alpha)}{\partial \alpha} \frac{\partial A(\alpha)}{\partial \alpha} M(\alpha) + M(\alpha) \frac{\partial^2 A(\alpha)}{\partial \alpha^2} M(\alpha) + M(\alpha) \frac{\partial A(\alpha)}{\partial \alpha} \frac{\partial M(\alpha)}{\partial \alpha}$$

□

Consider the following definition utilized in the proofs of the upcoming lemmas.

$$M^{\pi_\alpha} = [I - \Phi \mathbb{P}^{\pi_\alpha}]^{-1} \quad (\text{B.7})$$

Lemma B.6. Recall the definition of the projected average reward value function v_Φ^π in Equation (B.2).

Value function v_Φ^π is $2C_m^2 C_p \kappa_r + 2C_m C_r$ -Lipschitz in Π , that is

$$\left| \left\langle \frac{\partial v_\Phi^\pi}{\partial \pi}, \pi' - \pi \right\rangle \right| \leq 2 \left(C_m^2 C_p \kappa_r + C_m C_r \right) \|\pi' - \pi\|_2, \quad \forall \pi, \pi' \in \Pi.$$

Proof.

$$\begin{aligned}
v_{\Phi}^{\pi_{\alpha}} &= M^{\pi_{\alpha}} \Phi r^{\pi_{\alpha}} \\
\Rightarrow \frac{\partial v_{\Phi}^{\pi_{\alpha}}}{\partial \alpha} &= \frac{\partial M^{\pi_{\alpha}}}{\partial \alpha} \Phi r^{\pi_{\alpha}} + M^{\pi_{\alpha}} \Phi \frac{\partial r^{\pi_{\alpha}}}{\partial \alpha} \\
&= M^{\pi_{\alpha}} \frac{\partial \Phi \mathbb{P}^{\pi_{\alpha}}}{\partial \alpha} M^{\pi_{\alpha}} \Phi r^{\pi_{\alpha}} + M^{\pi_{\alpha}} \Phi \frac{\partial r^{\pi_{\alpha}}}{\partial \alpha}, \quad (\text{from Lemma B.5}), \\
&= M^{\pi_{\alpha}} \Phi \mathbb{P}^u M^{\pi_{\alpha}} \Phi r^{\pi_{\alpha}} + M^{\pi_{\alpha}} \Phi r^u, \quad (\text{from (B.5) and (B.6)}). \\
\Rightarrow \left\| \frac{\partial v_{\Phi}^{\pi_{\alpha}}}{\partial \alpha} \right\|_{\infty} &= \|M^{\pi_{\alpha}} \Phi \mathbb{P}^u M^{\pi_{\alpha}} \Phi r^{\pi_{\alpha}} + M^{\pi_{\alpha}} \Phi r^u\|_{\infty} \\
&\leq \|M^{\pi_{\alpha}}\|_{\infty} \|\Phi\|_{\infty} \|\mathbb{P}^u\|_{\infty} \|M^{\pi_{\alpha}}\|_{\infty} \|\Phi r^{\pi_{\alpha}}\|_{\infty} + \|M^{\pi_{\alpha}}\|_{\infty} \|\Phi r^u\|_{\infty} \\
&\leq 2C_m^2 C_p \kappa_r + 2C_m C_r.
\end{aligned}$$

The constants C_m, C_p, C_r and κ_r are characterized in Table 1 with their respective bounds in Lemma B.10. □

We can now build on the previous lemma to prove the smoothness of the average reward value function.

Lemma B.7. The value function v_{Φ}^{π} is $8(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r)$ -smooth in Π . That is,

$$\left\langle \pi' - \pi, \frac{\partial^2 v_{\Phi}^{\pi}(s)}{\partial \pi^2} (\pi' - \pi) \right\rangle \leq 8 \left(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r \right) \|\pi' - \pi\|_2^2 \quad \forall \pi', \pi \in \Pi, s \in \mathcal{S}$$

Proof. From Lemma B.6, it is true that

$$\begin{aligned}
\frac{\partial v_{\Phi}^{\pi_{\alpha}}}{\partial \alpha} &= M^{\pi_{\alpha}} \Phi \mathbb{P}^u M^{\pi_{\alpha}} \Phi r^{\pi_{\alpha}} + M^{\pi_{\alpha}} \Phi r^u \\
\Rightarrow \frac{\partial^2 v_{\Phi}^{\pi_{\alpha}}}{\partial \alpha^2} &= \frac{\partial}{\partial \alpha} \left[M^{\pi_{\alpha}} \Phi \mathbb{P}^u M^{\pi_{\alpha}} \Phi r^{\pi_{\alpha}} + M^{\pi_{\alpha}} \Phi r^u \right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{\partial M^{\pi_\alpha}}{\partial \alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi \mathbb{P}^u \frac{\partial M^{\pi_\alpha}}{\partial \alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi \frac{\partial r^{\pi_\alpha}}{\partial \alpha} + \frac{\partial M^{\pi_\alpha}}{\partial \alpha} \Phi r^u \\
&= M^{\pi_\alpha} \frac{\partial \Phi \mathbb{P}^{\pi_\alpha}}{\partial \alpha} M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \frac{\partial \Phi \mathbb{P}^{\pi_\alpha}}{\partial \alpha} M^{\pi_\alpha} \Phi r^{\pi_\alpha} \\
&\quad + M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi \frac{\partial r^{\pi_\alpha}}{\partial \alpha} + M^{\pi_\alpha} \frac{\partial \Phi \mathbb{P}^{\pi_\alpha}}{\partial \alpha} M^{\pi_\alpha} \Phi r^u, \quad (\text{from Lemma B.5}), \\
&= M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} \\
&\quad + M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^u + M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^u, \quad (\text{from (B.5) and (B.6)}).
\end{aligned}$$

Considering the L_∞ norm,

$$\begin{aligned}
\left\| \frac{\partial^2 v_\Phi^{\pi_\alpha}}{\partial \alpha^2} \right\|_\infty &= 2 \| M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^u \|_\infty \\
&\leq 2 \| M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} \|_\infty + \| M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^u \|_\infty \\
&\leq 8(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r).
\end{aligned}$$

Hence, we obtain,

$$\left\langle \pi' - \pi, \frac{\partial^2 v_\Phi^\pi(s)}{\partial \pi} (\pi' - \pi) \right\rangle \leq 8 \left(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r \right) \|\pi' - \pi\|_2^2 \quad \forall \pi', \pi \in \Pi, s \in \mathcal{S}$$

□

B.1.4 Lipschitzness of the Average Reward ρ^π – Lemma 3.3

The Lipschitzness and smoothness of the projected value function v_Φ^π is leveraged through the average reward Bellman equation to prove the Lipschitzness and smoothness of the infinite horizon average reward.

Lemma B.8. Recall the average reward Bellman Equation corresponding to a policy π and projected

value function v_Φ^π in Equation (B.1). The average reward ρ^π is L_1^Π -Lipschitz.

$$\left| \left\langle \frac{\partial \rho^\pi}{\partial \pi}, \pi' - \pi \right\rangle \right| \leq L_1^\Pi \|\pi' - \pi\|_2, \quad \forall \pi, \pi' \in \Pi,$$

where $L_1^\Pi = 2(C_r + C_p C_m \kappa_r + 2(C_m^2 C_p \kappa_r + C_m C_r))$

Proof. From Equation (B.1),

$$\rho^\pi \mathbb{1} = r^\pi + \mathbb{P}^\pi v_\Phi^\pi - v_\Phi^\pi.$$

Taking derivative with respect to α ,

$$\begin{aligned} \frac{\partial \rho^{\pi_\alpha}}{\partial \alpha} \mathbb{1} &= \frac{\partial r^{\pi_\alpha}}{\partial \alpha} + \frac{\partial \mathbb{P}^{\pi_\alpha}}{\partial \alpha} v_\Phi^{\pi_\alpha} + \mathbb{P}^{\pi_\alpha} \frac{\partial v_\Phi^{\pi_\alpha}}{\partial \alpha} - \frac{\partial v_\Phi^{\pi_\alpha}}{\partial \alpha} \\ &= \Phi r^\mathbf{u} + \Phi \mathbb{P}^\mathbf{u} v^{\pi_\alpha} + (\mathbb{P}^{\pi_\alpha} - \mathbf{I})(M^{\pi_\alpha} \Phi \mathbb{P}^\mathbf{u} M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi r^\mathbf{u}), \\ &\quad \text{(from Lemma B.6)} \\ &= \Phi r^\mathbf{u} + \Phi \mathbb{P}^\mathbf{u} M^{\pi_\alpha} \Phi r^{\pi_\alpha} + (\mathbb{P}^{\pi_\alpha} - \mathbf{I})(M^{\pi_\alpha} \Phi \mathbb{P}^\mathbf{u} M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi r^\mathbf{u}), \\ &\quad \text{(from (B.6) and (B.5)).} \end{aligned}$$

Considering the L_∞ norm of the above expression,

$$\begin{aligned} \left| \frac{\partial \rho^{\pi_\alpha}}{\partial \alpha} \right| &= \left\| \Phi r^\mathbf{u} + \Phi \mathbb{P}^\mathbf{u} M^{\pi_\alpha} \Phi r^{\pi_\alpha} + (\mathbb{P}^{\pi_\alpha} - \mathbf{I})(M^{\pi_\alpha} \Phi \mathbb{P}^\mathbf{u} M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi r^\mathbf{u}) \right\|_\infty, \\ &\leq \|\Phi r^\mathbf{u}\|_\infty + \|\Phi \mathbb{P}^\mathbf{u} M^{\pi_\alpha} \Phi r^{\pi_\alpha}\|_\infty + \\ &\quad \|\mathbb{P}^{\pi_\alpha} - \mathbf{I}\|_\infty (\|M^{\pi_\alpha} \Phi \mathbb{P}^\mathbf{u} M^{\pi_\alpha} \Phi r^{\pi_\alpha}\|_\infty + \|M^{\pi_\alpha} \Phi r^\mathbf{u}\|_\infty), \\ &\leq 2\|r^\mathbf{u}\|_\infty + 2\|\mathbb{P}^\mathbf{u} M^{\pi_\alpha} \Phi r^{\pi_\alpha}\|_\infty + \|\mathbb{P}^{\pi_\alpha} - \mathbf{I}\|_\infty (\|M^{\pi_\alpha} \Phi \mathbb{P}^\mathbf{u} M^{\pi_\alpha} \Phi r^{\pi_\alpha}\|_\infty + \|M^{\pi_\alpha} \Phi r^\mathbf{u}\|_\infty), \\ &\leq 2C_r + 2C_p C_m \kappa_r + 2(2C_m^2 C_p \kappa_r + 2C_m C_r). \end{aligned}$$

□

B.1.5 Smoothness of the Average Reward ρ^π – Lemma 3.4

Lemma B.9. The average reward ρ^π is L_2^Π -smooth.

$$\left| \left\langle \pi' - \pi, \frac{\partial^2 \rho^\pi}{\partial \pi^2} (\pi' - \pi) \right\rangle \right| \leq \frac{L_2^\Pi}{2} \|\pi' - \pi\|_2^2 \quad \forall \pi, \pi' \in \Pi,$$

where $L_2^\Pi = 4(C_p^2 C_m^2 \kappa_r + C_p C_m C_r + (C_p + 1)(C_m^2 C_p \kappa_r + C_m C_r) + 4(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r))$.

Proof. From Lemma B.8, we have

$$\frac{\partial \rho^{\pi_\alpha}}{\partial \alpha} \mathbb{1} = \Phi r^u + \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + (\mathbb{P}^{\pi_\alpha} - I)(M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi r^u).$$

Taking the derivative again, and repeatedly invoking Equations (B.5), (B.6) and Lemma B.5, it follows that,

$$\begin{aligned} \frac{\partial^2 \rho^{\pi_\alpha}}{\partial \alpha^2} \mathbb{1} &= 0 + \frac{\partial}{\partial \alpha} (\Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha}) + \frac{\partial}{\partial \alpha} ((\mathbb{P}^{\pi_\alpha} - I)(M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi r^u)) \\ &= \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^u + (\mathbb{P}^u)(M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi r^u) \\ &\quad + (\mathbb{P}^{\pi_\alpha} - I) (M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} \\ &\quad + M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^u + M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^u), \\ &= \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^u + (\mathbb{P}^u)(M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi r^u) \\ &\quad + 2(\mathbb{P}^{\pi_\alpha} - I) (M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^u). \end{aligned}$$

Considering the L_∞ norm of the above expression,

$$\begin{aligned} \left| \frac{\partial^2 \rho^{\pi_\alpha}}{\partial \alpha^2} \right| &\leq \|\Phi \mathbb{P}^u M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha}\|_\infty + \|\Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^u\|_\infty \\ &\quad + \|(\mathbb{P}^u)(M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi r^u)\|_\infty \\ &\quad + 2\|(\mathbb{P}^{\pi_\alpha} - I) (M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^{\pi_\alpha} + M^{\pi_\alpha} \Phi \mathbb{P}^u M^{\pi_\alpha} \Phi r^u)\|_\infty \\ &\leq 4(C_p^2 C_m^2 \kappa_r + C_p C_m C_r + (C_p + 1)(C_m^2 C_p \kappa_r + C_m C_r) + 4(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r)). \end{aligned}$$

□

Remark B.1. The smoothness and Lipschitz constant analysis of both the average reward value functions and the infinite horizon average reward are constrained to all directions u , such that every $u = \pi - \pi'$ can be expressed as a difference of any two policies $\pi, \pi' \in \Pi$. Hence the smoothness and Lipschitz constants derived are restricted to the directions that can be expressed as this difference and hence are referred to as restricted smoothness/Lipschitzness.

B.1.6 Table of constants capturing MDP complexity

We restate the table of constants and their description here for the sake of convenience.

Table B.1: Constants capturing the MDP Complexity

	Definition	Range	Remark
C_m	$\max_{\pi \in \Pi} \ (I - \Phi P^\pi)^{-1}\ $	$\frac{2C_\epsilon S }{1-\lambda}$	Lowest rate of mixing
C_p	$\max_{\pi, \pi' \in \Pi} \frac{\ P^{\pi'} - P^\pi\ }{\ \pi' - \pi\ _2}$	$[0, \sqrt{A}]$	Diameter of transition kernel
C_r	$\max_{\pi, \pi'} \frac{\ r^{\pi'} - r^\pi\ _\infty}{\ \pi' - \pi\ _2}$	$[0, \sqrt{A}]$	Diameter of reward function
κ_r	$\max_\pi \ \Phi r^\pi\ _\infty$	$[0, 2]$	Variance of reward function
$\frac{L_1^\Pi}{2}$	$C_r + C_p C_m \kappa_r + 2(C_m^2 C_p \kappa_r + C_m C_r)$	$[0, k_1 \sqrt{A} C_m^2]$	Restricted Lipschitz constant
$\frac{L_2^\Pi}{4}$	$C_p^2 C_m^2 \kappa_r + C_p C_m C_r + (C_p + 1)(C_m^2 C_p \kappa_r + C_m C_r) + 4(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r)$	$[0, k_2 A C_m^3]$	Restricted smoothness constant

Lemma B.10. The constants C_p, C_r, C_m, κ_r in Table B.1 and other operator norms are bounded as below:

1. $\|\Phi\| := \max_{\|v\|_\infty \leq 1} \|\Phi v\|_\infty \leq 2.$
2. $\|\mathbb{P}^\pi\| = \max_{\|v\|_\infty \leq 1} \|\mathbb{P}^\pi v\|_\infty \leq \max_{\|v\|_\infty \leq 1} \|v\|_\infty = 1.$
3. $\kappa_r = \max_{\pi} \|\Phi r^\pi\|_\infty \leq 2$
4. $C_m \leq \frac{2C_e S}{1-\lambda}$
5. $C_p = \max_{u=\frac{\pi'-\pi}{\|\pi'-\pi\|_2}, \pi', \pi \in \Pi} \max_{\|v\|_\infty \leq 1} \|\mathbb{P}^u v\|_\infty \leq \sqrt{A}.$
6. $C_r = \max_{u=\frac{\pi'-\pi}{\|\pi'-\pi\|_2}, \pi', \pi \in \Pi} \|\mathbb{R}^u\|_\infty \leq \sqrt{A}.$

Proof. 1. Consider the projection matrix Φ ,

$$\begin{aligned}
\|\Phi\|_\infty &= \max_{\|v\|_\infty \leq 1} \|\Phi v\|_\infty \leq \max_{\|v\|_\infty} \\
&= \max_{s \in \mathcal{S}} \left| v(s) - \frac{\sum_{s \in \mathcal{S}} v(s)}{S} \right| \\
&\leq \max_{s \in \mathcal{S}} |v(s)| + \left| \frac{\sum_{s \in \mathcal{S}} v(s)}{S} \right| \\
&\leq 2\|v\|_\infty = 2
\end{aligned}$$

2. The operator norm of $\mathbb{P}^\pi$ is bounded as below:

$$\begin{aligned}
\|\mathbb{P}^\pi\| &= \max_{\|v\|_\infty \leq 1} \|\mathbb{P}^\pi v\|_\infty \\
&\leq \max_{\|v\|_\infty \leq 1} \|v\|_\infty \\
&\leq 1.
\end{aligned}$$

Equality is attained by the vector $v = \mathbb{1}$.

3. κ_r is bounded as below:

$$\begin{aligned}
\kappa_r &= \max_{\pi \in \Pi} \|\Phi r^\pi\|_\infty \\
&= \max_{\pi \in \Pi} \left\| r^\pi - \frac{\sum_{s \in \mathcal{S}} r^\pi(s)}{|\mathcal{S}|} \mathbb{1} \right\|_\infty \\
&\leq \max_{\pi \in \Pi} \|r^\pi\|_\infty + \left\| \frac{\sum_{s \in \mathcal{S}} r^\pi(s)}{|\mathcal{S}|} \mathbb{1} \right\|_\infty \\
&\leq 2
\end{aligned}$$

κ_r , in some sense, captures the variance of the single step reward function across the class of policies. Greater the variation of the r across different actions, greater the value of κ_r .

4. C_m is the maximum of the operator norm of the matrix $(I - \Phi \mathbb{P}^\pi)^{-1}$ across all policies $\pi \in \Pi$. It is determined as follows:

$$\begin{aligned}
(I - \Phi \mathbb{P}^\pi)^{-1} &= \sum_{k=0}^{\infty} (\Phi \mathbb{P}^\pi)^k = \sum_{k=0}^{\infty} \Phi (\mathbb{P}^\pi)^k, \quad (\text{from Lemma B.4}), \\
&\stackrel{(a)}{=} \sum_{k=0}^{\infty} \Phi ((\mathbb{P}^\pi)^k - \mathbb{1} (d^\pi)^\top), \quad (\text{as } \Phi \mathbb{1} (d^\pi)^\top = 0)
\end{aligned}$$

Let $v \in \mathbb{R}^{|\mathcal{S}|}$ such that $\|v\|_\infty \leq 1$. Then,

$$\begin{aligned}
\Rightarrow \left\| (I - \Phi \mathbb{P}^\pi)^{-1} v \right\|_\infty &\leq \sum_{k=0}^{\infty} \left\| \Phi ((\mathbb{P}^\pi)^k - \mathbb{1} (d^\pi)^\top) v \right\|_\infty \\
&\leq \sum_{k=0}^{\infty} \|\Phi\|_\infty \left\| ((\mathbb{P}^\pi)^k - \mathbb{1} (d^\pi)^\top) v \right\|_\infty \\
&\leq \sum_{k=0}^{\infty} \|\Phi\|_\infty |\mathcal{S}| \left\| ((\mathbb{P}^\pi)^k - \mathbb{1} (d^\pi)^\top) \right\|_\infty \|v\|_\infty \\
&\stackrel{(b)}{\leq} \sum_{k=0}^{\infty} 2|\mathcal{S}| C_e \lambda^k \|v\|_\infty, \\
&= \frac{2C_e |\mathcal{S}|}{1 - \lambda}
\end{aligned}$$

where d^π represents the stationary measure associated with the transition kernel $\mathbb{P}^\pi$,

(a) follows from the fact that the projection matrix Φ projects vectors onto a subspace orthogonal to the subspace spanned by the all ones vector $\mathbb{1}$ and (b) is a consequence of the irreducibility and aperiodicity assumption of the Markov chain induced under all policies. More precisely, for any irreducible and aperiodic stochastic matrix A , it is true that:

$$\left\| A^n - \mathbb{1}d^\top \right\|_\infty \leq C_e \lambda^n,$$

for some constants $\lambda \in [0, 1)$, $C_e < \infty$, where d is stationary distribution of A . λ is the coefficient of mixing and captures the rate of geometric mixing of the Markov Chain. Hence, higher the value of λ , lower the rate of mixing.

5. C_p represents the diameter of the transition kernel as a function of the policy class and can be bound as below.

$$\begin{aligned} C_p &:= \max_{u = \frac{\pi' - \pi}{\|\pi' - \pi\|_2}, \pi', \pi \in \Pi, \|\mathbf{v}\|_\infty \leq 1} \|\mathbb{P}^u \mathbf{v}\|_\infty, \\ &= \max_{\pi', \pi \in \Pi, \|\mathbf{v}\|_\infty \leq 1} \frac{\left\| (\mathbb{P}^{\pi' - \pi}) \mathbf{v} \right\|_\infty}{\|\pi' - \pi\|_2}, \\ &= \max_{\pi', \pi \in \Pi, \|\mathbf{v}\|_\infty \leq 1} \max_{s \in \mathcal{S}} \frac{\left| (\mathbb{P}^{\pi'} \mathbf{v})(s) - (\mathbb{P}^\pi \mathbf{v})(s) \right|}{\|\pi' - \pi\|_2} \\ &\leq \max_{\pi', \pi \in \Pi, \|\mathbf{v}\|_\infty \leq 1} \max_{s \in \mathcal{S}} \frac{\left| (\mathbb{P}^{\pi'} \mathbf{v})(s) - (\mathbb{P}^\pi \mathbf{v})(s) \right|}{\|\pi'_s - \pi_s\|_2}, \quad (\text{since } \|\pi'_s - \pi_s\|_2 \leq \|\pi' - \pi\|_2), \\ &= \max_{\pi', \pi \in \Pi, \|\mathbf{v}\|_\infty \leq 1} \max_{s \in \mathcal{S}} \frac{\left| (\pi'_s)^\top \mathbb{P}_s \mathbf{v} - (\pi_s)^\top \mathbb{P}_s \mathbf{v} \right|}{\|\pi'_s - \pi_s\|_2}, \\ &\quad (\text{where } \mathbb{P}_s(a, s') = P(s'|s, a), \pi_s(a) := \pi(s, a)), \\ &= \max_{\pi', \pi \in \Pi, \|\mathbf{v}\|_\infty \leq 1} \max_{s \in \mathcal{S}} \frac{\left| (\pi'_s - \pi_s)^\top \mathbb{P}_s \mathbf{v} \right|}{\|\pi'_s - \pi_s\|_2}, \\ &\leq \max_{\pi', \pi \in \Pi, \|\mathbf{v}\|_\infty \leq 1} \max_{s \in \mathcal{S}} \frac{\|\pi'_s - \pi_s\|_1 \|\mathbb{P}_s \mathbf{v}\|_\infty}{\|\pi'_s - \pi_s\|_2}, \quad (\text{from Holder's inequality}) \\ &= \max_{\pi', \pi \in \Pi} \max_{s \in \mathcal{S}} \frac{\|\pi'_s - \pi_s\|_1}{\|\pi'_s - \pi_s\|_2}, \\ &\leq \sqrt{|\mathcal{A}|}. \end{aligned}$$

6. C_r represents the diameter of the single step reward function as a function of the policy class and can be bound as below.

$$\begin{aligned}
C_r &:= \max_{u = \frac{\pi' - \pi}{\|\pi' - \pi\|_2}, \pi', \pi \in \Pi} \|\mathbf{r}^u\|_\infty \\
&= \max_{\pi', \pi \in \Pi} \frac{\|\mathbf{r}^{\pi'} - \mathbf{r}^\pi\|_\infty}{\|\pi' - \pi\|_2} \\
&= \max_{\pi', \pi \in \Pi} \max_{s \in \mathcal{S}} \frac{|\mathbf{r}^{\pi'}(s) - \mathbf{r}^\pi(s)|}{\|\pi' - \pi\|_2} \\
\Rightarrow C_r &\leq \max_{\pi', \pi \in \Pi} \max_{s \in \mathcal{S}} \frac{|\mathbf{r}^{\pi'}(s) - \mathbf{r}^\pi(s)|}{\|\pi'_s - \pi_s\|_2}, \quad (\text{as } \|\pi'_s - \pi_s\|_2 \leq \|\pi' - \pi\|_2), \\
&= \max_{\pi', \pi \in \Pi} \max_{s \in \mathcal{S}} \frac{|(\pi'_s)^\top \mathbf{r}_s - (\pi_s)^\top \mathbf{r}_s|}{\|\pi'_s - \pi_s\|_2}, \quad (\text{where } \mathbf{r}_s(\mathbf{a}) = \mathbf{r}(s, \mathbf{a}), \pi_s(\mathbf{a}) = \pi(s, \mathbf{a})), \\
C_r &= \max_{\pi', \pi \in \Pi} \max_{s \in \mathcal{S}} \frac{|(\pi'_s - \pi_s)^\top \mathbf{r}_s|}{\|\pi'_s - \pi_s\|_2}, \\
&\leq \max_{\pi', \pi \in \Pi} \max_{s \in \mathcal{S}} \frac{\|\pi'_s - \pi_s\|_1 \|\mathbf{r}_s\|_\infty}{\|\pi'_s - \pi_s\|_2}, \quad (\text{from Holder's inequality}) \\
&= \max_{\pi', \pi \in \Pi} \max_{s \in \mathcal{S}} \frac{\|\pi'_s - \pi_s\|_1}{\|\pi'_s - \pi_s\|_2}, \\
&\leq \sqrt{|\mathcal{A}|}.
\end{aligned}$$

Since the directional derivatives considered are all within the policy class, the analysis gives rise to constants such as C_p and C_r , which are functions of the underlying policy class. These constants capture the MDP complexity by the virtue of their definition and are an artifact of this proof technique. $\square$

B.2 Convergence of average reward projected policy gradient algorithm

Lemma B.11. For any convex set $\mathcal{X} \subseteq \mathbb{R}^d$, any point $\mathbf{a} \in \mathcal{X}$, and any update direction $\mathbf{u} \in \mathbb{R}^d$, let $\mathbf{b} = \mathbf{Proj}_{\mathcal{X}}(\mathbf{a} + \mathbf{u})$ be the projection of $\mathbf{a} + \mathbf{u}$ onto $\mathcal{X}$. It is true that

1. $\langle \mathbf{u}, \mathbf{b} - \mathbf{a} \rangle \geq \|\mathbf{b} - \mathbf{a}\|_2^2$.
2. $\langle \mathbf{c} - \mathbf{b}, \mathbf{u} - (\mathbf{b} - \mathbf{a}) \rangle \leq 0, \quad \forall \mathbf{c} \in \mathcal{X}$.

Proof. The formal proof can be found in [152].

However, the proof follows trivially from the geometrical representation of projection (see Figure B.1,[65]), and the fact that the hyperplane separates a convex set from a point not in the set.

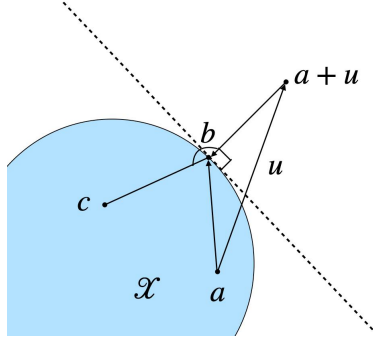


Figure B.1: Convex Projection

Intuitively, the proof of the lemma can be interpreted as below.

1. Since the angle between vectors $(\mathbf{a} - \mathbf{b})$ and $((\mathbf{a} + \mathbf{u}) - \mathbf{b})$ is greater than 90 degrees, it is true that $\langle \mathbf{a} - \mathbf{b}, (\mathbf{a} + \mathbf{u}) - \mathbf{b} \rangle \leq 0$, which then directly implies $\|\mathbf{b} - \mathbf{a}\|_2^2 \leq \langle \mathbf{u}, \mathbf{b} - \mathbf{a} \rangle$.
2. The angle between vectors $(\mathbf{c} - \mathbf{b})$ and $((\mathbf{a} + \mathbf{u}) - \mathbf{b})$ is greater than 90 degrees $\forall \mathbf{c} \in \mathcal{X}$, therefore $\langle \mathbf{c} - \mathbf{b}, \mathbf{u} - (\mathbf{b} - \mathbf{a}) \rangle \leq 0$.

□

B.2.1 Proof of Lemma 3.5

Lemma B.12. The average reward iterates ρ^{π_k} generated from projected policy gradient satisfy the following,

$$\rho^{\pi_{k+1}} - \rho^{\pi_k} \geq \frac{L_2^\Pi}{2} \|\pi_{k+1} - \pi_k\|^2, \quad \forall k \geq 0.$$

where L_2^Π is the restricted smoothness constant associated with average reward ρ^π .

Proof. From the restricted smoothness of the average cost, we have

$$\begin{aligned} \rho^{\pi_{k+1}} &\geq \rho^{\pi_k} + \left\langle \frac{d\rho^\pi}{d\pi} \Big|_{\pi=\pi_k}, \pi_{k+1} - \pi_k \right\rangle - \frac{L_2^\Pi}{2} \|\pi_{k+1} - \pi_k\|^2, \\ &= \rho^{\pi_k} + L_2^\Pi \left\langle \frac{1}{L_2^\Pi} \frac{d\rho^\pi}{d\pi} \Big|_{\pi=\pi_k}, \pi_{k+1} - \pi_k \right\rangle - \frac{L_2^\Pi}{2} \|\pi_{k+1} - \pi_k\|^2, \\ &\geq \rho^{\pi_k} + L_2^\Pi \|\pi_{k+1} - \pi_k\|^2 - \frac{L_2^\Pi}{2} \|\pi_{k+1} - \pi_k\|^2. \end{aligned}$$

The last inequality follows from the projected gradient ascent policy update rule and item 1 of Lemma B.11. Note that the proof only relies on the convexity of the projection set Π and the smoothness of the objective function. $\square$

B.2.2 Proof of Lemma 3.7

Lemma B.13. The suboptimality of a policy π can be bounded from above as:

$$\rho^* - \rho^\pi \leq C_{PL} \max_{\pi' \in \Pi} \left\langle \pi' - \pi, \frac{\partial \rho^\pi}{\partial \pi} \right\rangle, \quad \forall \pi \in \Pi,$$

where $C_{PL} = \max_{\pi, s} \frac{\partial \pi^*(s)}{\partial \pi(s)}$ and ρ^* is the optimal average reward.

Proof. Average Reward Performance Difference Lemma states that

$$\begin{aligned}
\rho^* - \rho^\pi &= \sum_{s \in \mathcal{S}} d^{\pi^*}(s) Q^\pi(s, \mathbf{a}) [\pi^*(\mathbf{a}|s) - \pi(\mathbf{a}|s)] \\
&\leq \max_{\pi'} \sum_{s \in \mathcal{S}} d^{\pi^*}(s) Q^\pi(s, \mathbf{a}) [\pi'(\mathbf{a}|s) - \pi(\mathbf{a}|s)] \\
&= \sum_{s \in \mathcal{S}} \frac{d^{\pi^*}(s)}{d^\pi(s)} d^\pi(s) \max_{\pi'_s} Q^\pi(s, \mathbf{a}) [\pi'(\mathbf{a}|s) - \pi(\mathbf{a}|s)] \\
&= \sum_{s \in \mathcal{S}} \frac{d^{\pi^*}(s)}{d^\pi(s)} \underbrace{d^\pi(s) \max_{\pi'_s} Q^\pi(s, \mathbf{a}) [\pi'(\mathbf{a}|s) - \pi(\mathbf{a}|s)]}_{\geq 0, \quad (= 0 \text{ when } \pi'_s = \pi_s)} \\
&\leq \sum_{s \in \mathcal{S}} \left(\max_{\pi, s} \frac{d^{\pi^*}(s)}{d^\pi(s)} \right) d^\pi(s) \max_{\pi'_s} Q^\pi(s, \mathbf{a}) [\pi'(\mathbf{a}|s) - \pi(\mathbf{a}|s)] \\
&= C_{\text{PL}} \max_{\pi'} \sum_{s \in \mathcal{S}} d^\pi(s) Q^\pi(s, \mathbf{a}) [\pi'(\mathbf{a}|s) - \pi(\mathbf{a}|s)] \\
&\stackrel{(a)}{=} C_{\text{PL}} \max_{\pi'} \left\langle \frac{d\rho^\pi}{d\pi}, \pi' - \pi \right\rangle,
\end{aligned}$$

where (a) follows from the average reward policy gradient theorem. $\square$

B.2.3 Proof of Lemma 3.8

Lemma B.14. Let π_{k+1} represent the policy iterates obtained through projected policy gradient. For any policy $\pi' \in \Pi$, it is true that,

$$\left\langle \frac{\partial \rho^{\pi_{k+1}}}{\partial \pi_{k+1}}, \pi' - \pi_{k+1} \right\rangle \leq 4\sqrt{|\mathcal{S}|} L_2^\Pi \|\pi_{k+1} - \pi_k\|_2,$$

Proof. For all $x, y \in \mathbb{C}$, we have:

$$\begin{aligned}
\left\langle \frac{\partial \rho^{\pi_{k+1}}}{\partial \pi_{k+1}}, \pi' - \pi_{k+1} \right\rangle &= \left\langle \frac{\partial \rho^{\pi_{k+1}}}{\partial \pi_{k+1}} - \frac{\partial \rho^{\pi_k}}{\partial \pi_k} + \frac{\partial \rho^{\pi_k}}{\partial \pi_k}, \pi' - \pi_{k+1} \right\rangle \\
&= \left\langle \frac{\partial \rho^{\pi_{k+1}}}{\partial \pi_{k+1}} - \frac{\partial \rho^{\pi_k}}{\partial \pi_k}, \pi' - \pi_{k+1} \right\rangle + \left\langle \frac{\partial \rho^{\pi_k}}{\partial \pi_k}, \pi' - \pi_{k+1} \right\rangle \\
&\leq \left\| \frac{\partial \rho^{\pi_{k+1}}}{\partial \pi_{k+1}} - \frac{\partial \rho^{\pi_k}}{\partial \pi_k} \right\| \|\pi' - \pi_{k+1}\| + \left\langle \frac{\partial \rho^{\pi_k}}{\partial \pi_k}, \pi' - \pi_{k+1} \right\rangle \\
&\stackrel{(a)}{\leq} L_2^\Pi \|\pi_{k+1} - \pi_k\| \|\pi' - \pi_{k+1}\| + \left\langle \frac{\partial \rho^{\pi_k}}{\partial \pi_k}, \pi' - \pi_{k+1} \right\rangle, \tag{B.8}
\end{aligned}$$

where (a) uses smoothness of average reward. Thus, we may continue the chain of inequalities as

$$\begin{aligned}
(\text{B.8}) &= L_2^\Pi \|\pi_{k+1} - \pi_k\| \|\pi' - \pi_{k+1}\| + \left\langle \frac{\partial \rho^{\pi_k}}{\partial \pi_k} - L_2^\Pi(\pi_{k+1} - \pi_k), \pi' - \pi_{k+1} \right\rangle \\
&\quad + L_2^\Pi \langle \pi_{k+1} - \pi_k, \pi' - \pi_{k+1} \rangle \\
&\leq 2L_2^\Pi \|\pi_{k+1} - \pi_k\| \|\pi' - \pi_{k+1}\| + \left\langle \frac{\partial \rho^{\pi_k}}{\partial \pi_k} - L_2^\Pi(\pi_{k+1} - \pi_k), \pi' - \pi_{k+1} \right\rangle \\
&\leq 2L_2^\Pi \|\pi_{k+1} - \pi_k\| \|\pi' - \pi_{k+1}\| + L_2^\Pi \underbrace{\left\langle \frac{1}{L_2^\Pi} \frac{\partial \rho^{\pi_k}}{\partial \pi_k} - (\pi_{k+1} - \pi_k), \pi' - \pi_{k+1} \right\rangle}_{\leq 0, \quad (\text{From item 2 of Lemma B.11})} \\
&\leq 2L_2^\Pi \|\pi_{k+1} - \pi_k\| \|\pi' - \pi_{k+1}\| \\
&\leq 2L_2^\Pi \|\pi_{k+1} - \pi_k\| \mathbf{diam}(\Pi).
\end{aligned}$$

The diameter of the policy class Π , can be upper bounded as

$$\mathbf{diam}(\Pi)^2 = \max_{\pi', \pi} \sum_s \|\pi'_s - \pi_s\|_2^2 \leq \max_{\pi', \pi} \sum_s \|\pi'_s - \pi_s\|_1^2 \leq 4S.$$

This yields the result. □

Lemma B.15. The scaled sub-optimality $\mathbf{a}_k := \rho^ - \rho_k$ follows the recursion*

$$c\mathbf{a}_{k+1}^2 + \mathbf{a}_{k+1} - \mathbf{a}_k \leq 0,$$

where $c = \frac{1}{32L_2^\Pi |S| C_{PL}^2}$.

Proof. From Lemma B.13, we know that,

$$\rho^* - \rho^{\pi_{k+1}} \leq C_{PL} \left\langle \pi' - \pi_{k+1}, \frac{\partial \rho^{\pi_{k+1}}}{\partial \pi_{k+1}} \right\rangle, \quad \forall \pi' \in \Pi,$$

From Lemma B.14, we know that,

$$\left\langle \frac{\partial \rho^{\pi_{k+1}}}{\partial \pi_{k+1}}, \pi' - \pi_{k+1} \right\rangle \leq 4\sqrt{|S|} L_2^\Pi \|\pi_{k+1} - \pi_k\|_2.$$

From Lemma B.12, we know that,

$$\|*\| \pi_{k+1} - \pi_{k2} \leq \sqrt{\frac{2(\rho^{\pi_{k+1}} - \rho^{\pi_k})}{L_2^\Pi}}, \quad \forall k \geq 0.$$

Combining the above equations yields,

$$\rho^* - \rho^{\pi_{k+1}} \leq \sqrt{32C_{\text{PL}}^2 L_2^\Pi |\mathcal{S}| (\rho^{\pi_{k+1}} - \rho^{\pi_k})}$$

This thus yields,

$$\frac{(\rho^* - \rho^{\pi_{k+1}})^2}{32C_{\text{PL}}^2 L_2^\Pi |\mathcal{S}|} + (\rho^* - \rho^{\pi_{k+1}}) - (\rho^* - \rho^{\pi_k}) \leq 0.$$

□

A more detailed interpretation of this Lemma can be found in [65].

B.2.4 Recursion Bound

In this subsection, we consider the sequence defined as

$$a_k - a_{k+1} \geq a_k^2,$$

where $p \geq 0$ and $0 \leq a_0 \leq 1$. Let f be linear interpolation of the sequence $\{a_k\}_{k \geq 0}$, formally defined as

$$f(x) := (1 - \alpha)a_k + \alpha a_{k+1}, \quad \text{where } k = \lfloor x \rfloor \text{ and } \alpha = x - \lfloor x \rfloor.$$

Let $g(0) := a_0$ and

$$\frac{dg(x)}{dx} = -g(x)^2, \quad \text{and} \quad \tau_k := g^{-1}(a_k).$$

Observe that g is a strictly decreasing function.

Proposition B.1. If $\tau_k \geq k$ then

$$g(x) \geq f(x), \quad \forall x \in [k, k+1].$$

Proof. We have $f(k) = g(\tau_k) = a_k$ and for $\alpha \in (0, 1)$

$$\begin{aligned} \frac{dg(\tau_k + \alpha)}{dx} &= -g(\tau_k + \alpha)^2, && \text{(by definition)} \\ &\geq -g(\tau_k)^2, && \text{(g is a decreasing function)} \\ &= -f(k)^2, && \text{(as } f(k) = g(\tau_k)) \\ &= -a_k^2 && \text{(by definition of f)} \\ &\geq a_{k+1} - a_k, && \text{(by definition of } a_{k+1}) \\ &= \frac{df(k + \alpha)}{dx}, && \text{(by definition of f).} \end{aligned}$$

Above together with continuity of f and g , for all $\alpha \in [0, 1]$, we have

$$\begin{aligned} f(k + \alpha) &\leq g(\tau_k + \alpha), \\ &\leq g(k + \alpha), && \text{(as } \tau_k \geq k \text{ and } g \text{ is a decreasing function).} \end{aligned}$$

Hence claim is proved. □

Proposition B.2. For all $k \geq 0$, we have

$$\tau_k \geq k.$$

Proof. Note that $\tau_0 = 0$ by definition $g(0) = a_0$. Now let $\tau_k \geq k$, then from Proposition B.1, we have

$$\begin{aligned} g(x) &\geq f(x), && \forall x \in [k, k+1] \\ \implies g(k+1) &\geq f(k+1) \\ &= a_{k+1} \\ \implies \tau_{k+1} &\geq k+1, && \text{(as } g \text{ is a decreasing function).} \end{aligned}$$

Hence, by induction the claim is established. $\square$

Lemma B.16. [Recursion Upper Bound] For $p \geq 2$, and $0 \leq a_0 \leq 1$, sequence $\{a_k\}_{k \geq 0}$ satisfying the recursion $a_k - a_{k+1} \geq a_k^2$, follows

$$a_k \leq \frac{1}{\frac{1}{a_0} + k}, \quad \forall k \geq 1.$$

Proof. From Proposition B.2, we get $\tau_k \geq k$. Combining it with Proposition B.1, we get

$$g(x) \geq f(x), \quad \forall x \geq 0.$$

Now, we solve the o.d.e. to get

$$\begin{aligned} \frac{dg(x)}{dx} &= -g(x)^2 \\ \implies dx \Big|_{x=0}^{x=k} &= \int_{x=0}^k -\frac{dg(x)}{g^2(x)} \\ \implies k &= \frac{1}{g(k)} - \frac{1}{g(0)} \\ \implies g(k) &= \frac{g(0)}{1 + g(0)k} \\ \implies f(k) &\leq \frac{a_0}{1 + a_0 k}, \quad (\text{as } f(x) \leq g(x)) \\ \implies a_k &\leq \frac{1}{\frac{1}{a_0} + k}. \end{aligned}$$

This proves the claim. $\square$

Lemma B.17. If $a_k - a_{k+1} \geq ca_k^2$ then

$$a_k \leq \frac{1}{\frac{1}{a_0} + ck}.$$

Proof. We have

$$\begin{aligned}
& a_k - a_{k+1} \geq ca_k^2 \\
\implies ca_k - ca_{k+1} & \geq (ca_k)^2 \\
\implies ca_k & \leq \frac{1}{\frac{1}{ca_0} + k}, \quad (\text{from Lemma B.16}) \\
\implies a_k & \leq \frac{1}{\frac{1}{a_0} + ck}.
\end{aligned}$$

□

This subsection (proving the recursion upper bound) is inspired by a technique from [153]. However, while their result is similar, it is not applicable to our case. Their result assumes that c is upper bounded by a constant, which does not hold in our setting. In our case, the smoothness constant (or hardness coefficient) can approach zero, causing the constant c to diverge to infinity. Our result is more general, and the proof technique we use is distinct.

Lemma B.18. Given $a_k - a_{k+1} \geq ca_{k+1}^2$, we have

$$a_k \leq \frac{1}{\frac{1}{a_0} + ck},$$

where $\gamma = c(1 + \frac{8c}{1-\gamma})^{-\frac{3}{2}}$ and $c = \frac{1}{32C_{p_L}^2|S|L_2^{\frac{1}{p_L}}}$.

Proof. We have

$$\begin{aligned}
& ca_{k+1}^2 + a_{k+1} - a_k \leq 0 \\
\implies a_{k+1} & \leq \frac{-1 + \sqrt{1 + 4ca_k}}{2c}, \\
& = \frac{-1 + f(0) + f'(0)4ca_k + f''(b)\frac{(4ca_k)^2}{2}}{2c}, \quad (\text{where } f(x) = \sqrt{1+x}, b \in [0, 4ca_k]) \\
& = \frac{2ca_k - 2c^2a_k^2(1+b)^{-\frac{3}{2}}}{2c}, \quad (\text{putting } f(0) = 1, f'(0) = \frac{1}{2}, f''(a) = \frac{(1+b)^{-\frac{3}{2}}}{4})
\end{aligned}$$

$$\begin{aligned}
a_{k+1} &\leq \frac{2ca_k - 2c^2a_k^2(1+4ca_k)^{-\frac{3}{2}}}{2c}, \\
&\quad (\text{as } b \leq 4ca_k \text{ and } -(1+y)^{-\frac{3}{2}} \text{ is an increasing function}) \\
&\leq a_k - \frac{ca_k^2}{(1+4ca_k)^{\frac{3}{2}}}, \\
&\leq a_k - \nu a_k^2, \quad (\text{as } a_k \leq 1 \text{ and } \nu := c(1+4c)^{-\frac{3}{2}}).
\end{aligned}$$

We get the desired result from Lemma B.17. $\square$

Lemma B.19. If $\frac{1}{c} = 32C_{PL}^2|\mathcal{S}|L_2^\Pi < 1$ that is MDP is very easy (i.e. $L_2^\Pi \ll 1$) then the policy gradient converges exponentially fast, that is

$$a_k \leq \left(\frac{1}{c}\right)^{\frac{k}{2}} a_0^{-2^k}.$$

Proof. From the above discussion, we have

$$\begin{aligned}
ca_{k+1}^2 &\leq a_k - a_{k+1} \\
&\leq a_k, \quad (\text{as } a_{k+1} \geq 0 \text{ by definition}) \\
\Rightarrow a_{k+1} &\leq \sqrt{\frac{a_k}{c}} \\
&\leq \left(\frac{1}{c}\right)^{\frac{k+1}{2}} a_0^{\frac{1}{2^{k+1}}}
\end{aligned}$$

$\square$

B.2.5 Proof of Theorem 3.1

We restate the theorem for the sake of convenience.

Theorem B.1. Let ρ^{π_k} be the average reward corresponding to the policy iterates π_k , obtained through the policy gradient update (3.5). Let ρ^ represent the optimal average reward, that is, $\rho^* = \max_{\pi \in \Pi} \rho^\pi$. There exist constants L_2^Π and C_{PL} which are determined by the underlying MDP such that:*

- For all MDPs it is true that,

$$\rho^* - \rho^{\pi_k} \leq \frac{1}{\frac{1}{\rho^* - \rho^{\pi_0}} + \nu k}, \quad \forall k \geq 0.$$

$$\text{where } \nu := \left(\frac{1}{32C_{PL}^2 |S| L_2^\Pi} \right) \left(1 + 4 \left(\frac{1}{32C_{PL}^2 |S| L_2^\Pi} \right) \right)^{-\frac{3}{2}}$$

- For simple MDPs (i.e. $L_2^\Pi \ll 1$) we obtain exponential convergence, that is

$$\rho^* - \rho^{\pi_k} \leq c^{-\frac{k}{2}} \left(\rho^* - \rho^{\pi_0} \right)^{\frac{1}{2k}}, \quad \forall k \geq 0$$

$$\text{where } \frac{1}{c} = 32|S|L_2^\Pi C_{PL}^2 < 1.$$

Proof. Using Gradient Domination Lemma and Sufficient Increase Lemma as shown in Lemma B.15, we get the following recursion

$$a_k - a_{k+1} \geq c a_{k+1}^2,$$

where $a_k = J^* - J^{\pi_k}$ and $c = \frac{1}{32C_{PL}^2 |S| L_2^\Pi}$ is a small constant. Then we get the desired result by solving the above recursion in Lemma B.18 and Lemma B.19 for complex MDPs and simple MDPs respectively. $\square$

B.3 Simulation Details

B.3.1 Convergence with different action and state space size

In the first experiment, we compare the convergence of PG for a tabular MDP with $(S, A) \in \{(3, 3), (9, 9), (81, 81)\}$. We set the reward kernel to be with maximal variance as described above. For the transition kernel, we use the following matrix:

$$P(\cdot | s, \cdot) = \frac{1}{2} \left(1_{S \times A} + \frac{1}{S} \right)$$

so $P(i|s, i) = \frac{1+\frac{1}{S}}{2}$ and $P(i | s, j) = \frac{1}{2S}$ for $i \neq j$. For the reward kernel we set the rewards of half the actions to 1 and the rest to -1 , for every state.

B.3.2 Convergence with different reward functions

In the second experiment, we compare the convergence of PG for a tabular MDP with $S = 16$ and $A = 16$. We set $r(s, a) = 0$ for any s and a except for one state which we denote by s_0 . We use the same randomly generated transition kernel and use the following procedure to generate the reward function:

- No variance: We assign each (s_0, a) pair a reward of 1.
- Low variance: We assign $\frac{1}{8}$ of the actions for s_0 a reward of -1 , and 1 otherwise.
- High variance: We assign $\frac{1}{4}$ of the actions for s_0 a reward of -1 , and 1 otherwise.
- Max variance: We assign $\frac{1}{2}$ of the actions for s_0 a reward of -1 , and 1 otherwise.

B.3.3 Convergence with different transition kernels

In the third experiment, we compare the convergence of PG for a tabular MDP with $S = 16$ and $A = 16$. We create three different MDPs with the same (S, A) values and the same reward function that is generated according to the process described above for high variance reward function. We then generate three different transition kernels:

- Uniform: We assign for all values of s, a, s' , $P(s' | s, a) = \frac{1}{S}$.
- Non-uniform: We assign $P(i | s, i) = \frac{1}{2S} + \frac{1}{2}$ and $P(i | s, j) = \frac{1}{2S}$ for $i \neq j$.
- Deterministic: We look at $P(\cdot | s, \cdot)$ as an $S \times A$ matrix, and assign it a random permutation of the identity matrix. In this way the result MDP is deterministic but not trivial (so every state leads to a different one).

B.4 Additional Discussion and Possible Extensions

Extension to Discounted Reward Setting. In the discounted reward setting, the return ρ^π , and the value function v^π is defined as

$$\rho^\pi = \mu^\top (I - \gamma P^\pi) R^\pi, \quad v^\pi = (I - \gamma P^\pi) R^\pi,$$

where $\gamma \in [0, 1)$ is the discount factor [61]. The return ρ^π is proven to be $\frac{8}{(1-\gamma)^3}$ -smooth [46]. Note that it is an MDP-agnostic bound. We can achieve an MDP instance-dependent bound with a very minor change in the smoothness analysis of the average-reward case.

Let us define,

$$M^{\pi_\alpha} := (I - \gamma P^{\pi_\alpha})^{-1}.$$

Observe that Lemma B.5 holds for $A(\alpha) = \gamma P^{\pi_\alpha}$, which yields us

$$\frac{\partial^2 M(\alpha)}{\partial \alpha^2} = \frac{\partial M(\alpha)}{\partial \alpha} \frac{\partial A(\alpha)}{\partial \alpha} M(\alpha) + M(\alpha) \frac{\partial^2 A(\alpha)}{\partial \alpha^2} M(\alpha) + M(\alpha) \frac{\partial A(\alpha)}{\partial \alpha} \frac{\partial M(\alpha)}{\partial \alpha},$$

where $M(\alpha)$ is shorthand for M^{π_α} .

Table B.2: Constants capturing the MDP Complexity for Discounted reward

	Definition	Range	Remark
C_m	$\max_{\pi \in \Pi} \ (I - \gamma P^\pi)^{-1}\ $	$\frac{1}{1-\gamma}$	
C_p	$\gamma \max_{\pi, \pi' \in \Pi} \frac{\ p^{\pi'} - p^\pi\ }{\ \pi' - \pi\ _2}$	$[0, \gamma\sqrt{A}]$	Diameter of transition kernel
C_r	$\max_{\pi, \pi'} \frac{\ r^{\pi'} - r^\pi\ _\infty}{\ \pi' - \pi\ _2}$	$[0, \sqrt{A}]$	Diameter of reward function
κ_r	$\max_{\pi} \ r^\pi\ _\infty$	$[0, 1]$	Variance of reward function

Lemma B.20. The value function v_Φ^π is $8(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r)$ -smooth in Π . That is,

$$\left\langle \pi' - \pi, \frac{\partial^2 v_\Phi^\pi(s)}{\partial \pi} (\pi' - \pi) \right\rangle \leq 8 \left(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r \right) \|\pi' - \pi\|_2^2 \quad \forall \pi', \pi \in \Pi, s \in \mathcal{S}$$

Proof.

$$\begin{aligned} \frac{\partial^2 v^{\pi_\alpha}}{\partial \alpha^2} = & M^{\pi_\alpha} \gamma \mathbb{P}^u M^{\pi_\alpha} \gamma \mathbb{P}^u M^{\pi_\alpha} r^{\pi_\alpha} + M^{\pi_\alpha} \gamma \mathbb{P}^u M^{\pi_\alpha} \gamma \mathbb{P}^u M^{\pi_\alpha} r^{\pi_\alpha} \\ & + M^{\pi_\alpha} \gamma \mathbb{P}^u M^{\pi_\alpha} r^u + M^{\pi_\alpha} \gamma \mathbb{P}^u M^{\pi_\alpha} r^u, \quad (\text{from (B.5) and (B.6)}). \end{aligned}$$

Considering the L_∞ norm,

$$\begin{aligned} \left\| \frac{\partial^2 v^{\pi_\alpha}}{\partial \alpha^2} \right\|_\infty &= 2 \|M^{\pi_\alpha} \gamma \mathbb{P}^u M^{\pi_\alpha} \gamma \mathbb{P}^u M^{\pi_\alpha} r^{\pi_\alpha} + M^{\pi_\alpha} \gamma \mathbb{P}^u M^{\pi_\alpha} r^u\|_\infty \\ &\leq 2 \|M^{\pi_\alpha} \gamma \mathbb{P}^u M^{\pi_\alpha} \gamma \mathbb{P}^u M^{\pi_\alpha} r^{\pi_\alpha}\|_\infty + \|M^{\pi_\alpha} \gamma \mathbb{P}^u M^{\pi_\alpha} r^u\|_\infty \\ &\leq 8(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r). \end{aligned}$$

Hence, we obtain,

$$\left\langle \pi' - \pi, \frac{\partial^2 v^\pi(s)}{\partial \pi} (\pi' - \pi) \right\rangle \leq 8 \left(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r \right) \|\pi' - \pi\|_2^2 \quad \forall \pi', \pi \in \Pi, s \in \mathcal{S}$$

□

The above result implies the return ρ^π is $L_2^\Pi = 8(C_m^3 C_p^2 \kappa_r + C_m^2 C_p C_r)$ -smooth. This establishes the convergence of the projected policy gradient algorithm with an iteration complexity similar to that stated in Theorem 3.1 for the average reward case.

Follow up/concurrent work in discounted reward case: The work [154] improved the iteration complexity of the policy gradient (discounted reward case) method to $O(\frac{\Lambda}{\epsilon})$ from the previous state-of-the-art iteration complexity of $O(\frac{S\Lambda}{\epsilon})$ [62], [63]. [154] does not use smoothness of the return to establish sufficient increase lemma, instead leverages the performance difference lemma in a novel way.

- Our instance-dependent bound for the MDP stems from the smoothness of the return. As a result, it is unclear how the two approaches can be effectively combined to achieve tighter bounds, leaving this as an avenue for future research.
- However, the bound in [154] can be asymptotically improved by a factor of $\frac{1}{1-\gamma}$, and more meaningful bounds for initial iterates can be obtained using the enhanced recursion-solving techniques presented in [153].

Our work can provide improved, alternative, or suboptimal results (depending on the parameters) for the discounted reward case. However, it remains the first to establish the global convergence of policy gradient methods for the average reward setting.

All existing works [46], [53], [62], [63] on the discounted reward case have an iteration complexity that scales with the cardinalities of the state space S and the action space A , with the exception of the recent work by [154], which depends only on A . This dependence on the size of the action space poses significant challenges when attempting to generalize to infinite state-action spaces.

Infinite Action Space. Our work (for both average and discounted reward case) has the MDP instance bounded bound of $\frac{SL_2^\Pi}{\epsilon}$, where L_2^Π is the smoothness constant that encodes the hardness of the MDP. This hardness coefficients that makes up L_2^Π , may be small/finite for even large/infinite action-space MDPs. However, it requires more careful study to determine the conditions for this to happen, which we leave for the future work.

Infinite State Space For the discounted reward setting, [154] provides a state-independent bound of $O(\frac{A}{\epsilon})$. The approach taken in their work is fundamentally different from ours, and extending this technique to the average reward case presents an intriguing direction for future research.

In our work, which continues the line of research from [46], [53], [62], [63], the bound exhibits state dependence. This dependence arises from the diameter of the policy class, defined as $\text{diam}(\Pi)^2 = \sum_{\pi, \pi'} \|\pi - \pi'\|_2^2 \leq S$. While this quantity is inherently tied to the state space, it can potentially be bounded for infinite state spaces under certain structures, such as low-rank policy classes. Exploring this direction in greater detail is an intriguing

avenue for future research. Another challenge is to circumvent the dependence on C_{PL} constant which captures the suboptimality of a policy. C_{PL} can be ∞ when the state space is infinite, necessitating a different approach to characterizing the suboptimality of a policy.

B.4.1 Parametrized policy class lower bounds

Parametrized policy class. Our work can also be extended to parameterized policy classes, such as softmax policies. For parameterized policy classes, the smoothness coefficient can be derived by augmenting our analysis with the chain and product rules, which is a straightforward extension. However, this may result in different hardness coefficients, making it an interesting direction for further exploration. We leave this investigation for future work.

Lower Bounds for policy gradient for average reward case. For the discounted reward setting, [62] establishes a lower bound of $O(\epsilon^{-1})$ for policy gradient methods. Specifically, Theorem 9 of [62] derives this lower bound using a bandit problem as a counterexample. Since a bandit is a special case of an MDP with a single state, it serves as an example for both discounted reward and average reward MDPs. Consequently, the same lower bound of $O(\epsilon^{-1})$ also applies to average reward MDPs, as implied by Theorem 9 of [62].

Linear Rates of Policy Gradient with aggressively increasing step sizes. Policy gradient can be interpreted as a form of soft policy iteration, assuming all states are visited or updated by the policy. Specifically, as the learning rate increases, policy gradient behavior increasingly resembles policy iteration. Since policy iteration is known to converge linearly, which is significantly faster than the typical convergence rates of policy gradient methods, it is natural to expect linear convergence bounds for policy gradient with aggressively increasing learning rates. This has been established in several works, including [63], [154], [155].

In most cases, the model is not known, and the exact gradient cannot be computed, requiring the use of stochastic gradient descent. In such noisy settings, using aggressive step sizes can lead to instability in the algorithm.

However, in Theorem 3.1, we demonstrated linear convergence rates for simple MDPs with constant step sizes. It may also be possible to achieve similar rates for the average reward case by employing aggressively increasing step sizes, which we leave as an interesting direction for future work.

C

PROOFS: NATURAL POLICY GRADIENT AND MIRROR DESCENT FOR AVERAGE-REWARD MDPs

C.1 Performance Difference Lemma – Lemma 4.1

Lemma C.1. (Performance Difference Lemma) Recall the definition of advantage function in Equation (4.3). Let J^π and $J^{\pi'}$ be the average rewards associated with policies π and π' respectively. Then it is true that,

$$J^\pi - J^{\pi'} = \sum_{s \in \mathcal{S}} d^\pi(s) A^{\pi'}(s, \pi(s)), \quad (\text{C.1})$$

where d^π is the stationary distribution induced over $\mathcal{S}$ by policy π .

Proof. The proof can be found in [110].

$$J^\pi - J^{\pi'} = r(s, a) + \sum_{s' \in \mathcal{S}} \mathbb{P}(s'|s, a) V^\pi(s') - Q^\pi(s, a) - J^{\pi'}.$$

Weighting with $\sum_{a \in \mathcal{A}} \pi'(a|s)$ on both sides,

$$J^\pi - J^{\pi'} = \sum_{a \in \mathcal{A}} \pi'(a|s) r(s, a) - J^{\pi'} + \sum_{\substack{s' \in \mathcal{S} \\ a \in \mathcal{A}}} \mathbb{P}(s'|s, a) \pi'(a|s) V^\pi(s') - \sum_{a \in \mathcal{A}} \pi'(a|s) Q^\pi(s, a).$$

Weighting with $\sum_{s \in \mathcal{S}} d^{\pi'}(s)$ on both sides,

$$\begin{aligned} J^\pi - J^{\pi'} &= \sum_{s \in \mathcal{S}} d^{\pi'}(s) \sum_{a \in \mathcal{A}} \pi'(a|s) r(s, a) - J^{\pi'} + \sum_{s \in \mathcal{S}} d^{\pi'}(s) \sum_{\substack{s' \in \mathcal{S} \\ a \in \mathcal{A}}} \mathbb{P}(s'|s, a) \pi'(a|s) V^\pi(s') \\ &\quad - \sum_{s \in \mathcal{S}} d^{\pi'}(s) \sum_{a \in \mathcal{A}} \pi'(a|s) Q^\pi(s, a). \end{aligned}$$

Since $J^{\pi'}$ can be expressed as the ergodic sum of $r(s, a)$,

$$J^\pi - J^{\pi'} = \sum_{s \in \mathcal{S}} d^{\pi'}(s) \sum_{\substack{s' \in \mathcal{S} \\ a \in \mathcal{A}}} \mathbb{P}(s'|s, a) \pi'(a|s) V^\pi(s') - \sum_{s \in \mathcal{S}} d^{\pi'}(s) \sum_{a \in \mathcal{A}} \pi'(a|s) Q^\pi(s, a).$$

On expanding $V^\pi(s')$,

$$\begin{aligned} J^\pi - J^{\pi'} &= \sum_{s \in \mathcal{S}} d^{\pi'}(s) \sum_{s' \in \mathcal{S}} \mathbb{P}(s'|s, \pi'(s)) Q^\pi(s', \pi'(s')) \\ &\quad - \sum_{s \in \mathcal{S}} d^{\pi'}(s) \sum_{s' \in \mathcal{S}} \mathbb{P}(s'|s, \pi'(s)) A^\pi(s', \pi'(s')) - \sum_{s \in \mathcal{S}} d^{\pi'}(s) \sum_{a \in \mathcal{A}} \pi'(a|s) Q^\pi(s, a). \end{aligned}$$

Since,

$$\sum_{s \in \mathcal{S}} d^{\pi'}(s) \sum_{s' \in \mathcal{S}} \mathbb{P}(s'|s, \pi'(s)) Q^\pi(s', \pi'(s)) = \sum_{s' \in \mathcal{S}} d^{\pi'}(s') Q^\pi(s', \pi'(s')),$$

we obtain

$$J^{\pi'} - J^\pi = \sum_{s \in \mathcal{S}} d^{\pi'}(s) \sum_{s' \in \mathcal{S}} \mathbb{P}(s'|s, \pi'(s)) A^\pi(s', \pi'(s')).$$

$$J^\pi - J^{\pi'} = \sum_{s \in \mathcal{S}} d^\pi(s) A^{\pi'}(s, \pi(s)). \quad (\text{C.2})$$

□

C.2 Proof of Lemma 4.2

Lemma C.2. Given π_k which are generated according to Equation (4.10), the corresponding sequence of average rewards J^{π_k} are increasing, that is,

$$J^{\pi_{k+1}} - J^{\pi_k} \geq 0. \quad (\text{C.3})$$

Proof. From the MDM update Equation (4.10) we have the following,

$$A^{\pi_k}(s, a) = \frac{1}{\beta} \log \left(\frac{z_k(s) \pi_{k+1}(a|s)}{\pi_k(a|s)} \right).$$

From Lemma 4.1, we know that

$$\begin{aligned} J^{\pi_{k+1}} - J^{\pi_k} &= \sum_{s \in \mathcal{S}} d^{\pi_{k+1}}(s) \sum_{a \in \mathcal{A}} \pi_{k+1}(a|s) A^{\pi_k}(s, a) \\ &= \frac{1}{\beta} \sum_{s \in \mathcal{S}} d^{\pi_{k+1}}(s) \sum_{a \in \mathcal{A}} \pi_{k+1}(a|s) \log \left(\frac{z_k \pi_{k+1}(a|s)}{\pi_k(a|s)} \right) \\ &= \frac{1}{\beta} \sum_{s \in \mathcal{S}} d^{\pi_{k+1}}(s) \underbrace{\sum_{a \in \mathcal{A}} \pi_{k+1}(a|s) \log \left(\frac{\pi_{k+1}(a|s)}{\pi_k(a|s)} \right)}_{D_{\text{KL}}(\pi_{k+1}(\cdot|s) \parallel \pi_k(\cdot|s)) \geq 0} \\ &\quad + \frac{1}{\beta} \sum_{s \in \mathcal{S}} d^{\pi_{k+1}}(s) \sum_{a \in \mathcal{A}} \pi_{k+1}(a|s) \log(z_k(s)) \end{aligned}$$

$$\begin{aligned} J^{\pi_{k+1}} - J^{\pi_k} &\geq \frac{1}{\beta} \sum_{s \in \mathcal{S}} d^{\pi_{k+1}}(s) \sum_{a \in \mathcal{A}} \pi_{k+1}(a|s) \log \left(\sum_{a' \in \mathcal{A}} \pi_k(a'|s) e^{\beta A^{\pi_k}(s, a')} \right) \\ &\stackrel{(a)}{\geq} \frac{1}{\beta} \sum_{s \in \mathcal{S}} d^{\pi_{k+1}}(s) \underbrace{\sum_{a \in \mathcal{A}} \pi_{k+1}(a|s)}_{=1} \sum_{a' \in \mathcal{A}} \pi_k(a'|s) \beta A^{\pi_k}(s, a') \\ &= \sum_{s \in \mathcal{S}} d^{\pi_{k+1}}(s) \sum_{a' \in \mathcal{A}} \pi_k(a'|s) A^{\pi_k}(s, a'). \end{aligned}$$

where (a) is due to Jensen's inequality and concavity of the log function. Substituting for $A^{\pi_k}(s, a')$,

$$J^{\pi_{k+1}} - J^{\pi_k} \geq \sum_{s \in \mathcal{S}} d^{\pi_{k+1}}(s) \sum_{a' \in \mathcal{A}} \pi_k(a'|s) (Q^{\pi_k}(s, a') - V^{\pi_k}(s))$$

Since $V^{\pi_k}(s) = \sum_{a' \in \mathcal{A}} \pi_k(a'|s) Q^{\pi_k}(s, a')$, we obtain

$$J^{\pi_{k+1}} - J^{\pi_k} \geq 0.$$

□

C.3 Proof of Lemma 4.3

Lemma C.3. Consider the sequence of policies π_k obtained from Algorithm 4. It is true that

$$W(\pi_{k+1}) - W(\pi_k) = \sum_{s \in \mathcal{S}} d^{\pi^*}(s) \log(z_k(s)) - \beta (J^{\pi^*} - J^{\pi_k}),$$

where $z_k(s) = \sum_{a' \in \mathcal{A}} \pi_k(a'|s) e^{\beta A^{\pi_k}(s, a')}$.

Proof. From Equation (4.14) we know that,

$$W(\pi_{k+1}) - W(\pi_k) = \sum_{s \in \mathcal{S}} d^{\pi^*}(s) \sum_{a \in \mathcal{A}} \pi^*(a|s) \log \left(\frac{\pi_k(a|s)}{\pi_{k+1}(a|s)} \right).$$

Using the update Equation (4.10),

$$\begin{aligned} W(\pi_{k+1}) - W(\pi_k) &= \sum_{s \in \mathcal{S}} d^{\pi^*}(s) \sum_{a \in \mathcal{A}} \pi^*(a|s) \log \left(\frac{z_k(s)}{e^{\beta A^{\pi_k}(s, a)}} \right) \\ &= \sum_{s \in \mathcal{S}} d^{\pi^*}(s) \sum_{a \in \mathcal{A}} \pi^*(a|s) \log(z_k(s)) - \beta \sum_{s \in \mathcal{S}} d^{\pi^*}(s) \sum_{a \in \mathcal{A}} \pi^*(a|s) A^{\pi_k}(s, a) \\ &= \sum_{s \in \mathcal{S}} d^{\pi^*}(s) \log(z_k(s)) - \beta (J^{\pi^*} - J^{\pi_k}), \end{aligned}$$

where the last step follows from Lemma 4.1.

□

C.4 Proof of Theorem 4.1

Theorem C.1. The sequence of policies π_k generated through the update rule in Equation (4.10), correspond to MDPs with average reward J^{π_k} , which approach the optimal average reward as below:

$$J^{\pi^*} - J^{\pi_T} \leq \frac{W(\pi_0) - W(\pi_T)}{\beta T} + \frac{J^{\pi_T} - J^{\pi_0}}{\Delta T}.$$

Proof. From Lemma 4.1,

$$\begin{aligned} J^{\pi_{k+1}} - J^{\pi_k} &= \sum_{s \in \mathcal{S}} d^{\pi_{k+1}}(s) A^{\pi_k}(s, \pi_{k+1}(s)) \\ &= \frac{1}{\beta} \left(\sum_{s \in \mathcal{S}} d^{\pi_{k+1}}(s) \underbrace{\sum_{a \in \mathcal{A}} \pi_{k+1}(a|s) \log \left(\frac{\pi_{k+1}(a|s)}{\pi_k(a|s)} \right)}_{D_{\text{KL}}(\pi_{k+1}(\cdot|s) \parallel \pi_k(\cdot|s)) \geq 0} \right. \\ &\quad \left. + \sum_{s \in \mathcal{S}} d^{\pi_{k+1}}(s) \underbrace{\sum_{a \in \mathcal{A}} \pi_{k+1}(a|s) \log(z_k(s))}_{=1} \right) \\ &\geq \frac{1}{\beta} \sum_{s \in \mathcal{S}} d^{\pi^*}(s) \frac{d^{\pi_{k+1}}(s)}{d^{\pi^*}(s)} \log(z_k(s)) \\ &\stackrel{(a)}{\geq} \frac{\Delta}{\beta} \sum_{s \in \mathcal{S}} d^{\pi^*}(s) \log(z_k(s)). \end{aligned}$$

where (a) follows from Assumption 4.1. From Lemma 4.3,

$$J^{\pi_{k+1}} - J^{\pi_k} \geq \frac{\Delta}{\beta} \left(W(\pi_{k+1}) - W(\pi_k) + \beta (J^{\pi^*} - J^{\pi_k}) \right).$$

Iterating the above expression for $k = 0, \dots, T$ yields,

$$J^{\pi_T} - J^{\pi_0} \geq \frac{\Delta}{\beta} \left(W(\pi_T) - W(\pi_0) + \beta \left(T J^{\pi^*} - \sum_{i=0}^{T-1} J^{\pi_i} \right) \right).$$

$$J^{\pi^*} - \frac{\sum_{i=0}^{T-1} J^{\pi_i}}{T} \leq \frac{W(\pi_0) - W(\pi_T)}{\beta T} + \frac{J^{\pi_T} - J^{\pi_0}}{\Delta T}.$$

From Lemma 4.2, we know that the sequence of policies obtained correspond to MDPs with monotonically increasing average rewards, that is $J^{\pi_{k+1}} \geq J^{\pi_k}$. Hence, we obtain,

$$J^{\pi^*} - J^{\pi_T} \leq \frac{W(\pi_0) - W(\pi_T)}{\beta T} + \frac{J^{\pi_T} - J^{\pi_0}}{\Delta T}.$$

□

D

PROOFS: REINFORCEMENT LEARNING IN COUNTABLE STATE SPACES WITH UNBOUNDED COSTS

D.1 Proof of Section 5.4.1

The proof of Step 1 can be found in Chapter 4 of [111].

D.2 Proof of Section 5.4.2

The following lemmas are a consequence of Assumptions 5.1 and 5.2.

Lemma D.1. Let $\mathbb{P}_\pi$ be an irreducible transition matrix on the countable state space $\mathcal{S}$. Suppose that (5.9) is satisfied. Then the corresponding homogenous Markov Chain is positive recurrent. Consequently, the stationary distribution $\mathbf{d}_\pi$ corresponding to $\mathbb{P}_\pi$ exists and is unique [156].

Lemma D.2. Suppose Assumptions 5.1 and 5.2 hold. Let $\mathbf{q}_{ss}$ be a random variable on $\mathcal{S}$ distributed according to $\mathbf{d}_\pi$. Then, there exists a positive constant α such that $\mathbb{E}_{\mathbf{d}_\pi}[e^{\alpha f(\mathbf{q}_{ss})}] < \infty$. [95], [157].

This lemma ensures that for all policies $\pi \in \Pi$, all moments of $f(\mathbf{q})$ exist. The first and second moments are particularly important since final regret depends on them.

As in [95], Lemmas D.1 and D.2 can be used to establish the following policy independent

upper bound on the infinite-horizon average-cost. We also present a proof for completeness.

Lemma D.3. Given Assumptions 5.1 and 5.2, for all policies $\pi \in \Pi$ it is true that,

$$J_\pi \leq \frac{g}{\epsilon} \quad (\text{D.1})$$

where $J_\pi = \mathbb{E}_{d_\pi} [c_\pi(\mathbf{q})]$ is the average cost associated with policy π and constants g, ϵ are the drift parameters in Equation 5.9.

Proof. From Assumption 5.2, it follows that for any policy $\pi \in \Pi$, the following drift inequality is satisfied $\forall \mathbf{q} \in \mathcal{S}$,

$$\mathbb{E}_\pi \left[f^2(\mathbf{q}_{k+1}) - f^2(\mathbf{q}_k) | \mathbf{q}_k = \mathbf{q} \right] \leq -\epsilon \bar{c}(\mathbf{q}) + g. \quad (\text{D.2})$$

Recall that d_π represents the stationary measure associated with policy π . Given that we assume all policies induce irreducible Markov chains and, based on Lemma D.1, the drift equation (5.9) ensures the Markov chain's positive recurrence, the stationary distribution d_π exists and is unique. Since $d_\pi \geq 0$, consider the following weighted drift inequality:

$$\sum_{\mathbf{q} \in \mathcal{S}} d_\pi(\mathbf{q}) \left(\mathbb{E}_\pi \left[f^2(\mathbf{q}_{k+1}) - f^2(\mathbf{q}_k) | \mathbf{q}_k = \mathbf{q} \right] \right) \leq -\epsilon \sum_{\mathbf{q} \in \mathcal{S}} d_\pi(\mathbf{q}) \bar{c}(\mathbf{q}) + g \quad (\text{D.3})$$

From Lemma D.2, recall that the second moment of $f(\mathbf{q})$ is defined and exists for all policies $\pi \in \Pi$. Hence the left hand summation in Equation D.3 is well defined. Since the expectation is taken with respect to $\mathbb{P}_\pi$ and since $d_\pi \mathbb{P}_\pi = d_\pi$, the left hand summation in Equation D.3 is 0. Hence the expected drift in stationarity is zero. We thus obtain the following:

$$\epsilon \sum_{\mathbf{q} \in \mathcal{S}} d_\pi(\mathbf{q}) \bar{c}(\mathbf{q}) \leq g$$

From definition of $\bar{c}(\mathbf{q})$, it follows that

$$\epsilon \sum_{\mathbf{q} \in \mathcal{S}} d_\pi(\mathbf{q}) c_\pi(\mathbf{q}) \leq g$$

Since $J_\pi = \sum_{\mathbf{q} \in \mathcal{S}} d_\pi(\mathbf{q}) c_\pi(\mathbf{q})$, we thus obtain,

$$J_\pi \leq \frac{g}{\epsilon} \quad (\text{D.4})$$

Equation D.4 is true for all policies π . Hence, the average cost is uniformly upper bounded by $\frac{g}{\epsilon}$. $\square$

Since Q_π is related to V_π through Equation (5.5), in order to bound Q_π , we first bound V_π . We now derive an upper bound on the value function V_π utilizing the drift equation 5.9 and the uniform upper bound on J_π in Equation D.4. First we leverage Assumptions 5.1 and 5.2 to establish policy dependent upper bounds on the value function as elaborated in the following subsection.

D.2.1 Policy Dependent Upper Bound on the State Value Function

Lemma D.4. Consider a set B defined in Equation 5.11. Let $V_\pi(\mathbf{q})$ represent the state value function associated with state $\mathbf{q} \in \mathcal{S}$ and policy $\pi \in \Pi$. Under Assumptions 5.1 and 5.2, for all $\mathbf{q} \in \mathcal{S}$ and for all policies $\pi \in \Pi$, there exists policy independent constants $K > 0$ and $C_B > 0$ such that,

$$V_\pi(\mathbf{q}) \leq \frac{2}{\epsilon} f^2(\mathbf{q}) + (K + C_B) \left(\mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi} \mathbb{I}(\mathbf{q}_k \in B) \mid \mathbf{q}_0 = \mathbf{q} \right] \right), \quad (\text{D.5})$$

where $\tau_{\mathbf{q}^*}^\pi$ is the time to hit a fixed state $\mathbf{q}^* \in B$ when starting at $\mathbf{q}$.

Proof. The key idea behind the proof is to apply [99, Theorem 2.1] to an appropriately defined drift inequality. We note that the theorem cannot be directly applied to 5.9 because it does not satisfy the conditions of the theorem. Define the following set:

$$A_\pi := \left\{ \mathbf{q} \in \mathcal{S} : \bar{c}(\mathbf{q}) \leq \frac{2g}{\epsilon} - \mathbb{E}_{d_\pi} [c_\pi(\mathbf{q})] \right\} \quad (\text{D.6})$$

Since $J_\pi \leq \frac{g}{\epsilon}$, it follows from Equation (5.11) in Assumption 5.2 that A_π is a finite, non-

empty set. Multiplying (D.2) throughout by $\frac{2}{\epsilon}$, we obtain the following:

$$\mathbb{E}_\pi \left[\frac{2}{\epsilon} f^2(\mathbf{q}_{k+1}) - \frac{2}{\epsilon} f^2(\mathbf{q}_k) \middle| \mathbf{q}_k = \mathbf{q} \right] \leq -2\bar{c}(\mathbf{q}) + \frac{2g}{\epsilon} \quad (\text{D.7})$$

Consider a $\mathbf{q} \in A_\pi^c$. Then, from definition it is true that $-\bar{c}(\mathbf{q}) \leq -\frac{2g}{\epsilon} + \mathbb{E}_{d_\pi} [c_\pi(\mathbf{q})]$. Bounding $-\bar{c}(\mathbf{q})$ from above, we obtain,

$$\mathbb{E}_\pi \left[\frac{2}{\epsilon} f^2(\mathbf{q}_{k+1}) - \frac{2}{\epsilon} f^2(\mathbf{q}_k) \middle| \mathbf{q}_k = \mathbf{q} \right] \leq -\bar{c}(\mathbf{q}) + \mathbb{E}_{d_\pi} [c_\pi(\mathbf{q})] \quad (\text{D.8})$$

Recall the definition of set B in Equation (5.11). Since the instantaneous costs $c(\mathbf{q}, \mathbf{a})$ are non-negative for all state-action pairs $(\mathbf{q}, \mathbf{a})$, the average cost J_π is also non-negative for all policies $\pi \in \Pi$. Hence, we obtain that $A_\pi \subset B$ for all $\pi \in \Pi$.

Since $B^c \in \mathcal{A}_\pi^c$, we thus obtain for all $\mathbf{q} \in B^c$, it is true that,

$$\mathbb{E}_\pi \left[\frac{2}{\epsilon} f^2(\mathbf{q}_{k+1}) - \frac{2}{\epsilon} f^2(\mathbf{q}_k) \middle| \mathbf{q}_k = \mathbf{q} \right] \leq -\bar{c}(\mathbf{q}) + \mathbb{E}_{d_\pi} [c_\pi(\mathbf{q})] \quad (\text{D.9})$$

Since $\bar{c}(\mathbf{q}) \geq c_\pi(\mathbf{q})$ for all $\mathbf{q} \in \mathcal{S}$,

$$\mathbb{E}_\pi \left[\frac{2}{\epsilon} f^2(\mathbf{q}_{k+1}) - \frac{2}{\epsilon} f^2(\mathbf{q}_k) \middle| \mathbf{q}_k = \mathbf{q} \right] \leq -c_\pi(\mathbf{q}) + \mathbb{E}_{d_\pi} [c_\pi(\mathbf{q})] \quad (\text{D.10})$$

Recall from our Assumption 5.3, the set B is finite and single step transitions are uniformly bounded. Therefore consider the following definition:

$$K := \max_{\substack{\mathbf{q}' : \mathbb{P}(\mathbf{q}' | \mathbf{q}, \mathbf{a}) > 0 \\ \mathbf{q} \in B, \mathbf{a} \in \mathcal{A}}} \frac{2}{\epsilon} f^2(\mathbf{q}') \quad (\text{D.11})$$

Hence for all $\mathbf{q} \in \mathcal{S}$, it is true that,

$$\mathbb{E}_\pi \left[\frac{2}{\epsilon} f^2(\mathbf{q}_{k+1}) - \frac{2}{\epsilon} f^2(\mathbf{q}_k) \middle| \mathbf{q}_k = \mathbf{q} \right] \leq \left(-c_\pi(\mathbf{q}) + \mathbb{E}_{d_\pi} [c_\pi(\mathbf{q})] \right) \mathbb{I}(\mathbf{q} \in B^c) + K \mathbb{I}(\mathbf{q} \in B) \quad (\text{D.12})$$

Let $\tilde{c}_\pi(\mathbf{q}) := c_\pi(\mathbf{q}) - \mathbb{E}_{d_\pi} [c_\pi(\mathbf{q})]$. From definition of B in Equation (5.11), it is true that for all $\mathbf{q} \in B$, $\underline{c}(\mathbf{q}) \leq \frac{2g}{\epsilon}$. Hence for all $\mathbf{q} \in B^c$, $c_\pi(\mathbf{q}) > \frac{2g}{\epsilon}$. And since from Lemma D.3 it is true that $\mathbb{E}_{d_\pi} [c_\pi(\mathbf{q})] \leq \frac{g}{\epsilon}$, we obtain $\tilde{c}_\pi(\mathbf{q}) > 0$, for all $\mathbf{q} \in B^c$.

Define

$$\mathbf{q}^* := \underset{\mathbf{q} \in \mathbf{B}}{\operatorname{argmin}} \underline{c}(\mathbf{q}), \quad (\text{D.13})$$

and let $\tau_{\mathbf{q}^*}^\pi$ be the first time to hit $\mathbf{q}^*$ under policy π starting from some state $\mathbf{q}_0 = \mathbf{q}$. Now, applying [99, Theorem 2.1], we get

$$\mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} (\tilde{c}_\pi(\mathbf{q}_k)) (1 - \mathbb{I}(\mathbf{q}_k \in \mathbf{B})) \mid \mathbf{q}_0 = \mathbf{q} \right] \leq \frac{2}{\epsilon} f^2(\mathbf{q}) + K \mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} \mathbb{I}(\mathbf{q}_k \in \mathbf{B}) \mid \mathbf{q}_0 = \mathbf{q} \right] \quad (\text{D.14})$$

$$\mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} (\tilde{c}_\pi(\mathbf{q}_k)) \mid \mathbf{q}_0 = \mathbf{q} \right] \leq \frac{2}{\epsilon} f^2(\mathbf{q}) + K \mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} \mathbb{I}(\mathbf{q}_k \in \mathbf{B}) \mid \mathbf{q}_0 = \mathbf{q} \right] + \quad (\text{D.15})$$

$$\mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} (\tilde{c}_\pi(\mathbf{q}_k) \mathbb{I}(\mathbf{q}_k \in \mathbf{B})) \mid \mathbf{q}_0 = \mathbf{q} \right] \quad (\text{D.16})$$

Since J_π is non negative,

$$\tilde{c}_\pi(\mathbf{q}_k) \mathbb{I}(\mathbf{q}_k \in \mathbf{B}) \leq \max_{\mathbf{q} \in \mathbf{B}} \bar{c}(\mathbf{q}) =: C_B \quad (\text{D.17})$$

Thus,

$$\mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} (\tilde{c}_\pi(\mathbf{q}_k)) \mid \mathbf{q}_0 = \mathbf{q} \right] \leq \frac{2}{\epsilon} f^2(\mathbf{q}) + (K + C_B) \left(\mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} \mathbb{I}(\mathbf{q}_k \in \mathbf{B}) \mid \mathbf{q}_0 = \mathbf{q} \right] \right) \quad (\text{D.18})$$

From Equation (5.4), we thus obtain the following bound on the value function for all

$\mathbf{q} \in \mathcal{S}$,

$$V_\pi(\mathbf{q}) \leq \frac{2}{\epsilon} f^2(\mathbf{q}) + (K + C_B) \left(\mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} \mathbb{I}(\mathbf{q}_k \in B) \mid \mathbf{q}_0 = \mathbf{q} \right] \right) \quad (\text{D.19})$$

□

In order to invoke the connection of NPG to prediction through expert advice, we need policy independent bounds on the estimate $\hat{Q}_\pi$. As a step towards achieving that, we first need to establish policy independent bounds on the exact value function Q_π and therefore on V_π . Since the drift provides us with a policy dependent upper bound alone, we exploit the structure of queuing systems in order to obtain a policy independent lower bound and upper bound.

D.2.2 Policy Independent Bounds on the State Value Function

The structural assumption 5.3 aids in obtaining policy independent bounds by providing an uniform upper bound on the time spent in B till state $\mathbf{q}^*$ is reached starting from any state $\mathbf{q} \in \mathcal{S}$.

Lemma D.5. Consider the set B in Equation (5.11). Define τ_B^{bound} as

$$\tau_B^{\text{bound}} = \max_{\substack{\mathbf{q} \in \mathcal{S} \\ \pi \in \Pi}} \mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} \mathbb{I}(\mathbf{q}_k \in B) \mid \mathbf{q}_0 = \mathbf{q} \right] \quad (\text{D.20})$$

Then under Assumption 5.3, for any policy $\pi \in \Pi$, τ_B^{bound} satisfies

$$\tau_B^{\text{bound}} \leq \frac{T_B}{p_B^2}. \quad (\text{D.21})$$

Proof. Since the $\mathbb{I}(\mathbf{q}_k \in B)$ is non-zero only when $\mathbf{q}_k \in B$,

$$\mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} \mathbb{I}(\mathbf{q}_k \in B) \mid \mathbf{q}_0 = \mathbf{q} \right] \leq \max_{\mathbf{q} \in B} \mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} \mathbb{I}(\mathbf{q}_k \in B) \mid \mathbf{q}_0 = \mathbf{q} \right] \quad (\text{D.22})$$

Thus, we can assume $\mathbf{q}_0 \in B$. Let τ_n denote the time at which the Markov chain $\mathbf{q}_k$ enters the set B for the n -th time. Let $\tilde{\mathbf{q}}_k = \mathbf{q}_{\tau_k}$. Then, from strong Markov property we know that $\tilde{\mathbf{q}}_k$ is also a Markov chain over B . Let $\tilde{\tau}_{\mathbf{q}^*}^\pi$ denote the time at which the state $\mathbf{q}^* \in B$ is first reached under policy π in the Markov chain $\tilde{\mathbf{q}}_k$. Then,

$$\mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} \mathbb{I}(\mathbf{q}_k \in B) \mid \mathbf{q}_0 = \mathbf{q} \right] \leq \mathbb{E}_\pi \left[\tilde{\tau}_{\mathbf{q}^*}^\pi \mid \mathbf{q}_0 = \mathbf{q} \right]. \quad (\text{D.23})$$

Denoting the transition kernel of $\tilde{\mathbf{q}}$ by $\tilde{\mathbb{P}}$, we have

$$\mathbb{E}_\pi \left[\tilde{\tau}_{\mathbf{q}^*}^\pi \right] = \sum_{k=1}^{\infty} k \tilde{\mathbb{P}}_\pi \left(\tilde{\tau}_{\mathbf{q}^*}^\pi = k \mid \mathbf{q}_0 = \mathbf{q} \right) \quad (\text{D.24})$$

$$\leq \sum_{k=1}^{\infty} k T_B \tilde{\mathbb{P}}_\pi \left((k-1)T_B < \tilde{\tau}_{\mathbf{q}^*}^\pi \leq kT_B \mid \mathbf{q}_0 = \mathbf{q} \right) \quad (\text{D.25})$$

$$\leq \sum_{k=1}^{\infty} k T_B \tilde{\mathbb{P}}_\pi \left(\tilde{\tau}_{\mathbf{q}^*}^\pi > (k-1)T_B \mid \mathbf{q}_0 = \mathbf{q} \right) \quad (\text{D.26})$$

Note that Assumption 5.3 also holds true in the context of Markov chain $\tilde{\mathbf{q}}_k$. Thus,

$$\mathbb{E}_\pi \left[\tilde{\tau}_{\mathbf{q}^*}^\pi \right] \leq \sum_{k=1}^{\infty} k T_B (1 - p_B)^{k-1} = \frac{T_B}{p_B^2}. \quad (\text{D.27})$$

Since the bound is independent of policy $\pi \in \Pi$ and state $\mathbf{q} \in B$, we obtain

$$\max_{\substack{\pi \in \Pi \\ \mathbf{q} \in B}} \mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi} \mathbb{I}(\mathbf{q}_k \in B) \mid \mathbf{q}_0 = \mathbf{q} \right] \leq \frac{T_B}{p_B^2}. \quad (\text{D.28})$$

□

Combining Lemma D.5 with Lemma D.4, we get the following policy independent upper bound on the value function in Equation (5.4).

$$V_\pi(\mathbf{q}) \leq \frac{2}{\epsilon} f^2(\mathbf{q}) + (K + C_B) \left(\frac{T_B}{p_B^2} \right) \quad (\text{D.29})$$

Lemma D.5 can be leveraged to further obtain a policy independent upper bound on the value function as below.

Lemma D.6. Let T_B, p_B be policy independent constants that satisfy Assumption 5.3 and g, ϵ be policy independent constants that satisfy Assumptions 5.2. Then, the value function $V_\pi(\mathbf{q})$ is lower bounded $\forall \mathbf{q} \in \mathcal{S}$ and for all policies $\pi \in \Pi$ as follows:

$$V_\pi(\mathbf{q}) \geq -\frac{g}{\epsilon} \frac{T_B}{p_B^2} \quad (\text{D.30})$$

Proof. Recall the definition of the state value function $V_\pi(\mathbf{q})$ in Equation 5.4. Consider any state $\mathbf{q} \in \mathcal{S}$ and policy $\pi \in \Pi$, such that $\tau_{\mathbf{q}^*}^\pi$ represents the time to hit state $\mathbf{q}^*$ when starting at $\mathbf{q}$. Then,

$$V_\pi(\mathbf{q}) = \mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} \left(c_\pi(\mathbf{q}_k) - \mathbb{E}_\pi [c_\pi(\mathbf{q})] \right) \middle| \mathbf{q}_0 = \mathbf{q} \right] \quad (\text{D.31})$$

$$= \mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} \left(\tilde{c}_\pi(\mathbf{q}_k) \right) (\mathbb{I}(\mathbf{q}_k \in B) + \mathbb{I}(\mathbf{q}_k \in B^c)) \middle| \mathbf{q}_0 = \mathbf{q} \right]. \quad (\text{D.32})$$

From definition of B in Equation (5.11), we know that $\tilde{c}_\pi(\mathbf{q}) \geq 0$ when $\mathbf{q} \in B^c$. Hence,

$$V_\pi(\mathbf{q}) \geq \mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} \left(\tilde{c}_\pi(\mathbf{q}_k) \right) (\mathbb{I}(\mathbf{q}_k \in B)) \middle| \mathbf{q}_0 = \mathbf{q} \right]. \quad (\text{D.33})$$

Since the instantaneous costs are non negative,

$$V_\pi(\mathbf{q}) \geq \mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} -J_\pi(\mathbb{I}(\mathbf{q}_k \in B)) \middle| \mathbf{q}_0 = \mathbf{q} \right]. \quad (\text{D.34})$$

From Lemma D.3,

$$V_\pi(\mathbf{q}) \geq -\frac{g}{\epsilon} \mathbb{E}_\pi \left[\sum_{k=0}^{\tau_{\mathbf{q}^*}^\pi - 1} (\mathbb{I}(\mathbf{q}_k \in B)) \middle| \mathbf{q}_0 = \mathbf{q} \right]. \quad (\text{D.35})$$

From Lemma D.5, we obtain the result,

$$V_\pi(\mathbf{q}) \geq -\frac{g}{\epsilon} \frac{T_B}{p_B^2} \quad (\text{D.36})$$

□

D.2.3 Policy Independent Bounds on the State-Action Value Function

In order to obtain policy independent bounds on the estimate $\widehat{Q}_\pi$ of the state action value function associated with some policy π , it is necessary to first obtain bounds on the exact state action value function Q_π . The following lemma provides with state-dependent, policy-independent bounds on the state action value function Q .

Lemma D.7. There exists constant $g_1 > 0$, such that under Assumptions 5.1, 5.2 and 5.3, the state action value function Q_π for all policies $\pi \in \Pi$ and for all $\mathbf{q} \in \mathcal{S}$ satisfies:

$$|Q_\pi(\mathbf{q}, \mathbf{a}) - Q_\pi(\mathbf{q}, \mathbf{a}')| \leq \frac{2}{\epsilon} f^2(\mathbf{q}) + \frac{4D}{\epsilon} f(\mathbf{q}) + \bar{c}(\mathbf{q}) + g_1 \quad \mathbf{a}, \mathbf{a}' \in \mathcal{A} \quad (\text{D.37})$$

where $\epsilon > 0$ is the drift parameter and $g_1 = \frac{2D^2}{\epsilon} + (K + C_B) \left(1 + \frac{T_B}{p_B^2}\right) + \frac{g}{\epsilon} \left(\frac{T_B}{p_B^2}\right)$.

Proof. Recall the Poisson Equation (5.5) corresponding to the state action value function Q_π :

$$Q_\pi(\mathbf{q}, \mathbf{a}) = c(\mathbf{q}, \mathbf{a}) + \mathbb{E}_{\mathbf{q}' \sim \mathbb{P}(\cdot | \mathbf{q}, \mathbf{a})} V_\pi(\mathbf{q}') - J_\pi \quad (\text{D.38})$$

For any pair of actions $\mathbf{a}, \mathbf{a}' \in \mathcal{A}$

$$\begin{aligned} Q_\pi(\mathbf{q}, \mathbf{a}) - Q_\pi(\mathbf{q}, \mathbf{a}') &= c(\mathbf{q}, \mathbf{a}) - c(\mathbf{q}, \mathbf{a}') + \mathbb{E}_{\mathbf{q}' \sim \mathbb{P}(\cdot | \mathbf{q}, \mathbf{a})} V_\pi(\mathbf{q}') - \mathbb{E}_{\mathbf{q}'' \sim \mathbb{P}(\cdot | \mathbf{q}, \mathbf{a}')} V_\pi(\mathbf{q}'') \\ &\leq \bar{c}(\mathbf{q}) + \mathbb{E}_{\mathbf{q}' \sim \mathbb{P}(\cdot | \mathbf{q}, \mathbf{a})} \left(\frac{2}{\epsilon} f^2(\mathbf{q}') + (K + C_B) \left(1 + \frac{T_B}{p_B^2}\right) \right) \\ &\quad + \mathbb{E}_{\mathbf{q}'' \sim \mathbb{P}(\cdot | \mathbf{q}, \mathbf{a}')} \left(\frac{g}{\epsilon} \left(\frac{T_B}{p_B^2}\right) \right). \end{aligned} \quad (\text{D.39})$$

where the last inequality follows from Lemma D.6. Recall from Assumption 5.2, Equation (5.10), we know that $f(\mathbf{q}') \leq f(\mathbf{q}) + D$, for all $\mathbf{q}' : \mathbb{P}_\pi(\mathbf{q}' | \mathbf{q}) > 0$, for any policy

π .

Hence we obtain,

$$Q_\pi(\mathbf{q}, \mathbf{a}) - Q_\pi(\mathbf{q}, \mathbf{a}') \leq \frac{2}{\epsilon} f^2(\mathbf{q}) + \frac{4D}{\epsilon} f(\mathbf{q}) + \bar{c}(\mathbf{q}) + \frac{2D^2}{\epsilon} + (K + C_B) \left(1 + \frac{T_B}{p_B^2}\right) + \frac{g}{\epsilon} \left(\frac{T_B}{p_B^2}\right) \quad (\text{D.40})$$

Let $g_1 = \frac{2D^2}{\epsilon} + (K + C_B) \left(1 + \frac{T_B}{p_B^2}\right) + \frac{g}{\epsilon} \left(\frac{T_B}{p_B^2}\right)$, then we obtain the following:

$$Q_\pi(\mathbf{q}, \mathbf{a}) - Q_\pi(\mathbf{q}, \mathbf{a}') \leq \frac{2}{\epsilon} f^2(\mathbf{q}) + \frac{4D}{\epsilon} f(\mathbf{q}) + \bar{c}(\mathbf{q}) + g_1 \quad (\text{D.41})$$

Since the above inequality is true for all $\mathbf{a}, \mathbf{a}' \in \mathcal{A}$,

$$|Q_\pi(\mathbf{q}, \mathbf{a}) - Q_\pi(\mathbf{q}, \mathbf{a}')| \leq \frac{2}{\epsilon} f^2(\mathbf{q}) + \frac{4D}{\epsilon} f(\mathbf{q}) + \bar{c}(\mathbf{q}) + g_1 \quad (\text{D.42})$$

□

D.3 Proof of Section 5.4.3

The previous step provided us with bounds over the exact state action value function. Here we incorporate the policy evaluation error to obtain bounds over the state action value function estimate.

Lemma D.8. For all states $\mathbf{q} \in \mathcal{S}$, all pairs of actions $\mathbf{a}, \mathbf{a}' \in \mathcal{A}$, and all policies π , it is true that,

$$\left| \hat{Q}_\pi(\mathbf{q}, \mathbf{a}) - \hat{Q}_\pi(\mathbf{q}, \mathbf{a}') \right| \leq 2\delta(\mathbf{q}) + \frac{2}{\epsilon} f^2(\mathbf{q}) + \frac{4D}{\epsilon} f(\mathbf{q}) + \bar{c}(\mathbf{q}) + g_1$$

where $\hat{Q}_\pi(\mathbf{q}, \mathbf{a})$ is the estimate of $Q_\pi(\mathbf{q}, \mathbf{a})$ such that $\delta(\mathbf{q}) := \left| \hat{Q}_\pi(\mathbf{q}, \mathbf{a}) - Q_\pi(\mathbf{q}, \mathbf{a}) \right|, \forall \mathbf{a} \in \mathcal{A}$.

Proof.

$$\left| \widehat{Q}_\pi(\mathbf{q}, \mathbf{a}) - \widehat{Q}_\pi(\mathbf{q}, \mathbf{a}') \right| = \left| \widehat{Q}_\pi(\mathbf{q}, \mathbf{a}) - Q_\pi(\mathbf{q}, \mathbf{a}) + Q_\pi(\mathbf{q}, \mathbf{a}) \right| \quad (\text{D.43})$$

$$- \widehat{Q}_\pi(\mathbf{q}, \mathbf{a}') + Q_\pi(\mathbf{q}, \mathbf{a}') - Q_\pi(\mathbf{q}, \mathbf{a}') \right| \quad (\text{D.44})$$

$$\leq \left| \widehat{Q}_\pi(\mathbf{q}, \mathbf{a}) - Q_\pi(\mathbf{q}, \mathbf{a}) \right| + \left| \widehat{Q}_\pi(\mathbf{q}, \mathbf{a}') - Q_\pi(\mathbf{q}, \mathbf{a}') \right| \quad (\text{D.45})$$

$$+ \left| Q_\pi(\mathbf{q}, \mathbf{a}) - Q_\pi(\mathbf{q}, \mathbf{a}') \right| \quad (\text{D.46})$$

From Equation 5.30 and Lemma D.7, it follows that,

$$\left| \widehat{Q}_\pi(\mathbf{q}, \mathbf{a}) - \widehat{Q}_\pi(\mathbf{q}, \mathbf{a}') \right| \leq 2\delta(\mathbf{q}) + \frac{2}{\epsilon} f^2(\mathbf{q}) + \frac{4D}{\epsilon} f(\mathbf{q}) + \bar{c}(\mathbf{q}) + g_1 \quad (\text{D.47})$$

□

D.4 Proof of Main theorem – Section 5.4.4

The proof requires utilizing the performance difference lemma to establish a connection between the difference in average cost associated with a policy π and the optimal average cost in terms of the state-action value function Q_π .

Lemma D.9. Let J_π and $J_{\pi'}$ be the expected infinite horizon average cost associated with policies π and π' respectively. Let $\mathbf{d}_\pi$ be the stationary distribution over state space $\mathcal{S}$ associated with $\mathbb{P}_\pi$. Then it is true that,

$$J_\pi - J_{\pi'} = \sum_{\mathbf{q} \in \mathcal{S}} \mathbf{d}_\pi(\mathbf{q}) [Q_{\pi'}(\mathbf{q}, \pi(\mathbf{q})) - Q_{\pi'}(\mathbf{q}, \pi'(\mathbf{q}))] \quad (\text{D.48})$$

where $Q_{\pi'}(\mathbf{q}, \pi(\mathbf{q})) = \sum_{\mathbf{a} \in \mathcal{A}} \pi(\mathbf{a}|\mathbf{q}) Q_{\pi'}(\mathbf{q}, \mathbf{a})$ and $Q_{\pi'}(\mathbf{q}, \pi'(\mathbf{q})) = V_{\pi'}(\mathbf{q})$.

Proof. The proof can be found in [110].

□

We restate the theorem for convenience.

Theorem 5.1. Consider the sequence of policies $\pi_1, \pi_2, \dots, \pi_T$ obtained from Algorithm 6 with a

state-dependent step size $\eta_{\mathbf{q}} = \sqrt{\frac{8\log|\mathcal{A}|}{T}} \frac{1}{M_{\mathbf{q}}}$, where

$$M_{\mathbf{q}} = \left(2\delta(\mathbf{q}) + \frac{2}{\epsilon} f^2(\mathbf{q}) + \frac{4D}{\epsilon} + \bar{c}(\mathbf{q}) + g_1 \right)$$

and $\delta(\mathbf{q}) := \sup_{\pi \in \Pi} \left\| \hat{Q}_{\pi}(\mathbf{q}, \mathbf{a}) - Q_{\pi}(\mathbf{q}, \mathbf{a}) \right\|_{\infty}$. Let J_{π_k} be the average cost associated with policy π_k and let J_* be the minimum average cost across policy class Π . Let the learning error satisfy the following:

$$\mathbb{E}_{\pi} [\delta(\mathbf{q})] \leq \kappa(\mathbf{q}) \quad \forall \mathbf{q} \in \mathcal{S}, \pi \in \Pi \quad (5.13)$$

Then, under Assumptions 5.1, 5.2 and 5.3, there exist constants c', c'' not depending on T or $\pi_1, \pi_2, \dots, \pi_T$ such that:

$$\sum_{k=1}^T \mathbb{E} (J_{\pi_k} - J_*) \leq c' \sqrt{T} + c'' T \quad (5.14)$$

where $c' = \sqrt{\frac{\log|\mathcal{A}|}{2}} \left(2\beta + \beta_1 + \beta_2 + \frac{g}{\epsilon} + g_1 \right)$, $c'' = 2\beta$, $\beta := \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} [\kappa(\mathbf{q})]$, $\beta_1 = \frac{4D}{\epsilon} \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} [f(\mathbf{q})]$, $\beta_2 = \frac{2}{\epsilon} \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} [f^2(\mathbf{q})]$ and $g_1 = \frac{2D^2}{\epsilon} + (K + C_B + \frac{g}{\epsilon}) \left(\frac{T_B}{p_B^2} \right)$.

Proof. Let J^* be the optimal average cost. Let $\pi^* \in \Pi$ be the optimal policy. For any policy $\pi \in \Pi$, performance difference lemma provides the following,

$$J_{\pi} - J^* = -\mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} \left[Q_{\pi}(\mathbf{q}, \pi^*(\mathbf{q})) - Q_{\pi}(\mathbf{q}, \pi(\mathbf{q})) \right] \quad (D.49)$$

$$= -\mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} \left[Q_{\pi}(\mathbf{q}, \pi^*(\mathbf{q})) - \hat{Q}_{\pi}(\mathbf{q}, \pi^*(\mathbf{q})) + \hat{Q}_{\pi}(\mathbf{q}, \pi^*(\mathbf{q})) - Q_{\pi}(\mathbf{q}, \pi(\mathbf{q})) \right] \quad (D.50)$$

$$+ \hat{Q}_{\pi}(\mathbf{q}, \pi(\mathbf{q})) - \hat{Q}_{\pi}(\mathbf{q}, \pi(\mathbf{q})) \quad (D.51)$$

$$\leq \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} \left[\left| Q_{\pi}(\mathbf{q}, \pi^*(\mathbf{q})) - \hat{Q}_{\pi}(\mathbf{q}, \pi^*(\mathbf{q})) \right| \right] \quad (D.52)$$

$$+ \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} \left[\left| Q_{\pi}(\mathbf{q}, \pi(\mathbf{q})) - \hat{Q}_{\pi}(\mathbf{q}, \pi(\mathbf{q})) \right| \right] \quad (D.53)$$

$$+ \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} \left[\left| \hat{Q}_{\pi}(\mathbf{q}, \pi(\mathbf{q})) - \hat{Q}_{\pi}(\mathbf{q}, \pi^*(\mathbf{q})) \right| \right] \quad (D.54)$$

From Equation 5.30, we know that $\mathbb{E} \left[\left| Q_{\pi}(\mathbf{q}, \mathbf{a}) - \hat{Q}_{\pi}(\mathbf{q}, \mathbf{a}) \right| \right] \leq \kappa(\mathbf{q})$. Hence we obtain

the following:

$$\mathbb{E} (J_\pi - J^*) \leq 2\mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} (\kappa(\mathbf{q})) + \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} \left[\mathbb{E} \left(\hat{Q}_\pi(\mathbf{q}, \pi(\mathbf{q})) - \hat{Q}_\pi(\mathbf{q}, \pi^*(\mathbf{q})) \right) \right] \quad (\text{D.55})$$

The total expected regret across time horizon T can be expressed by summing the above inequality as follows,

$$\sum_{k=1}^T \mathbb{E} [J_{\pi_k} - J^*] \leq 2T\mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} (\kappa(\mathbf{q})) + \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} \left[\mathbb{E} \left(\sum_{k=1}^T \left(\hat{Q}_{\pi_k}(\mathbf{q}, \pi_k(\mathbf{q})) - \hat{Q}_{\pi_k}(\mathbf{q}, \pi^*(\mathbf{q})) \right) \right) \right] \quad (\text{D.56})$$

where π_k are policy iterates obtained through the NPG policy update below:

$$\pi_k(a|\mathbf{q}) = \frac{\pi_{k-1}(a|\mathbf{q}) \exp \left(-\eta_{\mathbf{q}} \hat{Q}_{\pi_{k-1}}(\mathbf{q}, a) \right)}{\sum_{l \in \mathcal{A}} \pi_{k-1}(l|\mathbf{q}) \exp \left(-\eta_{\mathbf{q}} \hat{Q}_{\pi_{k-1}}(\mathbf{q}, l) \right)} \quad (\text{D.57})$$

The above update is performed for all $\mathbf{q}$ and $a \in \mathcal{A}$. Let the update parameter $\eta_{\mathbf{q}} = \sqrt{\frac{8 \log |\mathcal{A}|}{T}} \frac{1}{M_{\mathbf{q}}}$, where $M_{\mathbf{q}} = 2\delta(\mathbf{q}) + \frac{2}{\epsilon} f^2(\mathbf{q}) + \frac{4D}{\epsilon} f(\mathbf{q}) + \bar{c}(\mathbf{q}) + g_1$. Then from Theorem 5.2, it follows that,

$$\sum_{k=1}^T \mathbb{E} [J_{\pi_k} - J^*] \leq 2T\mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} (\kappa(\mathbf{q})) + \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} \left[\sqrt{\frac{T \log |\mathcal{A}|}{2}} \mathbb{E} [M_{\mathbf{q}}] \right] \quad (\text{D.58})$$

$$= 2T\mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} (\kappa(\mathbf{q})) + \sqrt{\frac{T \log |\mathcal{A}|}{2}} \left(\mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} \left(2\kappa(\mathbf{q}) + \frac{2}{\epsilon} f^2(\mathbf{q}) + \frac{4D}{\epsilon} f(\mathbf{q}) + \bar{c}(\mathbf{q}) \right) + g_1 \right) \quad (\text{D.59})$$

$$\leq 2T\mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} (\kappa(\mathbf{q})) + \sqrt{\frac{T \log |\mathcal{A}|}{2}} \left(\mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} \left(2\kappa(\mathbf{q}) + \frac{2}{\epsilon} f^2(\mathbf{q}) + \frac{4D}{\epsilon} f(\mathbf{q}) \right) + \frac{g}{\epsilon} + g_1 \right) \quad (\text{D.60})$$

where the last inequality follows from the fact that $\bar{c}(\mathbf{q}) \leq \frac{g}{\epsilon}$ in Lemma D.3.

Let $\beta := \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} [\kappa(\mathbf{q})]$ be defined. From Lemma D.2, it is known that moments of $f(\mathbf{q})$

exist. Let $\beta_1 = \frac{4D}{\epsilon} \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} [f(\mathbf{q})]$ and $\beta_2 = \frac{2}{\epsilon} \mathbb{E}_{\mathbf{q} \sim d_{\pi^*}} [f^2(\mathbf{q})]$. Hence, we obtain,

$$\sum_{k=1}^T \mathbb{E} [J_{\pi_k} - J^*] \leq 2\beta T + \sqrt{T} \left(\sqrt{\frac{\log |\mathcal{A}|}{2}} \left(2\beta + \beta_1 + \beta_2 + \frac{g}{\epsilon} + g_1 \right) \right) \quad (\text{D.61})$$

Setting $c' = \sqrt{\frac{\log |\mathcal{A}|}{2}} \left(2\beta + \beta_1 + \beta_2 + \frac{g}{\epsilon} + g_1 \right)$ and $c'' = 2\beta$ yields the result in the theorem. □

E

PROOFS: MODIFIED POLICY ITERATION FOR RISK SENSITIVE MDPs

We present the lemmas and their proofs and also provide the main proof presented in the proof outline. Before presenting the proof, we first state the Perron-Frobenius theorem, which is extensively used in the paper.

Theorem E.1. Let A be an irreducible non-negative $N \times N$ matrix, which is aperiodic and has a spectral radius $\rho(A) = r$. Then the following statements are true:

- *The number $r \in \mathbb{R}^+$ is a positive real number and it is an eigenvalue of the matrix A , called the Perron-Frobenius eigenvalue.*
- *The Perron-Frobenius eigenvalue is of multiplicity 1. Both right and left eigenspaces associated with r are one-dimensional.*
- *A has both a right and a left eigenvector, respectively v and w , with eigenvalue r and whose components are all positive. Moreover these are the only eigenvectors whose components are all positive.*

Proof. The proof can be found in [140].

□

Recall the following notation. Let g_n be defined as

$$g_n(i) := \frac{\left(\mathcal{T} e^{V'_n} \right)(i)}{e^{V'_n(i)}}, \quad (\text{E.1})$$

and set u_n and ℓ_n as

$$u_n := \max_{i \in \mathcal{S}} (g_n(i)) \quad \ell_n := \min_{i \in \mathcal{S}} (g_n(i)). \quad (\text{E.2})$$

Define $\tilde{Q}_f(i, j)$ corresponding to some policy $f \in \Pi$, as follows:

$$\tilde{Q}_f(i, j) := e^{\alpha d_f(i)} Q_f(j|i) \quad (\text{E.3})$$

E.1 Proof of Section 6.5.1

Lemma E.1. Let $\tilde{\Lambda}^*$ be the optimal risk-sensitive average cost associated with the MDP considered in Algorithm 7. Then $\forall n > 0$:

$$\ell_n \leq e^{\tilde{\Lambda}^*} \leq e^{\tilde{\Lambda}_{f_{n+1}}} \leq u_n. \quad (\text{E.4})$$

Proof. • $u_n \geq e^{\tilde{\Lambda}_{f_{n+1}}}$.

Recall by definition in Equation (E.2) that

$$\begin{aligned} u_n &= \max_i \left(\frac{\left(\mathcal{T} e^{V'_n} \right)(i)}{e^{V'_n(i)}} \right) \\ &\geq \frac{\left(\mathcal{T} e^{V'_n} \right)(i)}{e^{V'_n(i)}} \\ &= \frac{\left(\mathcal{T}_{f_{n+1}} e^{V'_n} \right)(i)}{e^{V'_n(i)}} \quad \forall i \in \mathcal{S}. \end{aligned}$$

$$\begin{aligned}
\implies u_n e^{V'_n(i)} &\geq e^{\alpha d_{f_{n+1}}(i)} \sum_{j \in \mathcal{S}} Q_{f_{n+1}}(j | i) e^{V'_n(j)} \\
&\geq \left(\tilde{Q}_{f_{n+1}} e^{V'_n} \right)(i),
\end{aligned}$$

where $\tilde{Q}_{f_{n+1}}(i, j)$ is defined in Equation (E.3).

Consequently, it follows that

$$u_n \tilde{Q}_{f_{n+1}}^k \left(e^{V'_n} \right) \geq \left(\tilde{Q}_{f_{n+1}} \right)^{k+1} e^{V'_n}$$

$$\frac{u_n}{e^{\tilde{\Lambda}_{f_{n+1}}}} \left(\frac{\tilde{Q}_{f_{n+1}}}{e^{\tilde{\Lambda}_{f_{n+1}}}} \right)^k e^{V'_n} \geq \left(\frac{\tilde{Q}_{f_{n+1}}}{e^{\tilde{\Lambda}_{f_{n+1}}}} \right)^{k+1} e^{V'_n}.$$

$$\lim_{k \rightarrow \infty} \frac{u_n}{e^{\tilde{\Lambda}_{f_{n+1}}}} \left(\frac{\tilde{Q}_{f_{n+1}}}{e^{\tilde{\Lambda}_{f_{n+1}}}} \right)^k e^{V'_n} \geq \lim_{k \rightarrow \infty} \left(\frac{\tilde{Q}_{f_{n+1}}}{e^{\tilde{\Lambda}_{f_{n+1}}}} \right)^{k+1} e^{V'_n}.$$

Since $\tilde{Q}$ satisfies the conditions of the Perron-Frobenius Theorem, by definition, so does $\tilde{\tilde{Q}}$. The vector $e^{V'_n}$ consists of all positive elements and hence, in the limit of $k \rightarrow \infty$, due to power iteration, the following holds true:

$$\frac{u_n}{e^{\tilde{\Lambda}_{f_{n+1}}}} z \geq z.$$

As $\tilde{\tilde{Q}}_{f_{n+1}}$ and $e^{V'_n}$ are both non negative, the resulting z is a non-zero vector containing non negative elements. Hence we obtain $u_n \geq e^{\tilde{\Lambda}_{f_{n+1}}}$ as desired.

- $e^{\tilde{\Lambda}^*} \leq e^{\tilde{\Lambda}_{f_{n+1}}}.$

Recall that

$$\begin{aligned}
e^{\tilde{\Lambda}^*} e^{V^*(i)} &= \min_{f \in \Pi} e^{\alpha d_f(i)} \sum_{j \in \mathcal{S}} Q_f(j | i) e^{V^*(j)} \\
&\leq e^{\alpha d_{f_{n+1}}(i)} \sum_{j \in \mathcal{S}} Q_{f_{n+1}}(j | i) e^{V^*(j)}.
\end{aligned}$$

Therefore, we have

$$\begin{aligned}
e^{V^*(i)} &\leq \frac{e^{\alpha d_{f_{n+1}}(i)} \sum_{j \in \mathcal{S}} Q_{f_{n+1}}(j | i) e^{V^*(j)}}{e^{\tilde{\Lambda}^*}} \\
&= \frac{e^{\alpha d_{f_{n+1}}(i)} \mathbb{E} \left[e^{V^*(x_1)} \mid x_0 = i, f_{n+1} \right]}{e^{\tilde{\Lambda}^*}} \\
&\leq \frac{e^{\alpha d_{f_{n+1}}(i)}}{e^{\tilde{\Lambda}^*}} \cdot \mathbb{E} \left[\frac{e^{\alpha d_{f_{n+1}}(x_1)}}{e^{\tilde{\Lambda}^*}} \cdot \mathbb{E} \left[e^{V^*(x_2)} \mid x_1, f_{n+1} \right] \mid x_0 = i, f_{n+1} \right].
\end{aligned}$$

Iterating, we get

$$e^{V^*(x_0)} \leq \mathbb{E} \left[\frac{e^{\sum_{i=0}^{k-1} \alpha d_{f_{n+1}}(x_i)} e^{V^*(x_k)}}{(e^{\tilde{\Lambda}^*})^k} \mid x_0 \right].$$

Recall that e^{V^*} is the optimal value function vector defined in Equation (6.5). Since the state and policy spaces are finite, and we are dealing with irreducible policies the solution to the optimality equation exists. Thus, $V^*(x_k) \leq C$ for all $x_k \in \mathcal{S}$, where $C < \infty$ is some constant. Hence we obtain,

$$\begin{aligned}
k \cdot \tilde{\Lambda}^* + V^*(x_0) &\leq \ln \left(\mathbb{E}_{x_0} \left[e^{\alpha \sum_{i=0}^{k-1} d(x_i, f_{n+1}(x_i))} \right] \right) + C, \\
\Rightarrow \tilde{\Lambda}^* + \frac{V^*(x_0)}{k} &\leq \frac{1}{k} \ln \left(\mathbb{E}_{x_0} \left[e^{\alpha \sum_{i=0}^{k-1} d(x_i, f_{n+1}(x_i))} \right] \right) + \frac{C}{k}.
\end{aligned}$$

Letting $k \rightarrow \infty$, we get

$$\begin{aligned}
\tilde{\Lambda}^* &\leq \tilde{\Lambda}_{f_{n+1}} \\
\Rightarrow e^{\tilde{\Lambda}^*} &\leq e^{\tilde{\Lambda}_{f_{n+1}}}.
\end{aligned}$$

- $\ell_n \leq e^{\tilde{\Lambda}^*}$.

Recall that $e^{\tilde{\lambda}^*}$ satisfies the following equation:

$$e^{\tilde{\lambda}^*} e^{V^*} = \min_{f \in \Pi} e^{\alpha d_f} Q_f \left(e^{V^*} \right).$$

Let the minimising policy be f^* . We have

$$\begin{aligned} \ell_n &= \min_i \frac{T e^{V'_n}(i)}{e^{V'_n}(i)} \\ &\leq \frac{T e^{V'_n}(i)}{e^{V'_n}(i)} \\ &= \frac{T_{f_{n+1}} e^{V'_n}(i)}{e^{V'_n}(i)} \\ &\leq \frac{T_{f^*} e^{V'_n}(i)}{e^{V'_n}(i)}. \end{aligned}$$

$$\begin{aligned} \ell_n e^{V'_n(i)} &\leq e^{\alpha d_{f^*}(i)} \sum_{j \in \mathcal{S}} Q_{f^*}(j | i) e^{V'_n(j)} \\ &= \left(\tilde{Q}_{f^*} \left(e^{V'_n} \right) \right) (i) \end{aligned}$$

$$\implies \ell_n \tilde{Q}_{f^*}^k \left(e^{V'_n} \right) \leq \tilde{Q}_{f^*}^{k+1} e^{V'_n}.$$

$$\implies \frac{\ell_n}{e^{\tilde{\lambda}^*}} \left(\frac{\tilde{Q}_{f^*}}{e^{\tilde{\lambda}^*}} \right)^k e^{V'_n} \leq \left(\frac{\tilde{Q}_{f^*}}{e^{\tilde{\lambda}^*}} \right)^{k+1} e^{V'_n}.$$

As a consequence of Perron-Frobenius Theorem, it follows that

$$\ell_n \leq e^{\tilde{\lambda}^*},$$

as desired. This concludes the proof.

□

E.2 Proof of Section 6.5.2

Lemma E.2. The sequence u_n is non-increasing, i.e. $u_n \leq u_{n-1}$ for all n .

Proof. Recall that

$$u_n = \max_i \frac{(Te^{V'_n})(i)}{e^{V'_n(i)}}.$$

Let $\chi^* = \operatorname{argmax}_i \frac{(Te^{V'_n})(i)}{e^{V'_n(i)}}$. If the maximum is attained at more than one index, we break ties by selecting the smallest such i (without loss of generality). Then,

$$\begin{aligned} u_n &= \frac{(Te^{V'_n})(\chi^*)}{e^{V'_n(\chi^*)}} \\ &= \frac{(\tilde{Q}_{f_{n+1}} e^{V'_n})(\chi^*)}{e^{V'_n(\chi^*)}} \\ &\leq \frac{(\tilde{Q}_{f_n} e^{V'_n})(\chi^*)}{e^{V'_n(\chi^*)}}. \end{aligned}$$

Since $e^{V'_n(j)} = \frac{e^{V_n(j)}}{\sum_{j \in S} e^{V_n(j)}}$, it follows that

$$\begin{aligned} u_n &\leq \frac{(\tilde{Q}_{f_n} e^{V_n})(\chi^*)}{e^{V_n(\chi^*)}} \\ &= \frac{\tilde{Q}_{f_n} \tilde{Q}_{f_n}^{m_{n-1}} e^{V'_{n-1}}(\chi^*)}{\tilde{Q}_{f_n}^{m_{n-1}} e^{V'_{n-1}}(\chi^*)} \\ &= \frac{\left(\tilde{Q}_{f_n}^{m_{n-1}} \left(\tilde{Q}_{f_n} e^{V'_{n-1}} \right) \right)(\chi^*)}{\left(\tilde{Q}_{f_n}^{m_{n-1}} e^{V'_{n-1}} \right)(\chi^*)}. \end{aligned}$$

Let $e^{W'_{n-1}} = \tilde{Q}_{f_n} e^{V'_{n-1}}$ and $H_n = \tilde{Q}_{f_n}^{m_{n-1}}$. It then follows that

$$\begin{aligned} u_n &\leq \frac{\left(H_n e^{W'_{n-1}}\right)(x^*)}{\left(H_n e^{V'_{n-1}}\right)(x^*)} \\ &= \frac{\sum_{j \in \mathcal{S}} H(j | x^*) e^{W'_{n-1}(j)}}{\sum_{j \in \mathcal{S}} H(j | x^*) e^{V'_{n-1}(j)}}. \end{aligned}$$

Let $p = \operatorname{argmax}_i \frac{e^{W'_{n-1}(i)}}{e^{V'_{n-1}(i)}}$. Then, we have

$$\frac{e^{W'_{n-1}(p)}}{e^{V'_{n-1}(p)}} \geq \frac{e^{W'_{n-1}(i)}}{e^{V'_{n-1}(i)}} \quad \forall i \in \mathcal{S}.$$

This yields

$$u_n \leq \frac{\sum_{j \in \mathcal{S}} \left(H(j | x^*) e^{V'_{n-1}(j)} \cdot \frac{e^{W'_{n-1}(p)}}{e^{V'_{n-1}(p)}} \right)}{\sum_{j \in \mathcal{S}} H(j | x^*) e^{V'_{n-1}(j)}}.$$

Therefore,

$$\begin{aligned} u_n &\leq \frac{e^{W'_{n-1}(p)}}{e^{V'_{n-1}(p)}} \\ &= \frac{\left(\tilde{Q}_{f_n} e^{V'_{n-1}}\right)(p)}{e^{V'_{n-1}(p)}} \\ &= \max_i \frac{\left(T e^{V'_{n-1}}\right)(i)}{e^{V'_{n-1}(i)}} \\ &= u_{n-1}, \end{aligned}$$

which establishes the desired monotonicity. □

E.3 Proof of Section 6.5.3

A crucial component to the convergence of the algorithm is a source of contraction, which is obtained from any finite product of ergodic matrices.

Lemma E.3. There exists a finite natural number R such that $\forall R' > R$ for any sequence of policies $f_1, f_2, \dots, f_{R'} \in \Pi$, there exists a constant $\epsilon_{R'} > 0$.

$$\min_{i,j \in \mathcal{S}} Q_{f_1} Q_{f_2} \dots Q_{f_{R'}}(j|i) \geq \epsilon_{R'} > 0. \quad (\text{E.5})$$

Proof. Given finite state and action spaces, the total number of deterministic policies are given by $|\Pi| = |\mathcal{A}|^{|\mathcal{S}|}$. Since every policy induces an irreducible Markov chain, $\mathbb{P}_f^{|\mathcal{S}|}(j|i) > 0 \forall i, j \in \mathcal{S}, \forall f \in \Pi$. When $R = |\mathcal{A}|^{|\mathcal{S}|} + 1$, there exists a policy that is repeated at least twice in the sequence. Hence, if $R = (|\mathcal{S}| - 1) \cdot |\mathcal{A}|^{|\mathcal{S}|} + 1$, there exists a policy which is repeated at least $|\mathcal{S}|$ times. Since in the transformed model, under every policy the probability of staying in the same state is non-zero, there exists a non-zero probability of traversing from any state to any other state when $R \geq (|\mathcal{S}| - 1) \cdot |\mathcal{A}|^{|\mathcal{S}|} + 1$ under any sequence of policies. $\square$

Lemma E.4. Recall that $\max_k m_k < M$, where m_k corresponds to the number of fixed point iterations performed during partial policy evaluation during the k th execution of the algorithm. Then, there exists β such that $0 < \beta < 1$,

$$e^{V'_k(i)} > \beta > 0 \quad \forall i \in \mathcal{S} \quad \forall k \geq R. \quad (\text{E.6})$$

Proof. Since $e^{V'_k(i)} = \frac{e^{V_k(i)}}{\sum_{j \in \mathcal{S}} e^{V_k(j)}}$, it follows that $\sum_{j \in \mathcal{S}} e^{V'_k(j)} = 1$. We then have

$$e^{V_k(i)} = \left(T_{f_k}^{m_{k-1}} e^{V'_{k-1}} \right) (i).$$

Let

$$\underline{d} = \min_{\substack{f \in \Pi \\ i \in \mathcal{S}}} d_f(i)$$

and

$$\overline{d} = \max_{\substack{f \in \Pi \\ i \in \mathcal{S}}} d_f(i)$$

. Then,

$$\begin{aligned} \left(T_{f_k} e^{V'_{k-1}} \right) (i) &= e^{\alpha d_f(i)} \sum_{j \in \mathcal{S}} Q_{f_k}(j|i) e^{V'_{k-1}(j)} \\ &\geq e^{\alpha d} \left(Q_{f_k} e^{V'_{k-1}} \right) (i). \end{aligned}$$

Iterating,

$$\begin{aligned} \left(T_{f_k}^{m_{k-1}} e^{V'_{k-1}} \right) (i) &\geq e^{m_{k-1} \alpha d} \left(Q_{f_k}^{m_{k-1}} e^{V'_{k-1}} \right) (i) \\ &= \frac{e^{m_{k-1} \alpha d} \left(Q_{f_k}^{m_{k-1}} e^{V_{k-1}} \right) (i)}{\sum_{j \in \mathcal{S}} e^{V_{k-1}(j)}}, \\ e^{V_k(i)} &\geq \frac{e^{m_{k-1} \alpha d} (Q_{f_k}^{m_{k-1}} e^{V_{k-1}})(i)}{\sum_{j \in \mathcal{S}} e^{V_{k-1}(j)}}. \end{aligned}$$

We now prove that $V_R(i)$ is uniformly lower bounded from below. However the same uniform lower bound on $V_k(i)$, $\forall k > R$ can be obtained by considering the last $(k - R)$ iterations of the algorithm and utilizing the same analysis as in the proof below.

$$e^{V_R(i)} \geq \frac{e^{\sum_{l=1}^R m_{R-l} \alpha d} \left(Q_{f_R}^{m_{R-1}} Q_{f_{R-1}}^{m_{R-2}} \dots Q_{f_1}^{m_0} e^{V_0} \right) (i)}{\sum_{j \in \mathcal{S}} e^{V_{R-1}(j)} \sum_{j \in \mathcal{S}} e^{V_{R-2}(j)} \dots \sum_{j \in \mathcal{S}} e^{V_1(j)}}.$$

Consider the following definition:

$$H(j|i) := Q_{f_R}^{m_{R-1}} Q_{f_{R-1}}^{m_{R-2}} \dots Q_{f_1}^{m_0}(j|i) \quad (\text{E.7})$$

$$H(j|i) \geq \min_{i', j' \in \mathcal{S}} \min_{m_0, m_1, \dots, m_{R-1} \in \{1, 2, \dots, M\}} \min_{f_1, f_2, \dots, f_R \in \Pi} Q_{f_R}^{m_{R-1}} Q_{f_{R-1}}^{m_{R-2}} \dots Q_{f_1}^{m_0}(j'|i') \quad (\text{E.8})$$

Since Π is a finite set, the above minimum exists. Furthermore, since $R \leq \sum_{k=0}^{R-1} m_k \leq MR$,

from Lemma 6.3 it follows that,

$$H(j|i) \geq \min_{k \in \{R, \dots, MR\}} \epsilon_k := \epsilon \quad (\text{E.9})$$

$$e^{V_R(i)} \geq \frac{e^{\sum_{l=1}^R m_{R-l} \alpha \bar{d}} \sum_{j \in \mathcal{S}} \epsilon e^{V'_0(j)}}{\sum_{j \in \mathcal{S}} e^{V_{R-1}(j)} \sum_{j \in \mathcal{S}} e^{V_{R-2}(j)} \dots \sum_{j \in \mathcal{S}} e^{V_1(j)}} \quad (\text{E.10})$$

$$= \frac{e^{\sum_{l=1}^R m_{R-l} \alpha \bar{d}} \epsilon}{\sum_{j \in \mathcal{S}} e^{V_{R-1}(j)} \sum_{j \in \mathcal{S}} e^{V_{R-2}(j)} \dots \sum_{j \in \mathcal{S}} e^{V_1(j)}}. \quad (\text{E.11})$$

Similarly we obtain,

$$e^{V_R(i)} \leq \frac{e^{m_{R-1} \alpha \bar{d}} \left(Q_{f_R}^{m_{R-1}} e^{V_{R-1}} \right) (i)}{\sum_{j \in \mathcal{S}} e^{V_{R-1}(j)}}.$$

Further iterating,

$$\begin{aligned} e^{V_R(i)} &\leq \frac{e^{\sum_{l=1}^R m_{R-l} \alpha \bar{d}} \left(Q_{f_R}^{m_{R-1}} Q_{f_{R-1}}^{m_{R-2}} \dots Q_{f_1}^{m_0} e^{V'_0} \right) (i)}{\sum_{j \in \mathcal{S}} e^{V_{R-1}(j)} \sum_{j \in \mathcal{S}} e^{V_{R-2}(j)} \dots \sum_{j \in \mathcal{S}} e^{V_1(j)}} \\ &= \frac{e^{\sum_{l=1}^R m_{R-l} \alpha \bar{d}} \left(\sum_{j \in \mathcal{S}} \left(Q_{f_R}^{m_{R-1}} Q_{f_{R-1}}^{m_{R-2}} \dots Q_{f_1}^{m_0} \right) (j|i) e^{V'_0(j)} \right)}{\sum_{j \in \mathcal{S}} e^{V_{R-1}(j)} \sum_{j \in \mathcal{S}} e^{V_{R-2}(j)} \dots \sum_{j \in \mathcal{S}} e^{V_1(j)}} \\ &\leq \frac{e^{\sum_{l=1}^R m_{R-l} \alpha \bar{d}} \left(\sum_{j \in \mathcal{S}} e^{V'_0(j)} \right)}{\sum_{j \in \mathcal{S}} e^{V_{R-1}(j)} \sum_{j \in \mathcal{S}} e^{V_{R-2}(j)} \dots \sum_{j \in \mathcal{S}} e^{V_1(j)}} \\ &= \frac{e^{\sum_{l=1}^R m_{R-l} \alpha \bar{d}}}{\sum_{j \in \mathcal{S}} e^{V_{R-1}(j)} \sum_{j \in \mathcal{S}} e^{V_{R-2}(j)} \dots \sum_{j \in \mathcal{S}} e^{V_1(j)}}. \end{aligned}$$

This implies

$$\sum_{i \in \mathcal{S}} e^{V_R(i)} \leq \frac{e^{\sum_{l=1}^R m_{R-l} \alpha \bar{d}} |\mathcal{S}|}{\sum_{j \in \mathcal{S}} e^{V_{R-1}(j)} \sum_{j \in \mathcal{S}} e^{V_{R-2}(j)} \dots \sum_{j \in \mathcal{S}} e^{V_1(j)}}.$$

Therefore,

$$\frac{1}{\sum_{i \in \mathcal{S}} e^{V_R(i)}} \geq \frac{\sum_{j \in \mathcal{S}} e^{V_{R-1}(j)} \sum_{j \in \mathcal{S}} e^{V_{R-2}(j)} \dots \sum_{j \in \mathcal{S}} e^{V_1(j)}}{e^{\sum_{l=1}^R m_{R-l} \alpha \bar{d}} |\mathcal{S}|}. \quad (\text{E.12})$$

Combining (E.10) and (E.12), since $\forall l, m_l \geq 1$ and $m_l < M$,

$$\begin{aligned} \frac{e^{V_R(i)}}{\sum_{j \in S} e^{V_R(j)}} &\geq \frac{e^{\sum_{l=1}^R m_{R-l} \alpha \underline{d}} \epsilon}{e^{\sum_{l=1}^R m_{R-l} \alpha \bar{d}} |S|} \\ &> \frac{e^{R \alpha \underline{d}} \epsilon}{e^{RM \alpha \bar{d}} |S|} \\ &> 0. \end{aligned}$$

We conclude that $e^{V'_k(i)} > \beta > 0$ for all k , where $\beta = \frac{e^{R \alpha \underline{d}} \epsilon}{e^{RM \alpha \bar{d}} |S|}$. □

E.4 Proof of Section 6.5.4

Theorem E.2. Recall the definition of the optimal multiplicative Bellman operator in (6.3). Let $e^{V'_n}$ and f_n , for all $n \in \mathbb{N}$ be the outputs of Algorithm 7. Then there exists $\gamma \in (0, 1)$ such that for each n , the iterates of risk-sensitive MPI satisfy:

$$\left(\max_{i \in S} \frac{(\mathcal{T}e^{V'_n})(i)}{e^{V'_n(i)}} - e^{\tilde{\Lambda}^*} \right) \leq (1 - \gamma) \left(\max_{i \in S} \frac{(\mathcal{T}e^{V'_{n-R}})(i)}{e^{V'_{n-R}(i)}} - e^{\tilde{\Lambda}^*} \right), \quad (\text{E.13})$$

where $\gamma \geq \frac{\epsilon^2 e^{2R \alpha \underline{d}}}{e^{2MR \alpha \bar{d}} |S|}$, $e^{\tilde{\Lambda}^*}$ is defined in (6.6) and R is defined in Lemma 6.3. Consequently, the risk-sensitive average cost iterates converge to $e^{\tilde{\Lambda}^*}$, i.e.

$$\lim_{n \rightarrow \infty} e^{\tilde{\Lambda}_{f_n}} = e^{\tilde{\Lambda}^*}. \quad (\text{E.14})$$

Proof. By definition of g_n , we have

$$\begin{aligned}
g_n(i) &= \frac{\tau e^{V'_n(i)}}{e^{V'_n(i)}} \\
&= \frac{\left(\tilde{Q}_{f_{n+1}} e^{V'_n} \right)(i)}{e^{V'_n(i)}} \\
&\stackrel{(a)}{\leq} \frac{\left(\tilde{Q}_{f_n} e^{V'_n} \right)(i)}{e^{V'_n(i)}} \\
&\stackrel{(b)}{=} \frac{\left(\tilde{Q}_{f_n} e^{V_n} \right)(i)}{e^{V_n(i)}},
\end{aligned}$$

where $\tilde{Q}$ is defined in Equation (E.3), (a) follows from the fact that f_{n+1} is the minimizing policy, and (b) is due to $e^{V'_n(i)} = \frac{e^{V_n(i)}}{\sum_{j \in \mathcal{S}} e^{V_n(j)}}$. Using the definition of $e^{V_n(i)}$, we have

$$\begin{aligned}
g_n(i) &\leq \frac{\left(\left(\tilde{Q}_{f_n} \right) \left(\tilde{Q}_{f_n}^{m_{n-1}} \cdot e^{V'_{n-1}} \right) \right)(i)}{\left(\tilde{Q}_{f_n}^{m_{n-1}} e^{V'_{n-1}} \right)(i)} \\
&= \frac{\left(\tilde{Q}_{f_n}^{m_{n-1}} \tilde{Q}_{f_n} e^{V'_{n-1}} \right)(i)}{\left(\tilde{Q}_{f_n}^{m_{n-1}} e^{V'_{n-1}} \right)(i)} \\
&\leq \frac{\left(\tilde{Q}_{f_n}^{m_{n-1}} \cdot \tilde{Q}_{f_{n-1}} e^{V_{n-1}} \right)(i)}{\left(\tilde{Q}_{f_n}^{m_{n-1}} e^{V_{n-1}} \right)(i)} \\
&= \frac{\left(\tilde{Q}_{f_n}^{m_{n-1}} \tilde{Q}_{f_{n-1}}^{m_{n-2}} \tilde{Q}_{f_{n-1}} e^{V'_{n-2}} \right)(i)}{\left(\tilde{Q}_{f_n}^{m_{n-1}} \tilde{Q}_{f_{n-1}}^{m_{n-2}} e^{V'_{n-2}} \right)(i)}.
\end{aligned}$$

Continuing the above for R iterations, we get

$$g_n(i) \leq \frac{\left(\tilde{Q}_{f_n}^{m_{n-1}} \tilde{Q}_{f_{n-1}}^{m_{n-2}} \tilde{Q}_{f_{n-2}}^{m_{n-3}} \dots \tilde{Q}_{f_{n-R+1}}^{m_{n-R}} \tilde{Q}_{f_{n-R+1}} e^{V'_{n-R}} \right)(i)}{\left(\tilde{Q}_{f_n}^{m_{n-1}} \tilde{Q}_{f_{n-1}}^{m_{n-2}} \tilde{Q}_{f_{n-2}}^{m_{n-3}} \dots \tilde{Q}_{f_{n-R+1}}^{m_{n-R}} e^{V'_{n-R}} \right)(i)}.$$

Let $H_n := \tilde{Q}_{f_n}^{m_{n-1}} \tilde{Q}_{f_{n-1}}^{m_{n-2}} \tilde{Q}_{f_{n-2}}^{m_{n-3}} \dots \tilde{Q}_{f_{n-R+1}}^{m_{n-R}}$. Then for all $i, j \in \mathcal{S}$, it is true that,

$$\begin{aligned} H_n(j|i) &\geq e^{\sum_{l=1}^R m_{n-l} \alpha \bar{d}} Q_{f_n}^{m_{n-1}} Q_{f_{n-1}}^{m_{n-2}} Q_{f_{n-2}}^{m_{n-3}} \dots Q_{f_{n-R+1}}^{m_{n-R}}(j|i) \\ &\geq e^{R\alpha \bar{d}} Q_{f_n}^{m_{n-1}} Q_{f_{n-1}}^{m_{n-2}} Q_{f_{n-2}}^{m_{n-3}} \dots Q_{f_{n-R+1}}^{m_{n-R}}(j|i) \end{aligned} \quad (E.15)$$

Since $1 \leq m_{n-k} < M, \forall k \in \{1, 2, \dots, R\}$, from Equation (E.9) it follows that

$$H_n(j|i) \geq \epsilon e^{R\alpha \bar{d}} \quad \forall i, j \in \mathcal{S} \quad (E.16)$$

Since Equation (E.16) holds for all n , it is also true in the limit as $n \rightarrow \infty$. Let $e^{W'_{n-R}} := \tilde{Q}_{f_{n-R+1}}^{V'_{n-R}}$.

$$\begin{aligned} g_n(i) &\leq \frac{(H_n e^{W'_{n-R}})(i)}{H_n e^{V'_{n-R}}(i)} \\ &= \frac{\sum_{j \in \mathcal{S}} H_n(j|i) e^{W'_{n-R}(j)}}{\sum_{\ell \in \mathcal{S}} H_n(\ell|i) e^{V'_{n-R}(\ell)}} \\ &= \frac{\sum_{j \in \mathcal{S}} (H_n(j|i) e^{V'_{n-R}(j)}) \cdot \left(\frac{e^{W'_{n-R}(j)}}{e^{V'_{n-R}(j)}} \right)}{\sum_{\ell \in \mathcal{S}} H_n(\ell|i) e^{V'_{n-R}(\ell)}}. \end{aligned}$$

Define a transition probability matrix q_n as follows:

$$q_n(j|i) := \frac{H_n(j|i) e^{V'_{n-R}(j)}}{\sum_{\ell \in \mathcal{S}} H_n(\ell|i) e^{V'_{n-R}(\ell)}}.$$

Recall the definition of H_n . Then,

$$\begin{aligned} H_n(j|i) &\leq e^{\sum_{l=1}^R m_{n-l} \alpha \bar{d}} Q_{f_n}^{m_{n-1}} Q_{f_{n-1}}^{m_{n-2}} Q_{f_{n-2}}^{m_{n-3}} \dots Q_{f_{n-R+1}}^{m_{n-R}}(j|i) \\ &\leq e^{MR\alpha \bar{d}} \\ \implies \frac{1}{H_n(j|i)} &\geq \frac{1}{e^{MR\alpha \bar{d}}} \quad \forall i, j \in \mathcal{S} \quad \forall n \in \mathbb{N} \end{aligned} \quad (E.17)$$

Hence combining Equation (E.16) and Equation (E.17), we thus obtain,

$$q_n(j|i) \geq \frac{\epsilon e^{R\alpha \underline{d}} e^{V'_{n-R}(j)}}{\sum_{l \in \mathcal{S}} e^{MR\alpha \bar{d}} e^{V'_{n-R}(l)}}$$

From the normalization in Line 4 in Algorithm 7 we know that $\sum_{l \in \mathcal{S}} e^{V'_{n-R}(l)} = 1$ for all $n \in \mathbb{N}$. Hence,

$$q_n(j|i) \geq \frac{\epsilon e^{R\alpha \underline{d}}}{e^{MR\alpha \bar{d}}} e^{V'_{n-R}(j)} \quad (E.18)$$

From Lemma 6.4 we know that $e^{V'_{n-R}(j)} > \frac{\epsilon e^{R\alpha \underline{d}}}{e^{MR\alpha \bar{d}|\mathcal{S}|}} \quad \forall n > R, \forall j \in \mathcal{S}$. We thus obtain,

$$q_n(j|i) \geq \frac{\epsilon^2 e^{2R\alpha \underline{d}}}{e^{2MR\alpha \bar{d}|\mathcal{S}|}} \quad (E.19)$$

Since the above equation is true for all n , it is also true as $n \rightarrow \infty$. We thus obtain,

$$\begin{aligned} g_n(i) &\leq \sum_{j \in \mathcal{S}} q_n(j|i) \left(\frac{\left(\tilde{Q}_{f_{n-R+1}} e^{V'_{n-R}} \right)(j)}{e^{V'_{n-R}(j)}} \right) \\ &= \sum_{j \in \mathcal{S}} q_n(j|i) \left(\frac{T e^{V'_{n-R}(j)}}{e^{V'_{n-R}(j)}} \right). \end{aligned}$$

Let $\gamma := \min_{i,j} \inf_n q_n(j|i)$. From, Equation (E.19), we know that $\gamma \geq \frac{\epsilon^2 e^{2R\alpha \underline{d}}}{e^{2MR\alpha \bar{d}|\mathcal{S}|}}$. Hence it follows that,

$$g_n(i) \leq \gamma \ell_{n-R} + (1 - \gamma) u_{n-R} \quad \forall i \quad (E.20)$$

$$\implies u_n \leq \gamma \ell_{n-R} + (1 - \gamma) u_{n-R}. \quad (E.21)$$

Since $\ell_{n-R} \leq e^{\tilde{\Lambda}^*}$, we have

$$u_n \leq \gamma e^{\tilde{\Lambda}^*} + (1 - \gamma) u_{n-R} \quad (E.22)$$

$$\left(u_n - e^{\tilde{\lambda}^*}\right) \leq (1 - \gamma) \left(u_{n-k} - e^{\tilde{\lambda}^*}\right). \quad (\text{E.23})$$

Since $u_n \leq u_{n-1}$ from Lemma 6.2 and $u_n \geq e^{\tilde{\lambda}^*}$ from Lemma 6.1, it follows from (E.23) that $u_n \rightarrow e^{\tilde{\lambda}^*}$.

From (E.21), we obtain

$$\begin{aligned} e^{\tilde{\lambda}^*} &\leq \gamma \ell_n + (1 - \gamma) u_n, \quad e^{\tilde{\lambda}^*} - u_n \\ &\leq \gamma (\ell_n - u_n), \end{aligned}$$

and so we have that $0 \leq \gamma (u_n - \ell_n) \leq (u_n - e^{\tilde{\lambda}^*})$. Since $u_n \rightarrow e^{\tilde{\lambda}^*}$, we conclude that $\ell_n \rightarrow u_n \implies \ell_n \rightarrow e^{\tilde{\lambda}^*}$ as desired. $\square$

E.5 Simulations

In this section, we present simulation results to empirically investigate the convergence behavior of risk-sensitive modified policy iteration. Through these simulations, we aim to gain a deeper understanding of how modified policy iteration converges in various scenarios and settings. We primarily test for the following metrics:

1. The time taken for modified policy iteration to converge across an array of risk-sensitivity factors α . We plot these times together with the time taken by value iteration and policy iteration algorithms to show the better performance of modified policy iteration across all values of α . We also present these plots for varying values of m_k .
2. We study the empirical dependence of total time taken for convergence of the modified policy iteration algorithm as a function of m_k . We study this behaviour for various values of α as well.
3. We study the improvement in convergence time as a function of state space and action space cardinalities as a function of α , for a fixed value of m .

Setup

We consider a Markov decision process with $|\mathcal{S}|$ states and $|\mathcal{A}|$ actions. Transition kernels of dimension $|\mathcal{S}| \times |\mathcal{S}|$ are generated; each row in the transition kernel can take one of $|\mathcal{A}|$ possible values corresponding to the actions. A cost matrix of dimension $|\mathcal{S}| \times |\mathcal{A}|$ is also randomly generated with values in $[0, 1]$.

Convergence Metric: Three algorithms, namely value iteration, policy iteration and modified policy iteration are simulated. The convergence criteria for all three algorithms are identical. Specifically, the convergence is achieved when the absolute difference between two successive value function iterates falls below $1e-7$.

Results

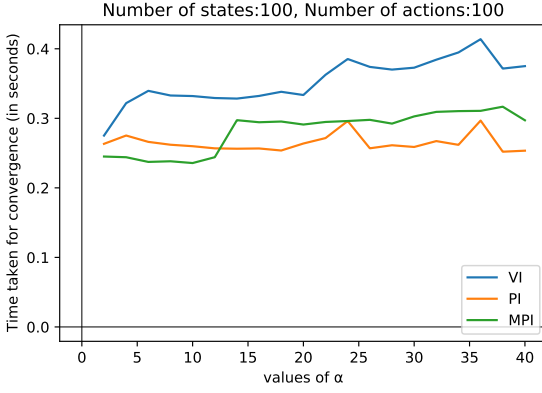
In Figure E.1 we present the time taken for each of the dynamic programming algorithms to converge as a function of the risk-sensitivity factor α . Each plot corresponds to a different value of the input parameter m , in Algorithm 1. Note that $m_k = m$ for all $k \in \mathbb{N}$.

In Figure E.2 represents the time taken for modified policy iteration to converge, for different values of risk-sensitivity parameter α , as a function of the algorithm input parameter m .

In Figure E.3 we present the performance of various dynamic programming algorithms as a function of different risk-sensitivity factors α as the cardinality of the state and action spaces increase.

Discussion

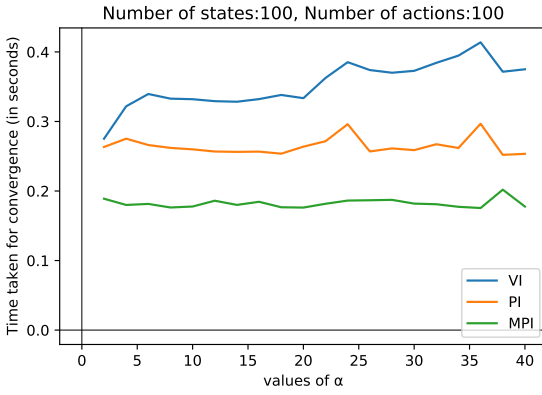
Figure E.1 shows that for very small values of m , policy iteration outperforms modified policy iteration but once you increase m , modified policy iteration quickly starts outperforming policy iteration. Value iteration seems to be consistently slower than the other two algorithms. From Figure E.2, one can see that the running time of modified policy iteration decreases with increasing m but becomes nearly a constant once m reaches a relatively small threshold value. We believe this is due to increased accuracy in policy evaluation being offset by the increased amount of time it takes to evaluate each policy. Figure E.3



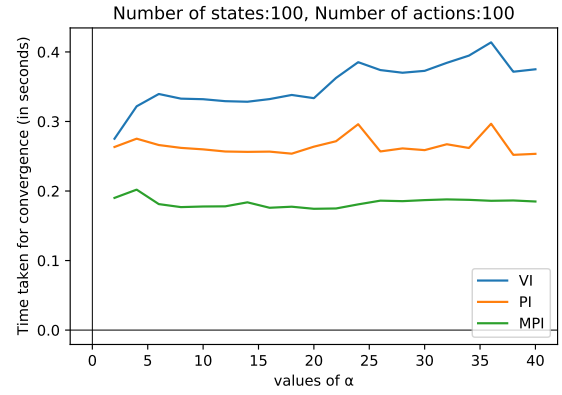
(a) $m = 2$



(b) $m = 5$



(c) $m = 10$



(d) $m = 15$

Figure E.1: Convergence performance of value iteration and policy iteration in comparison with modified policy iteration across various risk-sensitivity factors.

shows that the difference between modified policy iteration and the other two algorithms becomes more pronounced as the sizes of the state and action spaces increase.

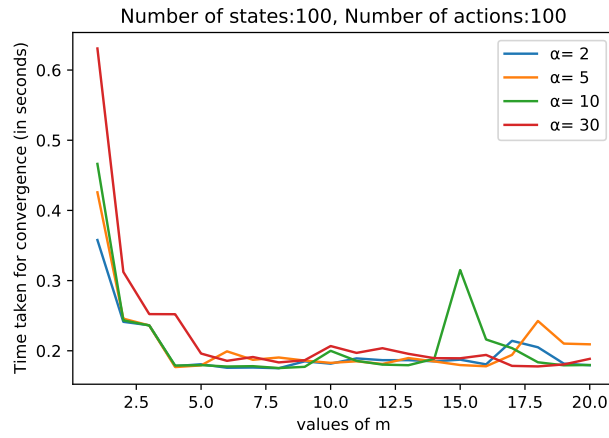
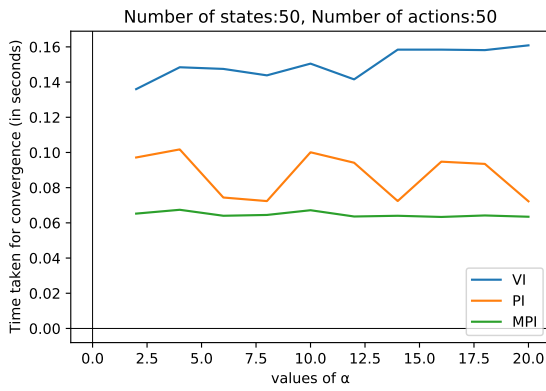
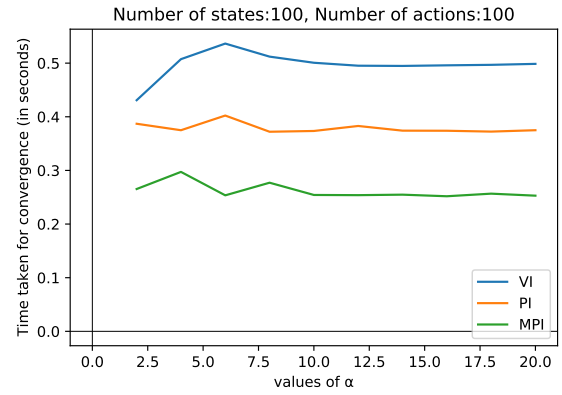


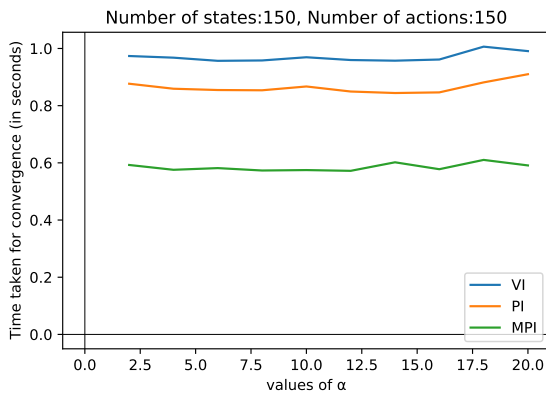
Figure E.2: Comparison of time for convergence as a function of m



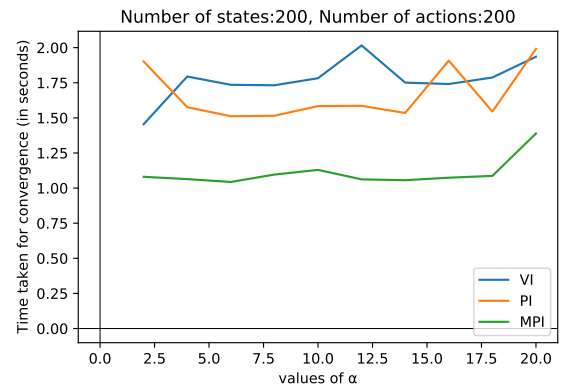
(a) $|\mathcal{S}| = |\mathcal{A}| = 50$



(b) $|\mathcal{S}| = |\mathcal{A}| = 100$



(c) $|\mathcal{S}| = |\mathcal{A}| = 150$



(d) $|\mathcal{S}| = |\mathcal{A}| = 200$

Figure E.3: Convergence performance of value iteration and policy iteration in comparison with modified policy iteration across different state and action space cardinalities.

F

PROOFS: APPROXIMATE POLICY ITERATION FOR RISK-SENSITIVE MDPs

In this appendix, we prove the convergence of risk sensitive approximate modified policy iteration for exponential cost MDPs subject to certain constraints on the approximation accuracy.

Define the Bellman Operator $T_f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ as follows:

$$\left(T_f e^h\right)(s, a) = e^{\alpha d_f(s)} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} Q(s'|s, a) f(a'|s') e^{h(s', a')}$$

F.1 Convergence Analysis

Recall the following definitions:

$$\begin{aligned} g_k(s, a) &= \frac{(T e^{Q_k})(s, a)}{e^{Q_k(s, a)}} \\ l_k &= \min_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} g_k(s, a) \\ u_k &= \max_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} g_k(s, a) \end{aligned}$$

In order to obtain the finite time and asymptotic performance of the iterates obtained as a consequence of the above algorithm, it is necessary to prove the lemma characterizing a single step of approximate policy improvement below.

Lemma F.1. *Given the iterates l_k, u_k obtained from Algorithm 8, $\forall k$ it is true that,*

$$l_k \leq e^{\tilde{\lambda}^*} \leq e^{\tilde{\lambda}_{f_{k+1}}} \leq u_k e^{\alpha \epsilon} \quad (\text{F.1})$$

Proof. Recall the definition of l_k ,

$$\begin{aligned} l_k &= \min_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} \frac{(\mathcal{T}e^{Q_k})(s, a)}{e^{Q_k(s, a)}} \\ &\leq \frac{(\mathcal{T}e^{Q_k})(s, a)}{e^{Q_k(s, a)}} \\ &\leq \frac{e^{\alpha d(s, a)} \sum_{s' \in \mathcal{S}} \mathbb{Q}(s'|s, a) \min_{a' \in \mathcal{A}} e^{Q_k(s', a')}}{e^{Q_k(s, a)}} \end{aligned}$$

Let f^* be the policy which yields the lowest risk sensitive average cost $e^{\tilde{\lambda}^*}$. Recall that $e^{\tilde{\lambda}^*}$ satisfies the following equation:

$$e^{\tilde{\lambda}^*} e^{Q_{f^*}} = e^{\alpha d} \left(\min_{f \in \Pi} \mathbb{Q}_f e^{Q_{f^*}} \right).$$

Then,

$$l_k \leq \frac{e^{\alpha d(s, a)} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} \mathbb{Q}(s'|s, a) f^*(a'|s') e^{Q_k(s', a')}}{e^{Q_k(s, a)}}$$

Therefore,

$$\begin{aligned} l_k e^{Q_k(s, a)} &\leq e^{\alpha d(s, a)} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} \mathbb{Q}(s'|s, a) f^*(a'|s') e^{Q_k(s', a')} \\ &= \left(\tilde{\mathbb{Q}}_{f^*} e^{Q_k} \right)(s, a) \end{aligned}$$

where $\tilde{Q}(s', f^*(s')|s, a) = e^{\alpha d(s, a)} Q(s', f^*(s')|s, a)$. It follows that,

$$l_k \left(\tilde{Q}_{f^*}^n e^{Q_k} \right) \leq \tilde{Q}_{f^*}^{n+1} e^{Q_k}.$$

Similarly, we have

$$\begin{aligned} \frac{l_k}{e^{\tilde{\lambda}^*}} \left(\frac{\tilde{Q}_{f^*}}{e^{\tilde{\lambda}^*}} \right)^n e^{Q_k} &\leq \left(\frac{\tilde{Q}_{f^*}}{e^{\tilde{\lambda}^*}} \right)^{n+1} e^{Q_k} \\ \lim_{n \rightarrow \infty} \frac{l_k}{e^{\tilde{\lambda}^*}} \left(\frac{\tilde{Q}_{f^*}}{e^{\tilde{\lambda}^*}} \right)^n e^{Q_k} &\leq \lim_{n \rightarrow \infty} \left(\frac{\tilde{Q}_{f^*}}{e^{\tilde{\lambda}^*}} \right)^{n+1} e^{Q_k} \end{aligned}$$

Since Q_f satisfies the conditions of the Perron-Frobenius theorem, by definition, so does $\tilde{Q}_f$ for all $f \in \Pi$. The vector e^{Q_k} consists of all positive elements and hence, in the limit of $n \rightarrow \infty$, due to power iteration we obtain the following:

$$\lim_{n \rightarrow \infty} \left(\frac{\tilde{Q}_{f^*}}{e^{\tilde{\lambda}^*}} \right)^n e^{Q_k} = z$$

where z is a vector with all non negative entries.

Hence we obtain the following,

$$\frac{l_k}{e^{\tilde{\lambda}^*}} z \leq z$$

which yields the following:

$$l_k \leq e^{\tilde{\lambda}^*}$$

as desired.

Recall that

$$\begin{aligned} e^{\tilde{\lambda}^*} e^{Q_{f^*}(s, a)} &= e^{\alpha d(s, a)} \sum_{s' \in \mathcal{S}} Q(s' | s, a) \min_{a' \in \mathcal{A}} e^{Q_{f^*}(s', a')} \\ &\leq e^{\alpha d(s, a)} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} Q(s' | s, a) f_{k+1}(a' | s') e^{Q_{f^*}(s', a')}. \end{aligned}$$

Therefore, we have

$$\begin{aligned}
e^{Q_{f^*}(s,a)} &\leq \frac{e^{\alpha d(s,a)} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} \mathbb{Q}(s' | s, a) f_{k+1}(a' | s') e^{Q_{f^*}(s', a')}}{e^{\tilde{\Lambda}^*}} \\
&= \frac{e^{\alpha d(s,a)} \mathbb{E} \left[e^{Q_{f^*}(s', a')} \mid s_0 = s, a_0 = a, f_{k+1} \right]}{e^{\tilde{\Lambda}^*}} \\
&\leq \frac{e^{\alpha d(s,a)}}{e^{\tilde{\Lambda}^*}} \cdot \mathbb{E} \left[\frac{e^{\alpha d(s', a')}}{e^{\tilde{\Lambda}^*}} \cdot \mathbb{E} \left[e^{Q_{f^*}(s'', a'')} \mid s', a', f_{k+1} \right] \mid s_0 = s, a_0 = a, f_{k+1} \right]
\end{aligned}$$

Iterating, we get

$$e^{Q_{f^*}(s,a)} \leq \mathbb{E}_{f_{k+1}} \left[\frac{e^{\sum_{i=0}^{m-1} \alpha d(s_i, a_i)} Q_{f^*}(s_m, a_m)}{(e^{\tilde{\Lambda}^*})^m} \mid s, a \right]$$

Since $Q_{f^*}(s_m, a_m) \leq M < \infty$ for all $s_k \in \mathcal{S}, a_k \in \mathcal{A}$, we have

$$m \cdot \tilde{\Lambda}^* + Q_{f^*}(s, a) \leq \ln \left(\mathbb{E}_{s,a,f_{k+1}} \left[e^{\alpha \sum_{i=0}^{m-1} d(s_i, a_i)} \right] \right) + \ln(M),$$

or equivalently

$$\tilde{\Lambda}^* + \frac{Q_{f^*}(s, a)}{m} \leq \frac{1}{m} \ln \left(\mathbb{E}_{s,a,f_{k+1}} \left[e^{\alpha \sum_{i=0}^{m-1} d(s_i, a_i)} \right] \right) + \frac{M}{m}.$$

As $m \rightarrow \infty$, we obtain

$$\tilde{\Lambda}^* \leq \tilde{\Lambda}_{f_{k+1}}.$$

By monotonicity of the exponential function, we conclude as desired that

$$e^{\tilde{\Lambda}^*} \leq e^{\tilde{\Lambda}_{f_{k+1}}}.$$

We now present the last part of the proof.

Recall the definition of u_k ,

$$\begin{aligned} u_k &= \max_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} \frac{(Te^{Q_k})(s, a)}{e^{Q_k(s, a)}} \\ &\geq \frac{(Te^{Q_k})(s, a)}{e^{Q_k(s, a)}} \end{aligned}$$

From the policy improvement step of the algorithm, we obtain,

$$u_k \geq \frac{(T_{f_{k+1}} e^{Q_k})(s, a)}{e^{\alpha \epsilon} e^{Q_k(s, a)}}$$

$$e^{\alpha \epsilon} u_k e^{Q_k(s, a)} \geq e^{\alpha d(s, a)} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} Q(s'|s, a) f_{k+1}(a'|s') e^{Q_k(s', a')}$$

Since $\tilde{Q}(s', f(s')|s, a) = e^{\alpha d(s, a)} Q(s', f(s')|s, a)$, it follows that,

$$u_k e^{\alpha \epsilon} \left(\tilde{Q}_{f_{k+1}}^n e^{Q_k} \right) \geq \tilde{Q}_{f_{k+1}}^{n+1} e^{Q_k}.$$

Similarly, we have

$$\begin{aligned} \frac{e^{\alpha \epsilon} u_k}{e^{\bar{\lambda}^{f_{k+1}}}} \left(\frac{\tilde{Q}_{f_{k+1}}}{e^{\bar{\lambda}^{f_{k+1}}}} \right)^n e^{Q_k} &\geq \left(\frac{\tilde{Q}_{f_{k+1}}}{e^{\bar{\lambda}^{f_{k+1}}}} \right)^{n+1} e^{Q_k} \\ \lim_{n \rightarrow \infty} \frac{e^{\alpha \epsilon} u_k}{e^{\bar{\lambda}^{f_{k+1}}}} \left(\frac{\tilde{Q}_{f_{k+1}}}{e^{\bar{\lambda}^{f_{k+1}}}} \right)^n e^{Q_k} &\geq \lim_{n \rightarrow \infty} \left(\frac{\tilde{Q}_{f_{k+1}}}{e^{\bar{\lambda}^{f_{k+1}}}} \right)^{n+1} e^{Q_k} \end{aligned}$$

Since Q_f satisfies the conditions of the Perron-Frobenius theorem, by definition, so does $\tilde{Q}_f$ for all $f \in \Pi$. The vector e^{Q_k} consists of all positive elements and hence, in the limit of $n \rightarrow \infty$, due to power iteration we obtain the following:

$$\lim_{n \rightarrow \infty} \left(\frac{\tilde{Q}_{f^*}}{e^{\bar{\lambda}^{f_{k+1}}}} \right)^n e^{Q_k} = b$$

where b is a vector with all non negative entries.

Hence we obtain the following,

$$\frac{e^{\alpha\epsilon}u_k}{e^{\tilde{\lambda}^{f_{k+1}}}}b \geq b$$

which yields the following:

$$u_k e^{\alpha\epsilon} \geq e^{\tilde{\lambda}^{f_{k+1}}}$$

as desired. Hence, we obtain,

$$l_k \leq e^{\tilde{\lambda}^*} \leq e^{\tilde{\lambda}^{f_{k+1}}} \leq u_k e^{\alpha\epsilon}$$

□

Given the above lemma, we now present the main theorem of this section that is the performance bound of policy obtained through repeated execution of Algorithm 8.

Theorem F.1. Let $\gamma \in (0, 1)$. Let $n \in \mathbb{N}$ such that $m_k + m_{k-1} + \dots + m_{k-n} \geq R$. Suppose the approximation errors $e^{\alpha\delta_1}$, $e^{\alpha\delta_2}$ and $e^{\alpha\epsilon}$ are such that

$$\left(\frac{e^{\alpha\delta_2}e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (1 - \gamma) < 1.$$

Then, the iterates Q_k generated by Algorithm 8 satisfy:

$$\left(e^{\tilde{\lambda}^{f_{k+1}}} - e^{\tilde{\lambda}^*} \right) \leq (\gamma')^{\frac{k}{n}} \left(u_0 e^{\alpha\epsilon} - e^{\tilde{\lambda}^*} \right) + \frac{\sigma e^{\tilde{\lambda}^*}}{1 - \gamma'},$$

where $\sigma = \left(\left(\frac{e^{\alpha\delta_2}e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (1 + (e^{\alpha\epsilon} - 1)\gamma) - 1 \right)$ and $\gamma' = \left(\frac{e^{\alpha\delta_2}e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (1 - \gamma)$.

Proof. Recall the definition of $g_k(i)$:

$$\begin{aligned} g_k(s, a) &\leq \frac{(Te^{Q_k})(s, a)}{e^{Q_k(s, a)}} \\ &\leq \frac{(T_{f_k}e^{Q_k})(s, a)}{e^{Q_k(s, a)}} \end{aligned}$$

Since T is a minimizing operator. From Algorithm 8, we know that

$$e^{\alpha\delta_1} \leq \left\| \frac{e^{Q'_{f_k}}}{e^{Q_k}} \right\|_{\infty} \leq e^{\alpha\delta_2}$$

Since $e^{Q_k(s,a)} \leq \frac{e^{Q'_{f_k}(s,a)}}{e^{\alpha\delta_1}}$, $\forall i \in \mathcal{S}$, we obtain the following:

$$g_k(s, a) \leq \frac{(T_{f_k} e^{Q'_{f_k}})(s, a)}{e^{\alpha\delta_1} e^{Q_k(s,a)}}$$

Since $\frac{1}{e^{Q_k(s,a)}} \leq \frac{e^{\alpha\delta_2}}{e^{Q'_{f_k}(s,a)}}$, we obtain the following:

$$g_k(s, a) \leq \frac{e^{\alpha\delta_2} T_{f_k} e^{Q'_{f_k}}(s, a)}{e^{\alpha\delta_1} e^{Q'_{f_k}(s,a)}}$$

An important property of $T_f \forall f \in \Pi$ is its scaling with scaled inputs as shown below. Let $\beta \in \mathbb{R}$ be a constant, then for all $s \in \mathcal{S}, a \in \mathcal{A}$

$$\begin{aligned} (T_f (\beta e^Q))(s, a) &= e^{\alpha d(s,a)} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} Q(s'|s, a) f(a'|s') \beta e^{Q(s', a')} \\ &= \beta e^{\alpha d(s,a)} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} Q(s'|s, a) f(a'|s') e^{Q(s', a')} \\ &= \beta (T_f e^Q)(s, a) \end{aligned}$$

Recall that $e^{Q'_{f_k}(s,a)} = \frac{e^{Q_{f_k}(s,a)}}{\sum_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} e^{Q_{f_k}(s,a)}}$. Due to the multiplicative scaling property of T_f , substituting for $e^{Q'_{f_k}(s,a)}$ yields the following,

$$g_k(s, a) \leq \frac{e^{\alpha\delta_2} T_{f_k} e^{Q_{f_k}}(s, a)}{e^{\alpha\delta_1} e^{Q_{f_k}(s,a)}}$$

From Algorithm 8, we know that $e^{Q_{f_k}(s,a)} = T_{f_k}^{m_{k-1}} e^{Q_{k-1}}(s, a)$,

$$\begin{aligned} g_k(s, a) &\leq \frac{e^{\alpha\delta_2} T_{f_k}^{m_{k-1}+1} e^{Q_{k-1}}(s, a)}{e^{\alpha\delta_1} T_{f_k}^{m_{k-1}} e^{Q_{k-1}}(s, a)} \\ &= \frac{e^{\alpha\delta_2} T_{f_k}^{m_{k-1}} (T_{f_k} e^{Q_{k-1}})(s, a)}{e^{\alpha\delta_1} T_{f_k}^{m_{k-1}} e^{Q_{k-1}}(s, a)} \end{aligned}$$

Since $T_{f_k} e^{Q_{k-1}} \leq e^{\alpha\epsilon} T e^{Q_{k-1}}$,

$$\begin{aligned} g_k(s, a) &\leq \frac{e^{\alpha\delta_2} e^{\alpha\epsilon} T_{f_k}^{m_{k-1}} (T e^{Q_{k-1}})(s, a)}{e^{\alpha\delta_1} T_{f_k}^{m_{k-1}} e^{Q_{k-1}}(s, a)} \\ &\leq \frac{e^{\alpha\delta_2} e^{\alpha\epsilon} T_{f_k}^{m_{k-1}} (T_{f_{k-1}} e^{Q_{k-1}})(s, a)}{e^{\alpha\delta_1} T_{f_k}^{m_{k-1}} e^{Q_{k-1}}(s, a)} \\ &\leq \frac{e^{\alpha\delta_2} e^{\alpha\epsilon} T_{f_k}^{m_{k-1}} (T_{f_{k-1}} e^{Q_{f_{k-1}}})(s, a)}{\delta_1^2 T_{f_k}^{m_{k-1}} e^{Q_{k-1}}(s, a)} \\ &= \frac{\delta_2^2 e^{\alpha\epsilon} T_{f_k}^{m_{k-1}} (T_{f_{k-1}} e^{Q'_{f_{k-1}}})(s, a)}{\delta_1^2 T_{f_k}^{m_{k-1}} e^{Q'_{f_{k-1}}}(s, a)} \\ &\leq \frac{\delta_2^2 e^{\alpha\epsilon} T_{f_k}^{m_{k-1}} (T_{f_{k-1}} e^{Q_{f_{k-1}}})(s, a)}{\delta_1^2 T_{f_k}^{m_{k-1}} e^{Q_{f_{k-1}}}(s, a)} \\ &= \frac{\delta_2^2 e^{\alpha\epsilon} T_{f_k}^{m_{k-1}} (T_{f_{k-1}}^{m_{k-2}+1} e^{Q_{k-2}})(s, a)}{\delta_1^2 T_{f_k}^{m_{k-1}} T_{f_{k-1}}^{m_{k-2}} e^{Q_{k-2}}(s, a)} \\ &= \frac{\delta_2^2 e^{\alpha\epsilon} T_{f_k}^{m_{k-1}} T_{f_{k-1}}^{m_{k-2}} (T_{f_{k-1}} e^{Q_{k-2}})(s, a)}{\delta_1^2 T_{f_k}^{m_{k-1}} T_{f_{k-1}}^{m_{k-2}} e^{Q_{k-2}}(s, a)} \\ &= \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^2 \frac{T_{f_k}^{m_{k-1}} T_{f_{k-1}}^{m_{k-2}} (T e^{Q_{k-2}})(s, a)}{T_{f_k}^{m_{k-1}} T_{f_{k-1}}^{m_{k-2}} e^{Q_{k-2}}(s, a)} \end{aligned}$$

Iterating similarly, we obtain,

$$g_k(s, a) \leq \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n \frac{T_{f_k}^{m_{k-1}} T_{f_{k-1}}^{m_{k-2}} \dots T_{f_{k-n+1}}^{m_{k-n}} (T e^{Q_{k-n}})(s, a)}{T_{f_k}^{m_{k-1}} T_{f_{k-1}}^{m_{k-2}} \dots T_{f_{k-n+1}}^{m_{k-n}} e^{Q_{k-n}}(s, a)}$$

Define the following:

$$\begin{aligned} H_{n,k} &= T_{f_k}^{m_{k-1}} T_{f_{k-1}}^{m_{k-2}} \dots T_{f_{k-n+1}}^{m_{k-n}} \\ e^{w_{n,k}(s,a)} &= T e^{Q_{k-n}}(s, a) \\ e^{v_{n,k}(s,a)} &= e^{Q_{k-n}}(s, a) \end{aligned}$$

$g_k(s, a)$ can hence be expressed as:

$$\begin{aligned} g_k(s, a) &\leq \left(\frac{e^{\alpha \delta_2} e^{\alpha \epsilon}}{e^{\alpha \delta_1}} \right)^n \frac{H_{n,k} e^{w_{n,k}}(s, a)}{H_{n,k} e^{v_{n,k}}(s, a)} \\ &= \left(\frac{e^{\alpha \delta_2} e^{\alpha \epsilon}}{e^{\alpha \delta_1}} \right)^n \frac{\sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} H_{n,k}(s', a'|s, a) e^{w_{n,k}}(s', a')}{\sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} H_{n,k}(s', a'|s, a) e^{v_{n,k}}(s', a')} \end{aligned}$$

where $H_{n,k}(s', a'|s, a)$ is the element corresponding to the (s, a) th row and (s', a') th column.

$$g_k(s, a) \leq \left(\frac{e^{\alpha \delta_2} e^{\alpha \epsilon}}{e^{\alpha \delta_1}} \right)^n \sum_{j \in \mathcal{S}} \left(\frac{H_{n,k}(s', a'|s, a) e^{v_{n,k}}(s', a')}{\sum_{\substack{s'' \in \mathcal{S} \\ a'' \in \mathcal{A}}} H_{n,k}(s'', a''|s, a) e^{v_{n,k}}(s'', a'')} \frac{e^{w_{n,k}}(s', a')}{e^{v_{n,k}}(s', a')} \right)$$

Define a probability measure $q_{n,k}$ as follows:

$$q_{n,k}(s', a'|s, a) = \frac{H_{n,k}(s', a'|s, a) e^{v_{n,k}}(s', a')}{\sum_{\substack{s'' \in \mathcal{S} \\ a'' \in \mathcal{A}}} H_{n,k}(s'', a''|s, a) e^{v_{n,k}}(s'', a'')}$$

Let $H_{n,k} := \tilde{Q}_{f_k}^{m_{k-1}} \tilde{Q}_{f_{k-1}}^{m_{k-2}} \tilde{Q}_{f_{k-2}}^{m_{k-3}} \dots \tilde{Q}_{f_{k-n+1}}^{m_{k-n}}$. From an argument similar to Lemma 6.3, we assume that $\mathbb{Q}$ induces an irreducible Markov chain for any sequence of policies, i.e.:

$$\exists R < \infty \text{ such that } \forall \pi_1, \dots, \pi_R \in \Pi: (\mathbb{Q}_{\pi_1} \mathbb{Q}_{\pi_2} \dots \mathbb{Q}_{\pi_R})(s', a'|s, a) > 0 \quad \forall s, s', a, a'.$$

The number of time steps n is determined such that $m_{k-1} + m_{k-2} + \dots + m_{k-n} \geq R$. This implies that $H_{n,k}(s', a'|s, a) > 0$ for all $s', a'|s, a$.

Note from Lemma 7.2, we obtain that $e^{v_{n,k}(s,a)} \geq c_1 \geq 0, \forall n, k$, where c_1 is a positive constant.

Hence the minimum value of the transition measure $q_{n,k}$ is nonzero and defined as below:

$$\gamma := \min_{\substack{s', s \in \mathcal{S} \\ a, a' \in \mathcal{A}}} q(s', a' | s, a)$$

Thus $g_k(s, a)$ can be bounded as below:

$$\begin{aligned} g_k(s, a) &\leq \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n \left(\gamma \min_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} \frac{e^{w_{n,k}(s', a')}}{e^{v_{n,k}(s', a')}} + (1 - \gamma) \max_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} \frac{e^{w_{n,k}(s', a')}}{e^{v_{n,k}(s', a')}} \right) \\ &= \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n \left(\gamma \min_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} \frac{T e^{Q_{k-n}(s', a')}}{e^{Q_{k-n}(s', a')}} + (1 - \gamma) \max_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} \frac{T e^{Q_{k-n}(s', a')}}{e^{Q_{k-n}(s', a')}} \right) \\ &\leq \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (\gamma l_{k-n} + (1 - \gamma) u_{k-n}) \end{aligned}$$

From Lemma 7.1,

$$g_k(s, a) \leq \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (\gamma e^{\tilde{\lambda}^*} + (1 - \gamma) u_{k-n})$$

Since the above relation is true for all $s \in \mathcal{S}, a \in \mathcal{A}$,

$$u_k \leq \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (\gamma e^{\tilde{\lambda}^*} + (1 - \gamma) u_{k-n})$$

$$u_k e^{\alpha\epsilon} \leq \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (\gamma e^{\alpha\epsilon} e^{\tilde{\lambda}^*} + (1 - \gamma) u_{k-n} e^{\alpha\epsilon})$$

$$\begin{aligned} (u_k e^{\alpha\epsilon} - e^{\tilde{\lambda}^*}) &\leq \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n \left(\gamma e^{\alpha\epsilon} e^{\tilde{\lambda}^*} + (1 - \gamma) (u_{k-n} e^{\alpha\epsilon} - e^{\tilde{\lambda}^*}) \right) + \\ &\quad \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (1 - \gamma) e^{\tilde{\lambda}^*} - e^{\tilde{\lambda}^*} \end{aligned}$$

$$\begin{aligned} (u_k e^{\alpha\epsilon} - e^{\tilde{\lambda}^*}) &\leq \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (1 - \gamma) (u_{k-n} e^{\alpha\epsilon} - e^{\tilde{\lambda}^*}) \\ &\quad + e^{\tilde{\lambda}^*} \left(\left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (1 + (e^{\alpha\epsilon} - 1)\gamma) - 1 \right) \end{aligned}$$

Define the following:

$$\begin{aligned} \sigma &= \left(\left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (1 + (e^{\alpha\epsilon} - 1)\gamma) - 1 \right) \\ \gamma' &= \left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n (1 - \gamma) \end{aligned}$$

Important Condition: For γ' to be a valid source of contraction, γ' should be less than 1. Therefore the m_i have to be chosen such that, $\left(\frac{e^{\alpha\delta_2} e^{\alpha\epsilon}}{e^{\alpha\delta_1}} \right)^n$ is small enough to ensure that γ' is a contraction. Note that $\gamma \leq \frac{1}{|\mathcal{S}|}$. Provided that γ' is less than 1, we have the following:

$$\begin{aligned} (u_k e^{\alpha\epsilon} - e^{\tilde{\lambda}^*}) &\leq \gamma' (u_{k-n} e^{\alpha\epsilon} - e^{\tilde{\lambda}^*}) + \sigma e^{\tilde{\lambda}^*} \\ &\leq \gamma' \left(\gamma' (u_{k-n} e^{\alpha\epsilon} - e^{\tilde{\lambda}^*}) + \sigma e^{\tilde{\lambda}^*} \right) + \sigma e^{\tilde{\lambda}^*} \end{aligned}$$

Note that from Lemma 7.1, we know that $u_k e^{\alpha\epsilon} \geq e^{\tilde{\lambda}^*}, \forall k$. Hence it is possible to iterate the above equation to obtain,

$$(u_k e^{\alpha\epsilon} - e^{\tilde{\lambda}^*}) \leq (\gamma')^{\frac{k}{n}} (u_0 e^{\alpha\epsilon} - e^{\tilde{\lambda}^*}) + \frac{\sigma e^{\tilde{\lambda}^*}}{1 - \gamma'}$$

Since from Lemma 7.1, we know that $e^{\tilde{\lambda}_{f_{k+1}}} \leq u_k e^{\alpha\epsilon}$, we obtain the following,

$$(e^{\tilde{\lambda}_{f_{k+1}}} - e^{\tilde{\lambda}^*}) \leq (\gamma')^{\frac{k}{n}} (u_0 e^{\alpha\epsilon} - e^{\tilde{\lambda}^*}) + \frac{\sigma e^{\tilde{\lambda}^*}}{1 - \gamma'}$$

□

Lemma F.2. Let $\max_k m_k < C_1$, where m_k corresponds to the number of fixed point iterations performed during partial policy evaluation during the k th execution of the algorithm 8. Then, there

exists τ' such that

$$e^{Q_k(s,a)} \geq \tau' > 0.$$

Proof. Recall from the algorithm,

$$e^{Q'_{f_{k+1}}(s,a)} = \frac{e^{Q_{f_{k+1}}(s,a)}}{\sum_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} e^{Q_{f_{k+1}}(s,a)}}$$

$$\implies \sum_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} e^{Q'_{f_{k+1}}(s,a)} = 1$$

Then,

$$\begin{aligned} e^{Q_{f_{k+1}}(s,a)} &= T_{f_{k+1}}^{m_k} e^{Q_k(s,a)} \\ &\geq \frac{T_{f_{k+1}}^{m_k} e^{Q'_{f_k}(s,a)}}{e^{\alpha \delta_2}} \end{aligned}$$

Let $\underline{d} = \min_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} d(s,a)$, then

$$\begin{aligned} \left(T_{f_{k+1}}^{m_k} e^{Q'_{f_k}(s,a)} \right) &= e^{\alpha d(s,a)} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} Q(s'|s,a) f_{k+1}(a'|s') e^{Q'_{f_k}(s',a')} \\ &\geq e^{\alpha \underline{d}} \left(Q_{f_{k+1}} e^{Q'_{f_k}} \right)(s,a) \end{aligned}$$

Repeating the above we obtain,

$$\left(T_{f_{k+1}}^{m_k} e^{Q'_{f_k}} \right)(s,a) \geq \frac{e^{m_k \alpha \underline{d}} \left(Q_{f_{k+1}} e^{Q'_{f_k}} \right)(s,a)}{e^{\alpha \delta_2}}$$

Substituting for Q'_{f_k} ,

$$\left(T_{f_{k+1}}^{m_k} e^{Q'_{f_k}} \right)(s,a) \geq \frac{e^{m_k \alpha \underline{d}} \left(Q_{f_{k+1}} e^{Q_{f_k}} \right)(s,a)}{e^{\alpha \delta_2} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_k}(s',a')}}}$$

$$e^{Q_{f_{k+1}}(s,a)} \geq \frac{e^{m_k \alpha \underline{d}} \left(Q_{f_{k+1}} e^{Q_{f_k}} \right) (s, a)}{e^{\alpha \delta_2} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_k}(s',a')}} \quad (F.1)$$

Iterating the above equation we obtain,

$$e^{Q_{f_{k+1}}(s,a)} \geq \frac{e^{\alpha \underline{d} \sum_{p=0}^k m_{k-p}} \left(Q_{f_{k+1}}^{m_k} Q_{f_k}^{m_{k-1}} \dots Q_{f_1}^{m_0} e^{Q_0} \right) (s, a)}{(e^{\alpha \delta_2})^{k+1} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_k}(s',a')} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_{k-1}}(s',a')} \dots \sum_{j \in \mathcal{S}} e^{Q_{f_1}(s',a')}} \quad (F.2)$$

From Algorithm 8, for a sufficiently large k , we have $\sum_{l=0}^{k-1} m_l > R$.

Defining $H_k = Q_{f_{k+1}}^{m_k} Q_{f_k}^{m_{k-1}} \dots Q_{f_1}^{m_0}$, Lemma 6.3 yields $\lambda = \min_{\substack{s, s' \in \mathcal{S} \\ a, a' \in \mathcal{A}}} H_k(s', a' | s, a) > 0$, we continue the above sequence of inequalities:

$$e^{Q_{f_{k+1}}(s,a)} \geq \frac{e^{\alpha \underline{d} \sum_{p=0}^k m_{k-p}} \lambda \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_0(s',a')}}{(e^{\alpha \delta_2})^{k+1} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_k}(s',a')} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_{k-1}}(s',a')} \dots \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_1}(s',a')}} \quad (F.3)$$

Since we know that $\sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_0(s',a')} = 1$, we obtain the following relation,

$$e^{Q_{f_{k+1}}(s,a)} \geq \frac{e^{\alpha \underline{d} \sum_{p=0}^k m_{k-p}} \lambda}{(e^{\alpha \delta_2})^{k+1} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_k}(s',a')} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_{k-1}}(s',a')} \dots \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_1}(s',a')}} \quad (F.4)$$

Similarly,

$$e^{Q_{f_{k+1}}(s,a)} \leq \frac{T_{f_{k+1}}^{m_k} e^{Q'_{f_k}}(s, a)}{e^{\alpha \delta_1}}$$

Let $\bar{d} = \max_{f \in \Pi} d_f$, then

$$\begin{aligned} \left(T_{f_{k+1}} e^{Q'_{f_k}}(s, a) \right) &= e^{\alpha d_{f_{k+1}}(s,a)} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} Q(s', a' | s, a) e^{Q'_{f_k}(s', a')} \\ &\leq e^{\alpha \bar{d}} \left(Q_{f_{k+1}} e^{Q'_{f_k}} \right) (s, a) \end{aligned}$$

Repeating the above we obtain,

$$\left(T_{f_{k+1}}^{m_k} e^{Q'_{f_k}} \right) (s, a) \leq \frac{e^{m_k \alpha \bar{d}} \left(Q_{f_{k+1}} e^{Q'_{f_k}} \right) (s, a)}{e^{\alpha \delta_1}}$$

Substituting for Q'_{f_k} ,

$$\left(T_{f_{k+1}}^{m_k} e^{Q'_{f_k}} \right) (s, a) \leq \frac{e^{m_k \alpha \bar{d}} \left(Q_{f_{k+1}} e^{Q_{f_k}} \right) (s, a)}{e^{\alpha \delta_1} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_k}(s', a')}}}$$

$$e^{Q_{f_{k+1}}(s, a)} \leq \frac{e^{m_k \alpha \bar{d}} \left(Q_{f_{k+1}} e^{Q_{f_k}} \right) (s, a)}{e^{\alpha \delta_1} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_k}(s', a')}}}$$

Iterating the above equation we obtain,

$$\begin{aligned} e^{Q_{f_{k+1}}(s, a)} &\leq \frac{e^{\alpha \bar{d} \sum_{p=0}^k m_{k-p}} \left(Q_{f_{k+1}}^{m_k} Q_{f_k}^{m_{k-1}} \dots Q_{f_1}^{m_0} e^{Q_0} \right) (s, a)}{(e^{\alpha \delta_1})^{k+1} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_k}(s', a')} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_{k-1}}(s', a')} \dots \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_1}(s', a')}}} \\ \sum_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} e^{Q_{f_{k+1}}(s, a)} &\leq \frac{|\mathcal{S}| e^{\alpha \bar{d} \sum_{p=0}^k m_{k-p}}}{(e^{\alpha \delta_1})^{k+1} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_k}(s', a')} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_{k-1}}(s', a')} \dots \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_1}(s', a')}}} \end{aligned}$$

We thus obtain,

$$\frac{1}{\sum_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} e^{Q_{f_{k+1}}(s, a)}} \geq \frac{(e^{\alpha \delta_1})^{k+1} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_k}(s', a')} \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_{k-1}}(s', a')} \dots \sum_{\substack{s' \in \mathcal{S} \\ a' \in \mathcal{A}}} e^{Q_{f_1}(s', a')}}{|\mathcal{S}| e^{\alpha \bar{d} \sum_{p=0}^k m_{k-p}}} \quad (\text{F.3})$$

Thus from Appendix F.1 and Appendix F.1, we obtain the following,

$$\frac{e^{Q_{f_{k+1}}(s, a)}}{\sum_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} e^{Q_{f_{k+1}}(s, a)}} \geq \left(\frac{e^{\alpha \delta_1}}{e^{\alpha \delta_2}} \right)^{k+1} \frac{e^{\alpha(\underline{d} - \bar{d}) \sum_{p=0}^k m_{k-p} \lambda}}{|\mathcal{S}|}$$

Let $\tau = \frac{e^{\alpha(\underline{d}-\bar{d}) \sum_{p=0}^k m_k-p \lambda}}{|S|}$, then we have,

$$e^{Q'_{f_{k+1}}(s,a)} \geq \tau$$

From the Algorithm 8, we then obtain the following,

$$e^{Q_{k+1}(s,a)} \geq \frac{e^{Q'_{f_{k+1}}(s,a)}}{e^{\alpha\delta_2}} \geq \frac{\tau}{e^{\alpha\delta_2}}$$

Defining $\frac{\tau}{e^{\alpha\delta_2}} = \tau'$, we obtain the statement of the lemma,

$$e^{Q_{k+1}(s,a)} \geq \tau' > 0$$

□

F.2 Policy Improvement Errors

Lemma F.3. Let $d \in \mathbb{N}$, fix $\beta > 0$, and let $q = (q_1, \dots, q_d) \in \mathbb{R}^d$. For a probability vector $p = (p_1, \dots, p_d) \in \Delta_d := \{p \in [0, 1]^d : \sum_{i=1}^d p_i = 1\}$, define

$$q_{\min} := \min_{1 \leq i \leq d} q_i, \quad i_{\min} := \arg \min_{1 \leq i \leq d} q_i, \quad p_{\min} := p_{i_{\min}},$$

and assume $p_{\min} > 0$. Then

$$\frac{\sum_{i=1}^d p_i q_i e^{-\beta q_i}}{\sum_{i=1}^d p_i e^{-\beta q_i}} - q_{\min} \leq \frac{1}{\beta} \log\left(\frac{1}{p_{\min}}\right).$$

Proof. Fix $\beta > 0$ and define the log-moment-generating function

$$f_{\beta}(q) := \frac{1}{\beta} \log\left(\sum_{i=1}^d p_i e^{-\beta q_i}\right), \quad q \in \mathbb{R}^d.$$

Because f_{β} is the logarithm of a weighted moment-generating function, it is convex.

Step 1: An upper bound via the sub-gradient inequality. Convexity implies that for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$

$$f_\beta(\mathbf{y}) \geq f_\beta(\mathbf{x}) + \langle \nabla f_\beta(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle.$$

Choosing $\mathbf{y} = \mathbf{0}$ and $\mathbf{x} = \mathbf{q}$ yields $-f_\beta(\mathbf{q}) \geq \langle \nabla f_\beta(\mathbf{q}), -\mathbf{q} \rangle$. Since $\frac{\partial f_\beta}{\partial q_j}(\mathbf{q}) = -\frac{p_j e^{-\beta q_j}}{\sum_k p_k e^{-\beta q_k}}$, we obtain

$$-f_\beta(\mathbf{q}) \geq \sum_{j=1}^d \frac{p_j e^{-\beta q_j}}{\sum_k p_k e^{-\beta q_k}} q_j = \frac{\sum_j p_j q_j e^{-\beta q_j}}{\sum_k p_k e^{-\beta q_k}}.$$

Equivalently,

$$\frac{\sum_j p_j q_j e^{-\beta q_j}}{\sum_k p_k e^{-\beta q_k}} - q_{\min} \leq -f_\beta(\mathbf{q}) - q_{\min}. \quad (\text{F.4})$$

Step 2: A lower bound on $f_\beta(\mathbf{q})$. Because $p_{\min} = p_{i_{\min}} > 0$ and $q_{i_{\min}} = q_{\min}$, the sum inside f_β satisfies

$$\sum_{i=1}^d p_i e^{-\beta q_i} \geq p_{\min} e^{-\beta q_{\min}},$$

which implies

$$f_\beta(\mathbf{q}) = \frac{1}{\beta} \log\left(\sum_i p_i e^{-\beta q_i}\right) \geq \frac{1}{\beta} \log(p_{\min}) - q_{\min}. \quad (\text{F.5})$$

Step 3: Combine the bounds. Substituting (F.5) into (F.4) gives

$$\frac{\sum_j p_j q_j e^{-\beta q_j}}{\sum_k p_k e^{-\beta q_k}} - q_{\min} \leq \frac{1}{\beta} \log\left(\frac{1}{p_{\min}}\right),$$

which is precisely the claimed inequality. □

F.2.1 Mirror-Descent Update

Lemma F.4 (Mirror-descent policy error). Fix $\beta > 0$ and let $\{Q_k(s, a)\}_{s,a}$ be the risk-sensitive Q -values at iteration k . Define the mirror-descent policy update¹

$$f_{k+1}(a | s) = \frac{f_k(a | s) \exp(-\beta Q_k(s, a))}{\sum_{a' \in \mathcal{A}} f_k(a' | s) \exp(-\beta Q_k(s, a'))}. \quad (\text{F.6})$$

Let $(T_{f_{k+1}} e^{Q_k})(s, a)$ and $(Te^{Q_k})(s, a)$ denote, respectively, the controlled and optimal risk-sensitive Bellman operators²

For every $(s, a) \in \mathcal{S} \times \mathcal{A}$,

$$0 \leq \frac{(T_{f_{k+1}} e^{Q_k})(s, a)}{(Te^{Q_k})(s, a)} - 1 \leq \frac{1}{\beta} \max_{j \in \mathcal{S}} \frac{\log(1/f_k(a^*(j) | j))}{\min_{a'} e^{Q_k(j, a')}}, \quad (\text{F.7})$$

where $a^(j) \in \arg \min_{a'} q_k(j, a')$.*

If, in addition, the stage-costs are bounded ($0 \leq c(s, a) \leq C_{\max}$) and the current policy is $e^{\alpha\epsilon}$ -randomised ($f_k(a | s) \geq e^{\alpha\epsilon} > 0$ for all s, a), then there exists a constant $\kappa = \kappa(e^{\alpha\epsilon}, C_{\max}) > 0$ such that

$$\frac{(T_{f_{k+1}} e^{Q_k})(s, a)}{(Te^{Q_k})(s, a)} \leq 1 + \underbrace{\frac{1}{\kappa \beta} \max_{j \in \mathcal{S}} \log\left(\frac{1}{e^{\alpha\epsilon}}\right)}_{=: e^{\alpha\epsilon_k}}, \quad \forall (s, a). \quad (\text{F.8})$$

Hence the mirror-descent step incurs a policy-improvement error bounded by $e^{\alpha\epsilon_k}$ as defined above.

Proof. Recall that for fixed (s, a)

$$\left((T_{f_{k+1}} e^{Q_k}) - (Te^{Q_k}) \right)(s, a) = e^{\alpha d(s, a)} \sum_j \mathbb{Q}(j | s, a) \underbrace{\left[\sum_{a'} f_{k+1}(a' | j) e^{Q_k(j, a')} - \min_{a'} e^{Q_k(j, a')} \right]}_{=: \Delta_j}.$$

¹The update can also be interpreted as one step of entropic mirror descent or soft-min regularisation with stepsize β^{-1} .

²Their definitions are the ones used in Sections 7.2–7.3; in particular $(T_f e^Q)(s, a) = e^{\alpha d(s, a)} \sum_j \mathbb{Q}(j | s, a) \sum_{a'} f(a' | j) e^{Q(j, a')}$ and $(Te^Q)(s, a) = e^{\alpha d(s, a)} \sum_j \mathbb{Q}(j | s, a) \min_{a'} e^{Q(j, a')}$.

Focus on the inner gap Δ_j . With the shorthand $z_{a'} := e^{Q_k(j, a')}$ and $p_{a'} := f_{k+1}(a' | j)$, update (F.6) gives $p_{a'} \propto f_k(a' | j) e^{-\beta z_{a'}}$. Applying Lemma 7.3 (with $q_i = z_{a'}$ and $p_i = p_{a'}$) yields

$$\Delta_j = \sum_{a'} p_{a'} z_{a'} - \min_{a'} z_{a'} \leq \frac{1}{\beta} \log(1/f_k(a^*(j) | j)),$$

where $a^*(j)$ minimises $z_{a'}$. Therefore,

$$(T_{f_{k+1}} e^{Q_k})(s, a) - (Te^{Q_k})(s, a) \leq e^{\alpha d(s, a)} \sum_j Q(j | s, a) \frac{1}{\beta} \log(1/f_k(a^*(j) | j)).$$

Dividing by $(Te^{Q_k})(s, a) = e^{\alpha d(s, a)} \sum_j Q(j | s, a) \min_{a'} e^{Q_k(j, a')}$ gives the first bound (F.7).

Bounding the denominator. If $d(s, a) \leq C_{\max}$ and $f_k(a | s) \geq e^{\alpha \epsilon}$ for every state–action pair, then the multiplicative value iterates satisfy $e^{q_k(j, a')} \geq \kappa := h(e^{\alpha \epsilon}, C_{\max}) > 0$. Substituting $\min_{a'} e^{Q_k(j, a')} \geq \kappa$ into (F.7) yields (F.8). $\square$

F.2.2 Softmax Update

Lemma F.5 (Soft-max policy error). Fix $\beta > 0$ and let $\{Q_k(s, a)\}_{s, a}$ be the current risk–sensitive Q -values. The soft-max (a.k.a. Boltzmann) policy update is

$$f_{k+1}(a | s) = \frac{\exp(-\beta Q_k(s, a))}{\sum_{a' \in \mathcal{A}} \exp(-\beta Q_k(s, a'))}, \quad \forall(s, a). \quad (\text{F.9})$$

With the Bellman operators $T_{f_{k+1}}$ and T defined as in Lemma 7.4,

$$0 \leq \frac{(T_{f_{k+1}} e^{Q_k})(s, a)}{(Te^{Q_k})(s, a)} - 1 \leq \frac{1}{\beta \min_{a'} e^{Q_k(j, a')}} \log |\mathcal{A}|, \quad \forall(s, a). \quad (\text{F.10})$$

If the stage-costs are bounded ($0 \leq d(s, a) \leq C_{\max}$) so that $e^{Q_k(j, a')} \geq \kappa > 0$ for some $\kappa = \kappa(C_{\max})$, then

$$\frac{(T_{f_{k+1}} e^{Q_k})(s, a)}{(Te^{Q_k})(s, a)} \leq 1 + \underbrace{\frac{1}{\kappa \beta} \log |\mathcal{A}|}_{=: e^{\alpha \epsilon_k}}, \quad \forall(s, a). \quad (\text{F.11})$$

Thus the soft-max step incurs a policy-improvement error bounded by $e^{\alpha \epsilon_k}$.

Proof. For fixed (s, a) write

$$(T_{f_{k+1}} e^{Q_k})(s, a) - (Te^{Q_k})(s, a) = e^{c(s, a)} \sum_j Q(j | s, a) \left[\sum_{a'} f_{k+1}(a' | j) e^{Q_k(j, a')} - \min_{a'} e^{Q_k(j, a')} \right].$$

Denote $z_{a'} := e^{Q_k(j, a')}$ and let $p_{a'} := f_{k+1}(a' | j)$ defined by (F.9). Clearly $\sum_{a'} p_{a'} = 1$ and $p_{a'} \propto e^{-\beta z_{a'}}$. Applying Lemma 7.3 gives

$$\sum_{a'} p_{a'} z_{a'} - \min_{a'} z_{a'} \leq \frac{1}{\beta} \log\left(\frac{1}{p_{\min}}\right), \quad p_{\min} := \max_{a'} p_{a'}.$$

Since for every j , $p_{\max} = \frac{e^{-\beta z_{\min}}}{\sum_{a''} e^{-\beta z_{a''}}} \geq \frac{e^{-\beta z_{\min}}}{|\mathcal{A}| e^{-\beta z_{\min}}} = \frac{1}{|\mathcal{A}|}$, we have $\log(1/p_{\min}) \leq \log |\mathcal{A}|$. Substituting into the previous display and dividing by $(Te^{Q_k})(s, a)$ produces (F.10).

If the costs are bounded so that $\min_{a'} e^{Q_k(j, a')} \geq \kappa > 0$, then the right-hand side of (F.10) is further bounded by $\kappa^{-1} \beta^{-1} \log |\mathcal{A}|$, which is exactly (F.11). $\square$